

# **Stat 637: Generalized Linear Models**

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# Preface

This resource is provided to you and only you. It is compiled of many different resources: professors and colleagues, books, articles, etc. Where possible, I have cited these sources. Some resources are used throughout – both in content and ideas – and should be given credit upfront:

Books:

- “D&B” [An Introduction to Generalized Linear Models, 4th ed.](#) by Dobson and Barnett [not available through BYU library online but I have a few copies you can borrow; see Dobson and Barnett (2018)].
- “M&N” [Generalized Linear Models](#) by McCullagh and Nelder (see McCullagh and Nelder 1989).
- “A” [Categorical Data](#) by Agresti (see Agresti 2013)

People:

- [Dr. Peter Craigmile](#)
- [Dr. Natalie Blades](#)
- [Dr. David Dunson](#)
- [Dr. Andrew Gelman](#)
- [Dr. Annette Dobson](#)

I hope you will find this resource valuable and are able to use it for your own use for years to come. Please do not share or reproduce any of these resources without contacting me first. Thank you and may the GLM’s be ever in your favor.

# 1 Introduction

## 1.1 Types of Response Variables

Regression is a tool to relate explanatory variables/predictors/covariates to a response variable/dependent variable/outcome. Previous courses have covered extensively how to account for the many types of variables that exist (e.g., continuous, binary, categorical) as covariates, but generally restrict response variables to be continuous (most often normally distributed). However, the regression framework can be extended to account for *many*(!!) types of different response variables.

Table 1.1: Using this table, identify different types of response variables that may arise and a reasonable corresponding likelihood distribution.

| Type of Response Variable | Examples           | Possible Likelihood |
|---------------------------|--------------------|---------------------|
| 1. continuous & symmetric | weight f.t. babies | normal              |
| 2. binary                 | cannibalist/not    | binomial/Bernoulli  |
| 3.                        |                    |                     |
| 4.                        |                    |                     |
| 5.                        |                    |                     |
| 6.                        |                    |                     |
| 7.                        |                    |                     |
| 8.                        |                    |                     |
| 9. ...                    |                    |                     |

## 1.2 Regression: A Unifying Framework

### **i** Note

First, let's do a group activity to see if we can identify a regression framework given the models with which you are already familiar.

Consider a set of observations

$$\{Y_i, \mathbf{x}_i\}, i = 1, \dots, n,$$

where  $Y_i$  represents the response variable, and  $\mathbf{x}_i = (x_{i1}, \dots, x_{ik})'$  is a  $k \times 1$  vector of covariates. Omitting a few details for now, the generalized linear model (GLM) can be written using the following framework:

### **The Generalized Linear Model**

#### **random component**

$$Y_i \sim f(\mu_i),$$

where  $f(\cdot)$  is a probability distribution (in the exponential family), and  $\mu_i = E(Y_i)$ .

#### **link component**

$$g(\mu_i) = \eta_i,$$

where  $g(\cdot)$  is called the *link* function. We will discuss properties and requirements of the link function later, and for now consider just those link functions with which we are familiar.

The link component connects the random component to the systematic component.

#### **systematic component**

$$\eta_i = \mathbf{x}_i' \beta,$$

where  $\beta$  is a  $k \times 1$  vector of unknown coefficients.

*Some questions to consider:*

1. What is the benefit of having a unifying framework for a GLM when all the specific models (such as the normal linear model, logistic regression, and Poisson regression) are different?

2. Why bother with a GLM at all? Why not just transform the variables as necessary?
3. How are the data sets and corresponding models similar and different?

## 1.3 Dataset 1: Normal Linear Regression

Many analyses in statistics assume that random variables are continuous and normally distributed. When this is the case, the normal linear regression model can be used:

$$Y_i = \mathbf{x}_i' \beta + \epsilon_i, \quad (1.1)$$

$$\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2), \quad (1.2)$$

where  $\beta$  is a  $k \times 1$  vector of regression coefficients, and  $\sigma^2$  is the residual variance.

This model assumes:

- 1.
- 2.
- 3.
- 4.

*Example:*

Researchers may be interested in modeling how weight changes in children by their age and gender. We have means and standard deviations for the following data collected by the Environment Agency in the UK to monitor human health risks due to contaminated soil:

- $Y_i$  = Weight (in kg), for  $i = 1, \dots, n$
- $X_{i1}$  = Age (in years, ages 1–18)
- $X_{i2}$  = Gender (1 if Male)
- $X_{i3}$  = Age  $\times$  Gender
- $n = 15,313$  (8,104 Females; 7,209 Males)

Consider two models...

**Model 1:**

$$Y_i = X_{i1}\beta_1 + X_{i2}\beta_2 + X_{i3}\beta_3 + \epsilon_i$$
$$\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

Call:

```
lm(formula = Weight ~ Age * Gender)
```

Residuals:

| Min     | 1Q     | Median | 3Q    | Max    |
|---------|--------|--------|-------|--------|
| -54.948 | -9.729 | -0.063 | 9.168 | 58.854 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -7.26780 | 0.73112    | -9.941  | < 2e-16 ***  |
| Age         | 4.49773  | 0.04551    | 98.822  | < 2e-16 ***  |
| GenderM     | -6.03810 | 1.02018    | -5.919  | 3.31e-09 *** |
| Age:GenderM | 1.05191  | 0.06386    | 16.473  | < 2e-16 ***  |

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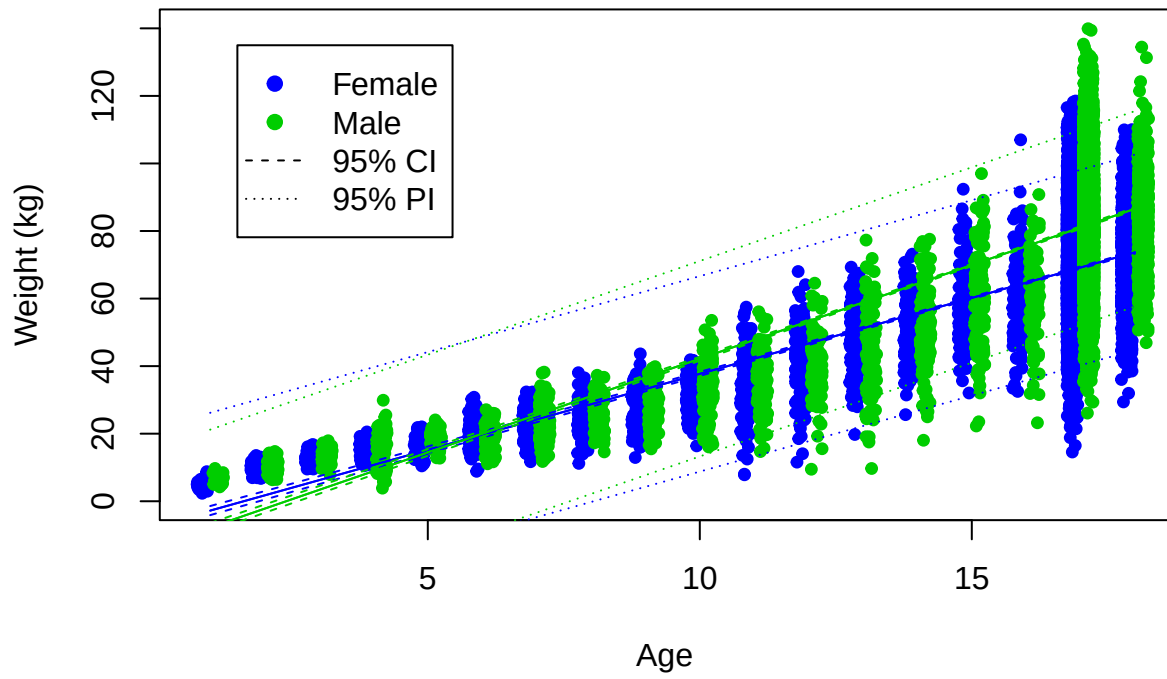
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.76 on 15309 degrees of freedom

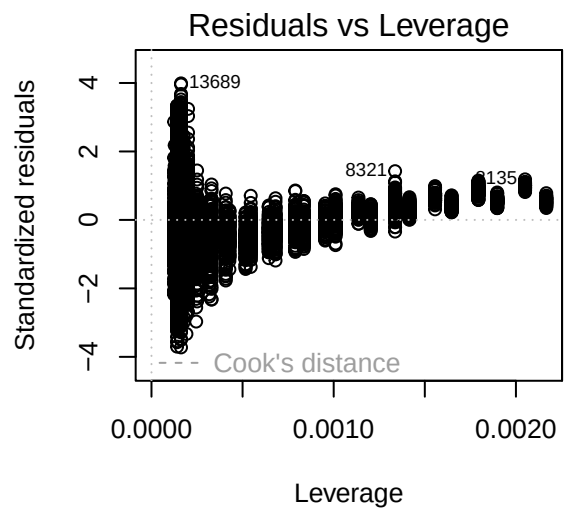
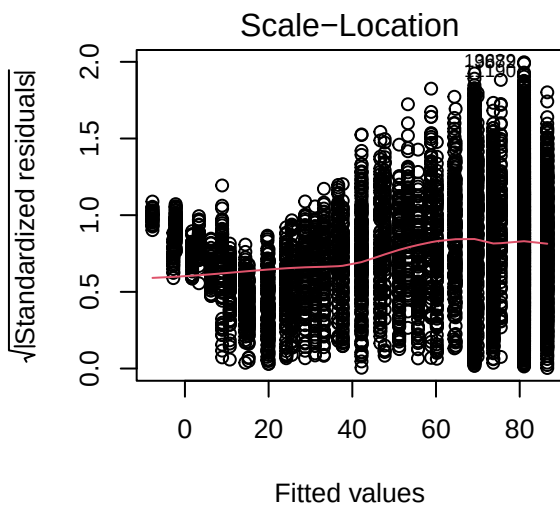
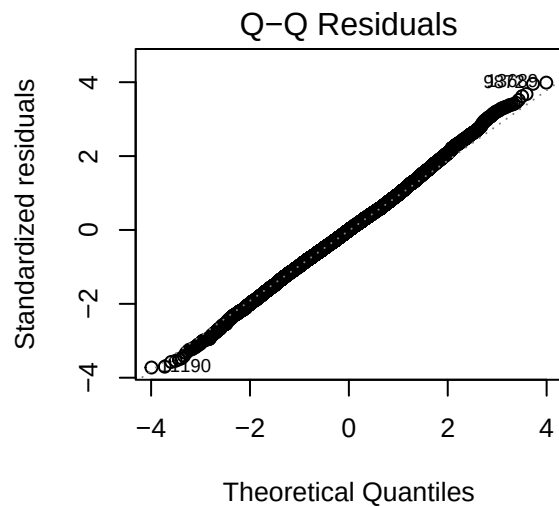
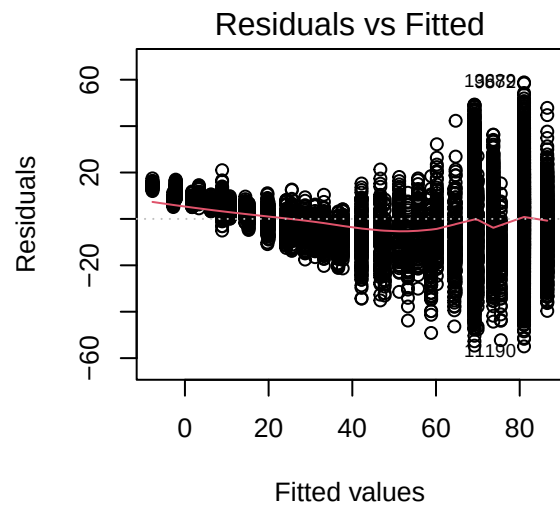
Multiple R-squared: 0.6343, Adjusted R-squared: 0.6342

F-statistic: 8850 on 3 and 15309 DF, p-value: < 2.2e-16

## Weight vs. Age of Children in the UK







What do you notice about this model?

What fits well?

What are some limitations?

## Model 2:

$$\log(Y_i) = X_{i1}\beta_1 + X_{i2}\beta_2 + X_{i3}\beta_3 + \epsilon_i$$
$$\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

Call:

```
lm(formula = log(Weight) ~ Age * Gender)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -1.57685 | -0.12830 | 0.02215 | 0.15404 | 0.73955 |

Coefficients:

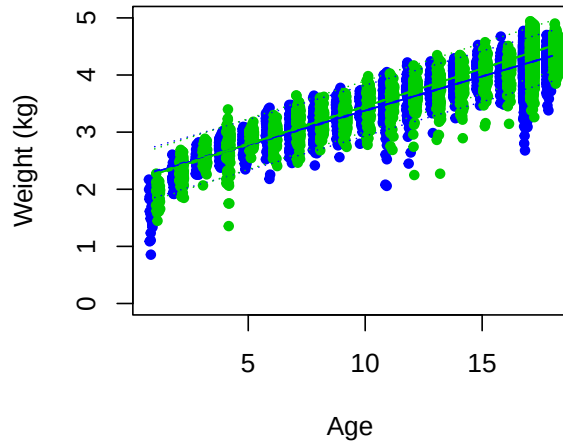
|             | Estimate   | Std. Error | t value | Pr(> t )    |
|-------------|------------|------------|---------|-------------|
| (Intercept) | 2.1806052  | 0.0112315  | 194.150 | < 2e-16 *** |
| Age         | 0.1196552  | 0.0006992  | 171.137 | < 2e-16 *** |
| GenderM     | -0.0515307 | 0.0156720  | -3.288  | 0.00101 **  |
| Age:GenderM | 0.0126447  | 0.0009810  | 12.890  | < 2e-16 *** |

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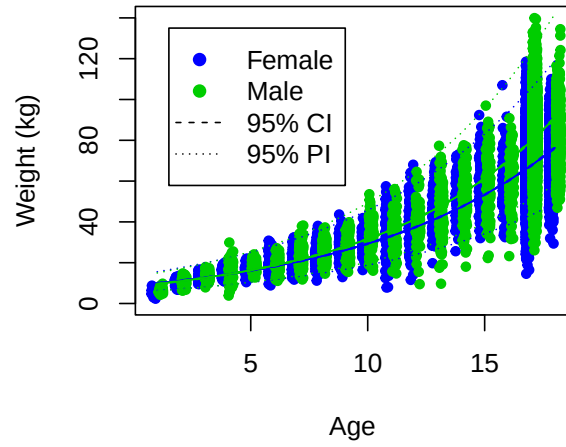
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

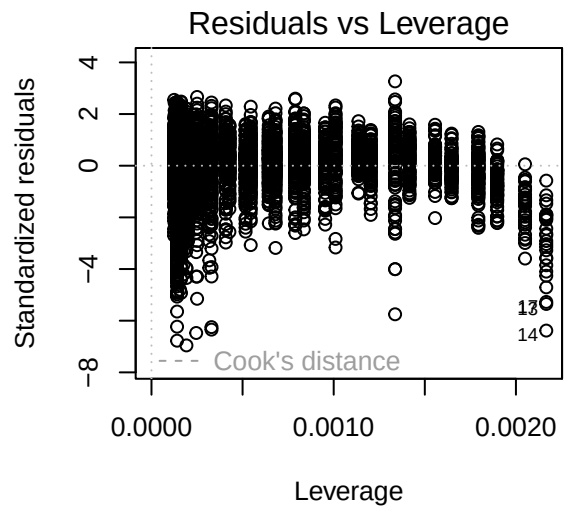
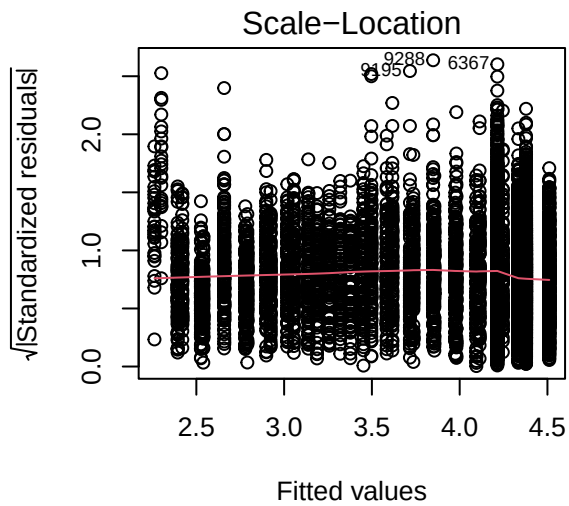
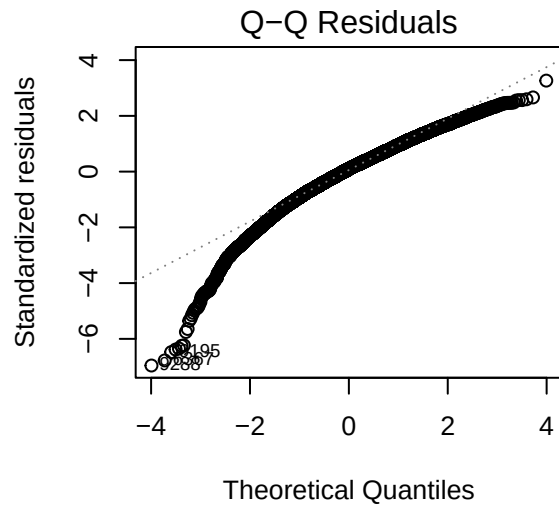
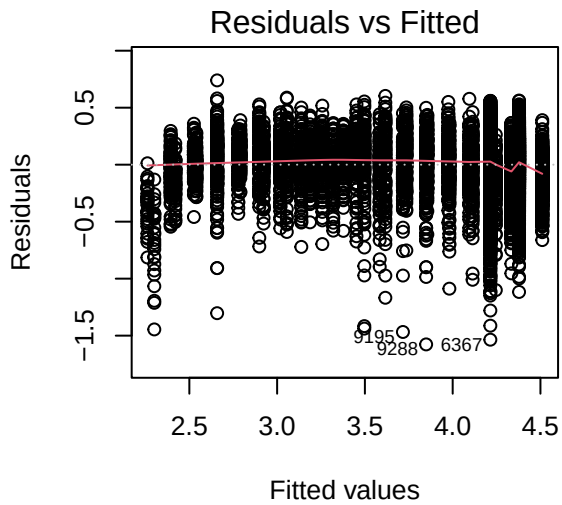
Residual standard error: 0.2267 on 15309 degrees of freedom  
Multiple R-squared: 0.8145, Adjusted R-squared: 0.8144  
F-statistic: 2.24e+04 on 3 and 15309 DF, p-value: < 2.2e-16

**log(Weight) vs. Age of UK Children**



**Weight vs. Age of UK Children**





What is better about this model?  
 What is worse?  
 How would you interpret this model?

## Model Purpose

We generally use a linear model for one (or both) of two purposes:

1. inference
2. prediction

When *inference* is the main goal, models are used to estimate relationships, often looking for correlational and causal relationships.

Ex:

When *prediction* is the main goal, models are used to estimate response values at various levels of the predictors.

Ex:

For what situations would you be comfortable using the models from our example?

## 1.4 Dataset 2: Logistic Regression

Logistic regression is often used when the response variable is binary (0 or 1). In this case, the logistic regression model can be written as:

$$Y_i \stackrel{ind}{\sim} \text{Bernoulli}(\pi_i), \quad (1.3)$$

$$\log \left( \frac{\pi_i}{1 - \pi_i} \right) = \mathbf{x}_i' \beta, \quad (1.4)$$

where  $E(Y_i) = \pi_i$  and represents the probability that  $Y_i = 1$ .

This model assumes:

- 1.
- 2.
- 3.

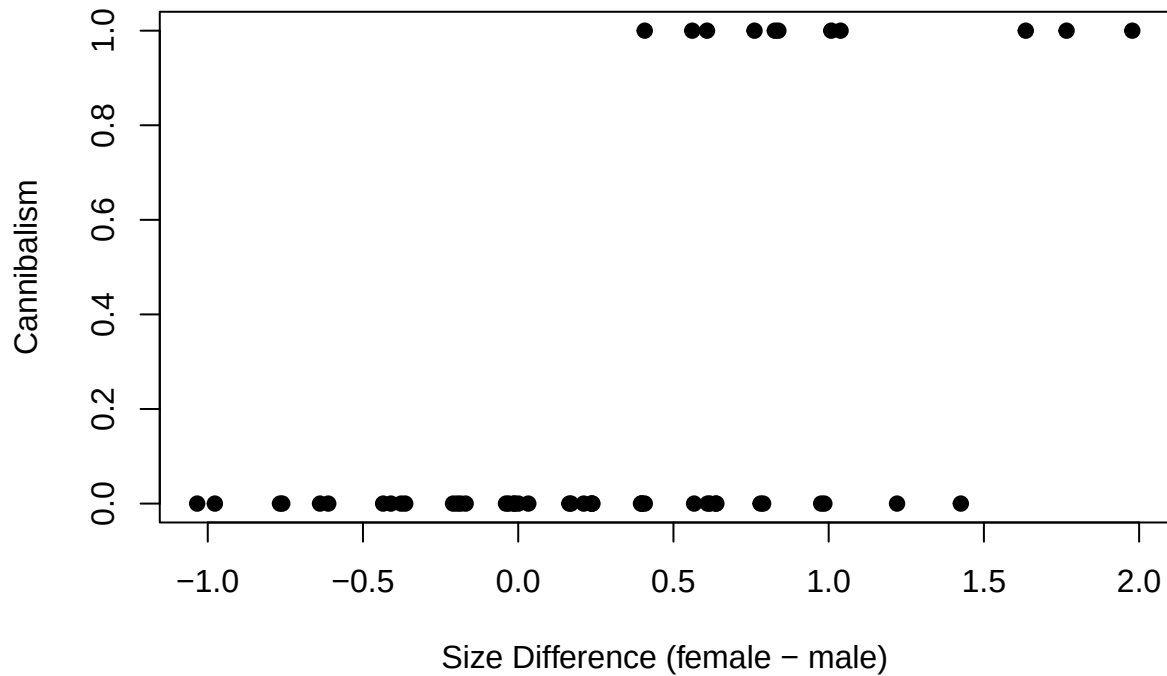
*Example: Cannibal Spiders*

What variables might be related to a female wolf spider's decision to kill and consume her partner during courtship or mating?

$$Y_i = \text{cannibalism} = \begin{cases} 0, & \text{no cannibalism} \\ 1, & \text{cannibalism} \end{cases}$$

$X_i$  = difference in body width (female - male)

$n$  = 52 pairs of courting wolf spiders



Model:

$$Y_i \overset{ind}{\sim} \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + x_i\beta_1$$

Summary:

Call:

```
glm(formula = cannib ~ sizediff, family = binomial)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | -3.0890  | 0.8288     | -3.727  | 0.000194 | *** |
| sizediff    | 3.0693   | 1.0041     | 3.057   | 0.002237 | **  |

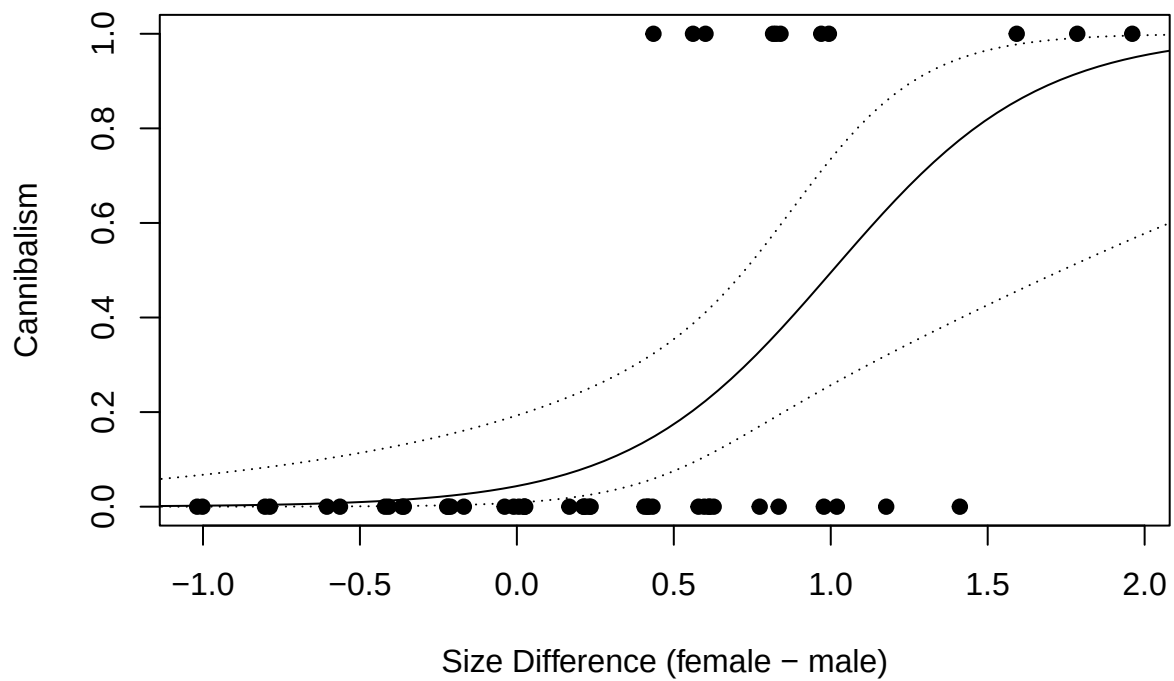
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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

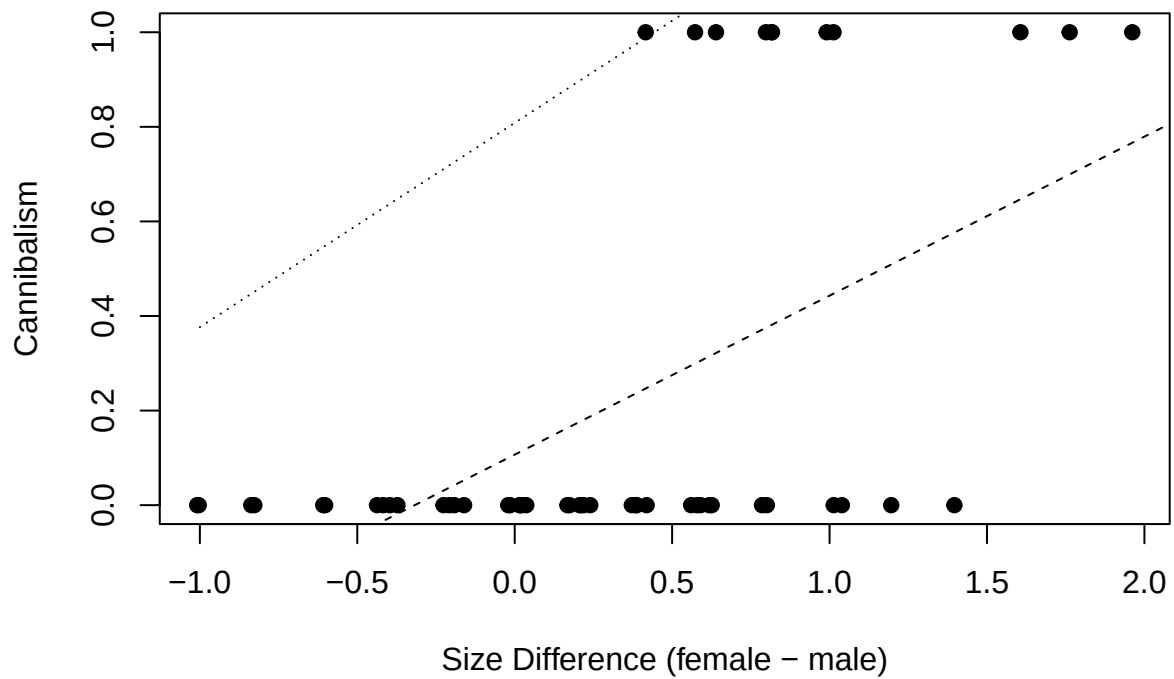
Null deviance: 53.663 on 51 degrees of freedom  
Residual deviance: 34.721 on 50 degrees of freedom  
AIC: 38.721

Number of Fisher Scoring iterations: 6





We can compare the predictions from the above model to those from a *simple linear regression model*:



Why would we prefer logistic regression to simple linear regression in this case?

How do the interpretations of the coefficients change for the logistic model?

How is this model the same as the normal linear regression model?

## 1.5 Dataset 3: Poisson Regression

For count data, Poisson regression is often used. The model follows this form:

$$Y_i \stackrel{ind}{\sim} \text{Poisson}(\mu_i), \quad (1.5)$$

$$\log(\mu_i) = \mathbf{x}_i' \beta, \quad (1.6)$$

where  $E(Y_i) = \mu_i$ .

This model assumes:

- 1.
- 2.
- 3.

How is this different from Model 2 in Section 1.3?

How are the interpretations of the coefficients different?

*Example: 2011 NBA Player Statistics*

We'll consider the relationship between games started by players and the number of field goals during the 2010-2011 season.

$Y_i$  = Number of Field Goals

$X_i$  = Number of Games Started

$n$  = 625 players

**Model:**

$$Y_i \stackrel{ind}{\sim} \text{Pois}(\mu_i)$$

$$\log(\mu_i) = \beta_0 + x_i \beta_1$$

Summary:

Call:

```
glm(formula = FG ~ GS, family = poisson, data = bball11)
```

Coefficients:

|             | Estimate  | Std. Error | z value | Pr(> z )   |
|-------------|-----------|------------|---------|------------|
| (Intercept) | 4.323e+00 | 5.402e-03  | 800.3   | <2e-16 *** |
| GS          | 2.403e-02 | 9.862e-05  | 243.7   | <2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

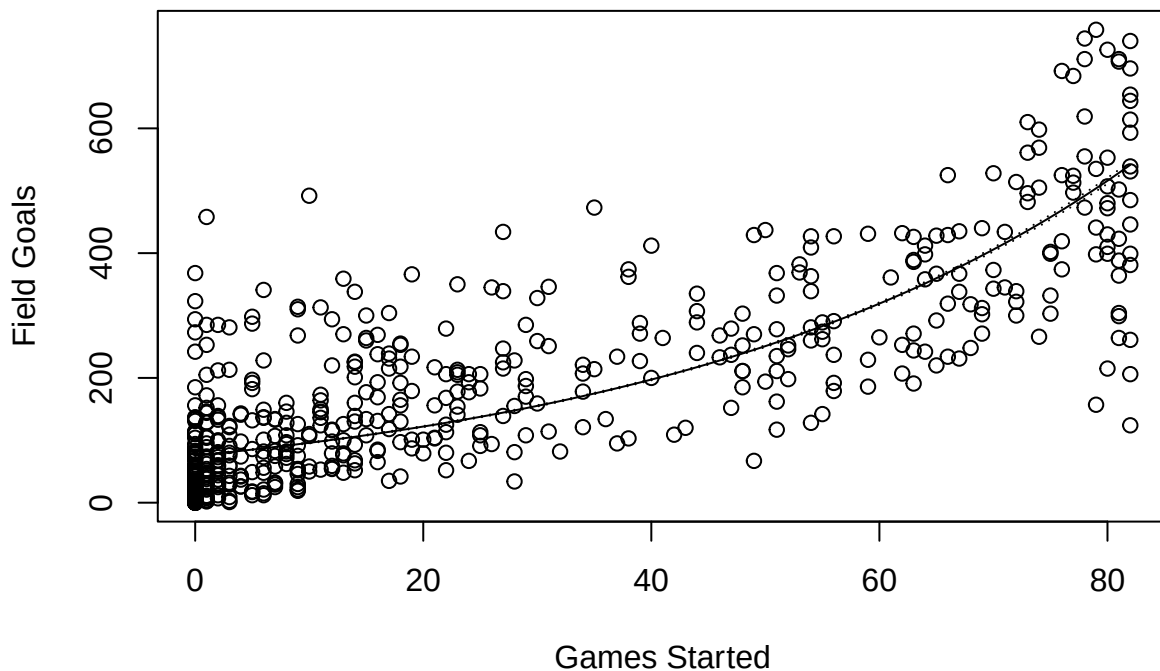
(Dispersion parameter for poisson family taken to be 1)

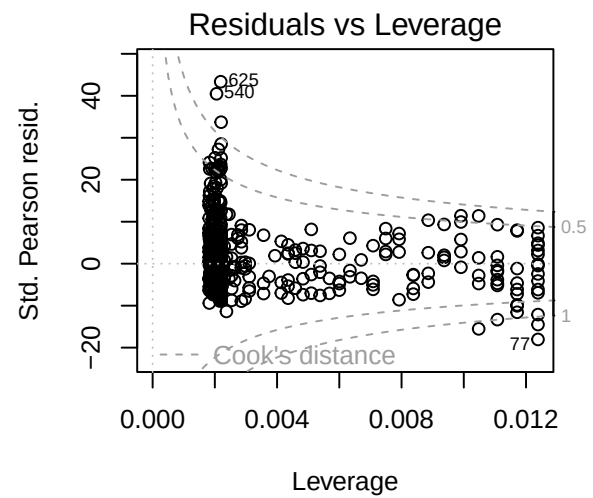
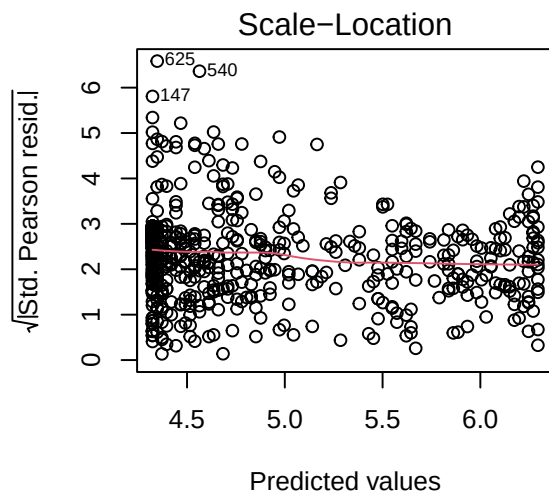
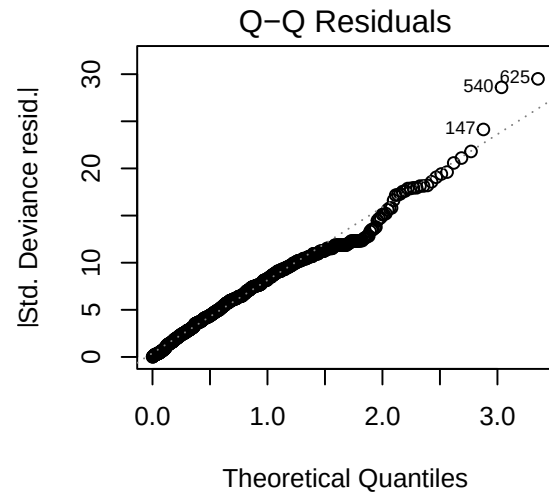
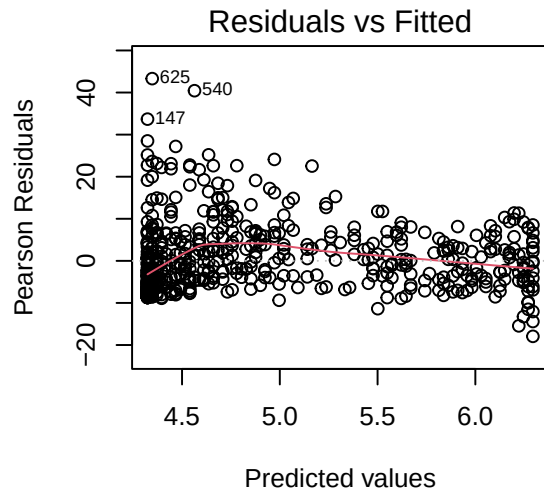
Null deviance: 97861 on 624 degrees of freedom

Residual deviance: 38992 on 623 degrees of freedom

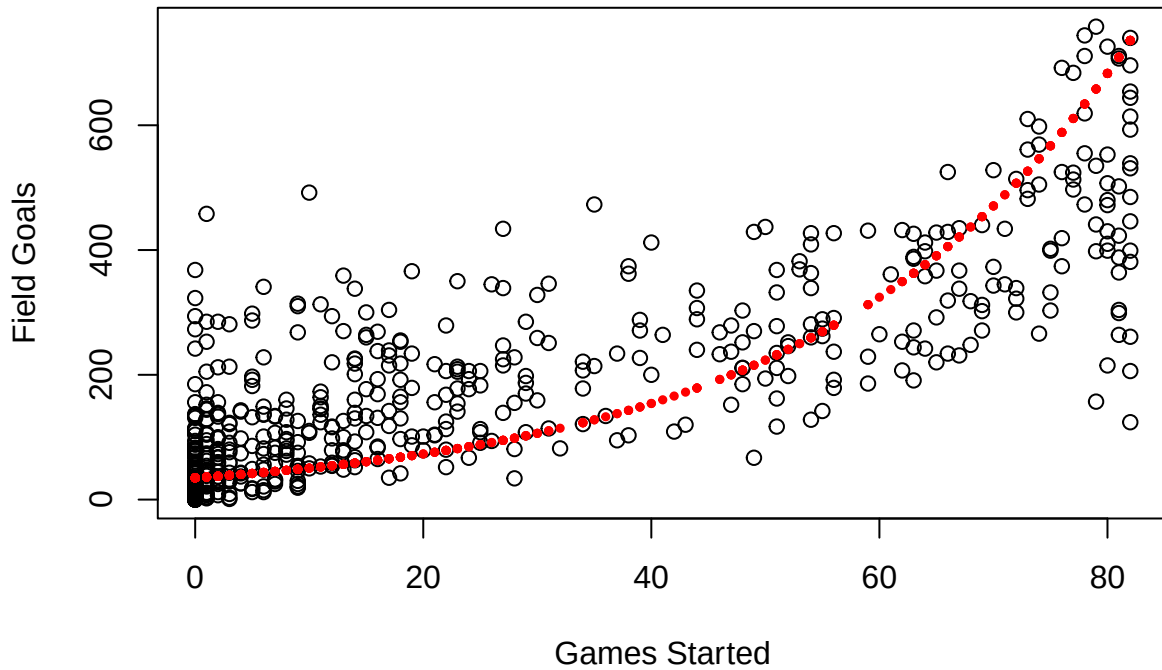
AIC: 42850

Number of Fisher Scoring iterations: 5





Compare the predictions from this model to the predictions from the normal linear regression model using a log transformation on the number of field goals (5 observations of 0 FG's were removed).



Which model would you prefer for prediction?

Which model for inference?

Why are the predictions different?

## Bringing it Together

Table 1.2: Side-by-side comparison of the models.

|            | GLM                                | Normal                                       | Logistic                                          | Poisson                                         |
|------------|------------------------------------|----------------------------------------------|---------------------------------------------------|-------------------------------------------------|
| Random     | $Y_i \overset{ind}{\sim} f(\mu_i)$ | $Y_i \overset{ind}{\sim} N(\mu_i, \sigma^2)$ | $Y_i \overset{ind}{\sim} \text{Bernoulli}(\pi_i)$ | $Y_i \overset{ind}{\sim} \text{Poisson}(\mu_i)$ |
| Link       | $g(\mu_i) = \eta_i$                | $g(\mu_i) = \mu_i = \eta_i$                  | $g(\pi_i) = \log(\pi_i / (1 - \pi_i)) = \eta_i$   | $g(\mu_i) = \log(\mu_i) = \eta_i$               |
| Systematic | $\eta_i = \mathbf{x}_i' \beta$     | $\eta_i = \mathbf{x}_i' \beta$               | $\eta_i = \mathbf{x}_i' \beta$                    | $\eta_i = \mathbf{x}_i' \beta$                  |

*Returning to the questions from the beginning of the chapter...*

1. What is the benefit of having a unifying framework for a GLM when all the specific models (such as the normal linear model, logistic regression, and Poisson regression) are different?
2. Why bother with a GLM at all? Why not just transform the variables as necessary?
3. How are the datasets and corresponding models similar and different?

## 1.6 Assignment

1. For each scenario below, propose a reasonable GLM. Clearly specify

$$Y_i \overset{ind}{\sim} f(\mu_i),$$

$$g(\mu_i) = \eta_i,$$

and  $\eta_i = \mathbf{x}_i' \beta$ .

Define all parameters and variables you introduce. You do not need to fit any models.

- a. A university records whether each admitted student enrolls (yes/no). Available predictors include distance from home, scholarship amount, undergraduate GPA, and whether the student attended an admitted-student event.
- b. An ecologist records the number of a particular bird species observed during a 10-minute survey at each of 200 sites. Predictors include elevation, forest cover, and temperature.
- c. An agricultural scientist measures the weight of tomatoes produced by individual plants. Predictors include fertilizer treatment, amount of water, and greenhouse.
- d. Choose one of the three scenarios and add an interaction that you think might be scientifically meaningful. Write the new systematic component and explain in words what question the interaction addresses.

2. A researcher previously fit normal linear regression models and is now fitting

$$Y_i \stackrel{ind}{\sim} \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i.$$

For each statement below, decide whether the idea carries over *essentially unchanged*, *carries over but requires modification*, or *does not carry over in the usual way*. Briefly explain.

- a.  $\beta_3$  represents an interaction between  $x$  and  $z$ .
  - b. We can compare a model containing  $x$ ,  $z$ ,  $xz$  to one containing only  $x$ ,  $z$ .
  - c.  $\beta_1$  is the change in the expected response associated with a one-unit increase in  $x$ , holding  $z$  and  $xz$  constant.
  - d. A fitted value represents an estimate of  $E(Y_i|\mathbf{x}_i)$ , where  $\mathbf{x}_i = (x_i, z_i)'$ .
  - e. The residuals,  $Y_i - \hat{\pi}_i$ , should be normally distributed.
  - f. The conditional variance of  $Y_i$  is assumed to be constant for all  $i$ .
  - g. Predictions should lie within the possible range of the response.
  - h. Confounding can make a regression coefficient misleading.
3. For each response variable below: (1) describe its support (possible values); (2) describe one important feature its conditional distribution might have; (3) decide whether ordinary normal linear regression seems reasonable; (4) if not, identify a GLM family that might be a reasonable starting point; (5) identify one additional piece of information you would want before committing to that model.
- a. Number of emergency-room visits during the next year.
  - b. Whether a customer defaults on a loan.
  - c. Proportion of 100 planted seeds that germinate.
  - d. Annual household income.
  - e. Number of goals scored by a soccer team in a match.
  - f. A student's score on a 20-question multiple choice test.
  - g. Number of days from infection until recovery.
  - h. Number of emails received by an employee during a workday.
4. Suppose  $Y_i$  is a count and a colleague proposes

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i,$$

$$\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2).$$

Another colleague proposes

$$Y_i \overset{ind}{\sim} \text{Poisson}(\mu_i),$$

$$\log(\mu_i) = \beta_0 + \beta_1 x_i.$$

- a. These equations both contain a log transformation. Explain why these are nevertheless fundamentally different models.
  - b. What happens to the first model if  $Y_i = 0$ ?
  - c. Under each model, identify the quantity that is assumed to be linear in  $x_i$ .
  - d. Under the Poisson model, what happens to the conditional variance as the conditional mean changes?
  - e. Give one reason you might nevertheless prefer the transformed-normal model for a particular data set.
5. For each model below, invent a plausible scientific study for which the model would make sense. Define  $Y_i$ ,  $x_i$ ,  $z_i$  in context and state the scientific question that could be addressed by  $\beta_1$ .

$$\text{Model A } Y_i \overset{ind}{\sim} N(\mu_i, \sigma^2), \quad \mu_i = \beta_0 + \beta_1 x_i + \beta_2 z_i.$$

$$\text{Model B } Y_i \overset{ind}{\sim} \text{Bernoulli}(\pi_i), \quad \text{logit}(\pi_i) = \beta_0 + \beta_1 x_i + \beta_2 z_i.$$

$$\text{Model C } Y_i \overset{ind}{\sim} \text{Poisson}(\mu_i), \quad \log(\mu_i) = \beta_0 + \beta_1 x_i + \beta_2 z_i.$$

- d. What modeling ideas are identical across all three examples?
  - e. What changes?
6. Find three response variables in the real world for which you might use a GLM: one binary response, one count response, and one other response for which normal linear regression might be questionable. For each response variable, state: the observable unit, 2-4 plausible explanatory variables, a candidate response distribution, a candidate link function, the resulting GLM, the scientific/business/policy question the model would answer, one reason your proposed GLM might turn out to be inadequate after seeing the data.
7. Your company wants to predict whether a customer will renew their subscription next year (yes/no) using customer characteristics and past usage. A colleague suggests using normal linear regression, but you recommend



logistic regression. In no more than two sentences, explain your recommendation to a business stakeholder who has no statistical training. Your explanation should communicate why logistic regression is better suited to the problem, not simply state that the response is binary.

## 2 Properties of the Exponential Family

The GLM requires the distribution of the response variable to be a member of the exponential family of distributions; therefore, we need to review some properties to estimate the parameters of the GLM. Specifically, notation, canonical parameters, sufficient statistics, the likelihood, mean, variance, score function, and Fisher information for the exponential family.

*Corresponding book sections:* D&B 3.2 and 3.3; A 4.4; M&N 2.2

### 2.1 Tool Kit for GLM Estimation

1. *Notation:* A family of pdf's or pmf's is called an exponential family if

$$f(y; \theta) = a(\theta)b(y) \exp \left\{ \sum_{j=1}^k w_j(\theta)t_j(y) \right\}, y \in A, \quad (2.1)$$

where we assume

- $A$  doesn't depend on  $\theta$ ,
- $a(\theta) > 0$  is a real-valued function that doesn't depend on  $y$ ,
- $w_j(\theta), j = 1, \dots, k$ , are real-valued functions that don't depend on  $y$ ,
- $b(y) \geq 0$  is a real-valued function that does not depend on  $\theta$ ,
- $t_j(y), j = 1, \dots, k$  are real-valued functions that do not depend on  $\theta$ .

2. *Canonical/Natural Parameter:* Instead of defining the distribution using the parameters  $\theta$ , we can let

$$\eta_j = w_j(\theta), j = 1, \dots, k,$$

be the *canonical* or *natural* parameters and the pdf/pmf becomes

$$f(y; \eta) = a^*(\eta)b(y) \exp \left\{ \sum_{j=1}^k \eta_j t_j(y) \right\}, \quad (2.2)$$

$$y \in A,$$

where  $\eta = (\eta_1, \dots, \eta_k)'$ . Note that:

- this equational form is *not* unique.
  - $\eta_j$  here is *not* the same as  $\eta_i$  introduced at the beginning of Section 1.2. Frequently used notation uses  $\eta$  to represent canonical parameters. Similarly,  $\eta$  is frequently used to represent the link between the mean and the systematic component of a GLM.
3. *Examples of Exponential Families:* Normal, Gamma, Beta, Inverse Gaussian, Poisson, and the Binomial and Negative Binomial under parameter restrictions.
4. *Likelihood:*

$$\begin{aligned} L(\theta; \mathbf{y}) &= \prod_{i=1}^n f(y_i; \theta) \\ &= \left( \prod_{i=1}^n b(y_i) \right) (a(\theta))^n \exp \left\{ \sum_{j=1}^k w_j(\theta) t_j(\mathbf{y}) \right\} \end{aligned}$$

and log likelihood:

$$\ell(\theta; \mathbf{y}) = \sum_{i=1}^n \log(b(y_i)) + n \log(a(\theta)) + \sum_{j=1}^k w_j(\theta) t_j(\mathbf{y}).$$

5. *Sufficient Statistic:* Suppose we have a set of independent random variables  $\{Y_i\}$  with  $Y_i \sim f(y_i; \theta)$ , then,

$$\begin{aligned} \prod_{i=1}^n f(y_i; \theta) &= \prod_{i=1}^n \left[ a(\theta) b(y_i) \exp \left\{ \sum_{j=1}^k w_j(\theta) t_j(y_i) \right\} \right] \\ &= \left( \prod_{i=1}^n b(y_i) \right) (a(\theta))^n \exp \left\{ \sum_{j=1}^k \left( w_j(\theta) \sum_{i=1}^n t_j(y_i) \right) \right\} \end{aligned}$$

Therefore, by the factorization theorem, a set of sufficient statistics for the exponential family is

$$\{t_j(\mathbf{y}) = \sum_{i=1}^n t_j(y_i); j = 1, \dots, k\}.$$

6. *Score Function:* Recall that the score function is the derivative of the log likelihood. We will simplify notation by letting  $n = 1$ . Note that Equation 2.1 can be written as

$$f(y; \theta) = \exp \left\{ \sum_{j=1}^k w_j(\theta) t_j(y) + \underbrace{\log(a(\theta))}_{c(\theta)} + \underbrace{\log(b(y))}_{d(y)} \right\}.$$

Suppose  $k = 1$  and  $\theta = \theta$ . Then, the log likelihood of  $y$  is

$$\ell(\theta; y) = w(\theta)t(y) + c(\theta) + d(y),$$

Therefore, the score function can be written as

$$\frac{\partial \ell(\theta; y)}{\partial \theta} = w'(\theta)t(y) + c'(\theta). \quad (2.3)$$

Furthermore, the score has an expected value of 0, or,

$$E \left( \frac{\partial \ell(\theta; y)}{\partial \theta} \right) = 0 \left\{ \begin{array}{l} = \int \left( \frac{\partial \ell(\theta; y)}{\partial \theta} \right) f(y; \theta) dy \\ = \int \left( \frac{\partial \log L(\theta; y)}{\partial \theta} \right) f(y; \theta) dy \\ = \int \frac{1}{L(\theta; y)} \frac{\partial L(\theta; y)}{\partial \theta} f(y; \theta) dy \\ = \int \frac{\partial L(\theta; y)}{\partial \theta} dy \\ = \frac{\partial}{\partial \theta} \underbrace{\int L(\theta; y) dy}_1 \\ = \frac{\partial}{\partial \theta} 1 = 0. \end{array} \right.$$

7. *Fisher Information:* The Fisher information is the variance of the score function. For the exponential family, this reduces to both

$$-E \left[ \frac{\partial^2 \ell(\theta; y)}{\partial \theta^2} \right] \text{ and } E \left[ \left( \frac{\partial \ell(\theta; y)}{\partial \theta} \right)^2 \right].$$

8. *Mean:* Using the score function (Equation 2.3), we take the expected value of both sides and solve for  $E(t(y))$ :

$$\begin{aligned} E\left(\frac{\partial \ell(\theta; y)}{\partial \theta}\right) &= w'(\theta)E(t(y)) + c'(\theta) \\ 0 &= w'(\theta)E(t(y)) + c'(\theta) \\ \Rightarrow E(t(y)) &= -\frac{c'(\theta)}{w'(\theta)}. \end{aligned}$$

9. *Variance:* For the variance, we will use both forms of the Fisher information:

$$\begin{aligned} -E\left[\frac{\partial^2 \ell(\theta; y)}{\partial \theta^2}\right] &= E\left[\left(\frac{\partial \ell(\theta; y)}{\partial \theta}\right)^2\right] \\ -E[w''(\theta)t(y) + c''(\theta)] &= E[(w'(\theta)t(y) + c'(\theta))^2] \\ -w''(\theta)E(t(y)) - c''(\theta) &= E\left[(w'(\theta))^2 \left(\frac{w'(\theta)t(y)}{w'(\theta)} + \frac{c'(\theta)}{w'(\theta)}\right)^2\right] \\ &= (w'(\theta))^2 E\left[\left(t(y) - \underbrace{-\frac{c'(\theta)}{w'(\theta)}}_{E(t(y))}\right)^2\right] \\ \Rightarrow -w''(\theta)E(t(y)) - c''(\theta) &= (w'(\theta))^2 \text{Var}(t(y)). \end{aligned} \tag{2.4}$$

Solving Equation 2.4 for  $\text{Var}(t(y))$ , we have,

$$\text{Var}(t(y)) = \frac{w''(\theta)c'(\theta) - w'(\theta)c''(\theta)}{(w'(\theta))^3}.$$

## Example: The Binomial Distribution

Here  $\theta = \pi$ ,  $k = 1$ , and  $m$  is fixed.

$$\begin{aligned} f(y; \pi) &= \binom{m}{y} \pi^y (1 - \pi)^{m-y} \\ &= \binom{m}{y} \exp\{y \log(\pi) + (m - y) \log(1 - \pi)\} \\ &= \binom{m}{y} \exp\{y(\log(\pi) - \log(1 - \pi)) + m \log(1 - \pi)\} \\ &= \binom{m}{y} (1 - \pi)^m \exp\left\{y \log\left(\frac{\pi}{1 - \pi}\right)\right\}. \end{aligned}$$

From Equation 2.1,

$$a(\theta) \equiv a(\pi) = (1 - \pi)^m$$

$$b(y) = \binom{m}{y}$$

$$w(\theta) \equiv w(\pi) = \log\left(\frac{\pi}{1 - \pi}\right)$$

$$t(y) = y$$

$$\text{and the canonical parameter is, } \eta = \log\left(\frac{\pi}{1 - \pi}\right).$$

To find the mean and variance:

$$f(y; \pi) = \exp\left\{y \log\left(\frac{\pi}{1 - \pi}\right) + m \log(1 - \pi) + \log\left(\binom{m}{y}\right)\right\}$$

$$\begin{aligned}
w'(\theta) &= \frac{1}{\pi/(1-\pi)} \frac{(1-\pi) + \pi}{(1-\pi)^2} = \frac{1-\pi}{\pi} \frac{1}{(1-\pi)^2} = \frac{1}{\pi(1-\pi)} \\
w''(\theta) &= \frac{-[(1-\pi) + -\pi]}{\pi^2(1-\pi)^2} = \frac{-(1-2\pi)}{\pi^2(1-\pi)^2} \\
c'(\theta) &= -\frac{m}{1-\pi} \\
c''(\theta) &= -\frac{m}{(1-\pi)^2}
\end{aligned}$$

Then,

$$\begin{aligned}
E(t(y)) = E(y) &= -\frac{c'(\theta)}{w'(\theta)} = -\frac{-\frac{m}{1-\pi}}{\frac{1}{\pi(1-\pi)}} = \frac{m\pi(1-\pi)}{1-\pi} = m\pi \\
\text{Var}(t(y)) = \text{Var}(y) &= \frac{\frac{-(1-2\pi)}{\pi^2(1-\pi)^2} \frac{-m}{(1-\pi)} - \frac{1}{\pi(1-\pi)} \frac{-m}{(1-\pi)^2}}{\left(\frac{1}{\pi(1-\pi)}\right)^3} \\
&= \left( \frac{m(1-2\pi)}{\pi^2(1-\pi)^3} + \frac{m}{\pi(1-\pi)^3} \right) \pi^3(1-\pi)^3 \\
&= m\pi(1-2\pi) + m\pi^2 \\
&= m\pi - 2m\pi^2 + m\pi^2 \\
&= m\pi(1-\pi)
\end{aligned}$$

## 2.2 Homework Assignment

1. Show that the following distributions belong to the exponential family. For each distribution, identify: a natural parameter, a sufficient statistic, the mean, and the variance using the equations shown in class.

- a. Gamma distribution with unknown rate parameter,  $\beta$ , and fixed shape parameter,  $\alpha$ :

$$f(y; \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-y\beta}.$$

- b. Poisson distribution:

$$f(y; \lambda) = \frac{\lambda^y}{y!} e^{-\lambda}.$$

- c. Negative Binomial distribution (fixed  $r$ ):

$$f(y; \pi) = \binom{y+r-1}{r-1} \pi^r (1-\pi)^y.$$

2. For the Poisson, binomial, and negative binomial distributions, write the variance as a function of the mean. Then answer:

- a. How does each distribution differ in its relationship between the mean and variance?
- b. What does this suggest about when a Poisson model might be inappropriate?
- c. Suppose your observed count data have substantially greater variability than would be expected from their mean under a Poisson model. What alternative model(s) might you consider, and why?
- d. Explain why comparing the sample mean and sample variance can be useful when beginning to choose a model, but is not by itself sufficient to determine the correct model.

3. Continuing from the previous problem,

- a. For the gamma distribution, write the variance as a function of the mean.



- b. For all four distributions, how does the variance change as the mean changes? (It may be helpful to make some plots of variance vs. mean.)
  - c. Which distributions have variance that increases linearly with the mean?
  - d. Which distributions have variance that increases faster than linearly?
  - e. Why is the relationship between the mean and variance important when choosing a probability model for a response variable?
4. For the Pareto distribution,  $f(y; \theta) = \theta y^{-\theta-1}$ , find the score statistic,  $U$ , and the information,  $\mathfrak{I} = \text{Var}(U)$ . Verify that  $E(U) = 0$ . What feature of this example makes it particularly easy to calculate the information? (Note: Let  $n = 1$ .)
5. You are working as a data scientist on a team that regularly builds models for continuous outcomes, yes/no outcomes, and counts. A colleague without formal statistical training notices that you have been reviewing exponential family distributions, sufficient statistics, means, variances, score functions, and information. They ask: “Why do you need all of this probability theory? Aren’t logistic regression and Poisson regression just different tools that you can ask software to fit?” In three sentences or less, respond to your colleague. Your response should:
  - explain at least one practical benefit of recognizing these models as members of a common family;
  - give at least one example of how the probability model affects what we can reasonably assume or learn from the data; and
  - avoid mathematical notation and statistical jargon that would require further explanation.

# 3 Estimation, Inference, and Model Diagnostics for the GLM

We'll first look at classical estimation for the GLM. Later in the semester we'll take a look at Bayesian estimation.

Recall:

## The Generalized Linear Model

### random component

$$Y_i \sim f(\mu_i),$$

where  $f(\cdot)$  is a probability distribution (in the exponential family), and  $\mu_i = E(Y_i)$ .

### link component

$$g(\mu_i) = \eta_i,$$

where  $g(\cdot)$  is called the *link* function. The link component connects the random component to the systematic component.

### systematic component

$$\eta_i = \mathbf{x}_i' \beta,$$

where  $\beta$  is a  $k \times 1$  vector of unknown coefficients.

Note that:

- $f(\cdot)$  must be a distribution in the exponential family.
- We also need to know more properties about the link function.

## 3.1 The Link Function

The *link function* is a function,  $g(\cdot)$ , such that

$$\eta_i = g(\mu_i),$$

where  $\mu_i = E(Y_i)$ .

→ Note that last time we showed that if  $t(y_i) = y_i$ , then  $\mu_i = \frac{-c'(\theta_i)}{w'(\theta_i)}$ .

→ The theory still works out otherwise, but this assumption keeps our inference just a little more

$g(\cdot)$  “links” the mean,  $\mu_i$  (or the canonical parameter), to  $\eta_i$  (and we set  $\eta_i = \mathbf{x}_i' \beta$ ).

### Note

For interpretation of the model, we generally assume  $g(\cdot)$  is **one-to-one, monotone, continuous, and differentiable**.

Refresher: What do these terms mean?

Why would these assumptions be valuable/necessary?

One-to-one means that if  $g(a) = g(b)$  then  $a = b$ .

monotone increasing means that if  $a > b$  then  $g(a) > g(b)$ ; monotone decreasing means that  $g(a) < g(b)$ .

continuous:  $\lim_{x \rightarrow c} = g(c)$ .

differentiable: means that the derivative exists at every value in its domain.

Assumptions are valuable for interpretation (e.g., how do we interpret  $\beta$  if the link has a discontinuity?). Needed for estimation properties (we’re going to take the derivative of the link, we’ll invert the link, etc.).

## Canonical Link

**Definition:** The *canonical link* is the function  $g^*(\cdot)$  such that the link is identity function on the canonical parameter,  $w(\theta_i)$ . Therefore,

$$g^*(w(\theta_i)) = w(\theta_i).$$

Find the canonical link for the normal, binomial, and Poisson distributions.

Normal:  $w(\mu_i) = \mu_i \rightarrow g(\mu_i) = \mu_i$  and  $g^*(w(\mu_i)) = \mu_i$ .

Binomial:  $w(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) \rightarrow g(\pi_i) = \log(\pi/(1-\pi))$  and  $g^*(w(\pi_i)) = w(\pi_i)$

Poisson:  $w(\lambda_i) = \log(\lambda_i)$ ...

Using the canonical link can simplify some of the estimation procedures. However, although this may make sense mathematically and statistically, this link may not make sense for our data. Thus, we proceed with estimation for any link function,  $g(\theta)$ .

## 3.2 Example to Motivate Estimation

```
library(faraway) #Load package for the data
data(orings)
y <- orings$damage #define a generic response vector "y"
y[which(orings$damage>0)] <- 1 #1 if any of the o-rings failed
temp <- orings$temp #easier to write "temp" #single covariate
X <- cbind(rep(1, length(temp)), temp) #X matrix

#Starting values for IWLS
b0 <- 0
b1 <- 0
b <- c(b0, b1)
epsilon <- .00001
diff <- 1
iter <- 0

#IWLS Loop
while(diff>epsilon){
```

```

# #W and Z in terms of beta only (not g'(mu), etc.)
# #w <- exp(b0 + temp*b1)/((1+exp(b0 + temp*b1))^2)
# w <- exp(X%*%b)/((1+exp(X%*%b))^2) #using vector notation
# W <- diag(c(w)) #create the diagonal matrix
# #z <- b0 + temp*b1 + (y - exp(b0 + temp*b1)/(1+exp(b0 + temp*b1)))*((1+exp(
# z <- X%*%b + (y-exp(X%*%b)/(1+exp(X%*%b)))*(1+exp(X%*%b))^2/exp(X%*%b) #aga

#Z and W as general as possible
mui <- exp(X%*%b)/(1+exp(X%*%b))
varyi <- mui*(1-mui)
gprimemui <- 1/(mui*(1-mui))
z <- X%*%b + (y-mui)*gprimemui
W <- diag(c(1/(varyi*gprimemui^2)))

b <- solve(t(X)%*%W%*%X)%*%t(X)%*%W%*%z
diff <- (b[1]-b0)^2 + (b[2]-b1)^2
b0 <- b[1]
b1 <- b[2]
iter <- iter + 1
print(paste("Iteration: ", iter))
print(paste("Estimate: ", b))
}

```

```

[1] "Iteration: 1"
[1] "Estimate: 9.61904761904759" "Estimate: -0.149523809523809"
[1] "Iteration: 2"
[1] "Estimate: 13.6557379084885" "Estimate: -0.211246980623623"
[1] "Iteration: 3"
[1] "Estimate: 14.93828914006" "Estimate: -0.230600099155813"
[1] "Iteration: 4"
[1] "Estimate: 15.0422911376214" "Estimate: -0.232153686700272"
[1] "Iteration: 5"
[1] "Estimate: 15.0429016271303" "Estimate: -0.232162743914735"

```

```
#Compare to the fitted values from the GLM
mod.glm <- glm(y~temp, family=binomial)
coefficients(mod.glm)
```

```
(Intercept)      temp
 15.0429016  -0.2321627
```

```
mod.glm$iter
```

```
[1] 5
```

**i** The same code, but in python:

```

# Import libraries
import numpy as np
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
from scipy import stats
import matplotlib.pyplot as plt
from ucimlrepo import fetch_ucirepo # used to load dataset

# Load dataset
orings = fetch_ucirepo(id=92).data.original.sort_values(by = 'launch_temp').re

# Prepare X matrix
temp = orings['launch_temp']
X = np.column_stack((np.ones(len(temp)), temp))

# Prepare y vector (1 if the o-ring failed else 0)
y = np.where(orings['num_thermal_distress'] > 0, 1, 0)

# Starting Values for IWLS
b0 = 0
b1 = 0
epsilon = 0.00001
diff = 1
iter_count = 0

b = np.array([0, 0])

# IWLS Loop
while diff > epsilon:

    ### W and z in terms of beta only (not g'(mu), etc.) ###
    # W = np.diag(np.exp(X @ b) / (1 + np.exp(X @ b))**2)
    # z = (X @ b) + (y - np.exp(X @ b)/(1 + np.exp(X @ b))) * (1 + np.exp(X @

    ### W and z as general as possible ###
    mui = np.exp(X @ b) / (1 + np.exp(X @ b))
    varyi = mui * (1 - mui)
    gprimemui = 1 / (mui * (1 - mui))
    z = X @ b + (y - mui) * gprimemui
    W = np.diag(1 / (varyi * gprimemui**2))

    b_new = np.linalg.inv(X.T @ W @ X) @ X.T @ W @ z
    diff = np.sum((b_new - b)**2)

```



```
Iteration: 1  
Estimate: (Intercept, Temp)  
[ 9.61904762 -0.14952381]
```

```
Iteration: 2  
Estimate: (Intercept, Temp)  
[13.65573791 -0.21124698]
```

```
Iteration: 3  
Estimate: (Intercept, Temp)  
[14.93828914 -0.2306001 ]
```

```
Iteration: 4  
Estimate: (Intercept, Temp)  
[15.04229114 -0.23215369]
```

```
Iteration: 5  
Estimate: (Intercept, Temp)  
[15.04290163 -0.23216274]
```

```
# Compare to the fitted values from the GLM  
data = pd.DataFrame({'y': y,  
                     'temp': temp})  
model = smf.glm(formula='y ~ temp', data=data, family=sm.families.Binomial()).  
print(model.params)
```

```
Intercept    15.042902  
temp         -0.232163  
dtype: float64
```

```
print(model.fit_history['iteration'])
```

5

### 3.3 Maximum Likelihood Estimation of $\beta$

Corresponding book sections: D&B Chapter 4; M&N Section 2.5, 4.4.2; A: Sections 4.4.4-5 & 4.6

Note that there are many unknown parameters in the GLM. However, all parameters can be determined when we know  $\beta$ .

! Important

Parameter of interest:  $\beta$

Recall:

$$Y_i \stackrel{ind}{\sim} f(y_i; \theta_i), \text{ for } i = 1, \dots, n.$$

Define:

$$\ell_i = \ell(\theta_i; y_i) = t(y_i)w(\theta_i) + c(\theta_i) + d(y_i).$$

We'll consider the simple case when  $t(y_i) = y_i$ , then,

$$E(Y_i) = \mu_i = \frac{-c'(\theta_i)}{w'(\theta_i)} \quad (3.1)$$

and

$$\text{Var}(Y_i) = \frac{w''(\theta_i)c'(\theta_i) - c''(\theta_i)w'(\theta_i)}{[w'(\theta_i)]^3}. \quad (3.2)$$

Finally, let  $g(\mu_i) = \mathbf{x}_i' \beta = \eta_i$ .

$$\Rightarrow \ell(\theta; \mathbf{y}) = \sum_{i=1}^n \ell_i = \sum_{i=1}^n y_i w(\theta_i) + \sum_{i=1}^n c(\theta_i) + \sum_{i=1}^n d(y_i)$$

To estimate  $\beta$ , define the score for the  $j$ th element of  $\beta$  to be,

$$U_j(\beta) = \frac{\partial \ell}{\partial \beta_j}, \text{ for } j = 1, \dots, k.$$

**i** Note

But  $\ell(\theta; \mathbf{y})$  does not depend on the parameter of interest,  $\beta$ !  
 $\Rightarrow$  Use the chain rule...

$$\begin{aligned} U_j(\beta) &= \frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^n \frac{\partial \ell_i}{\partial \beta_j} \\ &= \sum_{i=1}^n \frac{\partial \ell_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j} \end{aligned}$$

$$\begin{aligned} \frac{\partial \ell_i}{\partial \theta_i} &= y_i w'(\theta_i) + c'(\theta_i) = w'(\theta_i) \left( y_i + \frac{c'(\theta_i)}{w'(\theta_i)} \right) = w'(\theta_i)(y_i - \mu_i) \\ \frac{\partial \theta_i}{\partial \mu_i} &= \left( \frac{\partial \mu_i}{\partial \theta_i} \right)^{-1} \\ &= \left( \frac{-c''(\theta_i)w'(\theta_i) + w''(\theta_i)c'(\theta_i)}{[w'(\theta_i)]^2} \right)^{-1} \\ &= (\text{Var}(y_i)w'(\theta_i))^{-1} \\ \frac{\partial \mu_i}{\partial \eta_i} &= \left( \frac{\partial \eta_i}{\partial \mu_i} \right)^{-1} = (g'(\mu_i))^{-1} \\ \frac{\partial \eta_i}{\partial \beta_j} &= x_{ij} \end{aligned}$$

The score equation for  $\beta_j$ , for  $j = 1, \dots, k$ , is then,

$$\begin{aligned} U_j &= \sum_{i=1}^n w'(\theta_i)(y_i - \mu_i) \frac{1}{\text{Var}(y_i)w'(\theta_i)} \frac{1}{g'(\mu_i)} x_{ij} \\ &= \sum_{i=1}^n \frac{(y_i - \mu_i)}{\text{Var}(y_i)g'(\mu_i)} x_{ij} \end{aligned}$$

Covariance of  $U_j$  and  $U_q$  is

$$\begin{aligned}\mathfrak{J}_{jq} &= E(U_j U_q) \text{ (remember, } E(U_j) = 0\text{)} \\ &= E \left[ \sum_{i=1}^N \frac{(y_i - \mu_i)}{\text{Var}(y_i)g'(\mu_i)} x_{ij} \sum_{r=1}^n \frac{(y_r - \mu_r)}{\text{Var}(y_r)g'(\mu_r)} x_{rq} \right]\end{aligned}$$

{because the  $y_i$ 's are independent, we know that  $E((y_i - \mu_i)(y_r - \mu_r)) = 0, \forall i \neq r$  so that }

$$\begin{aligned}&= \sum_{i=1}^n \frac{E((y_i - \mu_i)^2)}{(\text{Var}(y_i))^2 (g'(\mu_i))^2} x_{ij} x_{iq} \\ &= \sum_{i=1}^n \frac{x_{ij} x_{iq}}{\text{Var}(y_i) (g'(\mu_i))^2}\end{aligned}\tag{3.3}$$

Using the Newton-Raphson algorithm, ([nice refresher here](#))

$$\mathbf{b}^{(t)} = \mathbf{b}^{(t-1)} + [\mathfrak{J}^{(t-1)}]^{-1} \mathbf{U}^{(t-1)},\tag{3.4}$$

(recall that  $\mathbf{U}$  would be the gradient and  $\mathfrak{J}$  would be the Hessian) where  $\mathbf{b}^{(t)} = \hat{\beta}^{(t)}$ , or the vector of estimates of the parameter  $\beta$  at the  $t^{th}$  iteration.

#### **i** Note

This is called: *the method of scoring*

We can multiply Equation 3.4 by  $\mathfrak{J}^{(t-1)}$  to get

$$\mathfrak{J}^{(t-1)} \mathbf{b}^{(t)} = \mathfrak{J}^{(t-1)} \mathbf{b}^{(t-1)} + \mathbf{U}^{(t-1)}.\tag{3.5}$$

From Equation 3.3, we know that

$$\mathfrak{J} = \mathbf{X}' \mathbf{W} \mathbf{X}$$

where  $\mathbf{W}$  is an  $n \times n$  diagonal matrix with

$$w_{ii} = \frac{1}{\text{Var}(y_i) (g'(\mu_i))^2}.$$

The  $j$ th element of  $\mathfrak{J}^{(t-1)}\mathbf{b}^{(t-1)} + \mathbf{U}^{(t-1)}$  is

$$\begin{aligned}
&= \sum_{r=1}^k \sum_{i=1}^n \frac{x_{ij}x_{ir}}{\text{Var}(y_i)(g'(\mu_i))^2} b_r^{(t-1)} + \sum_{i=1}^n \frac{(y_i - \mu_i)x_{ij}}{\text{Var}(y_i)g'(\mu_i)} \\
&= \sum_{r=1}^k \sum_{i=1}^n w_{ii}x_{ij}x_{ir}b_r^{(t-1)} + \sum_{i=1}^n w_{ii}g'(\mu_i)(y_i - \mu_i)x_{ij} \quad (\text{since } w_{ii} = \frac{1}{\text{Var}(Y_i)(g'(\mu_i))^2}) \\
&= \sum_{i=1}^n w_{ii}x_{ij} \underbrace{\left( \sum_{r=1}^k x_{ik}b_r^{(t-1)} + (y_i - \mu_i)g'(\mu_i) \right)}_{=z_i, \text{ where } \mu_i \text{ \& } g'(\mu_i) \text{ are evaluated at } \mathbf{b}^{(t-1)}} \\
&= \sum_{i=1}^n w_{ii}x_{ij}z_i
\end{aligned}$$

Therefore,

$$\mathfrak{J}^{(t-1)}\mathbf{b}^{(t-1)} + \mathbf{U}^{(t-1)} = \mathbf{X}'\mathbf{W}\mathbf{z}.$$

From Equation 3.5,

$$\mathbf{X}'\mathbf{W}\mathbf{X}\mathbf{b}^{(t)} = \mathbf{X}'\mathbf{W}\mathbf{z}$$

! Important

$$\mathbf{b}^{(t)} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1} \mathbf{X}'\mathbf{W}\mathbf{z} \quad (3.6)$$

Note this must be solved iteratively, since  $\mathbf{z}$  and  $\mathbf{W}$  depend on  $\mathbf{b}$ . This procedure is called **Iteratively (Re)Weighted Least Squares** (IWLS):

1. Choose a value for  $\mathbf{b}^{(0)}$ ;  $t = 0$
2. Set  $t = t + 1$
3. Evaluate  $\mathbf{z}$  and  $\mathbf{W}$  based on  $\mathbf{b}^{(t-1)}$
4. Solve for  $\mathbf{b}^{(t)}$  (using Equation 3.6)
5. Repeat steps 2–4 until  $\|\mathbf{b}^{(t)} - \mathbf{b}^{(t-1)}\|$  is sufficiently small

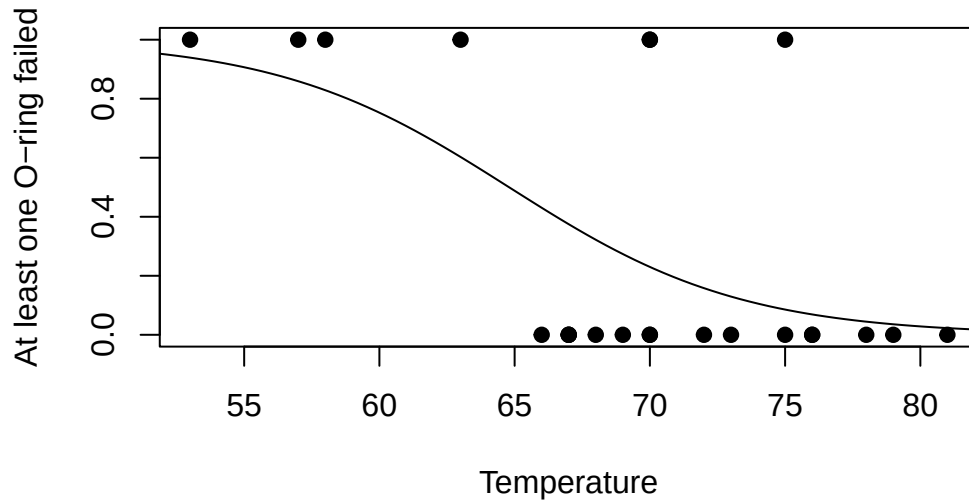
This is the default estimation procedure for GLM's in R.

## 3.4 Returning to the Motivating Example

### **i** Note

O-ring Failures: The space shuttle, *Challenger*, broke apart 73 seconds into its flight on January 28, 1986. The disaster is attributed to an O-ring failure, when the O-ring did not seal in the right solid rocket booster at liftoff. This failure is often attributed to the unusually low temperatures, at which the O-rings had not been confirmed to function properly.

Studies on the O-rings provided the following data (with our fitted line):



In Section 3.2 we fitted

$$Y_i \stackrel{ind}{\sim} \text{Bernoulli}(\pi_i), i = 1, \dots, n = 23$$
$$g(\pi_i) = \log \left( \frac{\pi_i}{1 - \pi_i} \right) = \eta_i$$
$$\eta_i = \beta_0 + x_i \beta_1,$$

where  $Y_i$  is equal to 0 if no o-rings failed and equal to 1 if at least one o-ring failed, and  $x_i$  is equal to the temperature at lift-off. Using both our own IWLS

code and R's glm function resulted in the same values of  $\mathbf{b} \equiv \hat{\beta}$ , the maximum likelihood estimate of  $\beta$ .

Using the code in Section 3.2, fit two “wrong” models to this data to examine how the IWLS algorithm can be adapted to fit other models. Change “mui”, “varyi”, and “gprimemui”, to accommodate these adjusted models. Then fit the corresponding model using R's glm function.

**Model A:**

$$\begin{aligned} Y_i &\overset{ind}{\sim} \mathcal{N}(\mu_i, \sigma^2 = 1), i = 1, \dots, n = 23 \\ g(\mu_i) &= \mu_i = \eta_i \\ \eta_i &= \beta_0 + x_i\beta_1, \end{aligned}$$

**Model B:**

$$\begin{aligned} Y_i &\overset{ind}{\sim} \text{Poisson}(\mu_i), i = 1, \dots, n = 23 \\ g(\mu_i) &= \log(\mu_i) = \eta_i \\ \eta_i &= \beta_0 + x_i\beta_1, \end{aligned}$$

```
#code to fit these models using the glm function
modA <- glm(y~temp)
coef(modA)

modB <- glm(y~temp, family=poisson)
coef(modB)
modB$iter
```

**i** The same code, but in python:

```
modA = smf.glm(formula='y ~ temp', data=data).fit()
print(modA.params)

modB = smf.glm(formula='y ~ temp', data=data, family=sm.families.Poisson()).fit()
print(modB.params)
print(f"\nModel B Iterations: {modB.fit_history['iteration']}")
```

**i** Note

Note that in our own IWLS algorithm, we left out  $\eta_i$ . We could likewise “leave out” other variables by putting all variables in terms of  $\beta$ . For the Bernoulli model, this looks like



$$\Rightarrow \mu_i = \pi_i = \frac{\exp\{\beta_0 + x_i\beta_1\}}{1 + \exp\{\beta_0 + x_i\beta_1\}}$$

$$\& 1 - \pi_i = \frac{1}{1 + \exp\{\beta_0 + x_i\beta_1\}}$$

$$\begin{aligned} g'(\mu_i) &= \frac{d g(\mu_i)}{d \mu_i} = \frac{d g(\pi_i)}{d \pi_i} \\ &= \frac{1}{\pi_i/(1 - \pi_i)} \left( \frac{(1 - \pi_i) + \pi_i}{(1 - \pi_i)^2} \right) \\ &= \frac{1}{\pi_i(1 - \pi_i)} \end{aligned}$$

$$= \frac{(1 + \exp\{\beta_0 + x_i\beta_1\})^2}{\exp\{\beta_0 + x_i\beta_1\}}$$

$$w_{ii} = \frac{1}{\text{Var}(Y_i)g'(\mu_i)^2}$$

$$= \frac{\pi_i^2(1 - \pi_i)^2}{\pi_i(1 - \pi_i)}$$

$$= \pi_i(1 - \pi_i)$$

$$= \frac{\exp\{\beta_0 + x_i\beta_1\}}{(1 + \exp\{\beta_0 + x_i\beta_1\})^2}$$

$$z_i = \sum_{j=0}^{(k-1)} x_{ij}b_j^{(t-1)} + (y_i - \mu_i)g'(\mu_i)$$

$$= b_o^{(t-1)} + x_i b_1^{(t-1)} + \left( y_i - \frac{\exp\{\beta_0 + x_i\beta_1\}}{1 + \exp\{\beta_0 + x_i\beta_1\}} \right) \frac{(1 + \exp\{\beta_0 + x_i\beta_1\})^2}{\exp\{\beta_0 + x_i\beta_1\}}$$

(Recall that in the algorithm we evaluate  $\beta$  at the current value of  $\mathbf{b}^{(t-1)}$ .)

## 3.5 Inference for $\beta$

**Corresponding book sections:** D&B: Sections 5.1-4; M&N: Section 4.4.4; A: Sections 4.4.6-7

### **i** Note

Recall that in a frequentist setting, we use *sampling distributions* to conduct inference on parameter coefficients.

Let  $\mathbf{U}(\beta) \equiv \mathbf{U}$  be the  $k \times 1$  vector of score statistics for  $\beta$ . Let  $\mathbf{b}$  be the maximum likelihood estimate of  $\beta$ . A few results for the sampling distributions:

- The sampling distribution for the score statistic:
  - **Result 1:** The Central Limit Theorem and relationship between the multivariate standard normal and  $\chi^2$  distributions tell us that

$$\mathbf{U}'\mathfrak{I}^{-1}\mathbf{U} \sim \chi^2(k)$$

for large samples ( $n$  large).

- The sampling distribution for  $\mathbf{b}$ :
  - **Result 2a:** Using a Taylor Series Approximation of  $\mathbf{U}$  around  $\mathbf{b}$ ,

$$\mathbf{U}(\beta) \approx \underbrace{\mathbf{U}(\mathbf{b})}_{=0} - \mathfrak{I}(\mathbf{b})(\beta - \mathbf{b}).$$

- **Result 2b:** Combining Results 1 and 2a,

$$\underbrace{(\mathbf{b} - \beta)' \mathfrak{I}(\mathbf{b})(\mathbf{b} - \beta)}_{\text{Wald Statistic}} \sim \chi^2(k)$$

(large  $n$ )

- **Result 2c:** when  $k = 1$ ,

$$\mathbf{b} \sim \mathcal{N}(\beta, \mathfrak{I}^{-1}).$$

If  $Y_i$  are independently and normally distributed, this distribution is exact; otherwise, this distribution is approximate for large  $n$ .

Using Result 2b, we can obtain confidence regions for  $\beta$ . Using Result 2c, we can perform univariate hypothesis tests and confidence intervals for  $\beta_j$ ,  $j = 1, \dots, k$ .

With the O-ring data, find the standard errors for each  $\beta_j$ , corresponding z-statistic (why not t-statistic?), and p-values (choose your model!).

I put my output in a data frame to resemble the coefficient table from the “summary” of a glm object:

|      | betahat    | se        | zstat     | pvals      |
|------|------------|-----------|-----------|------------|
|      | 15.0429016 | 7.3786364 | 2.038710  | 0.04147895 |
| temp | -0.2321627 | 0.1082365 | -2.144958 | 0.03195624 |

The official R glm output looks like:

```
summary(mod.glm)
```

Call:

```
glm(formula = y ~ temp, family = binomial)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | 15.0429  | 7.3786     | 2.039   | 0.0415 * |
| temp        | -0.2322  | 0.1082     | -2.145  | 0.0320 * |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 28.267 on 22 degrees of freedom

Residual deviance: 20.315 on 21 degrees of freedom

AIC: 24.315

Number of Fisher Scoring iterations: 5

**i** The same code, but in python:

```
print(model.summary())
```

```

              Generalized Linear Model Regression Results
=====
Dep. Variable:                  y      No. Observations:                  23
Model:                        GLM      Df Residuals:                    21
Model Family:                 Binomial  Df Model:                      1
Link Function:                 Logit    Scale:                        1.0000
Method:                       IRLS     Log-
Likelihood:                   -10.158
Date:                         Tue, 18 Aug 2026    Deviance:                    20.315
Time:                         13:02:45           Pearson chi2:                 23.2
No. Iterations:                5      Pseudo R-
squ. (CS):                     0.2923
Covariance Type:              nonrobust
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept      15.0429      7.379      2.039      0.041      0.581      29.505
temp          -0.2322      0.108     -2.145      0.032      -
0.444         -0.020
=====
```

**!** Important

How do we get a confidence interval for  $\mu_i$ ?

## 3.6 Inference for Model Fit

Corresponding book sections: D&B: 5.5-5.7, 7.5-7.6; M&N: 2.3-2.4; A: 4.5, 14.3

### 3.6.1 Assessing model fit through the likelihood ratio test (LRT)

Let  $m$  denote the maximum number of parameters that can be estimated. (Can be  $n$  if there are no repeated covariate combinations.) The model fitting all  $m$  parameters is called the *saturated*/*maximal*/full model. This is the most general model, having **a separate parameter for each observation** and the **perfect fit**  $\hat{\mu} = \mathbf{y}$ .

For example, the parameter estimates for the saturated model for the following distributions would be:

$$\begin{aligned} Y_i &\sim \text{Bern}(\pi_i), & \hat{\mu}_i &= \hat{\pi}_i = y_i \\ Y_i &\sim \text{Bin}(m_i, \pi_i), & \hat{\mu}_i &= m_i \hat{\pi}_i = y_i \\ Y_i &\sim \text{Pois}(\mu_i), & \hat{\mu}_i &= y_i \end{aligned}$$

Note: We are not simplifying our model at all; we consider a model with a perfect fit.

While a perfect fit sounds good, it's not a helpful model (remember: prediction and inference). We want a smoother, simpler model, but a model that still describes the data almost as well. For GLM's, we use the **deviance** to compare simpler models to the saturated model.

Let  $\beta_{max}$  denote the parameter vector for the saturated (*maximal*) model and let  $\mathbf{b}_{max}$  be the MLE of  $\beta_{max}$ . (Note: Simpler model will be a function of  $\beta$ , so we use  $\beta_{max}$  to represent the parameters in the saturated model, even though they may not technically be what we think of as  $\beta$ 's.)

Note that the likelihood function for the saturated model evaluated at  $\mathbf{b}_{max}$ ,  $L(\mathbf{b}_{max}; \mathbf{y})$ , will be larger than any other likelihood function for these observations – same distribution and link function – because it provides the most complete description of the data.

Let  $L(\mathbf{b}; \mathbf{y})$  denote the maximum value of the likelihood function for the model of interest (i.e.,  $\mathbf{b}$  is the MLE for the model of interest).

To compare these models, we might conduct a hypothesis test comparing

$H_0$  : model of interest fits as well as the saturated model

$H_1$  : model of interest does not fit as well as the saturated model.

Then, the likelihood ratio,  $\lambda$ , is

$$\lambda = \frac{L(\mathbf{b}_{max}; \mathbf{y})}{L(\mathbf{b}; \mathbf{y})}.$$

In practice, we use the log likelihood ratio:

$$\log \lambda = \ell(\mathbf{b}_{max}; \mathbf{y}) - \ell(\mathbf{b}; \mathbf{y}).$$

We know that  $2 \log \lambda$  has a  $\chi^2$  distribution. Because of this,

$$D = 2 \log \lambda = 2 [\ell(\mathbf{b}_{max}; \mathbf{y}) - \ell(\mathbf{b}; \mathbf{y})],$$

#eq-deviance is a more commonly used statistic and is called the **deviance**.

Distribution of  $D$ :

If  $\mathbf{b}$  is the mle of  $\beta$  (so  $\mathbf{U}(\beta) = \mathbf{0}$ ),

$$\begin{aligned} \ell(\beta) - \ell(\mathbf{b}) &\approx -\frac{1}{2}(\beta - \mathbf{b})' \mathfrak{I}(\mathbf{b})(\beta - \mathbf{b}) \\ \Rightarrow 2[\ell(\beta) - \ell(\mathbf{b})] &\approx (\beta - \mathbf{b})' \mathfrak{I}(\mathbf{b})(\beta - \mathbf{b}) \sim \chi^2(k) \end{aligned}$$

Therefore,

$$\begin{aligned} D &= 2[\ell(\mathbf{b}_{max}; \mathbf{y}) - \ell(\mathbf{b}; \mathbf{y})] \text{ (add and subtract } \ell(\beta_{max}; \mathbf{y}) \text{ and } \ell(\beta; \mathbf{y})) \\ &= \underbrace{2[\ell(\mathbf{b}_{max}; \mathbf{y}) - \ell(\beta_{max}; \mathbf{y})]}_{\sim \chi^2(m)} - \underbrace{2[\ell(\mathbf{b}; \mathbf{y}) - \ell(\beta; \mathbf{y})]}_{\sim \chi^2(k)} \\ &\quad + \underbrace{2[\ell(\beta_{max}; \mathbf{y}) - \ell(\beta; \mathbf{y})]}_{\text{constant, } v} \end{aligned}$$

• **Result 3:**

$$D \sim \chi^2(m - k; v)$$

(for large  $n$ ; where  $v$  is the non-centrality parameter)

! Important

Do we want  $D$  to be large or small?

Recall,  $\lambda$  close to 1 indicates that  $L(\mathbf{b}; \mathbf{y})$  is doing “as well as” the saturated model (which we want), so  $D = 2 \log \lambda \approx 2 \log 1 = 0$ .

How small is small enough?  $D \sim \chi^2(m - k; \nu)$ . We assume if the model of interest is fitting close to the fit of the saturated model,  $\nu \approx 0$ . Therefore, if  $D$  is small enough, it should **NOT** be significant compared to a  $\chi^2(m - k)$ .

Note that:

$H_0$  : model of interest fits as well as the saturated model

$H_1$  : model of interest does not fit as well as the saturated model

So  $D \sim \chi^2(m - k)$  is the distribution of  $D$  under the null hypothesis.

Deviance for O-ring logistic model

$$\begin{aligned}\ell(\beta; \mathbf{y}) &= \sum_{i=1}^n y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i) \\ \ell(\mathbf{b}_{max}; \mathbf{y}) &= \sum_{i=1}^n y_i \log y_i + (1 - y_i) \log(1 - y_i) \\ \ell(\mathbf{b}; \mathbf{y}) &= \sum_{i=1}^n y_i \log \hat{\pi}_i + (1 - y_i) \log(1 - \hat{\pi}_i) \\ \Rightarrow D = 2[\ell(\mathbf{b}_{max}; \mathbf{y}) - \ell(\mathbf{b}; \mathbf{y})] &= 2 \left[ \sum_{i=1}^n y_i \log \frac{y_i}{\hat{\pi}_i} + (1 - y_i) \log \left( \frac{1 - y_i}{1 - \hat{\pi}_i} \right) \right] \\ &= 2 \left[ \sum_{\{i: y_i=1\}} \log \frac{1}{\hat{\pi}_i} + \sum_{\{i: y_i=0\}} \log \frac{1}{1 - \hat{\pi}_i} \right]\end{aligned}$$

First consider the simple model with only 1 parameter:

$$\begin{aligned}\log \frac{\pi_i}{1 - \pi_i} &= \beta_0, \\ \Rightarrow \pi_i &= \frac{e^{\beta_0}}{1 + e^{\beta_0}}\end{aligned}$$

Compute  $D$  in R:

```
#Simple Model
b0 <- glm(y~1, family=binomial)$coeff
phat <- rep(exp(b0)/(1+exp(b0)), length(temp))
D <- 2*(-sum(log(phat[which(y==1)])) - sum(log(1-phat[which(y==0)])))
print(D)
```

```
[1] 28.26715
```

Compare  $D$  to  $\chi^2(m - k)$ . In our example here,  $m = n = 23$  and  $k = 1$ .

```
qchisq(.95, length(temp) - 1)
```

```
[1] 33.92444
```

```
pchisq(D, length(temp)-1)
```

```
[1] 0.8330044
```

### Conclusion?

Now consider the model that includes temperature as a covariate:

$$\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + x_i \beta_1,$$

$$\Rightarrow \pi_i = \frac{e^{\beta_0 + x_i \beta_1}}{1 + e^{\beta_0 + x_i \beta_1}}$$

Compute  $D$  in R:

```
#Model with Temperature
b <- glm(y~temp, family=binomial)$coeff
phat <- exp(X%*%b)/(1+exp(X%*%b))
D <- 2*(-sum(log(phat[which(y==1)])) - sum(log(1-phat[which(y==0)])))
print(D)
```



```
[1] 20.31519
```

Compare  $D$  to  $\chi^2(m - k)$ . In this model,  $m = n = 23$  and  $k = 2$ .

```
qchisq(.95, length(temp) - 2)
```

```
[1] 32.67057
```

```
pchisq(D, length(temp)-2)
```

```
[1] 0.4986173
```

### Conclusion?

The second model appears to fit “better” compared to the saturated model, **but is it statistically significantly better?**

Note: compare the computed values of  $D$  above to the summary from the glm objects in R.

```
summary(glm(y~1, family=binomial))
```

Call:

```
glm(formula = y ~ 1, family = binomial)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | -0.8267  | 0.4532     | -1.824  | 0.0681   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 28.267 on 22 degrees of freedom  
Residual deviance: 28.267 on 22 degrees of freedom  
AIC: 30.267

Number of Fisher Scoring iterations: 4

```
summary(glm(y~temp, family=binomial))
```

Call:

```
glm(formula = y ~ temp, family = binomial)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | 15.0429  | 7.3786     | 2.039   | 0.0415 * |
| temp        | -0.2322  | 0.1082     | -2.145  | 0.0320 * |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 28.267 on 22 degrees of freedom  
Residual deviance: 20.315 on 21 degrees of freedom  
AIC: 24.315

Number of Fisher Scoring iterations: 5

**i** The same code, but in python:

```
# Simple model
model_simple = smf.glm(formula='y ~ 1', data=data, family=sm.families.Binomial)
b0 = model_simple.params['Intercept']
phat = np.repeat(np.exp(b0) / (1 + np.exp(b0)), len(data))
D = -2 * (np.sum(y * np.log(phat) + (1 - y) * np.log(1 - phat)))
print(D)
```

28.267152734293504

```
critical_value = stats.chi2.ppf(0.95, len(temp) - 1)
print(critical_value)
```

33.92443847144381

```
p_val = stats.chi2.cdf(D, len(temp) - 1)
print(p_val)
```

0.8330043557407494

```
# Model with temperature
model_temp = smf.glm(formula='y ~ temp', data=data, family=sm.families.Binomial)
b = np.array(model_temp.params)
phat = np.exp(X @ b) / (1 + np.exp(X @ b))
D = -2 * (np.sum(y * np.log(phat) + (1 - y) * np.log(1 - phat)))
print(D)
```

20.315192687866826

```
critical_value = stats.chi2.ppf(0.95, len(temp) - 2)
print(critical_value)
```

```
32.67057334091731
```

```
p_val = stats.chi2.cdf(D, len(temp) - 2)
print(p_val)
```

```
0.49861734348054004
```

```
print(model_simple.summary())
```

#### Generalized Linear Model Regression Results

```
=====
```

|                  |                  |                   |        |
|------------------|------------------|-------------------|--------|
| Dep. Variable:   | y                | No. Observations: | 23     |
| Model:           | GLM              | Df Residuals:     | 22     |
| Model Family:    | Binomial         | Df Model:         | 0      |
| Link Function:   | Logit            | Scale:            | 1.0000 |
| Method:          | IRLS             | Log-              |        |
| Likelihood:      | -14.134          |                   |        |
| Date:            | Tue, 18 Aug 2026 | Deviance:         | 28.267 |
| Time:            | 13:02:45         | Pearson chi2:     | 23.0   |
| No. Iterations:  | 4                | Pseudo R-         |        |
| squ. (CS):       | -4.441e-16       |                   |        |
| Covariance Type: | nonrobust        |                   |        |

```
=====
```

|           | coef    | std err | z      | P> z  | [0.025 | 0.975] |
|-----------|---------|---------|--------|-------|--------|--------|
| Intercept | -0.8267 | 0.453   | -1.824 | 0.068 | -      |        |
| 1.715     | 0.062   |         |        |       |        |        |

```
=====
```

```
print(model_temp.summary())
```

```

                                Generalized Linear Model Regression Results
=====
Dep. Variable:                  y      No. Observations:                  23
Model:                          GLM      Df Residuals:                    21
Model Family:                   Binomial  Df Model:                        1
Link Function:                  Logit     Scale:                          1.0000
Method:                         IRLS     Log-
Likelihood:                     -10.158
Date:                           Tue, 18 Aug 2026    Deviance:                       20.315
Time:                           13:02:45           Pearson chi2:                   23.2
No. Iterations:                  5      Pseudo R-
squ. (CS):                       0.2923
Covariance Type:                nonrobust
=====
                                coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept      15.0429         7.379         2.039     0.041     0.581     29.505
temp           -0.2322         0.108        -2.145     0.032     -
0.444          -0.020
=====

```

### 3.6.2 Comparing nested models using deviance

Consider testing significance of model coefficients by comparing the goodness of fit of two models. Here we will consider a method for comparing *nested* GLM's, or models with the same probability distribution, same link function, and the linear component of the simpler model is a special case of the linear component of the more general model.

Hypotheses:

$$\begin{aligned}H_0 : \beta &= \beta_0 = (\beta_1, \dots, \beta_q)' \\H_1 : \beta &= \beta_1 = (\beta_1, \dots, \beta_q, \beta_{q+1}, \dots, \beta_k)'\end{aligned}$$

Here,  $q < k < n$ .

Let  $D_0$  represent the deviance of the null model and  $D_1$  the deviance of the alternative model. To compare these two models, use the **difference** of the deviance statistics:

$$\begin{aligned}\Delta D &= D_0 - D_1 \\&= 2[\ell(\mathbf{b}_{max}; \mathbf{y}) - \ell(\mathbf{b}_0; \mathbf{y})] - 2[\ell(\mathbf{b}_{max}; \mathbf{y}) - \ell(\mathbf{b}_1; \mathbf{y})] \\&= 2[\ell(\mathbf{b}_1; \mathbf{y}) - \ell(\mathbf{b}_0; \mathbf{y})]\end{aligned}$$

Since we are adding two  $\chi^2$  random variables,

- **Result 4:**  $\Delta D \sim \chi^2(k - q)$ 
  - Note 1: If data are not normal, the distributions of  $D_0$  and  $D_1$  hold only asymptotically (i.e., large  $n$ )
  - Note 2: The sampling distribution of the difference in deviances is usually better approximated by the  $\chi^2$  distribution than is the sampling distribution of a single deviance.

#### ! Important

If  $\Delta D$  is consistent with  $\chi^2(k - q)$ , then we fail to reject  $H_0$  and use the null model (since simpler models are better).

However, if the alternative model describes the data well, and the null model does not,  $\Delta D$  will be bigger than what is expected from a  $\chi^2(k - q)$ .

O-ring simple model vs. model *with temperature*

$$H_0 : \beta = \beta_0 \quad (\text{intercept only})$$

$$H_1 : \beta = (\beta_0, \beta_1)' \quad (\text{intercept and temperature coefficient})$$

We found  $D_0 = 28.267$ , and  $D_1 = 20.315$ , therefore,

$$\Delta D = D_0 - D_1 = 7.952.$$

Compare to  $\chi^2(k - q)$ , where  $k = 2$  and  $q = 1$ :

```
qchisq(.95, 2-1)
```

```
[1] 3.841459
```

```
pchisq(7.952, 2-1)
```

```
[1] 0.9951966
```

Conclusion?

O-ring simple model vs. model *with random noise covariate*

Suppose we include a random noise term in the model instead of temperature.

Let

$$x_i \sim \mathcal{N}(0, 1), \text{ for } i = 1, \dots, n.$$

Then,

$$H_0 : \beta = \beta_0 \quad (\text{intercept only})$$

$$H_1 : \beta = (\beta_0, \beta_1)' \quad (\text{intercept and coefficient for random noise covariate})$$

$D_0 = 28.267$ , but  $D_1$  must be recomputed using the new covariate.

```
set.seed(29)
x <- rnorm(length(temp))
b <- glm(y~x, family=binomial)$coeff
phat <- exp(b[1] + b[2]*x)/(1+exp(b[1] + b[2]*x))
D <- 2*(-sum(log(phat[which(y==1)])) - sum(log(1-phat[which(y==0)])))
print(D)
```

```
[1] 27.84686
```

```
qchisq(.95, length(x) - 2)
```

```
[1] 32.67057
```

```
pchisq(D, length(x)-2)
```

```
[1] 0.8554516
```

$D_1 = 27.8468555$ , therefore,

$$\Delta D = D_0 - D_1 = 0.42.$$

```
pchisq(0.42, 1)
```

```
[1] 0.483063
```

Conclusion?



**i** The same code, but in python:

```
print(stats.chi2.ppf(0.95, 2-1))
```

```
3.8414588206941205
```

```
print(stats.chi2.cdf(7.952, 2-1))
```

```
0.995196573531767
```

```
# NOTE: The resulting values will differ slightly from R due to RNG
np.random.seed(77)
x = np.random.normal(size=len(data))
data['x'] = x
model_x = smf.glm(formula='y ~ x', data=data, family=sm.families.Binomial()).f
b = np.array(model_x.params)
phat = np.exp(b[0] + b[1] * x) / (1 + np.exp(b[0] + b[1] * x))
D = -2 * (np.sum(y * np.log(phat) + (1 - y) * np.log(1 - phat)))
print(D)
```

```
27.95131510303315
```

```
print(stats.chi2.ppf(0.95, len(x) - 2))
```

```
32.67057334091731
```

```
print(stats.chi2.cdf(D, len(x) - 2))
```

```
0.8584624852738254
```

```
print(stats.chi2.cdf(0.42, 1))
```

```
0.48306295573551006
```

### 3.6.3 Assessing model fit using the generalized Pearson $X^2$ statistic

$$X^2 = \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\text{Var}(\hat{\mu}_i)},$$

where  $\text{Var}(\hat{\mu}_i)$  is the *estimated* variance of  $Y_i$  for the distribution under consideration.

- $X^2 \sim \chi^2(n - k)$  (for large  $n$ ).
- $X^2$  is more interpretable (based on standardized squared residuals), but not additive like deviance.

### 3.6.4 Assessing model fit using predicted values

Often we examine models based on their predictability, such as mean squared error, cross validation, etc. The generalized Pearson  $X^2$  statistic is one version of a mean square error. We might also want to look at confidence interval coverage or width.

Let's first consider,  $\hat{\mu}_i$ . Note that

$$\hat{\mu}_i = g^{-1}(\hat{\eta}_i) = g^{-1}(\mathbf{x}_i' \mathbf{b}),$$

where, as before,  $\mathbf{b}$  is the mle of  $\beta$ . We can get these predicted values from a glm object in R using the “fitted” or “predict” functions.

```
#Several ways to get mu-hat-i from a glm object:
preds <- data.frame(
  muhat=fitted(mod.glm),
  muhat2=predict(mod.glm, type="response"),
  etahat=predict(mod.glm), #predict type="link" is default
  coefterms=predict(mod.glm, type='terms') #returns x1*beta1, x2*beta2, etc.
)
head(preds, n=3)
```

|   | muhat     | muhat2    | etahat   | temp     |
|---|-----------|-----------|----------|----------|
| 1 | 0.9392478 | 0.9392478 | 2.738276 | 3.845826 |
| 2 | 0.8593166 | 0.8593166 | 1.809625 | 2.917175 |
| 3 | 0.8288448 | 0.8288448 | 1.577462 | 2.685013 |

What about a standard error for  $\hat{\mu}_i$ ? The “predict” function in R offers the option to return the standard error. (But where do these standard errors come from?)

```
preds.se <- data.frame(
  muhat.se = predict(mod.glm, type='response', se.fit=T)$se,
  etahat.se = predict(mod.glm, se.fit=T)$se,
  coefterms.se = predict(mod.glm, type='terms', se.fit=T)$se
)
head(preds.se, n=3)
```

|   | muhat.se   | etahat.se | temp     |
|---|------------|-----------|----------|
| 1 | 0.09775842 | 1.713216  | 1.792960 |
| 2 | 0.15848696 | 1.310984  | 1.360014 |
| 3 | 0.17218144 | 1.213733  | 1.251778 |

To get a confidence interval for  $\mu_i$ , we can consider two approaches:

- Since we have a sampling distribution for  $\beta$ , we can obtain a confidence interval for  $\mathbf{x}'_i\beta$  and translate this to a confidence interval for  $\mu_i$  (preserves the range of  $\mu_i$ ); or,

- Identify a sampling distribution for  $\hat{\mu}_i$  and compute a confidence interval using this sampling distribution.

### **i** Note

We know

$$(\mathbf{b} - \beta) \xrightarrow{D} \mathcal{N}(0, \mathfrak{J}).$$

Therefore,

$$(\mathbf{x}'_i \mathbf{b} - \mathbf{x}'_i \beta) = (\hat{\eta}_i - \eta_i) \xrightarrow{D} \mathcal{N}(0, \mathbf{x}'_i \mathfrak{J} \mathbf{x}_i).$$

By the delta method,

$$(g^{-1}(\hat{\eta}_i) - g^{-1}(\mu_i)) = (g^{-1}(\hat{\eta}_i) - \mu_i) \xrightarrow{D} \mathcal{N}\left(0, \left(\frac{d}{d\eta_i} g^{-1}(\eta_i)\right)^2 \mathbf{x}'_i \mathfrak{J} \mathbf{x}_i\right)$$

where,

$$\left(\frac{d}{d\eta_i} g^{-1}(\eta_i)\right)^2 = \left(\frac{d}{d\mu_i} g(\mu_i)\right)^{-2} = \frac{1}{(g'(\mu_i))^2}.$$

Thus, a sampling distribution for  $\hat{\mu}_i$  is

$$\mathcal{N}\left(\mu_i, \frac{\mathbf{x}'_i \mathfrak{J} \mathbf{x}_i}{(g'(\mu_i))^2}\right)$$

for large  $n$ . In practice, we plug in  $\hat{\mu}_i$  into  $g'(\mu_i)$ .

Using the predicted values and standard errors in the output above, compute a confidence interval for  $\mu_1$ , by

1. finding a confidence interval for  $\eta_1$  and transforming the bounds of that interval back to  $\mu_1$ .
2. using the standard error and sampling distribution of  $\mu_1$ . Check to make sure the standard error of  $\mu_1$  really is equal to the standard error of  $\eta_1$  divided by  $g'(\mu_1)$ . (Recall that for the logistic regression model,  $g'(\mu_i) = \frac{1}{\pi_i(1-\pi_i)}$ .)

**i** The same code, but in python:

```
preds_final = pd.DataFrame({
    'muhat': model.get_prediction(linear=False).predicted,
    'muhat2': model.predict(data),
    'etahat': model.get_prediction(linear=True).predicted,
    'temp': (temp - np.mean(temp)) * model.params['temp']
})

print(preds_final.head(3))
```

|   | muhat    | muhat2   | etahat   | temp     |
|---|----------|----------|----------|----------|
| 0 | 0.939248 | 0.939248 | 0.939248 | 3.845826 |
| 1 | 0.859317 | 0.859317 | 0.859317 | 2.917175 |
| 2 | 0.828845 | 0.828845 | 0.828845 | 2.685013 |

```
coefterms_se = np.abs(temp - np.mean(temp)) * model.bse['temp']
preds_se_final = pd.DataFrame({
    'muhat_se': model.get_prediction(linear=False).se,
    'etahat_se': model.get_prediction(linear=True).se,
    'temp': np.abs(temp - np.mean(temp)) * model.bse['temp']
})

print(preds_se_final.head(3))
```

|   | muhat_se | etahat_se | temp     |
|---|----------|-----------|----------|
| 0 | 0.097758 | 0.097758  | 1.792960 |
| 1 | 0.158487 | 0.158487  | 1.360014 |
| 2 | 0.172181 | 0.172181  | 1.251778 |

### 3.6.5 Assessing model fit using residuals

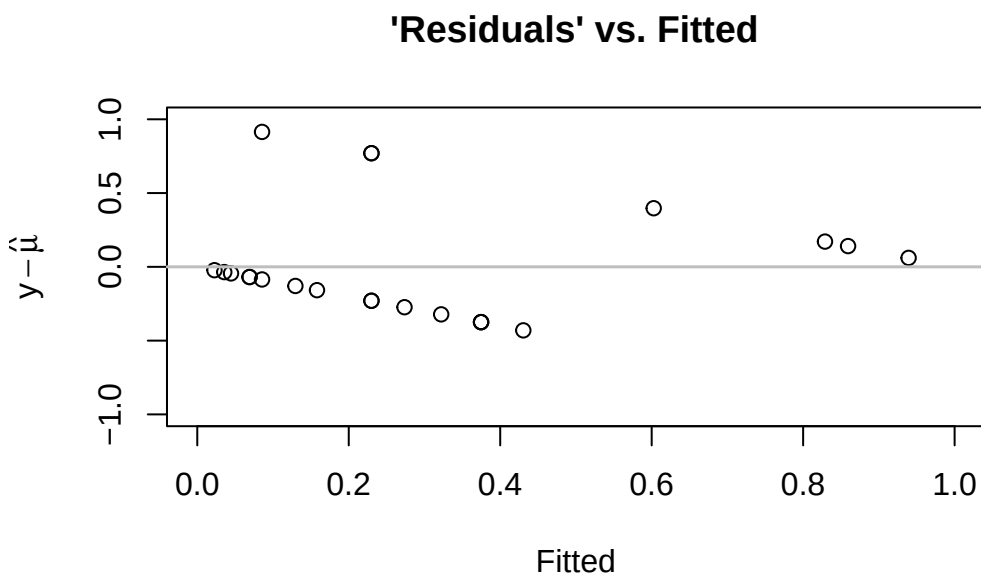
In the normal linear model, we used the residuals,  $r_i$ , to determine how well the model fit and to check model assumptions. We compute

$$r_i = y_i - \hat{y}_i = y_i - \widehat{E(Y_i)} = y_i - \hat{\mu}_i,$$

and look at residual plots.

We also want to look at residuals for GLM's, but is  $y_i - \hat{\mu}_i$  meaningful?

```
#Now let's look at a plot of y - muhat
plot(preds$muhat, y - preds$muhat, xlab="Fitted", ylab=expression(y - hat(mu)),
     abline(h=0, col='gray', lwd=1.5))
```



For binary data, there isn't much you can do (consider binning to get binomial residuals?). However, this extreme case shows the issues that can arise when we look at regular residuals. To combat some of the challenges, various scaled residuals have been introduced.

#### 1. Pearson Residuals:

$$r_{p_i} = \frac{y_i - \hat{\mu}_i}{\sqrt{\text{Var}(Y_i)}}$$

```

muhat <- fitted(mod.glm)
pearson <- data.frame(
  glmresid = resid(mod.glm, type='pearson'),
  byhandresid = (y - muhat)/sqrt(muhat*(1-muhat))
)
head(pearson, n=3)

```

```

      glmresid byhandresid
1 0.2543261    0.2543261
2 0.4046177    0.4046177
3 0.4544210    0.4544210

```

- $r_{p_i}$  are raw scaled residuals
- distribution of  $r_{p_i}$  for non-normal data is often skewed
- notice that  $X^2 = \sum_{i=1}^n r_{p_i}^2$

2. Anscombe residuals:

Idea: transform Pearson residuals to be less skewed (McCullagh and Nelder, pg. 38)

### 3. Deviance Residuals:

- Recall:

$$\begin{aligned}
 D &= 2[\ell(\mathbf{b}_{max}; \mathbf{y}) - \ell(\mathbf{b}; \mathbf{y})] \\
 &= 2\left[\sum_{i=1}^N \ell_i(\mathbf{b}_{max}; y_i) - \ell_i(\mathbf{b}; y_i)\right] \\
 &= \sum_{i=1}^N 2[\ell_i(\mathbf{b}_{max}; y_i) - \ell_i(\mathbf{b}; y_i)] \\
 &= \sum_{i=1}^N (d_i^+)^2
 \end{aligned}$$

- The deviance residual is then,

$$d_i = \text{sign}(y_i - \hat{\mu}_i) d_i^+$$

- The deviance residuals tend to be more symmetric
- Can also standardize:

$$r_{d_i} = \frac{d_i}{\sqrt{1 - h_i}},$$

where  $h_i$  is the leverage (diagonal of the hat matrix)... Recall:  $\mathbf{H} = \mathbf{W}^{1/2}\mathbf{X}(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}^{1/2}$

```
muhat <- fitted(mod.glm)
diplus <- sqrt(2*(dbinom(y, 1, y, log=T) - dbinom(y, 1, muhat, log=T)))
deviance <- data.frame(
  glmresid = resid(mod.glm), #type='deviance' is the default
  byhandresid = sign(y - muhat)*diplus
)
head(deviance, n=3)
```

|   | glmresid  | byhandresid |
|---|-----------|-------------|
| 1 | 0.3540506 | 0.3540506   |
| 2 | 0.5506685 | 0.5506685   |
| 3 | 0.6127353 | 0.6127353   |

#### 4. Working Residuals:

$$r_{w_i} = (y_i - \hat{\mu}_i)g'(\hat{\mu}_i)$$

- Residuals at final iteration of IWLS algorithm

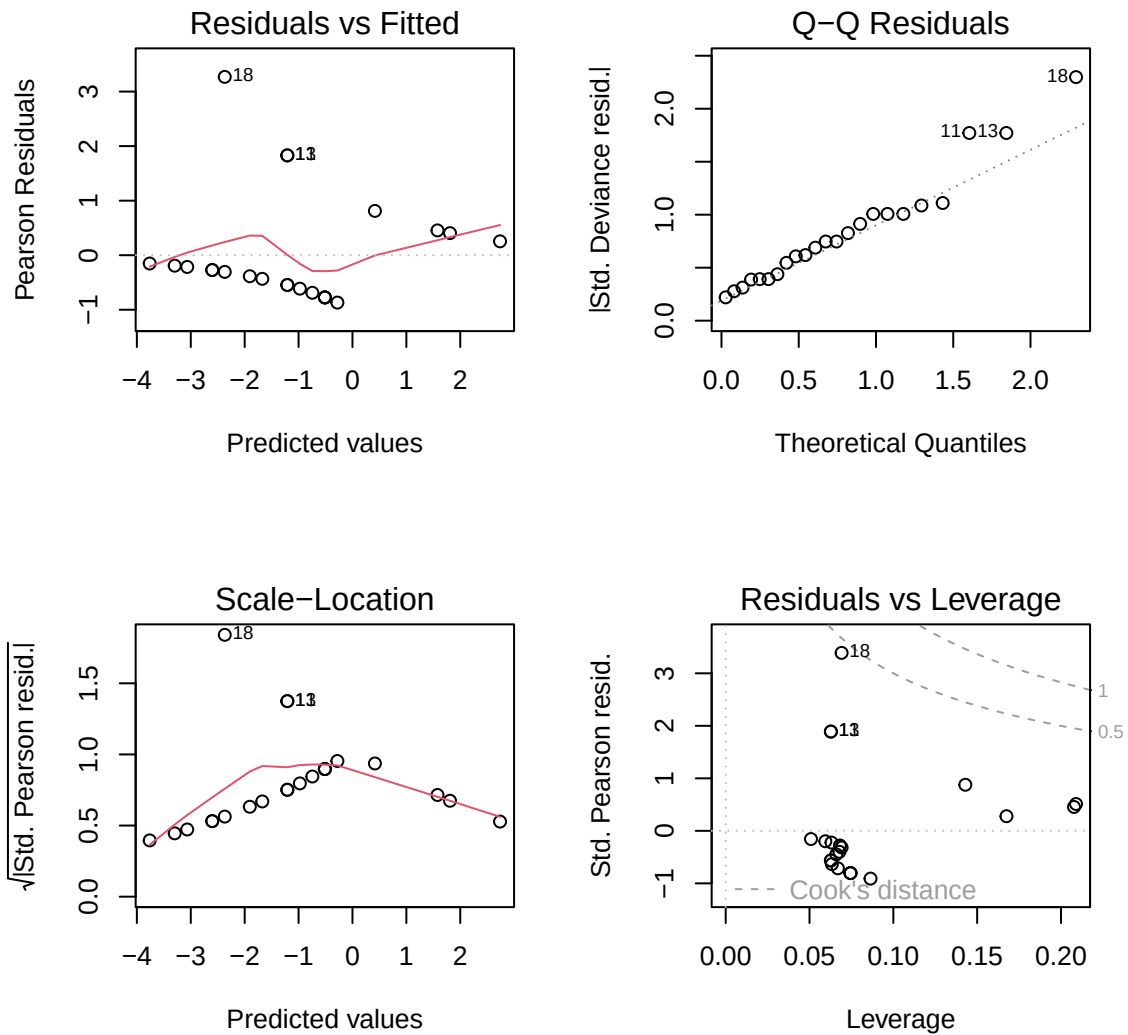
```
gprimemuhat <- 1/(muhat*(1-muhat))
working <- data.frame(
  glmresid1 = mod.glm$residuals, #residuals returned by the glm object
  glmresid2 = resid(mod.glm, type='working'),
  byhandresid = (y-muhat)*gprimemuhat
)
head(working, n=3)
```



|   | glmresid1 | glmresid2 | byhandresid |
|---|-----------|-----------|-------------|
| 1 | 1.064682  | 1.064682  | 1.064682    |
| 2 | 1.163715  | 1.163715  | 1.163715    |
| 3 | 1.206498  | 1.206498  | 1.206498    |

Plotting the glm object in R:

```
par(mfrow=c(2,2))
plot(mod.glm)
```



```

#note that there are 6 plots offered by plotting the lm/glm object. By default

par(mfrow=c(2,2))

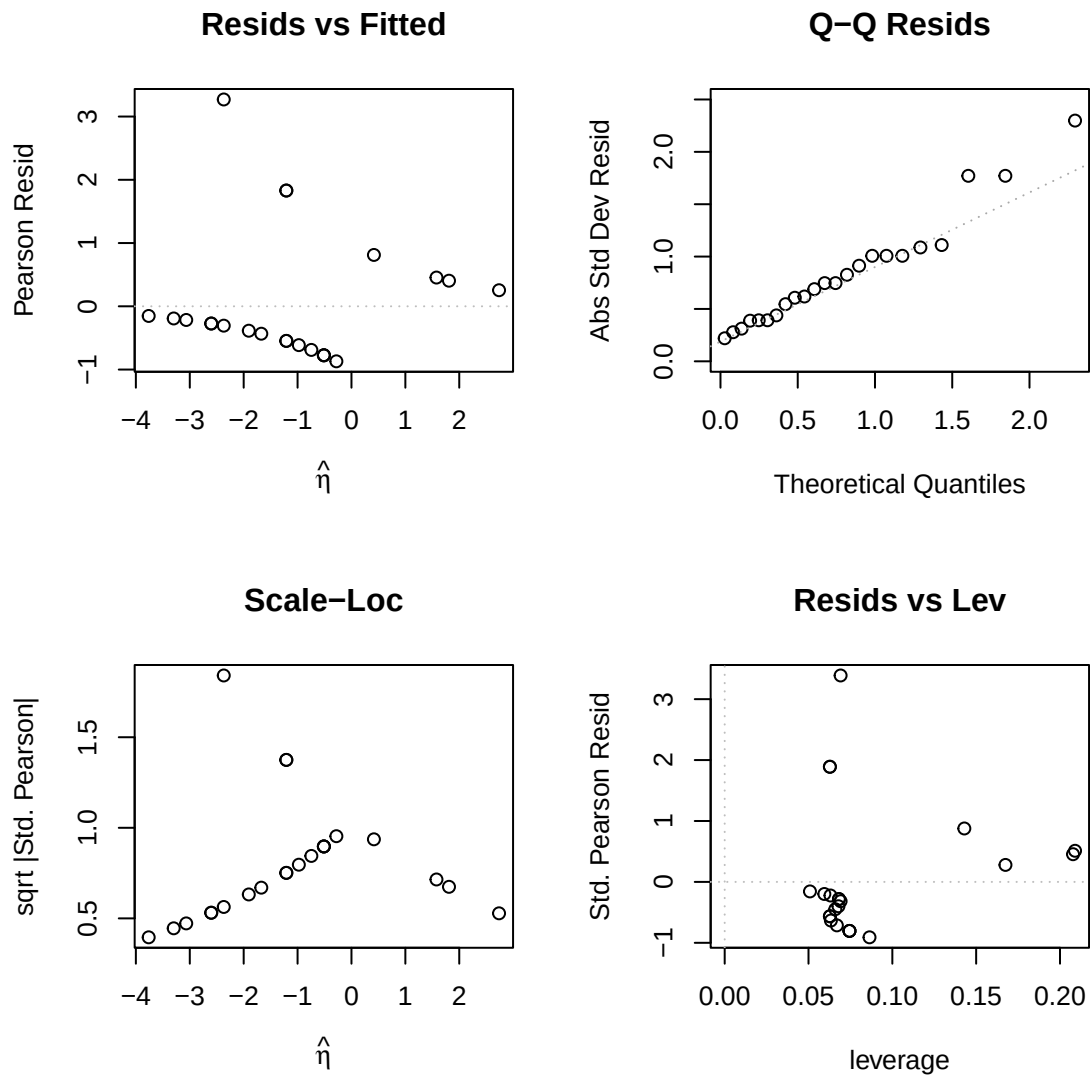
#resids vs fitted
plot(predict(mod.glm, type='link'), residuals(mod.glm, type='pearson'), ylab="P
abline(h=0, lty=3, col='gray')

#qqplot of standardized resids
#H <- sqrt(W)%*%X%*%solve(t(X)%*%W%*%X)%*%t(X)%*%sqrt(W) #By hand hat values
#h <- diag(H) #leverages
h <- hatvalues(mod.glm) #leverages
sdev <- residuals(mod.glm, type='deviance')/sqrt(1-h)
#Note: plot(glm object) creates a half normal qq plot;
# you can make just a regular qqnorm plot on the standardized resids;
# I've hacked a half-normal qqplot here
sortedsdev <- sort(abs(sdev))
probsfornorm <- ppoints(2*length(sdev))[length(sdev)+(1:length(sdev))] #(length
normqs <- qnorm(probsfornorm)
plot(normqs, sortedsdev, ylim=c(0, 2.5), xlab="Theoretical Quantiles", ylab="Ab
abline(lsfit(qnorm(.5+c(.25, .75)*.5), quantile(abs(sdev), prob=c(.25, .75))),

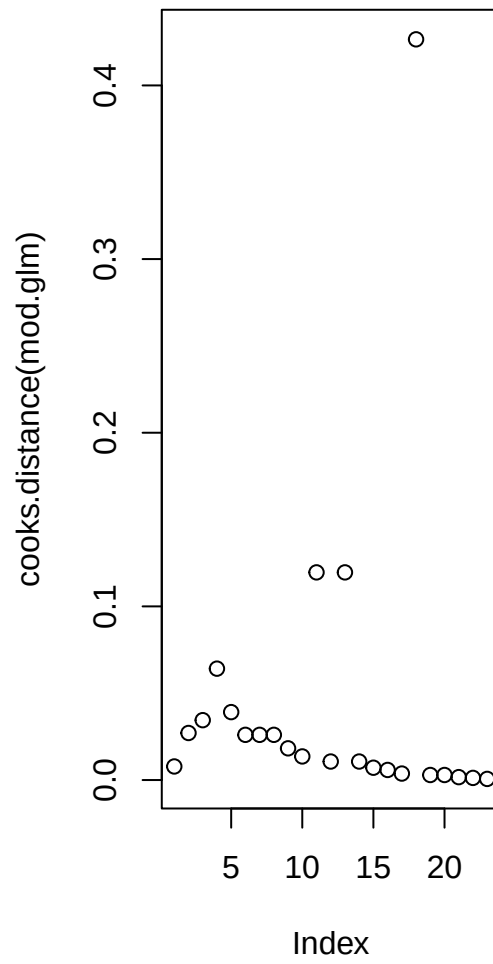
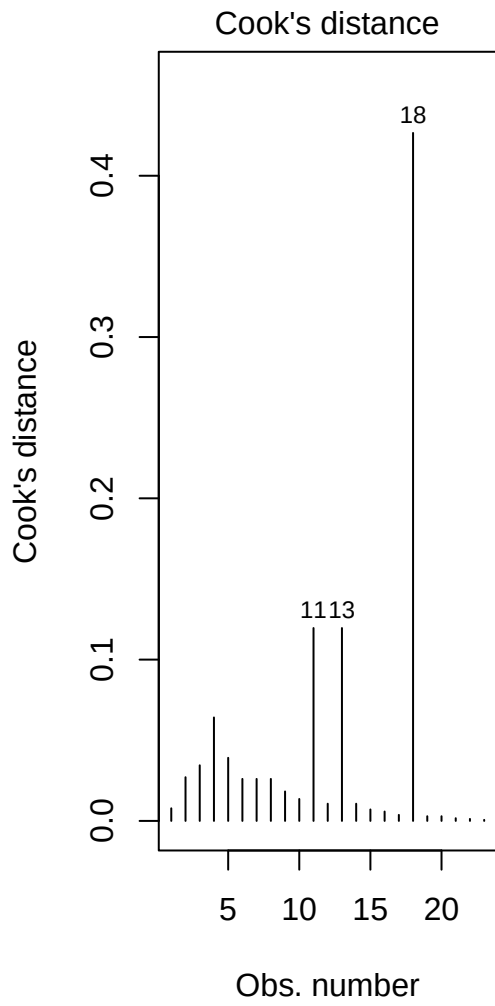
#scale-location
spears <- residuals(mod.glm, type='pearson')/sqrt(1-h) #standardized pearson re
plot(predict(mod.glm, type='link'), sqrt(abs(spears)), ylab="sqrt |Std. Pearson

#resids vs leverage
plot(h, spears, xlim=c(0, max(h)), xlab="leverage", ylab="Std. Pearson Resid",
abline(v=0, lty=3, col='gray')
abline(h=0, lty=3, col='gray')

```



```
par(mfrow=c(1,2))
#plot(mod, which=4) is not a default plot:
plot(mod.glm, which=4)
#Cook's distance for other (non-default) plots
plot(cooks.distance(mod.glm))
```



**i** The same code, but in python:

```
muhat = model.predict()
response_residuals = y - muhat

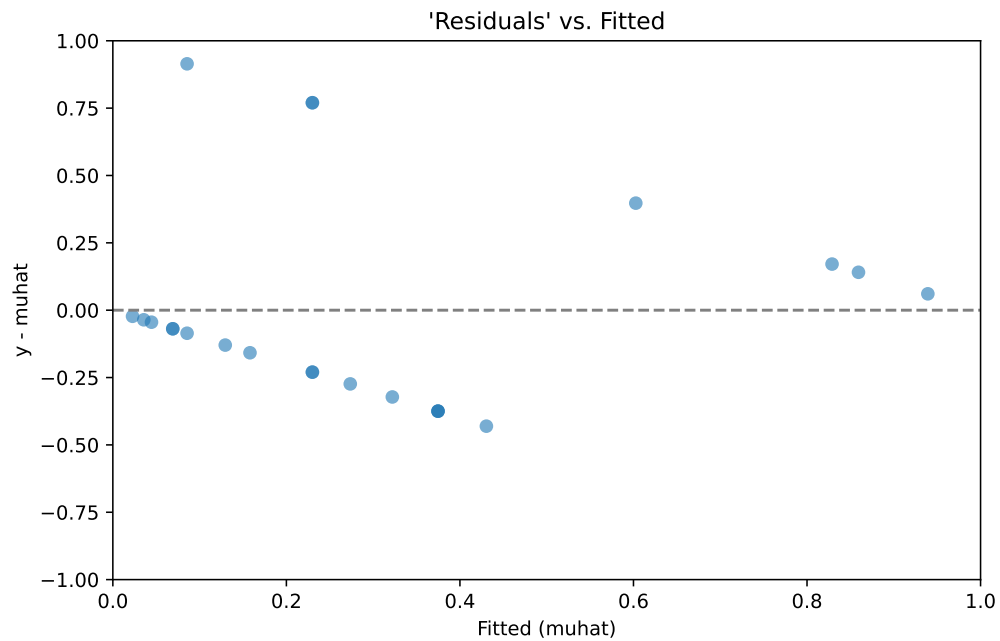
plt.figure(figsize=(8, 5))
plt.scatter(muhat, response_residuals, alpha=0.6)
plt.axhline(0, color='gray', linestyle='--', linewidth=1.5)
plt.xlabel("Fitted (muhat)")
plt.ylabel("y - muhat")
plt.title("'Residuals' vs. Fitted")
plt.ylim(-1, 1)
```

(-1.0, 1.0)

```
plt.xlim(0, 1)
```

(0.0, 1.0)

```
plt.show()
```



```
pearson = pd.DataFrame({
    'glmresid': model.resid_pearson,
    'byhandresid': (y - muhat) / np.sqrt(muhat * (1 - muhat))
})
print("Pearson Residuals:")
```

Pearson Residuals:

```
print(pearson.head(3))
```

|   | glmresid | byhandresid |
|---|----------|-------------|
| 0 | 0.254326 | 0.254326    |
| 1 | 0.404618 | 0.404618    |
| 2 | 0.454421 | 0.454421    |

```

ll_sat = stats.binom.logpmf(y, 1, y)
ll_mod = stats.binom.logpmf(y, 1, muhat)
diplus = np.sqrt(2 * (ll_sat - ll_mod))

deviance = pd.DataFrame({
    'glmresid': model.resid_deviance,
    'byhandresid': np.sign(y - muhat) * diplus
})
print(deviance.head(3))

```

```

    glmresid  byhandresid
0  0.354051    0.354051
1  0.550668    0.550668
2  0.612735    0.612735

```

```

g_prime_muhat = 1 / (muhat * (1 - muhat))

working = pd.DataFrame({
    'glmresid': model.resid_working,
    'byhandresid': (y - muhat) * g_prime_muhat
})

print("\nWorking Residuals:")

```

Working Residuals:

```

print(working.head(3))

```

```

    glmresid  byhandresid

```

|   |          |          |
|---|----------|----------|
| 0 | 1.064682 | 1.064682 |
| 1 | 1.163715 | 1.163715 |
| 2 | 1.206498 | 1.206498 |



```
#####
### Plotting ###
#####

# NOTE: There isn't a "one and done" function for glm diagnostics in python,
# each one needs to be calculated individually as shown below

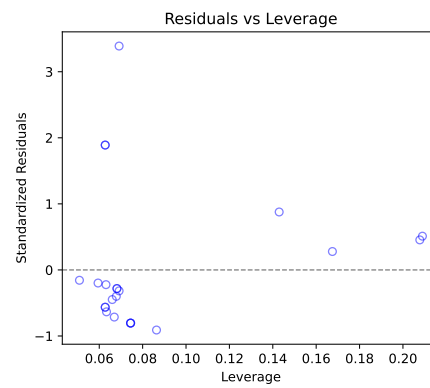
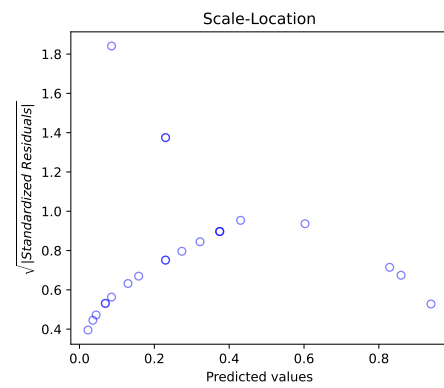
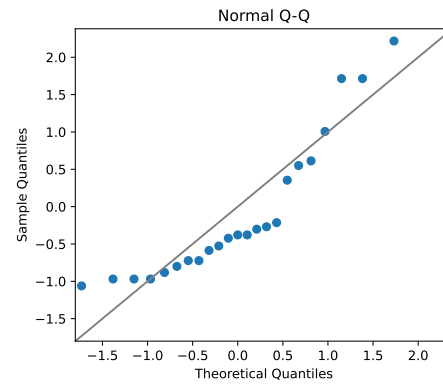
# Set up a 2x2 grid
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
fig.subplots_adjust(hspace=0.3, wspace=0.3)

# 1. Residuals vs Fitted
axes[0, 0].scatter(model.predict(),
                   model.resid_deviance,
                   alpha=0.5,
                   facecolors='none',
                   edgecolors='b')
axes[0, 0].axhline(0, color='grey', linestyle='--', lw=1)
axes[0, 0].set_title('Residuals vs Fitted')
axes[0, 0].set_xlabel('Predicted values')
axes[0, 0].set_ylabel('Deviance Residuals')

# 2. Normal Q-Q
sm.qqplot(model.resid_deviance,
           line='45',
           ax=axes[0, 1])
axes[0, 1].get_lines()[-1].set_color('grey')
axes[0, 1].set_title('Normal Q-Q')

# 3. Scale-Location
axes[1, 0].scatter(model.predict(),
                   np.sqrt(np.abs(model.get_influence().resid_studentized))),
                   alpha=0.5,
                   facecolors='none',
                   edgecolors='b')
axes[1, 0].set_title('Scale-Location')
axes[1, 0].set_xlabel('Predicted values')
axes[1, 0].set_ylabel(r'$\sqrt{|\text{Standardized Residuals}|}$')

# 4. Residuals vs Leverage
axes[1, 1].scatter(model.get_influence().hat_matrix_diag,
                   model.get_influence().resid_studentized,
                   alpha=0.5,
                   facecolors='none',
                   edgecolors='b')
```



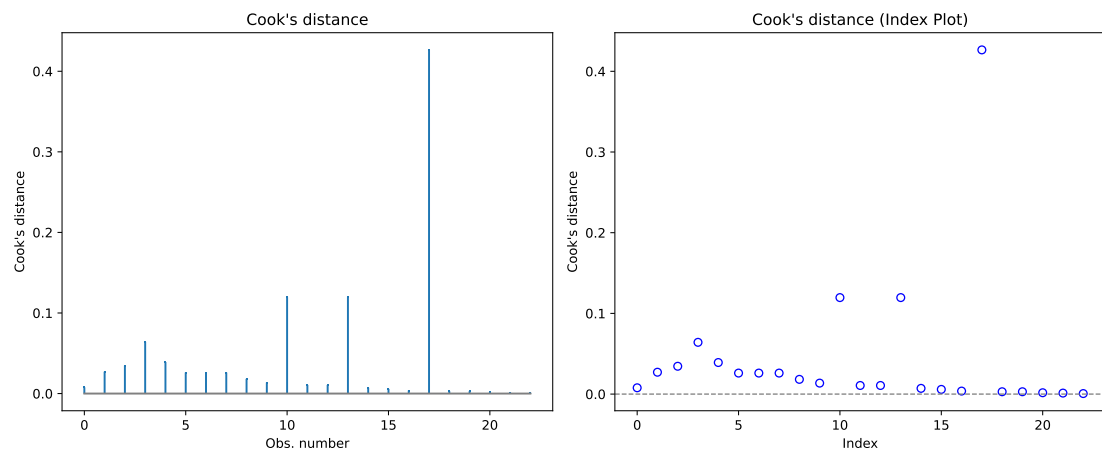
```
#####
### Other plots ###
#####

# Set up a 1x2 grid
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# 1. Cook's Distance Plot
axes[0].stem(np.arange(len(model.get_influence().cooks_distance[0])),
             model.get_influence().cooks_distance[0],
             markerfmt=",",
             basefmt = "grey")
axes[0].set_title("Cook's distance")
axes[0].set_xlabel("Obs. number")
axes[0].set_ylabel("Cook's distance")

# 2. Index Plot
axes[1].scatter(np.arange(len(model.get_influence().cooks_distance[0])),
               model.get_influence().cooks_distance[0],
               facecolors='none',
               edgecolors='b')
axes[1].axhline(0, color='grey', linestyle='--', lw=1)
axes[1].set_title("Cook's distance (Index Plot)")
axes[1].set_xlabel("Index")
axes[1].set_ylabel("Cook's distance")

plt.tight_layout()
plt.show()
```



## 3.7 Mini Project 1

Read the following seminal paper that introduces the theory of GLM's (available through BYU library).

Nelder, J. A. and Wedderburn, R. W. M. (1972) "Generalized Linear Models," *Journal of the Royal Statistical Society, Series A* 3(135), 370–384

Answer the following questions:

1. Identify three places in the paper where you recognize something we have discussed in class. For each, give the page number and write 1-2 sentences explaining the connection.
2. Identify one idea, computational strategy, presentation choice, or concern in the paper that seems dated or less important to a statistician working today. In 2-3 sentences, explain what has changed and why you think it matters less now.
3. Identify one place where some piece of statistical theory you have learned helps you understand what Nelder and Wedderburn are doing. Explain the connection in 2-3 sentences.
4. Identify one passage, equation, claim, or idea that you don't fully understand. Give the page number and briefly describe what is confusing.
5. Compare notes with at least two classmates. Discuss your answers to Questions 1–4. Record:
  - one connection that someone else noticed that you found particularly useful;
  - one "dated" feature one which your group disagreed or had different interpretations;
  - one question about the paper that your discussion helped resolve or that your group still cannot answer.
6. You will read many more technical papers during your career. Based on this experience, identify one strategy you would use the next time you encounter a methodology paper that you find difficult to read.

## 3.8 Homework Assignment

Table 3.1: Numbers of cases of AIDS in Australia for successive quarters from 1984 to 1985.

| Year | 1   | 2   | 3   | 4   |
|------|-----|-----|-----|-----|
| 1984 | 1   | 6   | 16  | 23  |
| 1985 | 27  | 39  | 31  | 30  |
| 1986 | 43  | 51  | 63  | 70  |
| 1987 | 88  | 97  | 91  | 104 |
| 1988 | 110 | 113 | 149 | 159 |

1. Table 3.1 provides data on number of cases of AIDS in Australia by the date of diagnosis for successive 3-month periods from 1984 to 1988. (Data from National Centre for HIV Epidemiology and Clinical Research 1994, as reported by Dobson and Barnett (2018).) In this early phase of the epidemic, the numbers of cases seemed to be increasing exponentially.

- a. Plot the number of cases  $y_i$  against time period  $i$  ( $i = 1, \dots, 20$ ).
- b. A possible model is the Poisson distribution with parameter  $\lambda_i = i^\theta$ , or equivalently,

$$\log \lambda_i = \theta \log i.$$

Plot  $\log y_i$  against  $\log i$  to examine this model.

- c. Fit a GLM to these data using the Poisson distribution, the log-link function, and the equation

$$g(\lambda_i) = \log \lambda_i = \beta_1 + \beta_2 x_i,$$

where  $x_i = \log i$ . Do this from first principles, working out expressions for the weight matrix  $W$  and other terms needed for the iterative equation

$$\mathbf{X}'\mathbf{W}\mathbf{X}\mathbf{b}^{(t)} = \mathbf{X}'\mathbf{W}\mathbf{z}$$

(and using a computer to carry out the computation).

- d. Fit the model described in (c) using statistical software which can perform Poisson regression. Compare the results with those obtained in (c).

- e. Compute and report the covariance matrix of  $\mathbf{b}$ . Is  $\log i$  significant in the model? (Support your answer.) Report 95% confidence intervals for  $\beta$ .
- f. Plot R's four default residual plots from the model using first principles (i.e., find the standardized Pearson residuals yourself, not using the "residuals" argument in R). You can compare your plots to those in R to check your plots.
- g. Find a 95% confidence interval for the average number of AIDS cases for the 21st time period (i.e., one time period outside the observed data). Do this (i) using the covariance of  $\mathbf{b}$  you computed in (e), and (ii) using the "predict" function in R (making sure to use the correct standard error).

2. Generate 50 normally distributed response variables using:

- (i) a linear model with a given covariate (which you may randomly generate or just give a set of values to) as the mean, and a constant variance (recommended: set  $\sigma^2 = 1$ ), i.e.,

$$Y_i \sim N(\beta_0 + x_i\beta_1, \sigma^2), \text{ for } i = 1, \dots, 50;$$

- (ii) a normal distribution with constant mean and variance (again, recommended: set  $\sigma^2 = 1$ ), i.e.,

$$Y_i \sim N(\beta_0, \sigma^2), \text{ for } i = 1, \dots, 50.$$

- a. For both sets of generated data, fit an intercept-only model and a model with the covariate.
- b. Write a paragraph stating how the four models compare to the corresponding standard models.
- c. For (a) and (b), compare the intercept-only model and the covariate model using the deviance information criterion.

3. A study examines whether a particular treatment is successful. Let  $Y_i$  be a continuous positive response variable. Suppose an exponential regression model is fit using treatment  $x_i$ :

$$Y_i \overset{ind}{\sim} \text{Exponential}(\lambda_i), E(Y_i) = \mu_i = 1/\lambda_i$$

$$\log(\mu_i) = \beta_0 + \beta_1 x_i.$$

The fitted model gives

$$\mathbf{b} = (-2.4, 0.8)'$$
$$\hat{\text{Cov}}(\mathbf{b}) = \begin{pmatrix} 0.34 & -0.15 \\ -0.15 & 0.10 \end{pmatrix}.$$

- a. At a treatment of  $x^* = 2$ , find  $\hat{\eta}^*$  and the estimated corresponding mean,  $\hat{\mu}^*$ .
- b. Find the standard error of  $\hat{\eta}^*$ . Show how you obtained it from the estimated covariance matrix of  $\mathbf{b}$ .
- c. Construct a 95 percent confidence interval for  $\eta^*$ . Then, transform the end points using  $g^{-1}(\cdot)$  to obtain a confidence interval for  $\mu^*$ .
- d. Use the delta method to approximate the standard error of  $\hat{\mu}^*$ . Use this standard error to construct a second 95 percent confidence interval for  $\mu$ .
- e. Compare the two confidence intervals from (c) and (d). Explain why they are not identical. Why is the confidence interval found in (c) preferable?



## 4 GLM's for Binary Data

### 4.1 Inference for a Proportion

Recall from Section 3.6.4 we discussed two ways to provide confidence intervals for  $\hat{\mu}_i$ : one using a transformation of a confidence interval on  $\eta_i$  or using the delta-method to get a standard error for  $\mu_i$  and computing the confidence interval “as usual.” Let’s step back from the GLM and just think about how we can get confidence intervals on proportions from a single sample of a binomial random variable (e.g., Stat 121).

#### **i** Note

##### *Confidence intervals for $\pi$*

Let  $Y \sim \text{Bin}(m, \pi)$  and let the sample proportion be  $\hat{\pi} = y/m$ . Then,  $\hat{\pi}$  is an unbiased estimator of  $\pi$  (proven in Stat 240) and

$$\text{Var}(\hat{\pi}) = \frac{\pi(1 - \pi)}{m}.$$

For large  $m$ ,  $\hat{\pi}$  is normally distributed. Thus, a  $100(1 - \alpha)\%$  confidence interval for  $\pi$  is

$$\hat{\pi} \pm z_{1-\alpha/2} \sqrt{\frac{\pi(1 - \pi)}{m}},$$

where  $z_{1-\alpha/2}$  is the  $(1 - \alpha/2)$ th quantile of a  $\mathcal{N}(0, 1)$  RV. The challenge, of course, is that in practice we don’t know the value of  $\pi$  for the variance.

Some solutions:

1. Use  $\pi = 0.5$  so that the estimated variance is as big as it can possibly be ( $1/(4m)$ ). These intervals will be conservative (coverage of a 95% confidence interval will tend to be more than 95%).

2. Estimate the variance by using  $\pi = \hat{\pi}$ :

$$\text{Var}(\hat{\pi}) = \frac{\hat{\pi}(1 - \hat{\pi})}{m}.$$

These intervals will be “anti-conservative” (coverage of a 95% confidence interval will tend to be *less* than 95%).

3. Add two “successes” and two “failures” to artificially make the estimated variance bigger. In other words, estimate the variance using  $\pi = \tilde{\pi} = (y + 2)/(m + 4)$  so that

$$\text{Var}(\hat{\pi}) = \frac{\tilde{\pi}(1 - \tilde{\pi})}{m}.$$

These intervals will be conservative, but less so than the first approach.

4. Ignore the normal limit and use the exact binomial distribution of  $Y$  instead. Recall from Berger and Berger (2002) Example 9.2.11 that when  $y$  is the observed number of successes,

- a 95% CI lower bound is the value of  $\pi$  such that

$$\sum_{x=y}^m \binom{m}{x} \pi^x (1 - \pi)^{m-x} = 0.025$$

and

- a 95% CI upper bound is the value of  $\pi$  such that

$$\sum_{x=0}^y \binom{m}{x} \pi^x (1 - \pi)^{m-x} = 0.025.$$

Tip: R computes these using “`binom.test(y, m)`”

5. Use the delta method to derive a confidence interval for transformation of  $\pi$ : Let

$$\eta = g(\pi) = \log \frac{\pi}{1 - \pi}.$$

Using the delta method,

$$\hat{\eta} = g(\hat{\pi}) \sim \mathcal{N} \left( g(\pi) = \eta, \frac{\pi(1 - \pi)}{m} \frac{1}{(g'(\pi))^2} \right).$$

Using the approach where we let  $\pi = \hat{\pi} = y/m$ , a  $100(1 - \alpha)\%$  confidence interval for  $\eta$  is

$$\log\left(\frac{y}{m-y}\right) \pm z_{1-\alpha/2} \sqrt{\frac{1}{y} + \frac{1}{m-y}},$$

resulting in the interval  $[\eta_L, \eta_U]$ . Then, a  $100(1 - \alpha)\%$  confidence interval for  $\pi$  is

$$[g^{-1}(\eta_L), g^{-1}(\eta_U)] = \left[ \frac{e^{\eta_L}}{1 + e^{\eta_L}}, \frac{e^{\eta_U}}{1 + e^{\eta_U}} \right].$$

Let's use these different approaches to find different confidence intervals for the following example:

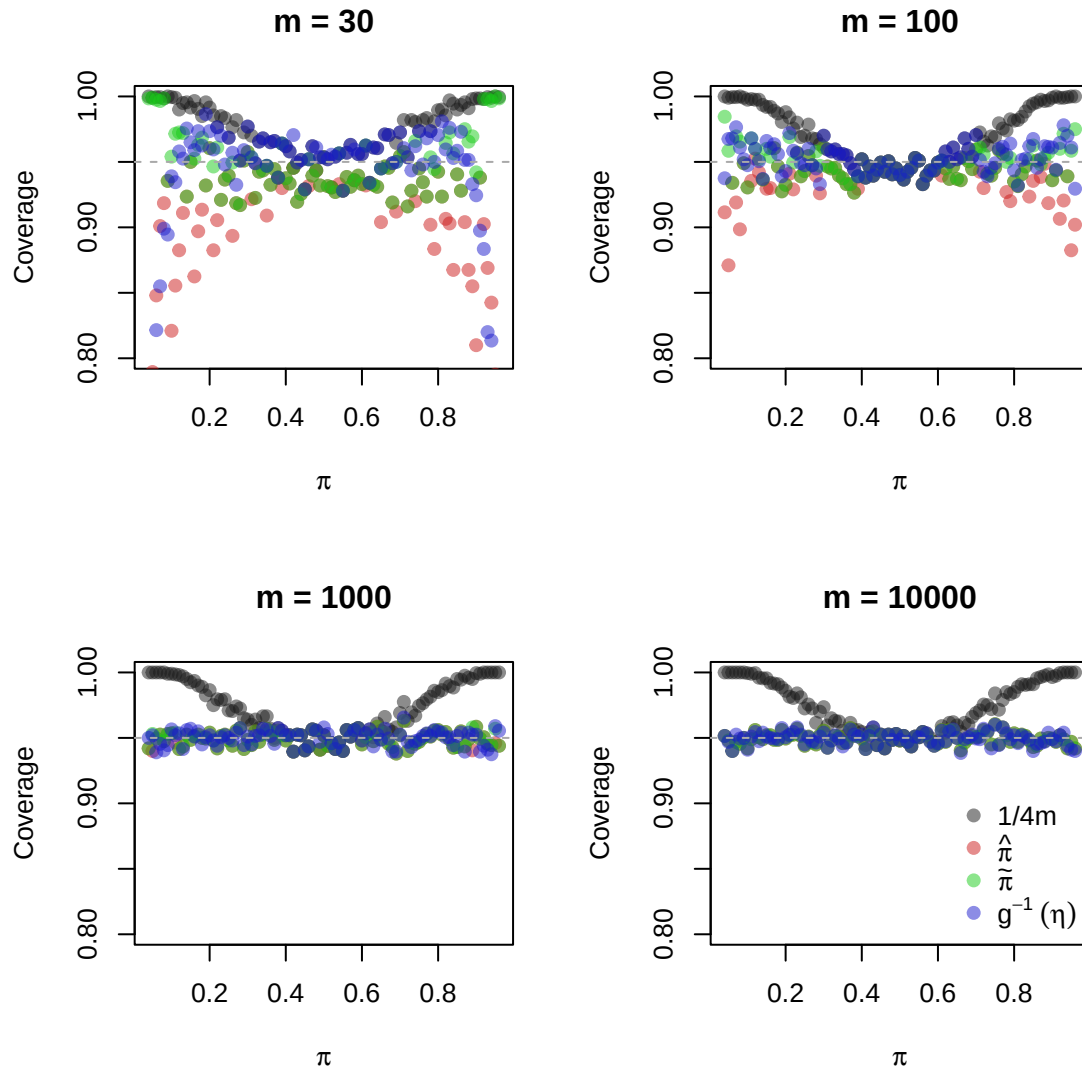
Suppose a marketing analyst carries out a study to determine the proportion of people in a city who drink coffee regularly. In their random sample of 50 people, the analyst finds that 2 regularly drink coffee. What is a 95% CI for the population proportion of those who regularly drink coffee?

```
y <- 2
m <- 50
pihat <- y/m
pitilde <- (y+2)/(m+4)
var1 <- 1/(4*m)
var2 <- pihat*(1-pihat)/m
var3 <- pitilde*(1-pitilde)/m
ci1 <- pihat + c(-1, 1)*qnorm(.975)*sqrt(var1)
ci2 <- pihat + c(-1, 1)*qnorm(.975)*sqrt(var2)
ci3 <- pihat + c(-1, 1)*qnorm(.975)*sqrt(var3)
ci4 <- binom.test(y, m)$conf.int
etaci <- log(y/(m-y)) + c(-1, 1)*qnorm(.975)*sqrt(1/y + 1/(m-y))
ci5 <- exp(etaci)/(1+exp(etaci))
ci <- data.frame(ci1, ci2, ci3, ci4, ci5)
print(ci)
```

|   | ci1         | ci2         | ci3         | ci4         | ci5        |
|---|-------------|-------------|-------------|-------------|------------|
| 1 | -0.09859038 | -0.01431612 | -0.03259126 | 0.004881433 | 0.01002561 |
| 2 | 0.17859038  | 0.09431612  | 0.11259126  | 0.137137626 | 0.14634358 |

What do you notice about these different confidence intervals?

Simulation study to examine coverage for the approximation intervals:



**i** Same code but in python:

```

# Import libraries
import numpy as np
import pandas as pd
from scipy import stats
import statsmodels.api as sm
import statsmodels.formula.api as smf
import matplotlib.pyplot as plt
import pyreadr
import requests
import os

# Initial variables
y = 2
m = 50

# Estimators
pihat = y / m
pitilde = (y + 2) / (m + 4)

# Variance calculations
var1 = 1 / (4 * m)
var2 = pihat * (1 - pihat) / m
var3 = pitilde * (1 - pitilde) / m

# Critical value (z-score for 95% confidence)
z = stats.norm.ppf(0.975)

# Confidence Interval Calculations
# CI 1: Wald-type with conservative variance
ci1 = pihat + np.array([-1, 1]) * z * np.sqrt(var1)

# CI 2: Standard Wald (Normal Approximation)
ci2 = pihat + np.array([-1, 1]) * z * np.sqrt(var2)

# CI 3: Agresti-Coull style / Adjusted Wald
ci3 = pihat + np.array([-1, 1]) * z * np.sqrt(var3)

# CI 4: Clopper-Pearson (Exact Binomial)
ci4_ob = stats.binomtest(y, m).proportion_ci(confidence_level=0.95, method='exact')
ci4 = np.array([ci4_ob.low, ci4_ob.high])

# CI 5: Logit Transformation
etaci = np.log(y / (m - y)) + np.array([-1, 1]) * z * np.sqrt(1/y + 1/(m - y))
ci5 = np.exp(etaci) / (1 + np.exp(etaci))

```

|   | ci1      | ci2       | ci3       | ci4      | ci5      |
|---|----------|-----------|-----------|----------|----------|
| 0 | -0.09859 | -0.014316 | -0.032591 | 0.004881 | 0.010026 |
| 1 | 0.17859  | 0.094316  | 0.112591  | 0.137138 | 0.146344 |

## 4.2 Link Function for the Binomial Distribution

Common link functions:

1. Logit (most used)

$$g_1(\pi) = \log\left(\frac{\pi}{1-\pi}\right)$$

2. Probit

$$g_2(\pi) = \Phi^{-1}(\pi)$$

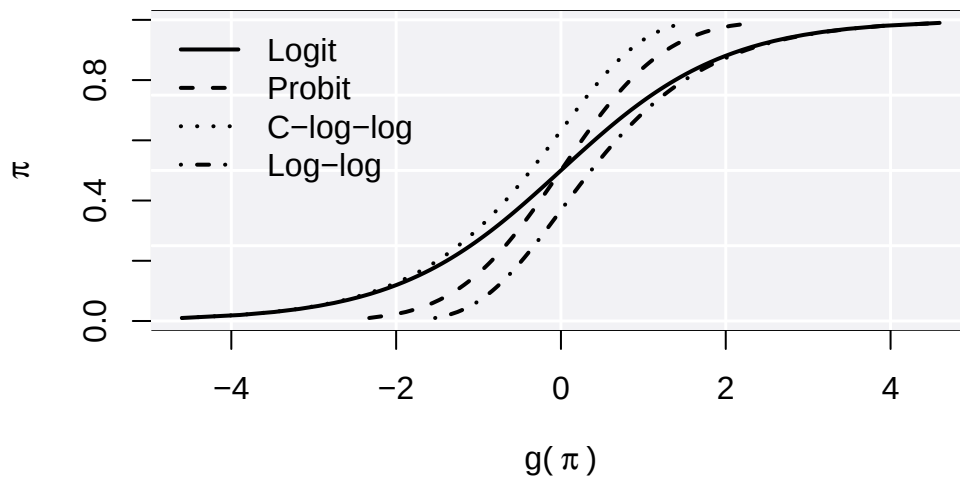
3. Complementary log-log

$$g_3(\pi) = \log(-\log(1-\pi))$$

4. Log-log (least used)

$$g_4(\pi) = -\log(-\log(\pi))$$

### Link Functions



What do you notice about these link functions?  
When might one link be more appropriate than another?

Notes about the history:



- Historically, the probit link was the first link function to be used in binomial regression.
- The probit link was used traditionally for bioassay experiments (experiments in which a biological organism is used to test for chemical toxicity) e.g., Finney (1971)
- Probit historically used for determining median lethal dose, LD(50) (dose that can be expected to kill half of the animals)

#### *Beetles and Carbon Disulphide*

- Number of beetles dead after 5 hours of exposure to gaseous carbon disulphide at various concentrations
- Concentrations are  $\log_{10} CS_2$  mg/l.

Table 4.1: Number of beetles given and died at each dose.

| Dose   | Total Beetles, $m_i$ | Number Dead, $y_i$ |
|--------|----------------------|--------------------|
| 1.6907 | 59                   | 6                  |
| 1.7242 | 60                   | 13                 |
| 1.7552 | 62                   | 18                 |
| 1.7842 | 56                   | 28                 |
| 1.8113 | 63                   | 52                 |
| 1.8369 | 59                   | 53                 |
| 1.8610 | 62                   | 61                 |
| 1.8839 | 60                   | 60                 |

Notice, this data can be written as either the grouped beetles or the individual beetles. Let  $Y_{ij}^* = 1$  if the  $j$ th beetle from the  $i$ th dose died (and 0 if it lived). Then, the total number of beetles that died for the  $i$ th dose is,

$$Y_i = \sum_{j=1}^{m_i} Y_{ij}^*.$$

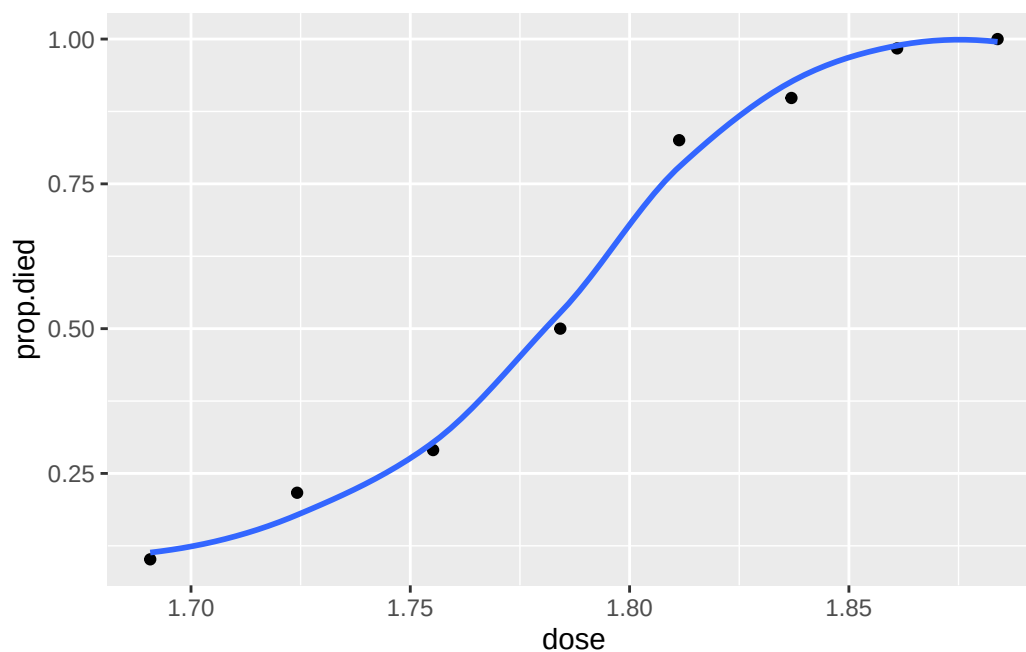
We can model either  $Y_{ij}^*$  or  $Y_i$ :

$$\begin{array}{ll} Y_{ij}^* \overset{ind}{\sim} \text{Bernoulli}(\pi_i) & Y_i \overset{ind}{\sim} \text{Binomial}(m_i, \pi_i) \\ g(\pi_i) = \beta_0 + \beta_1 x_i & g(\pi_i) = \beta_0 + \beta_1 x_i, \end{array}$$

where  $x_i$  is the measurement for the  $i$ th dose.

Warning: package 'dials' was built under R version 4.6.1

Warning: package 'dotwhisker' was built under R version 4.6.1



*What do you notice about the data? Can you tell which link function might fit best?*

1. Compare the following three fitted logistic regression models. *How are the summaries the same and how are they different?*

```
logit1 <- glm(cbind(y, m-y)~dose, family=binomial, data=beetles)
summary(logit1)
```

Call:

```
glm(formula = cbind(y, m - y) ~ dose, family = binomial, data = beetles)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -60.717  | 5.181      | -11.72  | <2e-16 *** |
| dose        | 34.270   | 2.912      | 11.77   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 284.202 on 7 degrees of freedom  
Residual deviance: 11.232 on 6 degrees of freedom  
AIC: 41.43

Number of Fisher Scoring iterations: 4

```
anova(logit1)
```

Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(y, m - y)

Terms added sequentially (first to last)

|        | Df     | Deviance | Resid. | Df | Resid. Dev | Pr(>Chi)      |
|--------|--------|----------|--------|----|------------|---------------|
| NULL   |        |          | 7      |    | 284.202    |               |
| dose 1 | 272.97 |          | 6      |    | 11.232     | < 2.2e-16 *** |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
logit2 <- glm(prop.died~dose, weights=m, family=binomial, data=beetles)
summary(logit2)
```

Call:

```
glm(formula = prop.died ~ dose, family = binomial, data = beetles,
     weights = m)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -60.717  | 5.181      | -11.72  | <2e-16 *** |
| dose        | 34.270   | 2.912      | 11.77   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 284.202 on 7 degrees of freedom
Residual deviance: 11.232 on 6 degrees of freedom
AIC: 41.43
```

```
Number of Fisher Scoring iterations: 4
```

```
anova(logit2)
```

```
Analysis of Deviance Table
```

```
Model: binomial, link: logit
```

```
Response: prop.died
```

```
Terms added sequentially (first to last)
```

|        | Df | Deviance | Resid. Df | Resid. Dev | Pr(>Chi)      |
|--------|----|----------|-----------|------------|---------------|
| NULL   |    |          | 7         | 284.202    |               |
| dose 1 | 1  | 272.97   | 6         | 11.232     | < 2.2e-16 *** |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
logit.long <- glm(y.star~dose, family=binomial, data=beetles.long)
summary(logit.long)
```

```
Call:
```

```
glm(formula = y.star ~ dose, family = binomial, data = beetles.long)
```

```
Coefficients:
```

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 60.717   | 5.181      | 11.72   | <2e-16 *** |

```
dose          -34.270      2.912  -11.77   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 645.44  on 480  degrees of freedom
Residual deviance: 372.47  on 479  degrees of freedom
AIC: 376.47

Number of Fisher Scoring iterations: 5
```

```
anova(logit.long)
```

Analysis of Deviance Table

Model: binomial, link: logit

Response: y.star

Terms added sequentially (first to last)

|        | Df     | Deviance | Resid. Df | Resid. Dev | Pr(>Chi)      |
|--------|--------|----------|-----------|------------|---------------|
| NULL   |        |          | 480       | 645.44     |               |
| dose 1 | 272.97 |          | 479       | 372.47     | < 2.2e-16 *** |

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

95% critical values for hypothesis tests on  $D$  and  $\Delta D$ :

```
qchisq(.95, c(1, 6, 7, 479, 480))
```

```
[1]  3.841459 12.591587 14.067140 531.022232 532.075374
```

2. Comparison of links: Compare deviances, residuals, predictions, interpretation. *Which model do you prefer?*

### Compare deviances:

```
anova(logit1)
```

Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(y, m - y)

Terms added sequentially (first to last)

|        | Df     | Deviance | Resid. Df | Resid. Dev | Pr(>Chi)      |
|--------|--------|----------|-----------|------------|---------------|
| NULL   |        |          | 7         | 284.202    |               |
| dose 1 | 272.97 |          | 6         | 11.232     | < 2.2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
probit.glm <- glm(prop.died~dose, weights=m, family=binomial(link="probit"), c  
summary(probit.glm)
```

Call:

```
glm(formula = prop.died ~ dose, family = binomial(link = "probit"),
```

```

data = beetles, weights = m)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -34.935      2.648  -13.19  <2e-16 ***
dose          19.728      1.487   13.27  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 284.20  on 7  degrees of freedom
Residual deviance:  10.12  on 6  degrees of freedom
AIC: 40.318

Number of Fisher Scoring iterations: 4

```

```
anova(probit.glm)
```

Analysis of Deviance Table

Model: binomial, link: probit

Response: prop.died

Terms added sequentially (first to last)

|        | Df     | Deviance | Resid. Df | Resid. Dev | Pr(>Chi)      |
|--------|--------|----------|-----------|------------|---------------|
| NULL   |        |          | 7         | 284.20     |               |
| dose 1 | 274.08 |          | 6         | 10.12      | < 2.2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1



```
cloglog.glm <- glm(prop.died~dose, weights=m, family=binomial(link="cloglog"),
summary(cloglog.glm)
```

Call:

```
glm(formula = prop.died ~ dose, family = binomial(link = "cloglog"),
    data = beetles, weights = m)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -39.572  | 3.240      | -12.21  | <2e-16 *** |
| dose        | 22.041   | 1.799      | 12.25   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 284.2024 on 7 degrees of freedom  
Residual deviance: 3.4464 on 6 degrees of freedom  
AIC: 33.644

Number of Fisher Scoring iterations: 4

```
anova(cloglog.glm)
```

Analysis of Deviance Table

Model: binomial, link: cloglog

Response: prop.died

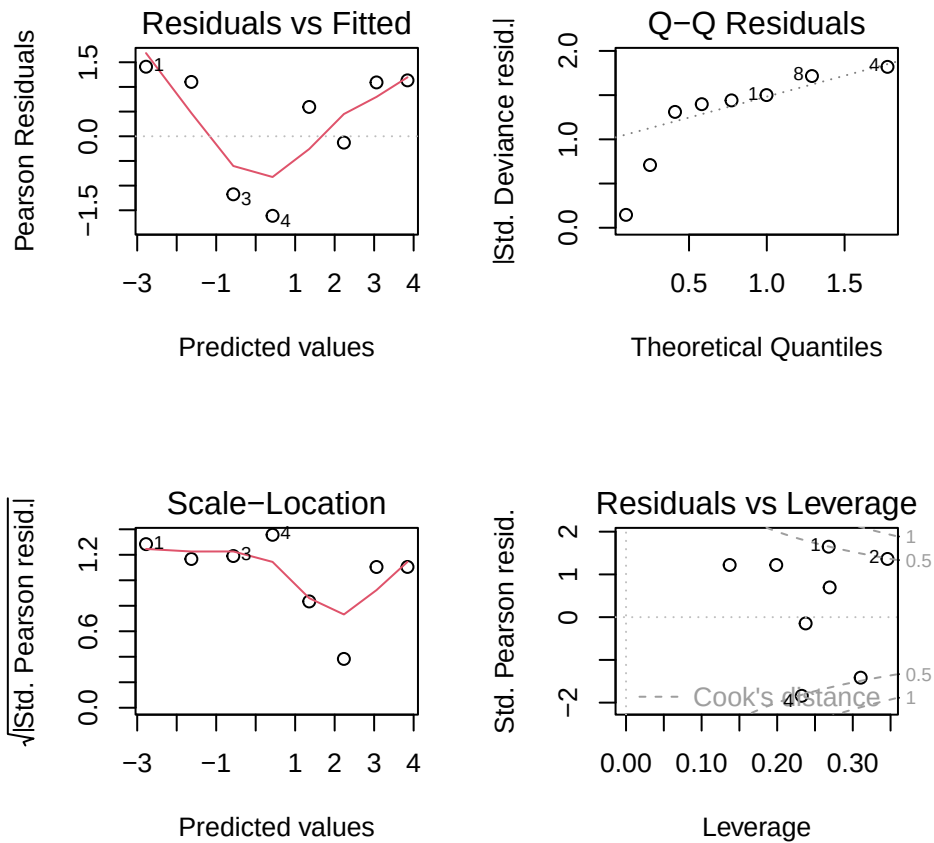
Terms added sequentially (first to last)

|      | Df | Deviance | Resid. | Df | Resid. Dev | Pr(>Chi)      |
|------|----|----------|--------|----|------------|---------------|
| NULL |    |          | 7      |    | 284.202    |               |
| dose | 1  | 280.76   | 6      |    | 3.446      | < 2.2e-16 *** |

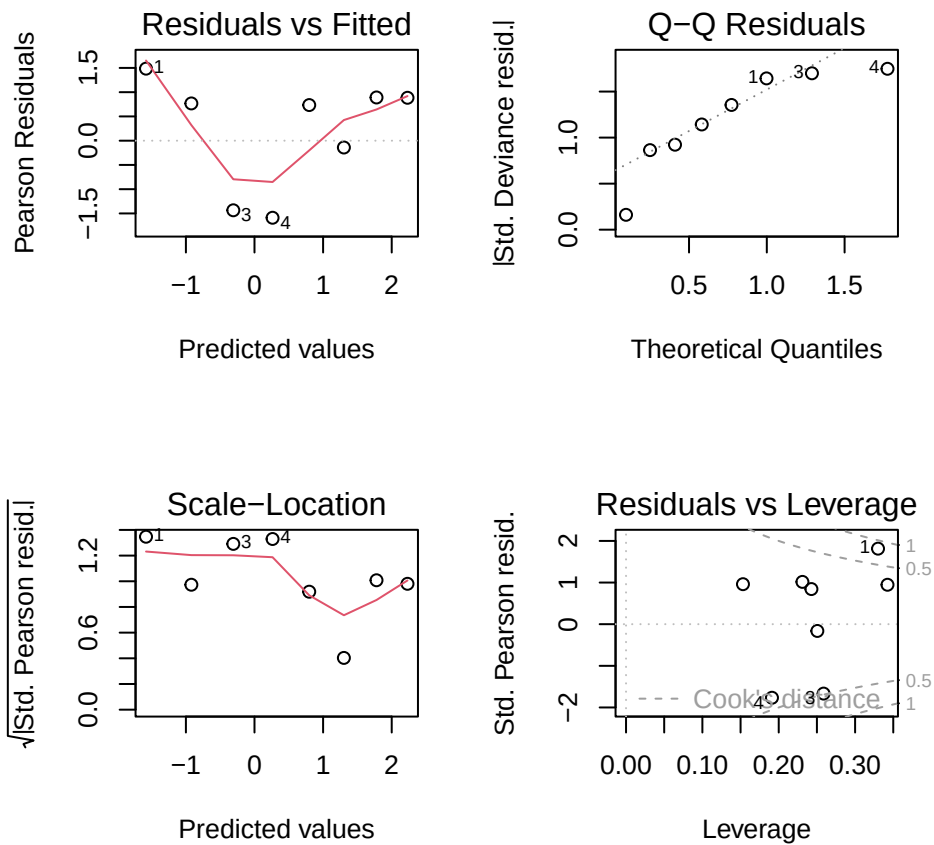
---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Compare residuals:

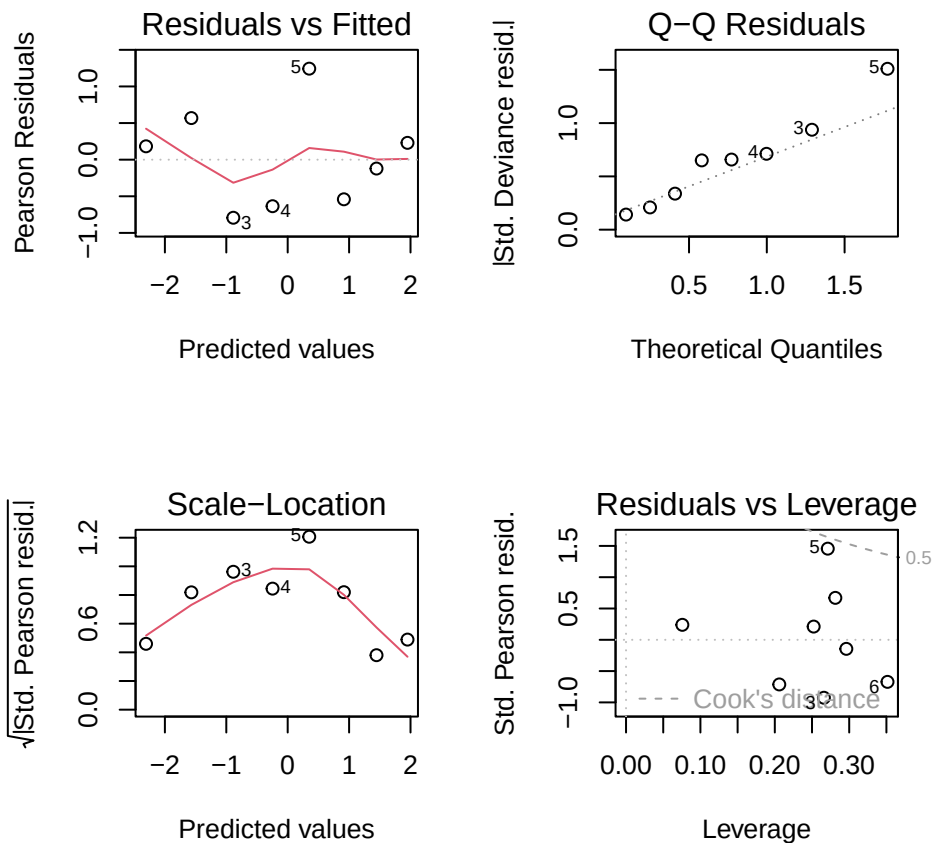
```
par(mfrow=c(2,2))
plot(logit1)
```



```
plot(probit.glm)
```

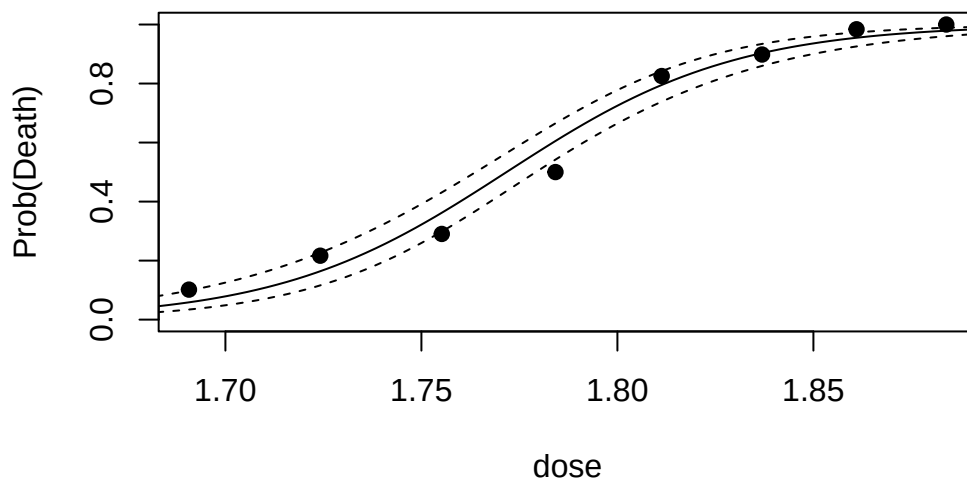


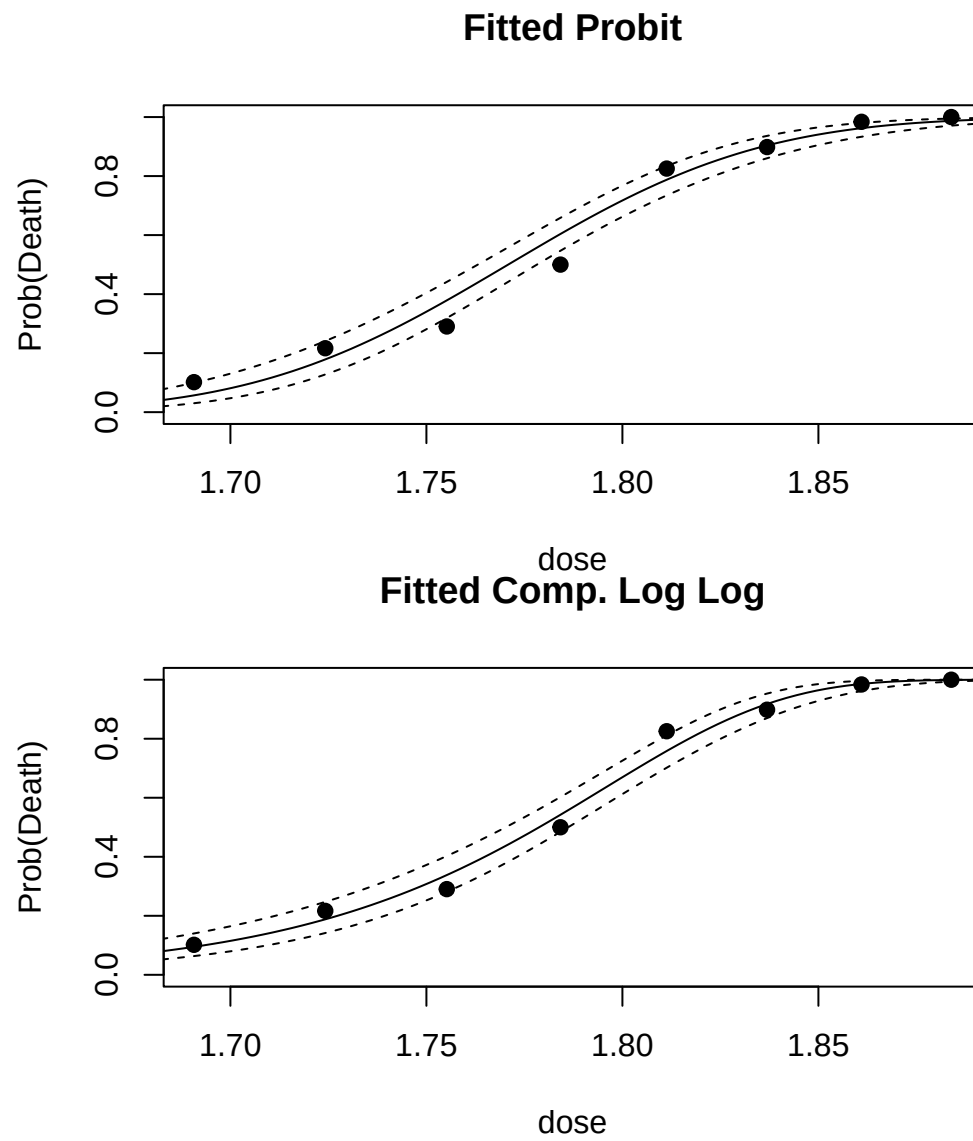
```
plot(cloglog.glm)
```



Compare predictions:

### Fitted Logit





## Compare interpretation:

Note: There are many tools for model comparison that you have learned in other classes that apply here. For example, AIC, ROC curves, five-fold cross validation, etc. For now, focus on these new approaches so you become familiar with them and can use them as frequently as you use other model comparison approaches.

### 3. Note: Predictions Using Tidy Models

```

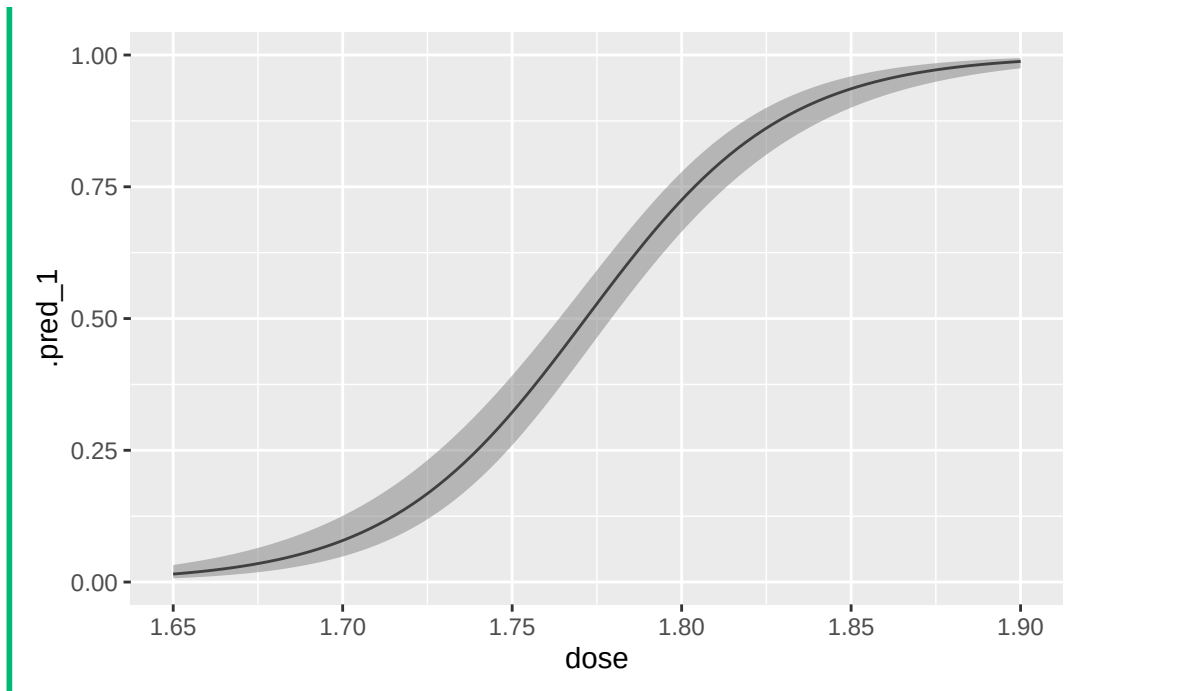
## TIDYMODELS
# For prediction, tidymodels wants the response variable to be a (binary) factor

logitmod <- logistic_reg() %>%
  set_engine("glm", family = stats::binomial(link = "logit")) %>%
  translate()
probitmod <- logistic_reg() %>%
  set_engine("glm", family = stats::binomial(link = "probit")) %>%
  translate()
cloglogmod <- logistic_reg() %>%
  set_engine("glm", family = stats::binomial(link = "cloglog")) %>%
  translate()

logitfit <- logitmod %>%
  fit(y.star~dose, data=beetles.long)
probitfit <- probitmod %>%
  fit(y.star~dose, data=beetles.long)
cloglogfit <- cloglogmod %>%
  fit(y.star~dose, data=beetles.long)

mean_pred <- predict(logitfit, new_data=new_points, type='prob')
conf_pred <- predict(logitfit, new_data = new_points, type="conf_int") #these
plotdat <- new_points %>%
  bind_cols(mean_pred) %>%
  bind_cols(conf_pred)
ggplot(plotdat, aes(x=dose)) +
  geom_line(aes(y=.pred_1)) +
  geom_ribbon(aes(ymin=.pred_lower_1, ymax=.pred_upper_1), fill=rgb(.5, .5, .5))

```





**i** The same code, but in python:

```

# Beetles Data
data = {
    'dose': [1.6907, 1.7242, 1.7552, 1.7842, 1.8113, 1.8369, 1.8610, 1.8839],
    'm': [59, 60, 62, 56, 63, 59, 62, 60],
    'y': [6, 13, 18, 28, 52, 53, 61, 60]
}
beetles = pd.DataFrame(data)

# Beetles Long Data
beetles_long = beetles.loc[beetles.index.repeat(beetles['m'])].copy()
y_star = []
for idx, row in beetles.iterrows():
    y_star.extend([1] * int(row['y'])) # Died
    y_star.extend([0] * int(row['m'] - row['y'])) # Survived
beetles_long['y_star'] = y_star
beetles_long = beetles_long.reset_index(drop=True)

# Creating ANOVA summary GLM function since we will use this code several times
def anova_glm(full_model, null_model):
    p_value = 1 - stats.chi2.cdf(null_model.deviance - logit1.deviance, 1)
    anova_table = pd.DataFrame({
        'Df': [None, 1],
        'Deviance': [None, null_model.deviance - full_model.deviance],
        'Resid. Df': [null_model.df_resid, full_model.df_resid],
        'Resid. Dev': [null_model.deviance, full_model.deviance],
        'Pr(>Chi)': [None, p_value]
    }, index=['NULL', 'dose'])

    print("Analysis of Deviance Table (Type I Sequential)\n")
    print(anova_table)

# First Logistic Regression Model
beetles['failures'] = beetles['m'] - beetles['y']
logit1 = smf.glm(formula="y + failures ~ dose", data=beetles, family=sm.families.Binomial())
print(logit1.summary())

```

| Generalized Linear Model Regression Results |                   |         |                   |       |        |        |
|---------------------------------------------|-------------------|---------|-------------------|-------|--------|--------|
| =====                                       |                   |         |                   |       |        |        |
| Dep. Variable:                              | ['y', 'failures'] |         | No. Observations: |       | 8      |        |
| Model:                                      | GLM               |         | Df Residuals:     |       | 6      |        |
| Model Family:                               | Binomial          |         | Df Model:         |       | 1      |        |
| Link Function:                              | Logit             |         | Scale:            |       | 1.0000 |        |
| Method:                                     | IRLS              |         | Log-              |       |        |        |
| Likelihood:                                 | -18.715           |         |                   |       |        |        |
| Date:                                       | Tue, 18 Aug 2026  |         | Deviance:         |       | 11.232 |        |
| Time:                                       | 13:02:58          |         | Pearson chi2:     |       | 10.0   |        |
| No. Iterations:                             | 6                 |         | Pseudo R-         |       |        |        |
| squ. (CS):                                  | 1.000             |         |                   |       |        |        |
| Covariance Type:                            | nonrobust         |         |                   |       |        |        |
| =====                                       |                   |         |                   |       |        |        |
|                                             | coef              | std err | z                 | P> z  | [0.025 | 0.975] |
| -----                                       |                   |         |                   |       |        |        |
| Intercept                                   | -60.7175          | 5.181   | -11.720           | 0.000 | -      |        |
| 70.871                                      | -50.563           |         |                   |       |        |        |
| dose                                        | 34.2703           | 2.912   | 11.768            | 0.000 | 28.563 | 39.978 |
| =====                                       |                   |         |                   |       |        |        |

```

null_logit1 = smf.glm(formula="y + failures ~ 1", data=beetles, family=sm.fami
anova_glm(logit1, null_logit1)

```

### Analysis of Deviance Table (Type I Sequential)

|      | Df  | Deviance   | Resid. Df | Resid. Dev | Pr(>Chi) |
|------|-----|------------|-----------|------------|----------|
| NULL | NaN | NaN        | 7         | 284.202449 |          |
| dose | 1.0 | 272.970218 | 6         | 11.232231  | 0.0      |

```
# Second Logistic Regression Model
beetles['prop_died'] = beetles['y'] / beetles['m']
logit2 = smf.glm(formula="prop_died ~ dose", data=beetles, family=sm.families.Binomial)
print(logit2.summary())
```

```

                                Generalized Linear Model Regression Results
=====
Dep. Variable:                  prop_died    No. Observations:                  8
Model:                          GLM          Df Residuals:                      6
Model Family:                   Binomial      Df Model:                        1
Link Function:                  Logit          Scale:                          1.0000
Method:                         IRLS          Log-likelihood:                   -129.83
Likelihood:                     -129.83
Date:                           Tue, 18 Aug 2026    Deviance:                        11.232
Time:                           13:02:58          Pearson chi2:                    10.0
No. Iterations:                  6              Pseudo R-squared:                0.975
squ. (CS):                      1.000
Covariance Type:                nonrobust
=====
                                coef    std err          z      P>|z|      [0.025      0.975]
-----
Intercept    -60.7175      5.181    -11.720      0.000      -70.871      -50.563
dose         34.2703      2.912     11.768      0.000      28.563      39.978
=====
```

```
null_logit2 = smf.glm(formula="prop_died ~ 1", data=beetles, family=sm.families.Binomial)
anova_glm(logit2, null_logit2)
```

Analysis of Deviance Table (Type I Sequential)

| Df | Deviance | Resid. Df | Resid. Dev | Pr(>Chi) |
|----|----------|-----------|------------|----------|
|----|----------|-----------|------------|----------|

|      |     |            |   |            |     |
|------|-----|------------|---|------------|-----|
| NULL | NaN | NaN        | 7 | 284.202449 |     |
| dose | 1.0 | 272.970218 | 6 | 11.232231  | 0.0 |

```
# Long Logistic Regression Model
logit_long = smf.glm(formula="y_star ~ dose", data=beetles_long, family=sm.fam
print(logit_long.summary())
```

#### Generalized Linear Model Regression Results

|                  |                  |                   |        |
|------------------|------------------|-------------------|--------|
| Dep. Variable:   | y_star           | No. Observations: | 481    |
| Model:           | GLM              | Df Residuals:     | 479    |
| Model Family:    | Binomial         | Df Model:         | 1      |
| Link Function:   | Logit            | Scale:            | 1.0000 |
| Method:          | IRLS             | Log-              |        |
| Likelihood:      | -186.24          |                   |        |
| Date:            | Tue, 18 Aug 2026 | Deviance:         | 372.47 |
| Time:            | 13:02:58         | Pearson chi2:     | 436.   |
| No. Iterations:  | 6                | Pseudo R-         |        |
| squ. (CS):       | 0.4331           |                   |        |
| Covariance Type: | nonrobust        |                   |        |

|           | coef     | std err | z       | P> z  | [0.025 | 0.975] |
|-----------|----------|---------|---------|-------|--------|--------|
| Intercept | -60.7175 | 5.181   | -11.720 | 0.000 | -      |        |
| 70.871    | -50.563  |         |         |       |        |        |
| dose      | 34.2703  | 2.912   | 11.768  | 0.000 | 28.563 | 39.978 |

```
null_logit_long = smf.glm(formula="y_star ~ 1", data=beetles_long, family=sm.f
anova_glm(logit_long, null_logit_long)
```

Analysis of Deviance Table (Type I Sequential)

|      | Df  | Deviance   | Resid. Df | Resid. Dev | Pr(>Chi) |
|------|-----|------------|-----------|------------|----------|
| NULL | NaN | NaN        | 480       | 645.441025 |          |
| dose | 1.0 | 272.970218 | 479       | 372.470807 | 0.0      |

```
# Critical Values
print(stats.chi2.ppf(0.95, [1, 6, 7, 479, 480]))
```

```
[ 3.84145882 12.59158724 14.06714045 531.0222321 532.07537415]
```

```
# Compare deviances
anova_glm(logit1, null_logit1)
```

Analysis of Deviance Table (Type I Sequential)

|      | Df  | Deviance   | Resid. Df | Resid. Dev | Pr(>Chi) |
|------|-----|------------|-----------|------------|----------|
| NULL | NaN | NaN        | 7         | 284.202449 |          |
| dose | 1.0 | 272.970218 | 6         | 11.232231  | 0.0      |

```
# Probit
probit_glm = smf.glm(formula="prop_died ~ dose", data=beetles, family=sm.famil
print(probit_glm.summary())
```

#### Generalized Linear Model Regression Results

|                |           |                   |        |
|----------------|-----------|-------------------|--------|
| Dep. Variable: | prop_died | No. Observations: | 8      |
| Model:         | GLM       | Df Residuals:     | 6      |
| Model Family:  | Binomial  | Df Model:         | 1      |
| Link Function: | Probit    | Scale:            | 1.0000 |
| Method:        | IRLS      | Log-              |        |
| Likelihood:    | -129.27   |                   |        |

Date: Tue, 18 Aug 2026 Deviance: 10.120  
Time: 13:02:58 Pearson chi2: 9.51  
No. Iterations: 6 Pseudo R-  
squ. (CS): 1.000  
Covariance Type: nonrobust

|           | coef     | std err | z       | P> z  | [0.025 | 0.975] |
|-----------|----------|---------|---------|-------|--------|--------|
| Intercept | -34.9353 | 2.648   | -13.194 | 0.000 | -      |        |
| dose      | 19.7279  | 1.487   | 13.265  | 0.000 | 16.813 | 22.643 |

```
null_probit_glm = smf.glm(formula="prop_died ~ 1", data=beetles, family=sm.fam
anova_glm(probit_glm, null_probit_glm)
```

#### Analysis of Deviance Table (Type I Sequential)

|      | Df  | Deviance   | Resid. Df | Resid. Dev | Pr(>Chi) |
|------|-----|------------|-----------|------------|----------|
| NULL | NaN | NaN        | 7         | 284.202449 |          |
| dose | 1.0 | 274.082691 | 6         | 10.119758  | 0.0      |

```
# C-log-log
cloglog_glm = smf.glm(formula="prop_died ~ dose", data=beetles, family=sm.fam
print(cloglog_glm.summary())
```

#### Generalized Linear Model Regression Results

|                |           |                   |        |
|----------------|-----------|-------------------|--------|
| Dep. Variable: | prop_died | No. Observations: | 8      |
| Model:         | GLM       | Df Residuals:     | 6      |
| Model Family:  | Binomial  | Df Model:         | 1      |
| Link Function: | CLogLog   | Scale:            | 1.0000 |

```

Method:                      IRLS    Log-
Likelihood:                  -125.94
Date:                        Tue, 18 Aug 2026    Deviance:                3.4464
Time:                        13:02:58    Pearson chi2:                3.29
No. Iterations:              6    Pseudo R-
squ. (CS):                   1.000
Covariance Type:             nonrobust

```

```

=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept    -39.5723      3.240     -12.213      0.000      -
45.923      -33.221
dose         22.0412      1.799      12.249      0.000      18.514      25.568
=====

```

```

null_cloglog = smf.glm(formula="prop_died ~ 1", data=beetles, family=sm.famili
anova_glm(cloglog_glm, null_cloglog)

```

#### Analysis of Deviance Table (Type I Sequential)

|      | Df  | Deviance   | Resid. Df | Resid. Dev | Pr(>Chi) |
|------|-----|------------|-----------|------------|----------|
| NULL | NaN | NaN        | 7         | 284.202449 |          |
| dose | 1.0 | 280.756011 | 6         | 3.446439   | 0.0      |



```

# Compare residuals
# NOTE: There isn't a native python function that gets glm diagnostics automatically
# Defining function that mimics the functionality of R's "plot(<glm_object>)"
def plot_glm_diagnostics(model):
    fig, axes = plt.subplots(2, 2, figsize=(12, 10))
    fig.subplots_adjust(hspace=0.3, wspace=0.3)

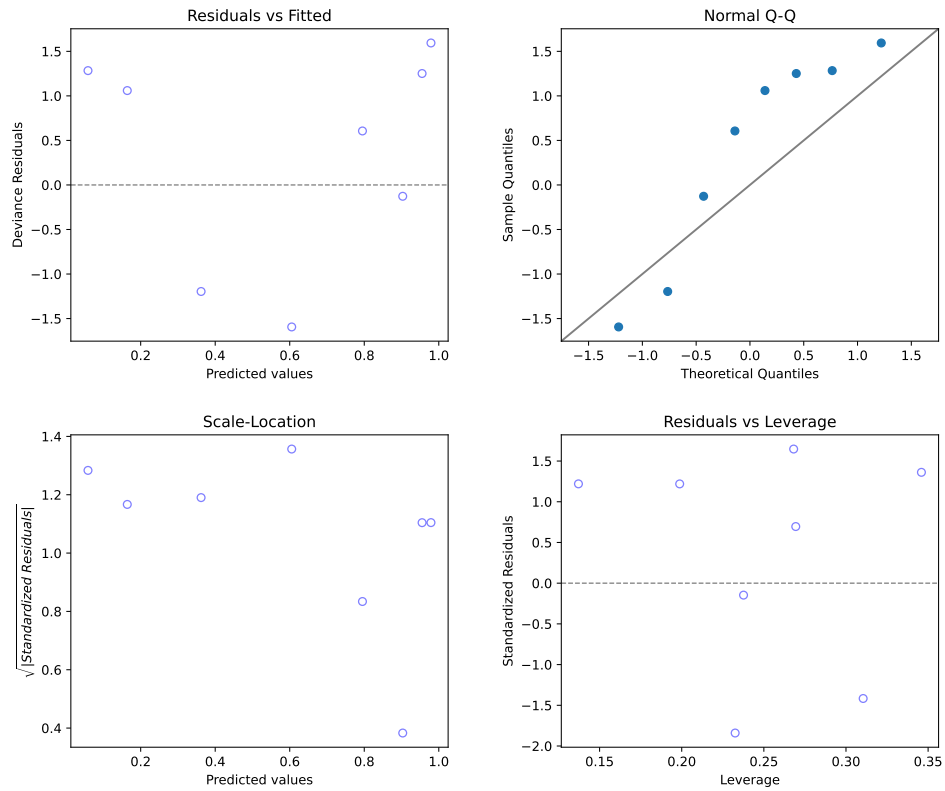
    # 1. Residuals vs Fitted
    axes[0, 0].scatter(model.predict(),
                       model.resid_deviance,
                       alpha=0.5,
                       facecolors='none',
                       edgecolors='b')
    axes[0, 0].axhline(0, color='grey', linestyle='--', lw=1)
    axes[0, 0].set_title('Residuals vs Fitted')
    axes[0, 0].set_xlabel('Predicted values')
    axes[0, 0].set_ylabel('Deviance Residuals')

    # 2. Normal Q-Q
    sm.qqplot(model.resid_deviance,
              line='45',
              ax=axes[0, 1])
    axes[0, 1].get_lines()[-1].set_color('grey')
    axes[0, 1].set_title('Normal Q-Q')

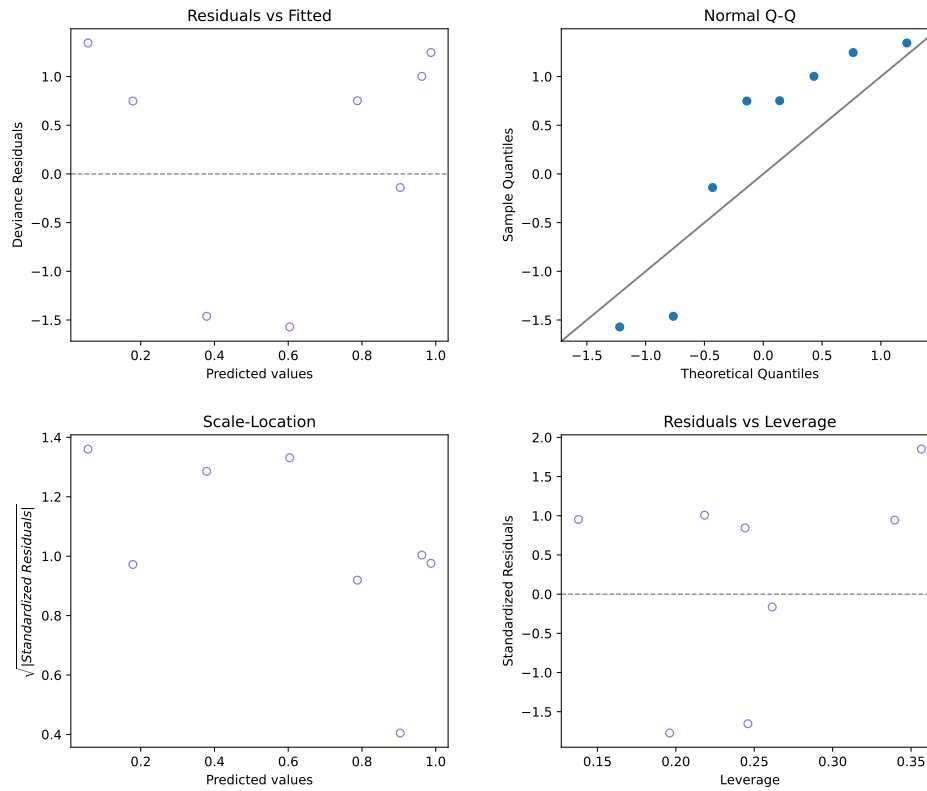
    # 3. Scale-Location
    axes[1, 0].scatter(model.predict(),
                      np.sqrt(np.abs(model.get_influence().resid_studentized)),
                      alpha=0.5,
                      facecolors='none',
                      edgecolors='b')
    axes[1, 0].set_title('Scale-Location')
    axes[1, 0].set_xlabel('Predicted values')
    axes[1, 0].set_ylabel(r'$\sqrt{|\text{Standardized Residuals}|}$')

    # 4. Residuals vs Leverage
    axes[1, 1].scatter(model.get_influence().hat_matrix_diag,
                      model.get_influence().resid_studentized,
                      alpha=0.5,
                      facecolors='none',
                      edgecolors='b')
    axes[1, 1].axhline(0, color='grey', linestyle='--', lw=1)
    axes[1, 1].set_title('Residuals vs Leverage')
    axes[1, 1].set_xlabel('Leverage')
    axes[1, 1].set_ylabel('Standardized Residuals')

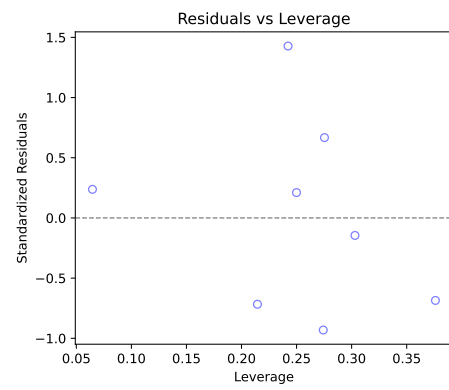
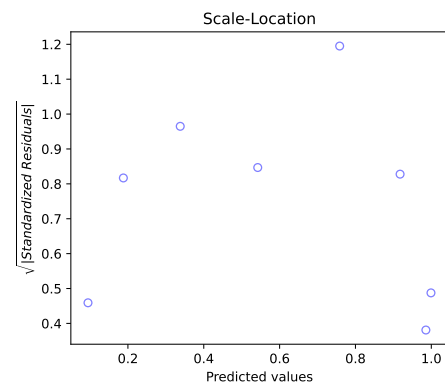
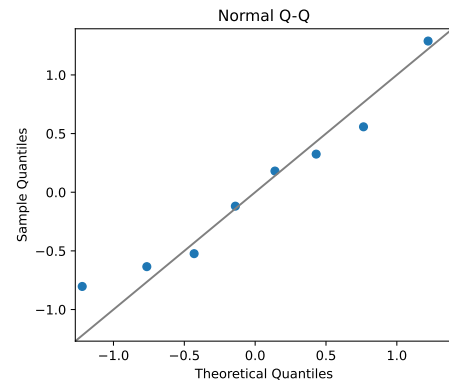
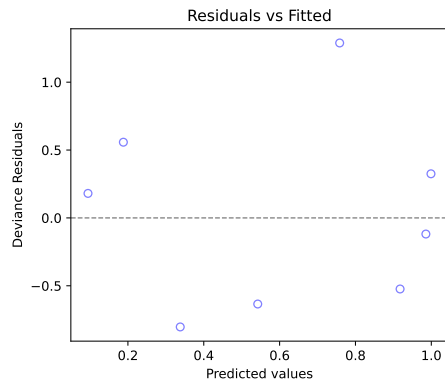
```



```
plot_glm_diagnostics(probit_glm)
```



```
plot_glm_diagnostics(cloglog_glm)
```



```

# Compare Predictions / Interpretations
def get_predictions(model, new_data):
    predictions = model.get_prediction(new_data)

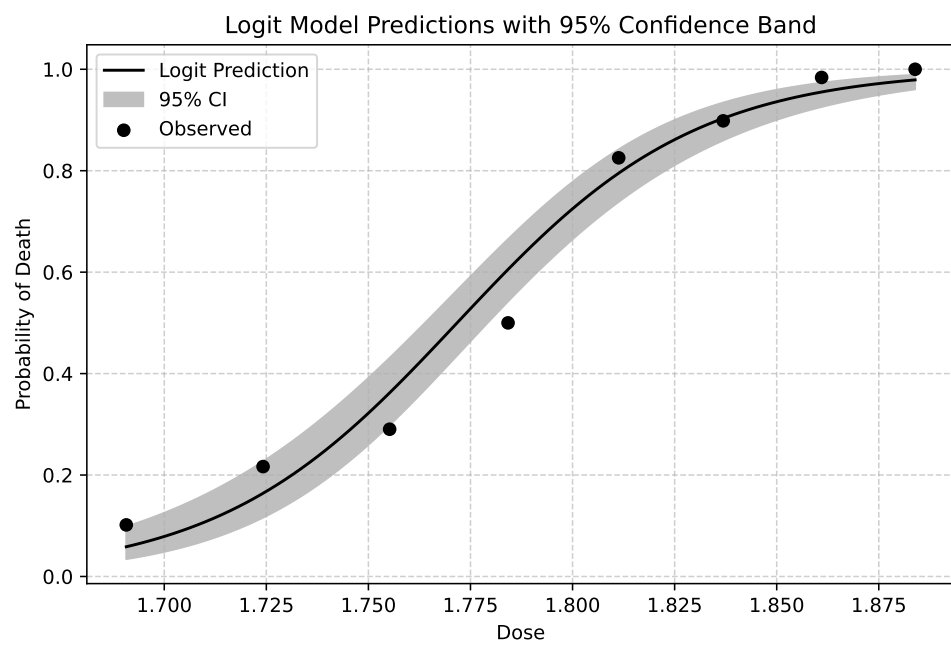
    # This automatically handles the back-transformation from the link scale
    frame = predictions.summary_frame(alpha=0.05)

    return pd.DataFrame({
        'dose': new_data['dose'],
        'prob': frame['mean'],
        'lower': frame['mean_ci_lower'],
        'upper': frame['mean_ci_upper']
    })

new_points = pd.DataFrame({'dose': np.linspace(beetles['dose'].min(), beetles[
plot_dat = get_predictions(logit_long, new_points)

plt.figure(figsize=(8, 5))
plt.plot(plot_dat['dose'], plot_dat['prob'], color='black', label='Logit Predi
plt.fill_between(plot_dat['dose'], plot_dat['lower'], plot_dat['upper'], color
plt.scatter(beetles['dose'], beetles['prop_died'], color='black', zorder=3, la
plt.xlabel('Dose')
plt.ylabel('Probability of Death')
plt.title('Logit Model Predictions with 95% Confidence Band')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()

```



## 4.3 Homework Assignment

1. The “sleep” data set in R contains data that provide the increase in hours of sleep compared to the control treatment of two sleep-inducing drugs for 10 individuals. Let  $\pi$  represent the probability that an individual increases their sleep when given the drug. For each drug type,
  - a. Create a 95% confidence interval for the true proportion of successes,  $\pi$ , using the following four methods: using  $\pi = 1/2$  to estimate the standard deviation
    - i. using  $\hat{\pi}$  to estimate the mean and standard deviation
    - ii. using  $\tilde{\pi} = (y + 2)/(m + 4)$  to estimate the standard deviation (i.e., for the standard deviation, use  $\sqrt{\tilde{\pi}(1 - \tilde{\pi})/m}$ ).
    - iii. using a logit link (i.e., estimate a confidence interval on  $\eta$ , then transform this interval to a confidence interval on  $\pi$ )
  - b. Comment on how these confidence intervals compare. If they are similar, comment on why you think this is so. If they are different, comment on which confidence interval you prefer and why.
  - c. Do any of the above interval types provide evidence that one of the drugs is significantly more likely to increase sleeping?
2. Suppose that  $Y \sim \text{Bin}(m, e^\lambda/(1+e^\lambda))$ . Show that  $m - Y$  also has a binomial distribution and that the induced parameter is  $\lambda' = -\lambda$ .
3. Table 4.2 shows the most important explanatory variable, a “labeling index” (LI), in a study to determine characteristics associated with remission in cancer patients. This index measures proliferative activity of cells after a patient receives an injection of tritiated thymidine, representing the percentage of cells that are “labeled”. The response,  $Y$ , measured whether the patient achieved remission ( $1 = \text{yes}$ ).
  - a. Determine the best link function for a binomial regression model for this data. Justify your choice using at least 3 model diagnostics.
  - b. Given your model, interpret the coefficient for LI. Is it statistically significant for predicting the probability of remission?
  - c. Find the value of LI for which  $\hat{\pi} = 0.5$  (LD(50)).
  - d. Construct (and interpret) a 95% CI for the chance of remission given an LI of 8.

Table 4.2: Number of patients (Cases) at each labeling index (LI) and the number that went into remission.

| LI | Cases | Remissions |
|----|-------|------------|
| 8  | 2     | 0          |
| 10 | 2     | 0          |
| 12 | 3     | 0          |
| 14 | 3     | 0          |
| 16 | 3     | 0          |
| 18 | 1     | 1          |
| 20 | 3     | 2          |
| 22 | 2     | 1          |
| 24 | 1     | 0          |
| 26 | 1     | 1          |
| 28 | 1     | 1          |
| 32 | 1     | 0          |
| 34 | 1     | 1          |
| 38 | 3     | 2          |



## 4.4 Odds Ratios

As we head into including factors as explanatory variables in logistic regression, it will be helpful to become familiar with various properties and interpretations of odds and odds ratios.

Table 4.3: Cross-tabulation representing the joint probabilities of exposure and outcome.

| Exposure ( $X$ ) | Outcome ( $Y$ ) |             |
|------------------|-----------------|-------------|
|                  | Success (1)     | Failure (0) |
| Present (1)      | $\pi_{11}$      | $\pi_{10}$  |
| Absent (0)       | $\pi_{01}$      | $\pi_{00}$  |

Examples of exposure and outcomes of interest:

Note:  $\pi_{XY}$  represents the joint probability for the exposure and outcome. Often we will consider the *marginal probability of success* ( $\pi_1 = \pi_{11} + \pi_{01}$ ) and *failure* ( $\pi_0 = \pi_{10} + \pi_{00}$ ).

### Properties of Odds

1. The **odds of success** are

$$O = \frac{\pi_1}{1 - \pi_1} = \frac{\pi_1}{\pi_0}.$$

2. Given exposure, the odds of success are

$$O_1 = \frac{P(Y = 1|X = 1)}{P(Y = 0|X = 1)} = \frac{\pi_{11}/(\pi_{11} + \pi_{10})}{\pi_{10}/(\pi_{11} + \pi_{10})} = \frac{\pi_{11}}{\pi_{10}}.$$

Given absence of exposure, the odds of success are

$$O_0 = \frac{P(Y = 1|X = 0)}{P(Y = 0|X = 0)} = \frac{\pi_{01}/(\pi_{01} + \pi_{00})}{\pi_{00}/(\pi_{01} + \pi_{00})} = \frac{\pi_{01}}{\pi_{00}}.$$

3.  $O$  and  $O_X$  are non-negative.
4.  $O$  and  $O_X$  are  $> 1$  when a success is more likely than a failure.
5. Suppose  $\pi_1 = 0.75$  and  $\pi_0 = 0.25$ . Then the odds of success is  $O = 0.75/0.25 = 3$ . Thus, a success is three times as likely as a failure. Notice too, when  $O = 1/3$  a failure will be three times as likely as a success.

## Properties of Odds Ratios

1. The ratio of the conditional odds  $O_1$  and  $O_0$  is called the **odds ratio**:

$$OR = \frac{O_1}{O_0} = \frac{\pi_{11}/\pi_{10}}{\pi_{01}/\pi_{00}} = \frac{\pi_{11}\pi_{00}}{\pi_{01}\pi_{10}}$$

2.  $OR \geq 0$ .
3. When  $O_1 = O_0$ , then the odds of success are the same no matter whether exposure is present or absent. *Therefore,  $X$  and  $Y$  are independent.* (Note: often we want to say that the likelihood of success is the same as a failure. Why is this statement not necessarily true?)
4. When  $OR > 1$  then those in the exposure group are more likely to have a success than those in the group where the exposure is absent. That is,

$$P(Y = 1|X = 1) > P(Y = 1|X = 0),$$

i.e.,

$$\frac{\pi_{11}}{\pi_{11} + \pi_{10}} > \frac{\pi_{01}}{\pi_{01} + \pi_{00}}.$$

Again, it does *not* necessarily mean that probability of success is greater than the probability of failure.

5. When, for example,  $OR = 4$ , then the odds of success for the exposed group are 4 times the odds of success for the unexposed group. An OR of 4 could be obtain with several different conditional odds:  $4 = 4/1 = 8/2 = 2/0.5 = 0.5/0.125$ .
6. It can be convenient to consider  $\log OR$  since it is symmetric about  $\log 1 = 0$ . Thus,  $\log 4 = -\log(1/4) = 1.39$ .

7. The value of the odds ratio does not change when the orientation of the table reverses (e.g.,  $Y$  becomes the explanatory variable and  $X$  becomes the response variable).

$$\begin{aligned}
OR &= \frac{P(Y = 1|X = 1)/P(Y = 0|X = 1)}{P(Y = 1|X = 0)/P(Y = 0|X = 0)} \\
&= \frac{P(Y = 1|X = 1)P(X = 1)/(P(Y = 0|X = 1)P(X = 1))}{P(Y = 1|X = 0)P(X = 0)/(P(Y = 0|X = 0)P(X = 0))} \\
&= \frac{P(Y = 1, X = 1)/P(Y = 0, X = 1)}{P(Y = 1, X = 0)/P(Y = 0, X = 0)} \\
&= \frac{P(Y = 1, X = 1)P(Y = 0, X = 0)}{P(Y = 0, X = 1)P(Y = 1, X = 0)} \\
&= \frac{P(Y = 1, X = 1)}{P(Y = 1, X = 0)} \frac{P(Y = 0, X = 0)}{P(Y = 0, X = 1)} \\
&= \left( \frac{P(X = 1|Y = 1)P(Y = 1)}{P(X = 0|Y = 1)P(Y = 1)} \right) \left( \frac{P(X = 0|Y = 0)P(Y = 0)}{P(X = 1|Y = 0)P(Y = 0)} \right) \\
&= \left( \frac{P(X = 1|Y = 1)}{P(X = 0|Y = 1)} \right) \left( \frac{P(X = 0|Y = 0)}{P(X = 1|Y = 0)} \right) \\
&= \frac{P(X = 1|Y = 1)/P(X = 0|Y = 1)}{P(X = 1|Y = 0)/P(X = 0|Y = 0)}
\end{aligned}$$

8. The estimated odds ratio can also be written in terms of cell counts:

$$OR = \frac{n_{11} n_{00}}{n_{10} n_{01}}.$$

For retrospective studies (observe outcome now and look back to see if exposure was present), we sample based on the outcomes – not on the exposure – and therefore, we are able to estimate  $P(X = 1|Y = 1)$  but NOT

$P(Y = 1|X = 1)$  (what we're actually interested in). This is due to features of the sampling design.

Suppose we are interested in factors related to developing diabetes. We sample 700 people with diabetes ( $Y = 1$ ) and 700 people without diabetes ( $Y = 0$ ). Of those with diabetes, 595 are overweight ( $X = 1$ ). Of those without diabetes, 469 are overweight ( $X = 1$ ).

Estimate the odds of being overweight conditional on diabetes being present and absent:

$$O_1 =$$

$$O_0 =$$

Estimate and interpret the odds ratio given the odds you just computed:

$$OR =$$

The interpretation above isn't actually what we are interested in. We want the odds of having diabetes when a patient is overweight compared to the odds of having diabetes when the patient is not overweight.

Therefore 2.79 can also be interpreted as the (estimated) odds of having diabetes when a patient is overweight times the odds of having diabetes when a patient is not overweight.

***Note that this (still) isn't a causal statement, just a relationship statement.***

## 4.5 More Complicated Binomial GLM Models

We want to consider models with multiple covariates, including factors and interactions. For each covariate, we need to consider whether to include it in the model, its coefficient's interpretation, and how it impacts probabilities and corresponding confidence intervals.

We'll motivate this exploration with a data set looking at whether a nesting female [horseshoe crab](#) has at least one satellite (male crab near the female). The data set is available to download through Box [here](#). This is an Rdata file with a data frame “crabs”.

- “satell” is the raw count of the number of satellites. “sat.bin” is the binary response of whether there is at least one satellite (1) or not (0).
- “color” is a 4-level factor describing the color of the main crab (light medium, medium, dark medium, or dark). “is.very.dark” is a 2-level factor of whether the main crab is “dark” (T) or not (F).
- “spine” is a 3-level factor describing the condition of the main crab's two spines (both good, one worn/broken, or both worn/broken).
- “width” contains the width of the main crab in centimeters (range of 21 to 33.5 cm).
- “weight” is the weight of the main crab in grams (range of 1200 to 5200).

Letting  $Y_i$  represent the binary random variable “sat.bin” for the  $i$ th female crab,

$$\begin{aligned} Y_i &\overset{ind}{\sim} \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \eta_i \\ \eta_i &= \mathbf{x}_i\beta \end{aligned}$$

where the covariate  $\mathbf{x}_i$  and the interpretation of the corresponding coefficient,  $\beta$ , will be explored in groups below.

Let's explore different considerations for this model in groups. Each group will teach us what they discover afterward.

```
load(url('https://byu.box.com/shared/static/ok29pvry4dep5hb0wilutz0e09n8nk2i.rd
```

**Group 1:** two-level factor as a covariate

1. Fit a logistic regression model using “is.very.dark” as a covariate.

We can write the linear component of our model a couple of different ways. Two popular ways include:

$$(a) \eta_{ij} = \alpha_0 + \gamma_j$$

$$(b) \eta_i = \beta_0 + I(x_i = \text{True})\beta_1$$

where in (a), we are explicitly grouping the crabs so that they are grouped by color. The approach in (a) requires a modeling constraint. The default in R is to assume  $\gamma_1 = 0$  (*options(contrasts=c(“contr.treatment”, “contr.poly”))*). This will be the same as the model defined by (b). To instead fit a sum-to-zero,  $\sum_j \gamma_j = 0$ , constraint, use *options(contrasts=c(“contr.sum”, “contr.poly”))*. Explore both options. What do you notice?

2. Using the approach defined by model (b), interpret the intercept,  $\beta_0$ , in terms of log odds and odds.
3. Interpret the added coefficients,  $\beta_0 + \beta_1$ , in terms of log odds and odds.

4. What is the estimate of the log odds ratio? How does it relate to  $\beta_1$  and a difference in logits for the two factor levels?
5. Find a point estimate and interpret (2 ways!) the odds ratio.

**Group 2:** four-level factor as a covariate

1. Conduct a test of deviance to determine whether “color” should be included in a logistic model with only color as a covariate (use “anova(mod)”). One way to write the linear component of this model is:

$$\eta_i = \beta_0 + I(x_i = \text{"dark med"})\beta_{dm} + I(x_i = \text{"light med"})\beta_{lm} + I(x_i = \text{"med"})\beta_{med}.$$

2. At what significance-level would “dark medium” *not* be statistically different from “dark” using a z-test on the appropriate coefficient?

3. Without refitting the model, determine if “medium” is statistically different from “light medium.” In other words, test:

$$H_0 : \beta_{med} - \beta_{lm} = 0$$

$$H_1 : \beta_{med} - \beta_{lm} \neq 0$$

(To get the estimated  $\text{Var}(\hat{\beta})$ , use “summary(mod)\$cov.unscaled”.)

4. Check your answer in 3 by refitting the model with “light medium” as the base category.
5. For the model fit in 1, what does *odds ratio* (not odds) does  $\exp(\beta_{med})$  represent?

### Group 3: collinearity

1. Fit a logistic model with width as the only covariate. Is there evidence that width should be included in the model (compared to the model



with only an intercept)?

2. Fit a logistic model with both width and weight as covariates. What happens to the estimate and standard error of the coefficient for width compared to the model fit in 1?
3. Compare the analysis of deviance for the models fit in 1 and 2. What do you notice about how deviance is compared as more covariates are added to the model?
4. Obviously, as a crab's width increases, so will its weight. Thus, there is a collinearity issue when we want to include both variables in the model. Which variable – width or weight – should be included in the model? Why did you make this decision?
5. Does the VIF that you learned about in Stat 330 still apply here? Why

or why not?

**Group 4:** multiple covariates

1. Identify the “best” main-effects logistic regression model by testing deviance. Keep track of your process so that you can share your process and tips you discovered with your classmates.
2. Interpret the final coefficients in your model as both odds and odds ratios.

**Group 5:** interactions

1. Fit a logistic regression model with main effects “width” and “is.very.dark”. Make four plots of fitted values (two of  $\hat{\pi}$  as the y-axis and two of  $\hat{\eta}$  as the y-axis) with confidence intervals: a plot with “width” as the x-axis when “is.very.dark” is false and another plot with “width” as the x-axis when “is.very.dark” is true.

2. Fit the logistic regression model from 1 but including an interaction between the two covariates. Make the same two plots as in 1.
3. How do the plots change?
4. How does interpretation of the coefficients change when the interaction is added?

#### **i** Loading the Rdata file into python

```
# load crabs data from rdata url
response = requests.get('https://byu.box.com/shared/static/ok29pvry4dep5hb0wil
with open("temp_data.rdata", "wb") as f:
    f.write(response.content)
```

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```
result = pyreadr.read_r("temp_data.rdata")
os.remove("temp_data.rdata")

crabs = result['crabs']
```

## 4.6 Readings for Logistic Regression and Rare Events

- [Discussion](#) with expertise offered by [Paul Allison](#) (second webpage for Dr. Allison [here](#)).
- Recommend profile penalized likelihood confidence intervals using the “logistf” R package.
- King and Zeng (2001) examine various correction factors for both predictions and coefficients.
- Some “rules of thumb” exist for when it is okay to use logistic regression such as “10 events per variable;” Smeden et al. (2016) discusses why this is *not* a useful rule of thumb.
- Extension of exploration in Section 4.1 is provided in Franco et al. (2019).
- [A summary/discussion](#) of various recommendations by [Richard Williams](#). (Note: he also has several other working papers that discuss other aspects of logistic regression, such as coefficient interpretation for added variables in logit/probit models.)
- Feng et al. (2023) provide a brief editorial on the holes that still exist in predicting rare event outcomes.
- Wang (2020) considers the computational problem of massive data sets and rare events, showing that undersampling non-events can result in the same asymptotic distributions as fitting the full data (and that over-sampling events does not).
- One challenge with rare events is “separation” of the classes (making the number of solutions for logistic regression parameter estimates infinite). Shen and Gao (2008) discuss this problem and suggest penalized likelihood with ridge penalization.
- Chang et al. (2023) is an “application” paper examining three logistic regression models to identify variables relating to fatalities and flash floods. (They have their data and code available!)
- Not a rare events paper, but somewhat related is the “large  $p$ ” (or large number of covariates) problem. Sur and Canès (2018) provide an overview of the challenges and solutions to this problem. Additionally, Loon et al. (2020) propose a stacked penalized regression for this type of data.

## 4.7 Mini Project 2

This is a group (size  $\in \{2, 3\}$ ) project.

Using the [Salt Lake City](#) and [San Francisco](#) Mormons Survey data from 1967–1969, choose (or define) a binary response variable. Select several possible covariates (some of *both* continuous and categorical) to possibly include in the model. Using model-building principles, determine a model for the binary outcome.

1. Write a report of your analysis (due on Learning Suite by February 11 at midnight). The report should include:
  - introduction and motivation
  - exploratory data analyses
  - summary of approach for variable selection and choice for link function
  - the final model (write out the formal model and include parameter estimates in a table)
  - parameter inference (interpretation of parameter estimates and confidence intervals on the proportion of successes at illustrative covariate values)
  - model checking (e.g., deviance, residuals, sensitivity/specificity, training/test error rates, etc.)
  - conclusion/summary (why are your results interesting in the context of the application; list remaining statistical and application questions). This should be 3-5 pages, including figures (include code as an appendix).
2. In class (Feb 11), give a short presentation of your results. This will include three slides (no more), one for each of the following items:
  - a. the (well-defined) final model,
  - b. one interesting statistical result (e.g., impact of link function, strengths/weaknesses of your model, variable definitions),
  - c. one interesting result for the application (e.g., relationships between variables, implications of the model, anecdotal predicted probabilities).

Additionally, each person in your group must present a portion of the work and demonstrate a clear understanding of the entire analysis.

## 4.8 Homework Assignment

Using a version of the model you used for Mini Project 2, provide a table that includes for at least two sets of given values of covariates, estimated:

- probabilities
- odds
- odds ratios

with corresponding confidence intervals for each of the above. Also include interpretations of each of the quantities and provide two different interpretations of the odds ratios (recall that the odds ratio estimates were the same whether we looked at “successes” for  $X$  or  $Y$ ). Note: if your model does not use the logit link, you are welcome to estimate these using the model you fit, or you can fit a logit link model for this assignment only and estimate the odds and odds ratios the way we did in class.

Be sure to write out the model (including the random, link, and systematic components) you use here in this assignment.

Table 4.4: Example of a table for this homework assignment.

| covariate/factor<br>levels | estimated prob. of<br>“success” (95% CI) | estimated odds for<br>“success” (95% CI) | estimated<br>odds ratio<br>(95% CI) |
|----------------------------|------------------------------------------|------------------------------------------|-------------------------------------|
| $x_1 = 1, x_2 = 3.5$       | 0.85 (low, up)                           | 5.67 (low, up)                           | 3.34                                |
| $x_1 = 0, x_2 = 3.5$       | 0.63 (low, up)                           | 1.70 (low, up)                           | (low, up)                           |

# 5 GLM's for Count Data

## 5.1 Loglinear Models

Loglinear models can be thought of as:

- Poisson approximations for a Binomial model with a “small” probability of success, or
- Poisson GLM when we (typically) only have covariates that are factors.

Although loglinear models have some nice features and properties, they aren't as common as they used to be. Agresti (2013) book for categorical data is the best general reference for these models.

### Contingency Tables

Consider two categorical variables,  $X_1$  and  $X_2$ , with  $I$  and  $J$  categories, respectively.

Example:

| $X_1$ | $X_2$    |          |
|-------|----------|----------|
|       | 1        | 2        |
| 1     | $y_{11}$ | $y_{12}$ |
| 2     | $y_{21}$ | $y_{22}$ |
| 3     | $y_{31}$ | $y_{32}$ |

Previously, for example,  $X_2$  was our response,  $X_1$  was a factor, and we modeled the probabilities of  $X_2 = j$ , for  $j = 1, \dots, J$ . We did this using binomial models and modeled the logit of the probabilities.



## Another Look at Contingency Tables

Consider the table above. Instead of modeling probabilities (e.g.,  $P(X_2 = 1)$ ), model the frequencies  $(y_{ij})$  in *each joint category*. We denote the expected frequencies as:

$$E(Y_{ij}) = \mu_{ij} = n\pi_{ij}$$

## Independence

The *marginal probabilities* are denoted by

$$P(X_1 = i) = \pi_{i.} \text{ and } P(X_2 = j) = \pi_{.j}.$$

If  $X_1$  and  $X_2$  are independent,

$$P(X_1 = i \text{ and } X_2 = j) = \pi_{i.}\pi_{.j}$$

and the expected frequencies can be written as

$$E(Y_{ij}) = \mu_{ij} = n\pi_{i.}\pi_{.j}.$$

### Models:

Notice

$$\log \mu_{ij} = \log n + \log \pi_{i.} + \log \pi_{.j} \quad (5.1)$$

To test independence, compare Equation 5.1 with

$$\log \mu_{ij} = \log n + \log \pi_{ij} \quad (5.2)$$

→ Using loglinear notation...

Write Equation 5.1 as the **Additive Model**:

$$\log \mu_{ij} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} \quad (5.3)$$

Write Equation 5.2 as the **Saturated Model**:

$$\log \mu_{ij} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} + \lambda_{ij}^{x_1 x_2} \quad (5.4)$$

If the overall sample size  $n$  is fixed (instead of the row or column totals being fixed), then we have the **Minimal Model**, and it can be written as:

$$\log \mu_{ij} = \lambda \quad (5.5)$$

### Testing for Independence:

Which model assumes independence of  $X_1$  and  $X_2$ ?

→ minimal model.

How, then, do we test for independence?

→ test if  $\lambda_{ij}^{x_1 x_2}$  is significant.

### Interpretation of Parameters

- *Independent Model* Let  $J = 2$  (so that our data is an  $I \times 2$  table) and fit the independent model:

$$\lambda + \lambda_i^{x_1} + \lambda_j^{x_2}.$$

Consider the log odds of level 2 to level 1 of the binary variable:  
 $P(X_2 = 2) = \pi$ .

$$\begin{aligned} \log \frac{\pi}{1 - \pi} &= \log \left( \frac{n\pi}{n(1 - \pi)} \right) = \log \left( \frac{\mu_{i2}}{\mu_{i1}} \right) \\ &= \log(\mu_{i2}) - \log(\mu_{i1}) \\ &= \lambda + \lambda_i^{x_1} + \lambda_2^{x_2} - (\lambda + \lambda_i^{x_1} + \lambda_1^{x_2}) \\ &= \lambda_2^{x_2} - \lambda_1^{x_2}. \end{aligned}$$

Notice:

- doesn't depend on  $i$  (logit for  $X_2$  doesn't depend on level of  $X_1$ )
- corresponds to the simple logit model  $\text{logit}(\pi) = \beta_0$
- chance of classification/prediction in a particular column is the same for all rows

- *Saturated Model* Now consider the log odds using the saturated model

$$\begin{aligned}
\log \frac{\pi}{1-\pi} &= \lambda + \lambda_i^X + \lambda_2^Y + \lambda_{i2}^{XY} - (\lambda + \lambda_i^X + \lambda_1^Y + \lambda_{i1}^{XY}) \\
&= (\lambda_2^Y - \lambda_1^Y) + (\lambda_{i2}^{XY} - \lambda_{i1}^{XY}) \\
&= \beta_0 + \beta_i
\end{aligned}$$

Notice:

- corresponds to a logit model fit to a factor covariate

- *Both Variables as Responses* Consider the saturated model for a  $2 \times 2$  table. Recall the odds ratio

$$OR = \frac{\mu_{11}/\mu_{12}}{\mu_{21}/\mu_{22}} = \frac{\mu_{11}/\mu_{21}}{\mu_{12}/\mu_{22}}.$$

(“odds of  $X_2 = 1$  vs 2 when  $X_1 = 1$  to odds of  $X_2 = 1$  vs 2 when  $X_1 = 2$ ” = “odds of  $X_1 = 1$  vs 2 when  $X_2 = 1$  to odds of  $X_1 = 1$  vs 2 when  $X_2 = 2$ ”.)  
Then

$$\begin{aligned}
\log OR &= \log \mu_{11} + \log \mu_{22} - \log \mu_{12} - \log \mu_{21} \\
&= \lambda + \lambda_1^{x_1} + \lambda_2^{x_2} + \lambda_{12}^{x_1 x_2} + \lambda + \lambda_2^{x_1} + \lambda_2^{x_2} + \lambda_{22}^{x_1 x_2} \\
&\quad - (\lambda + \lambda_1^{x_1} + \lambda_1^{x_2} + \lambda_{11}^{x_1 x_2}) - (\lambda + \lambda_2^{x_1} + \lambda_1^{x_2} + \lambda_{21}^{x_1 x_2}) \\
&= \lambda_{11}^{x_1 x_2} + \lambda_{22}^{x_1 x_2} - \lambda_{12}^{x_1 x_2} - \lambda_{21}^{x_1 x_2}
\end{aligned}$$

Notice:

- $\{\lambda_{ij}^{x_1 x_2}\}$  determine the log odds ratios
- $\lambda_{ij}^{x_1 x_2} = 0 \forall ij \Rightarrow \log \text{ odds ratio is } 0 \Rightarrow \text{odds ratio is } 1 \text{ and therefore } X_1 \text{ and } X_2 \text{ are independent.}$

#### **i** Note

Loglinear models are primarily useful for *relationships* among  $> 2$  categorical response variables.

## A Word on the Minimal Model...

When only the total number of observations is fixed/given, the minimal model is  $\log(\mu_{ij}) = \lambda$ .

However, when the number of observations for one of the margins is fixed, we must expand the minimal model to account for this. Consider a controlled trial for an influenza vaccine. Patients are randomly allocated to two groups (new vaccine vs. saline placebo). We also consider the levels of hemagglutinin inhibiting antibody measured 6 weeks after vaccination: small, medium, and large.

| Vaccine ( $X_1$ ) | hemagglutinin ( $X_2$ ) |        |       | Total |
|-------------------|-------------------------|--------|-------|-------|
|                   | small                   | medium | large |       |
| Placebo           | 25                      | 8      | 5     | 38    |
| Vaccine           | 6                       | 18     | 11    | 35    |

The minimal model in this case is:

$$\log(\mu_{ij}) = \lambda + \lambda_i^{x_1}.$$

## Extension to 3-way Tables

$$E(Y_{ijk}) = \mu_{ijk} = nP(X_1 = i \text{ and } X_2 = j \text{ and } X_3 = k)$$

| $X_3 = 1$ | $X_2$     |           |     |           |
|-----------|-----------|-----------|-----|-----------|
| $X_1$     | 1         | 2         | ... | J         |
| 1         | $y_{111}$ | $y_{121}$ |     | $y_{1J1}$ |
| 2         | $y_{211}$ | $y_{221}$ |     | $y_{2J1}$ |
| $\vdots$  |           |           |     |           |
| I         | $y_{I11}$ | $y_{I21}$ |     | $y_{IJ1}$ |
| $X_3 = 2$ |           |           |     |           |
| 1         | $y_{112}$ | $y_{122}$ |     | $y_{1J2}$ |
| 2         | $y_{212}$ | $y_{222}$ |     | $y_{2J2}$ |
| $\vdots$  |           |           |     |           |
| I         | $y_{I12}$ | $y_{I22}$ |     | $y_{IJ2}$ |

|           |           |           |     |           |
|-----------|-----------|-----------|-----|-----------|
| $X_3 = 1$ | $X_2$     |           |     |           |
| $X_1$     | 1         | 2         | ... | J         |
| $\vdots$  |           |           |     |           |
| $X_3 = K$ |           |           |     |           |
| 1         | $y_{11K}$ | $y_{12K}$ |     | $y_{1JK}$ |
| 2         | $y_{21K}$ | $y_{22K}$ |     | $y_{2JK}$ |
| $\vdots$  |           |           |     |           |
| I         | $y_{I1K}$ | $y_{I2K}$ |     | $y_{IJK}$ |

### Models for 3-way Tables

Consider the model

$$\log \mu_{ijk} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} + \lambda_k^{x_3} + \lambda_{ik}^{x_1x_3} + \lambda_{jk}^{x_2x_3}$$

- Since this model contains an  $X_1$ - $X_3$  two-factor term, it allows for an association between  $X_1$  and  $X_3$ , controlling for  $X_2$ . (Similarly, for  $X_2$  and  $X_3$ , controlling for  $X_1$ .)
- Since it does not contain a two-factor term for  $X_1$  and  $X_2$ , this model specifies *conditional independence* between  $X_1$  and  $X_2$ , controlling for  $X_3$ .
- Again, single factor terms represent marginal distributions and including  $\lambda_i^{x_1}$  in the model forces the fitted values to have the same totals at the various levels of  $X_1$ .
- Loglinear models primarily relate information regarding the higher-order parameters.

#### *Mutual Independence Model*

$$\log \mu_{ijk} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} + \lambda_k^{x_3}$$

$X_1, X_2, X_3$  are mutually independent.

#### *Homogeneous Association Model*

$$\log \mu_{ijk} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} + \lambda_k^{x_3} + \lambda_{ij}^{x_1x_2} + \lambda_{ik}^{x_1x_3} + \lambda_{jk}^{x_2x_3}$$

Why homogeneous? (Why “same”?)

Conditional OR's between any variables are identical at each level of the 3rd variable (i.e., no 3-factor interaction)

*Saturated Model*

$$\log \mu_{ijk} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} + \lambda_k^{x_3} + \lambda_{ij}^{x_1 x_2} + \lambda_{ik}^{x_1 x_3} + \lambda_{jk}^{x_2 x_3} + \lambda_{ijk}^{x_1 x_2 x_3}$$

This model allows OR between any 2 variables to var across levels of the 3rd variable  $\rightarrow$

$$OR_k = \frac{\mu_{11k}/\mu_{12k}}{\mu_{21k}/\mu_{22k}} \Rightarrow \text{diff } \forall k$$

## Interpreting Parameters

- Refer to their highest-order terms
- Main effect estimates can change dramatically depending on the parameter constraints of higher order terms.

*Examples of interpretations of relationships between variables*

- (Strongest Condition)  $X_1, X_2, X_3$  are mutually independent if  $\pi_{ijk} = \pi_{i..}\pi_{.j.}\pi_{..k}$

$$\Rightarrow \log \mu_{ijk} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} + \lambda_k^{x_3}$$

- $X_2$  is jointly independent of  $X_1$  and  $X_3$  when  $\pi_{ijk} = \pi_{i.k}\pi_{.j.}$

$$\Rightarrow \log \mu_{ijk} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} + \lambda_k^{x_3} + \lambda_{ik}^{x_1 x_3}$$

- (Weakest Condition)  $X_1$  and  $X_2$  conditionally independent given  $X_3$  when  $\pi_{ijk} = \pi_{i.|k}\pi_{.j|k}$

$$\Rightarrow \log \mu_{ijk} = \lambda + \lambda_i^{x_1} + \lambda_j^{x_2} + \lambda_k^{x_3} + \lambda_{ik}^{x_1 x_3} + \lambda_{jk}^{x_2 x_3}$$

*Conditional Odds Ratio*

Consider a  $2 \times 2 \times K$  table and the homogeneous model. The  $X_1$ - $X_2$  conditional odds ratio is

$$OR_{12(k)} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}} = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}$$

$\rightarrow$  The RHS doesn't depend on  $k \Rightarrow$  odds ratio is the same for all  $k$ . Does this make sense given the model?

*Alcohol-Cigarette-Marijuana Use*

In 1992, Wright State University School of Medicine and United Health Services in Dayton, OH conducted a survey asking students in their final year of high school in a non-urban area near Dayton whether or not they had ever used alcohol, cigarettes, or marijuana.

| Alcohol | Cigarettes | Marijuana |     |
|---------|------------|-----------|-----|
|         |            | Yes       | No  |
| Yes     | Yes        | 911       | 538 |
|         | No         | 44        | 456 |
| No      | Yes        | 3         | 43  |
|         | No         | 2         | 279 |



1. (A, C, M):  $\log \mu_{ijk} = \lambda + \lambda_i^A + \lambda_j^C + \lambda_k^M$
2. (AC, M):  $\log \mu_{ijk} = \lambda + \lambda_i^A + \lambda_j^C + \lambda_k^M + \lambda_{ij}^{AC}$
3. (AM, CM):  $\log \mu_{ijk} = \lambda + \lambda_i^A + \lambda_j^C + \lambda_k^M + \lambda_{ik}^{AM} + \lambda_{jk}^{CM}$
4. (AC, AM, CM):  $\log \mu_{ijk} = \lambda + \lambda_i^A + \lambda_j^C + \lambda_k^M + \lambda_{ij}^{AC} + \lambda_{ik}^{AM} + \lambda_{jk}^{CM}$
5. (ACM):  $\log \mu_{ijk} = \lambda + \lambda_i^A + \lambda_j^C + \lambda_k^M + \lambda_{ij}^{AC} + \lambda_{ik}^{AM} + \lambda_{jk}^{CM} + \lambda_{ijk}^{ACM}$

|              | Estimate   | Std. Error | z value   | Pr(> z )      |
|--------------|------------|------------|-----------|---------------|
| (Intercept)  | 4.1725378  | 0.06495836 | 64.234043 | 0.000000e+00  |
| alcoholYes   | 1.7851115  | 0.05975887 | 29.871911 | 4.559915e-196 |
| cigaretteYes | 0.6493063  | 0.04415087 | 14.706534 | 5.852911e-49  |
| marijuanaYes | -0.3154188 | 0.04244454 | -7.431316 | 1.075222e-13  |

|                           | Estimate  | Std. Error | z value    | Pr(> z )     |
|---------------------------|-----------|------------|------------|--------------|
| (Intercept)               | 5.633420  | 0.05970084 | 94.360822  | 0.000000e+00 |
| alcoholYes                | 0.487719  | 0.07576720 | 6.437073   | 1.217997e-10 |
| cigaretteYes              | -1.886669 | 0.16269698 | -11.596213 | 4.307172e-31 |
| marijuanaYes              | -5.309042 | 0.47519695 | -11.172299 | 5.571964e-29 |
| alcoholYes:cigaretteYes   | 2.054534  | 0.17406432 | 11.803304  | 3.752817e-32 |
| alcoholYes:marijuanaYes   | 2.986014  | 0.46467793 | 6.425987   | 1.310164e-10 |
| cigaretteYes:marijuanaYes | 2.847889  | 0.16383940 | 17.382200  | 1.125516e-67 |

|                         | Estimate   | Std. Error | z value     |
|-------------------------|------------|------------|-------------|
| (Intercept)             | 5.6312118  | 0.05986843 | 94.0597808  |
| alcoholYes              | 0.4912810  | 0.07600797 | 6.4635460   |
| cigaretteYes            | -1.8700117 | 0.16382931 | -11.4143901 |
| marijuanaYes            | -4.9380646 | 0.70963645 | -6.9585837  |
| alcoholYes:cigaretteYes | 2.0353774  | 0.17576052 | 11.5804014  |

|                                      |              |            |           |
|--------------------------------------|--------------|------------|-----------|
| alcoholYes:marijuanaYes              | 2.5997614    | 0.72698290 | 3.5760971 |
| cigaretteYes:marijuanaYes            | 2.2754768    | 0.92745511 | 2.4534630 |
| alcoholYes:cigaretteYes:marijuanaYes | 0.5895107    | 0.94236387 | 0.6255659 |
|                                      | Pr(> z )     |            |           |
| (Intercept)                          | 0.000000e+00 |            |           |
| alcoholYes                           | 1.022774e-10 |            |           |
| cigaretteYes                         | 3.543812e-30 |            |           |
| marijuanaYes                         | 3.437106e-12 |            |           |
| alcoholYes:cigaretteYes              | 5.180293e-31 |            |           |
| alcoholYes:marijuanaYes              | 3.487622e-04 |            |           |
| cigaretteYes:marijuanaYes            | 1.414881e-02 |            |           |
| alcoholYes:cigaretteYes:marijuanaYes | 5.315997e-01 |            |           |

|   | [,1]      | [,2]      | [,3]        | [,4]      | [,5] |
|---|-----------|-----------|-------------|-----------|------|
| 1 | 539.98258 | 611.17750 | 909.2395833 | 910.38317 | 911  |
| 2 | 740.22612 | 837.82250 | 438.8404255 | 538.61683 | 538  |
| 3 | 282.09123 | 210.89631 | 45.7604167  | 44.61683  | 44   |
| 4 | 386.70007 | 289.10369 | 555.1595745 | 455.38317 | 456  |
| 5 | 90.59739  | 19.40246  | 4.7604167   | 3.61683   | 3    |
| 6 | 124.19392 | 26.59754  | 142.1595745 | 42.38317  | 43   |
| 7 | 47.32880  | 118.52373 | 0.2395833   | 1.38317   | 2    |
| 8 | 64.87990  | 162.47627 | 179.8404255 | 279.61683 | 279  |

|   | [,1]       | [,2]       | [,3]        | [,4]        | [,5] |
|---|------------|------------|-------------|-------------|------|
| 1 | 14.521781  | 11.296737  | 0.05836282  | 0.02044112  | 0    |
| 2 | -7.816886  | -11.092036 | 4.57017346  | -0.02658329 | 0    |
| 3 | -17.682615 | -13.995751 | -0.26193377 | -0.09255965 | 0    |
| 4 | 3.425982   | 9.045028   | -4.34413277 | 0.02889876  | 0    |
| 5 | -12.439771 | -4.648031  | -0.86631165 | -0.33428452 | 0    |
| 6 | -8.436376  | 2.916827   | -9.77162435 | 0.09451934  | 0    |
| 7 | -8.831858  | -14.721399 | 2.22871517  | 0.49134221  | 0    |
| 8 | 19.639243  | 8.285643   | 6.83534950  | -0.03690149 | 0    |

We can continue and compare model deviances, odds ratios, confidence intervals for fitted values and/or odds ratios, and interpretations.

## 5.2 Poisson Regression

A GLM for Poisson-distributed response variables follows the standard GLM framework:

$$\begin{aligned}Y_i &\overset{ind}{\sim} \text{Poisson}(\mu_i), i = 1, \dots, n \\ \eta_i &= g(\mu_i) = \log(\mu_i) \\ \eta_i &= \mathbf{x}_i' \beta\end{aligned}$$

- For loglinear models, the individual covariates were factors, so we could include different  $\lambda$ 's for each level of the covariate. Here, we consider the case where covariates may be continuous.
- The log link is the most common, but the inverse link is also available through R's glm function.

### **i** Note

The biggest takeaway from this section (although we will consider/review model selection ideas), is the use of an “offset;” how the offset changes the model, its interpretation, its predictions, etc.

### Motivating Example

Top 25 passing quarterbacks (most pass yds) of the 2023 NFL season: season passing stats for each season of their NFL career.

*We're interested in how number of completed passes changes the more experienced the quarterback.*

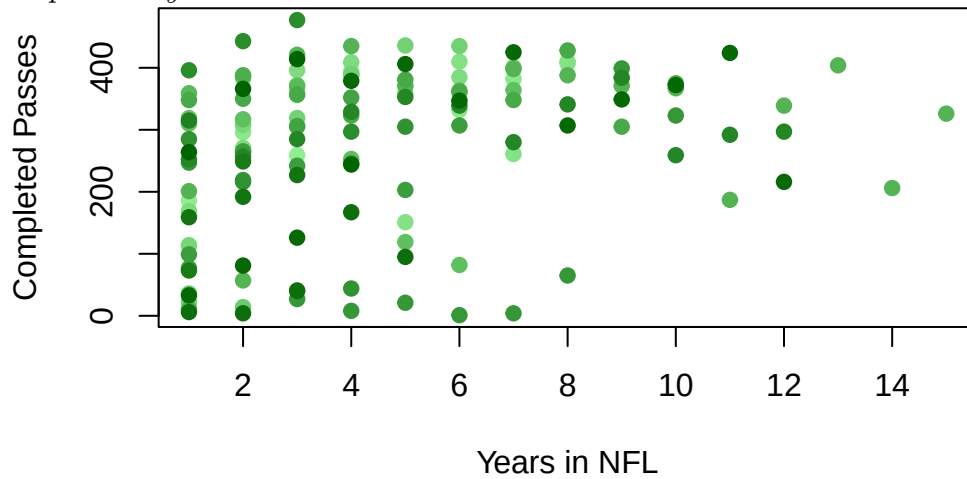
- Response variable:  $Y_i$  represents the number of completed passes during the season for each observation (variable name: “COMP”)
- Covariates: Factor by quarterback (“Player”), Number of years in the NFL (“SSNYR”)

```
qb <- read.csv(url("https://byu.box.com/shared/static/aivpn0ea03erb2c1oy7n715c"))
```

Note: For various reasons sometimes a player will not have attempted any

passes that year. In this data set there are no such cases, but if there were, we would need to remove years where the player did not attempt any passes.

### Exploratory Plot



### Model 1

Let  $Y_i$  be the number of completed passes for the  $i$ th observation. We model,

$$Y_i \stackrel{ind}{\sim} \text{Pois}(\mu_i)$$

$$\log(\mu_i) = \beta_0 + x_{1i}\beta_1 + I(x_{2i} = \text{"Burrow"})\beta_2 + \cdots + I(x_{2i} = \text{"Young"})\beta_{25}$$

where  $x_{1i}$  is the year in the NFL for the player corresponding to the  $i$ th observation and  $x_{2i}$  is the last name of the player corresponding to the  $i$ th observation.

Call:

```
glm(formula = COMP ~ SSNYR + Player, family = poisson, data = qb)
```

Coefficients:

|               | Estimate  | Std. Error | z value | Pr(> z )     |
|---------------|-----------|------------|---------|--------------|
| (Intercept)   | 5.669471  | 0.023448   | 241.789 | < 2e-16 ***  |
| SSNYR         | 0.037737  | 0.001900   | 19.861  | < 2e-16 ***  |
| PlayerBurrow  | 0.009847  | 0.035822   | 0.275   | 0.7834       |
| PlayerCarr    | -0.003479 | 0.028282   | -0.123  | 0.9021       |
| PlayerCousins | -0.257673 | 0.028837   | -8.935  | < 2e-16 ***  |
| PlayerDobbs   | -1.650542 | 0.060901   | -27.102 | < 2e-16 ***  |
| PlayerFields  | -0.484459 | 0.047349   | -10.232 | < 2e-16 ***  |
| PlayerGoff    | -0.037513 | 0.029722   | -1.262  | 0.2069       |
| PlayerHerbert | 0.234852  | 0.033568   | 6.996   | 2.63e-12 *** |

|                  |           |          |         |          |     |
|------------------|-----------|----------|---------|----------|-----|
| PlayerHowell     | -0.430441 | 0.054999 | -7.826  | 5.02e-15 | *** |
| PlayerHurts      | -0.243244 | 0.038818 | -6.266  | 3.70e-10 | *** |
| PlayerJackson    | -0.378678 | 0.035171 | -10.767 | < 2e-16  | *** |
| PlayerLawrence   | 0.173473  | 0.037522 | 4.623   | 3.78e-06 | *** |
| PlayerLove       | -0.799028 | 0.053678 | -14.886 | < 2e-16  | *** |
| PlayerMahomes    | 0.008196  | 0.030380 | 0.270   | 0.7873   |     |
| PlayerMayfield   | -0.301806 | 0.032791 | -9.204  | < 2e-16  | *** |
| PlayerMinshew    | -0.601200 | 0.040325 | -14.909 | < 2e-16  | *** |
| PlayerPrescott   | -0.061124 | 0.029873 | -2.046  | 0.0407   | *   |
| PlayerPurdy      | -0.374397 | 0.053743 | -6.966  | 3.25e-12 | *** |
| PlayerRidder     | -0.644851 | 0.060202 | -10.711 | < 2e-16  | *** |
| PlayerSmith      | -1.004648 | 0.035794 | -28.067 | < 2e-16  | *** |
| PlayerStafford   | -0.209247 | 0.028341 | -7.383  | 1.54e-13 | *** |
| PlayerStroud     | 0.057982  | 0.060516 | 0.958   | 0.3380   |     |
| PlayerTagovailoa | -0.151576 | 0.037673 | -4.023  | 5.73e-05 | *** |
| PlayerWilson     | -0.200739 | 0.028559 | -7.029  | 2.08e-12 | *** |
| PlayerYoung      | 0.045364  | 0.060844 | 0.746   | 0.4559   |     |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 11308.1 on 143 degrees of freedom  
Residual deviance: 6936.4 on 118 degrees of freedom  
AIC: 8020.9

Number of Fisher Scoring iterations: 5

Analysis of Deviance Table

Model: poisson, link: log

Response: COMP

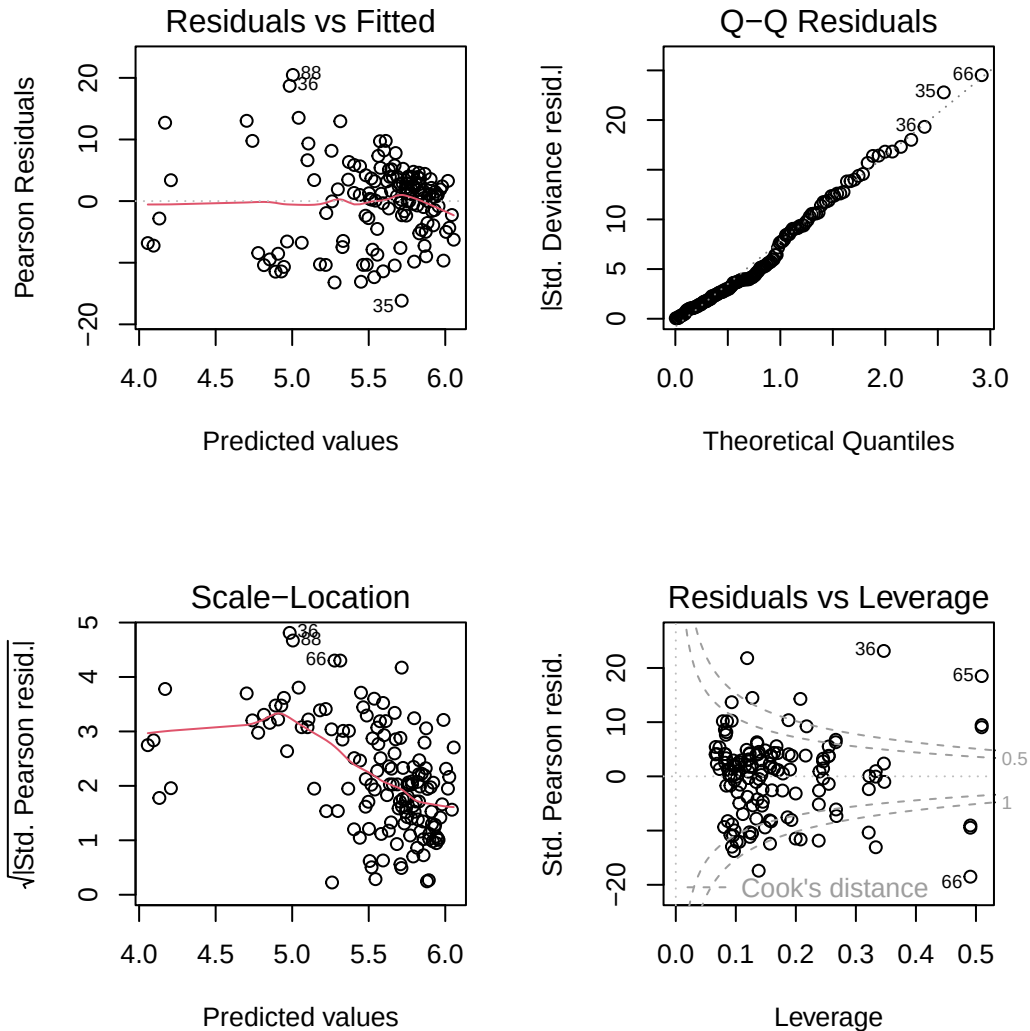
Terms added sequentially (first to last)

|  | Df | Deviance | Resid. Df | Resid. Dev | Pr(>Chi) |
|--|----|----------|-----------|------------|----------|
|--|----|----------|-----------|------------|----------|

|        |    |        |     |         |               |
|--------|----|--------|-----|---------|---------------|
| NULL   |    |        | 143 | 11308.1 |               |
| SSNYR  | 1  | 637.9  | 142 | 10670.2 | < 2.2e-16 *** |
| Player | 24 | 3733.8 | 118 | 6936.4  | < 2.2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1



*Parameter interpretation:*

- For every one year increase in season, the log expected number of completed passes per season for a given player increases by 0.0377. Thus, the expected number of completed passes increases by a factor of  $e^{0.0377} = 1.0385$ . So for someone that had 300

completed passes one year, their next year, on average they would have  $300 \times 1.0385 = 311.5375$  completed passes.

- Burrow's log expected number of completed passes is 0.0098 more(+)/less(-) than Allen's for a given year in the NFL (expected number is  $e^{0.0098} = 1.0099$  times Allen's).



## Model 2

Let  $Y_i$  be the number of completed passes for the  $i$ th observation. We model,

$$Y_i \overset{ind}{\sim} \text{Pois}(\mu_i)$$
$$\log(\mu_i) = \beta_0 + x_{1i}\beta_1 + I(x_{2i} = \text{"Burrow"})\beta_2 + \dots + I(x_{2i} = \text{"Young"})\beta_{25} + \log(x_{3i}),$$

where  $x_{1i}$  and  $x_{2i}$  are the same as in Model 1, and  $x_{3i}$  is the number of attempted passes for the  $i$ th observation.

→  $\log(x_{3i})$  is called the **offset**. Notice that if we subtract the offset from both sides,

$$\log\left(\frac{\mu_i}{x_{3i}}\right) = \beta_0 + x_{1i}\beta_1 + I(x_{2i} = \text{"Burrow"})\beta_2 + \dots + I(x_{2i} = \text{"Young"})\beta_{25}.$$

Thus, when we use an offset, we model the “rate” of an event. An offset can be a time or space interval, a maximum possible count (in this case, the rate will be a proportion/probability of the event), or another constraining variable.

We can fit an offset term in R using two approaches (as part of the formula or as a separate argument):

- `mod2 <- glm(COMP~SSNYR+Player, offset=log(ATT), data=qb, family=poisson)`
- `mod2 <- glm(COMP~SSNYR+Player + offset(log(ATT)), data=qb, family=poisson)`

Call:

```
glm(formula = COMP ~ SSNYR + Player + offset(log(ATT)), family = poisson,  
    data = qb)
```

Coefficients:

|              | Estimate  | Std. Error | z value | Pr(> z ) |     |
|--------------|-----------|------------|---------|----------|-----|
| (Intercept)  | -0.493062 | 0.023641   | -20.856 | <2e-16   | *** |
| SSNYR        | 0.009228  | 0.001976   | 4.670   | 3e-06    | *** |
| PlayerBurrow | 0.083895  | 0.035856   | 2.340   | 0.0193   | *   |
| PlayerCarr   | 0.010911  | 0.028179   | 0.387   | 0.6986   |     |

|                  |           |          |        |          |
|------------------|-----------|----------|--------|----------|
| PlayerCousins    | 0.023003  | 0.029052 | 0.792  | 0.4285   |
| PlayerDobbs      | -0.020155 | 0.060895 | -0.331 | 0.7407   |
| PlayerFields     | -0.031658 | 0.047370 | -0.668 | 0.5039   |
| PlayerGoff       | 0.012394  | 0.029733 | 0.417  | 0.6768   |
| PlayerHerbert    | 0.064191  | 0.033615 | 1.910  | 0.0562 . |
| PlayerHowell     | 0.016539  | 0.054973 | 0.301  | 0.7635   |
| PlayerHurts      | 0.010290  | 0.038807 | 0.265  | 0.7909   |
| PlayerJackson    | 0.019315  | 0.035171 | 0.549  | 0.5829   |
| PlayerLawrence   | 0.024914  | 0.037571 | 0.663  | 0.5073   |
| PlayerLove       | 0.017125  | 0.053632 | 0.319  | 0.7495   |
| PlayerMahomes    | 0.042361  | 0.030403 | 1.393  | 0.1635   |
| PlayerMayfield   | -0.020155 | 0.032775 | -0.615 | 0.5386   |
| PlayerMinshew    | -0.001557 | 0.040353 | -0.039 | 0.9692   |
| PlayerPrescott   | 0.050537  | 0.029843 | 1.693  | 0.0904 . |
| PlayerPurdy      | 0.102163  | 0.053750 | 1.901  | 0.0573 . |
| PlayerRidder     | 0.030670  | 0.060202 | 0.509  | 0.6104   |
| PlayerSmith      | -0.020936 | 0.035855 | -0.584 | 0.5593   |
| PlayerStafford   | -0.040345 | 0.027996 | -1.441 | 0.1496   |
| PlayerStroud     | 0.036419  | 0.060564 | 0.601  | 0.5476   |
| PlayerTagovailoa | 0.065817  | 0.037674 | 1.747  | 0.0806 . |
| PlayerWilson     | -0.002925 | 0.028431 | -0.103 | 0.9181   |
| PlayerYoung      | -0.030794 | 0.060892 | -0.506 | 0.6131   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 123.821 on 143 degrees of freedom  
 Residual deviance: 70.808 on 118 degrees of freedom  
 AIC: 1155.3

Number of Fisher Scoring iterations: 3

Analysis of Deviance Table

Model: poisson, link: log

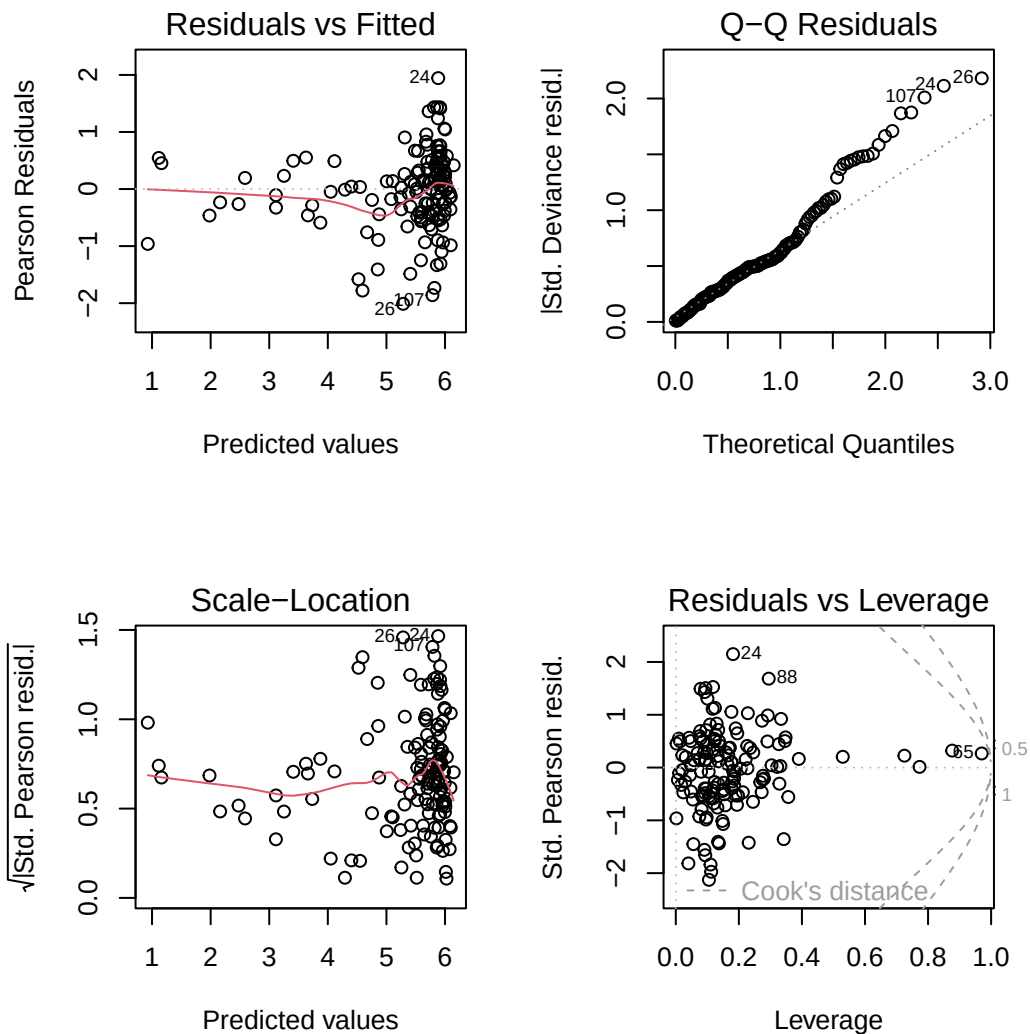
Response: COMP

Terms added sequentially (first to last)

|        | Df | Deviance | Resid. | Df  | Resid. | Dev     | Pr(>Chi)      |
|--------|----|----------|--------|-----|--------|---------|---------------|
| NULL   |    |          |        | 143 |        | 123.821 |               |
| SSNYR  | 1  | 13.813   |        | 142 |        | 110.008 | 0.0002019 *** |
| Player | 24 | 39.200   |        | 118 |        | 70.808  | 0.0260204 *   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

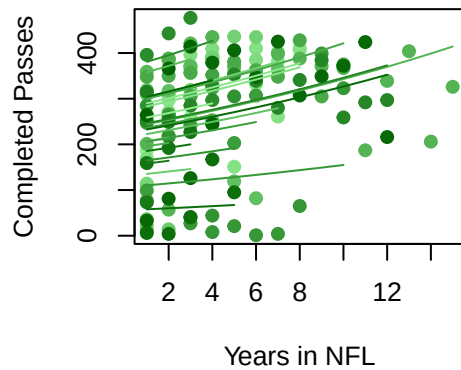


*Parameter interpretation:*

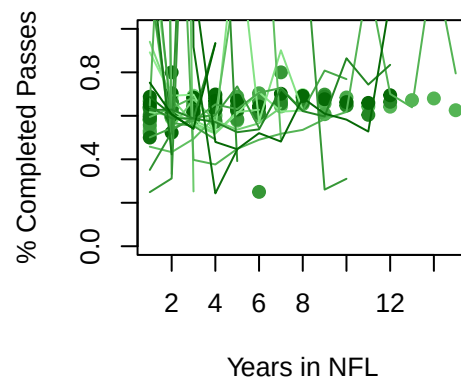
- For every one year increase in season for a given player, the log of the rate of completed passes per season (for a fixed number of attempts) increases by 0.0092. Thus, the expected rate of completed passes increases by a factor of  $e^{0.0092} = 1.0093$ . So for someone that in one year had a rate of 0.65 completed passes, the next year, on average their rate would be  $0.65 \times 1.0093 = 0.675$ .
- Burrow's log rate of completed passes is 0.0839 more(+)/less(-) than Allen's for a fixed year in the NFL (expected rate is  $e^{0.0839} = 1.0875$  times Allen's).

*Predictions of the counts and rates for Models 1 and 2*

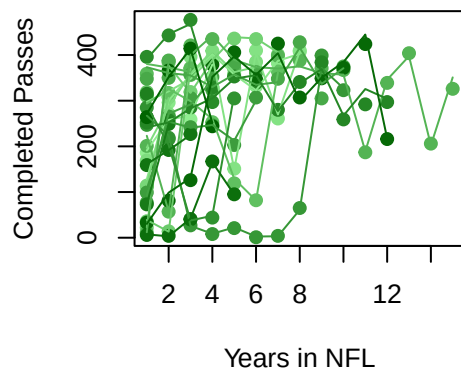
**Model 1**



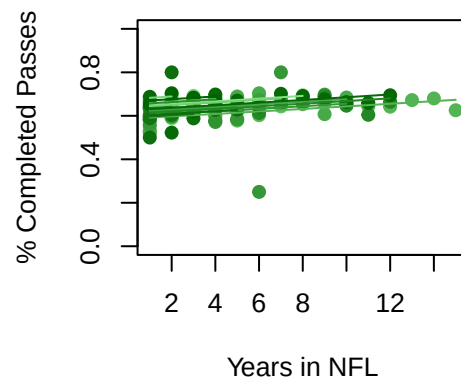
**Model 1**



**Model 2**



**Model 2**



### **Model 3**

*Why not give the offset a coefficient as well?*

$$Y_i \overset{ind}{\sim} \text{Pois}(\mu_i)$$

$$\log(\mu_i) = \beta_0 + x_{1i}\beta_1 + I(x_{2i} = \text{"Burrow"})\beta_2 + \dots + I(x_{2i} = \text{"Young"})\beta_{25} + \log(x_{3i})\beta_{26},$$

Call:

```
glm(formula = COMP ~ SSNYR + Player + log(ATT), family = poisson,
    data = qb)
```

Coefficients:

| Estimate | Std. Error | z value | Pr(> z ) |
|----------|------------|---------|----------|
|----------|------------|---------|----------|

|                  |           |          |        |          |     |
|------------------|-----------|----------|--------|----------|-----|
| (Intercept)      | -0.709220 | 0.105369 | -6.731 | 1.69e-11 | *** |
| SSNYR            | 0.008720  | 0.001992 | 4.377  | 1.21e-05 | *** |
| PlayerBurrow     | 0.086890  | 0.035886 | 2.421  | 0.0155   | *   |
| PlayerCarr       | 0.010875  | 0.028176 | 0.386  | 0.6995   |     |
| PlayerCousins    | 0.027936  | 0.029146 | 0.958  | 0.3378   |     |
| PlayerDobbs      | 0.020242  | 0.063807 | 0.317  | 0.7511   |     |
| PlayerFields     | -0.014782 | 0.048046 | -0.308 | 0.7583   |     |
| PlayerGoff       | 0.013174  | 0.029736 | 0.443  | 0.6577   |     |
| PlayerHerbert    | 0.058925  | 0.033709 | 1.748  | 0.0805   | .   |
| PlayerHowell     | 0.014498  | 0.054983 | 0.264  | 0.7920   |     |
| PlayerHurts      | 0.017503  | 0.038956 | 0.449  | 0.6532   |     |
| PlayerJackson    | 0.032603  | 0.035731 | 0.912  | 0.3615   |     |
| PlayerLawrence   | 0.021144  | 0.037615 | 0.562  | 0.5740   |     |
| PlayerLove       | 0.024380  | 0.053731 | 0.454  | 0.6500   |     |
| PlayerMahomes    | 0.040066  | 0.030423 | 1.317  | 0.1878   |     |
| PlayerMayfield   | -0.013359 | 0.032931 | -0.406 | 0.6850   |     |
| PlayerMinshew    | 0.011222  | 0.040796 | 0.275  | 0.7833   |     |
| PlayerPrescott   | 0.053295  | 0.029870 | 1.784  | 0.0744   | .   |
| PlayerPurdy      | 0.116789  | 0.054194 | 2.155  | 0.0312   | *   |
| PlayerRidder     | 0.050369  | 0.060917 | 0.827  | 0.4083   |     |
| PlayerSmith      | -0.008680 | 0.036305 | -0.239 | 0.8110   |     |
| PlayerStafford   | -0.038514 | 0.027998 | -1.376 | 0.1690   |     |
| PlayerStroud     | 0.037572  | 0.060568 | 0.620  | 0.5350   |     |
| PlayerTagovailoa | 0.073663  | 0.037858 | 1.946  | 0.0517   | .   |
| PlayerWilson     | 0.002749  | 0.028556 | 0.096  | 0.9233   |     |
| PlayerYoung      | -0.031534 | 0.060894 | -0.518 | 0.6046   |     |
| log(ATT)         | 1.034690  | 0.016474 | 62.806 | < 2e-16  | *** |

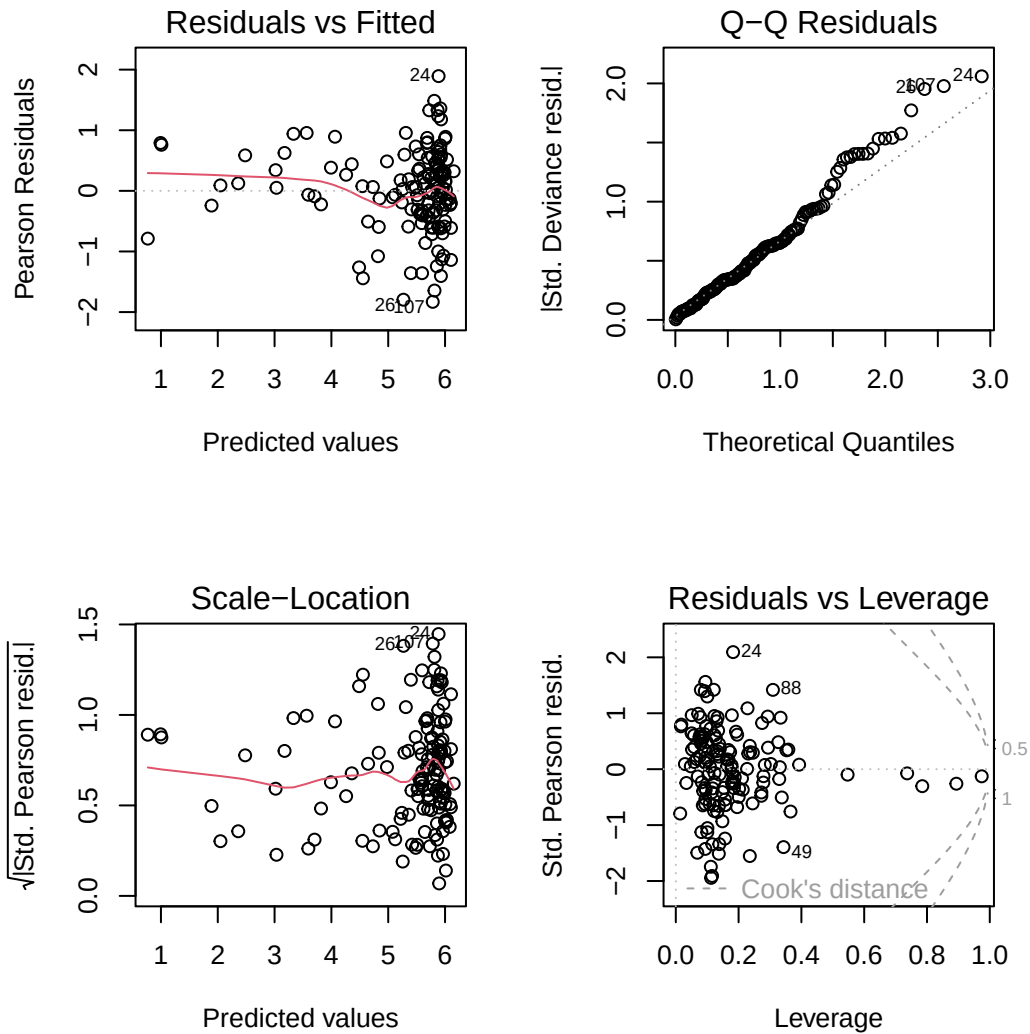
---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 11308.122 on 143 degrees of freedom  
Residual deviance: 66.304 on 117 degrees of freedom  
AIC: 1152.8

Number of Fisher Scoring iterations: 3



#### Model 4

What about a *logit model* for the proportion of completed passes?

$$Y_i \stackrel{\text{ind}}{\sim} \text{Binomial}(x_{3i}, \pi_i)$$

$$\log \left( \frac{\pi_i}{1 - \pi_i} \right) = \beta_0 + x_{1i}\beta_1 + I(x_{2i} = \text{"Burrow"})\beta_2 + \dots + I(x_{2i} = \text{"Young"})\beta_{25},$$

where  $x_{1i}$ ,  $x_{2i}$ , and  $x_{3i}$  are the same as before.

1. How is Model 2 different from a logit model?

## 2. How would you decide which likelihood to use?

Call:

```
glm(formula = CmpPercent ~ SSNYR + Player, family = binomial,
     data = qb, weights = ATT)
```

Coefficients:

|                  | Estimate  | Std. Error | z value | Pr(> z ) |     |
|------------------|-----------|------------|---------|----------|-----|
| (Intercept)      | 0.444886  | 0.039032   | 11.398  | < 2e-16  | *** |
| SSNYR            | 0.026009  | 0.003321   | 7.832   | 4.79e-15 | *** |
| PlayerBurrow     | 0.242750  | 0.061728   | 3.933   | 8.40e-05 | *** |
| PlayerCarr       | 0.031200  | 0.046864   | 0.666   | 0.50557  |     |
| PlayerCousins    | 0.068839  | 0.048813   | 1.410   | 0.15846  |     |
| PlayerDobbs      | -0.054805 | 0.099190   | -0.553  | 0.58059  |     |
| PlayerFields     | -0.080171 | 0.075898   | -1.056  | 0.29083  |     |
| PlayerGoff       | 0.034421  | 0.049479   | 0.696   | 0.48663  |     |
| PlayerHerbert    | 0.182362  | 0.056961   | 3.202   | 0.00137  | **  |
| PlayerHowell     | 0.046105  | 0.090664   | 0.509   | 0.61108  |     |
| PlayerHurts      | 0.028487  | 0.064089   | 0.444   | 0.65668  |     |
| PlayerJackson    | 0.053489  | 0.058624   | 0.912   | 0.36155  |     |
| PlayerLawrence   | 0.069180  | 0.062264   | 1.111   | 0.26654  |     |
| PlayerLove       | 0.047183  | 0.088969   | 0.530   | 0.59589  |     |
| PlayerMahomes    | 0.120915  | 0.051238   | 2.360   | 0.01828  | *   |
| PlayerMayfield   | -0.053438 | 0.053583   | -0.997  | 0.31862  |     |
| PlayerMinshew    | -0.003102 | 0.066190   | -0.047  | 0.96262  |     |
| PlayerPrescott   | 0.146074  | 0.050429   | 2.897   | 0.00377  | **  |
| PlayerPurdy      | 0.297832  | 0.094826   | 3.141   | 0.00168  | **  |
| PlayerRidder     | 0.085113  | 0.100211   | 0.849   | 0.39569  |     |
| PlayerSmith      | -0.056475 | 0.059132   | -0.955  | 0.33955  |     |
| PlayerStafford   | -0.110768 | 0.046049   | -2.405  | 0.01615  | *   |
| PlayerStroud     | 0.101339  | 0.100708   | 1.006   | 0.31429  |     |
| PlayerTagovailoa | 0.187861  | 0.064324   | 2.921   | 0.00349  | **  |
| PlayerWilson     | -0.007300 | 0.047191   | -0.155  | 0.87707  |     |
| PlayerYoung      | -0.074909 | 0.096661   | -0.775  | 0.43836  |     |

---

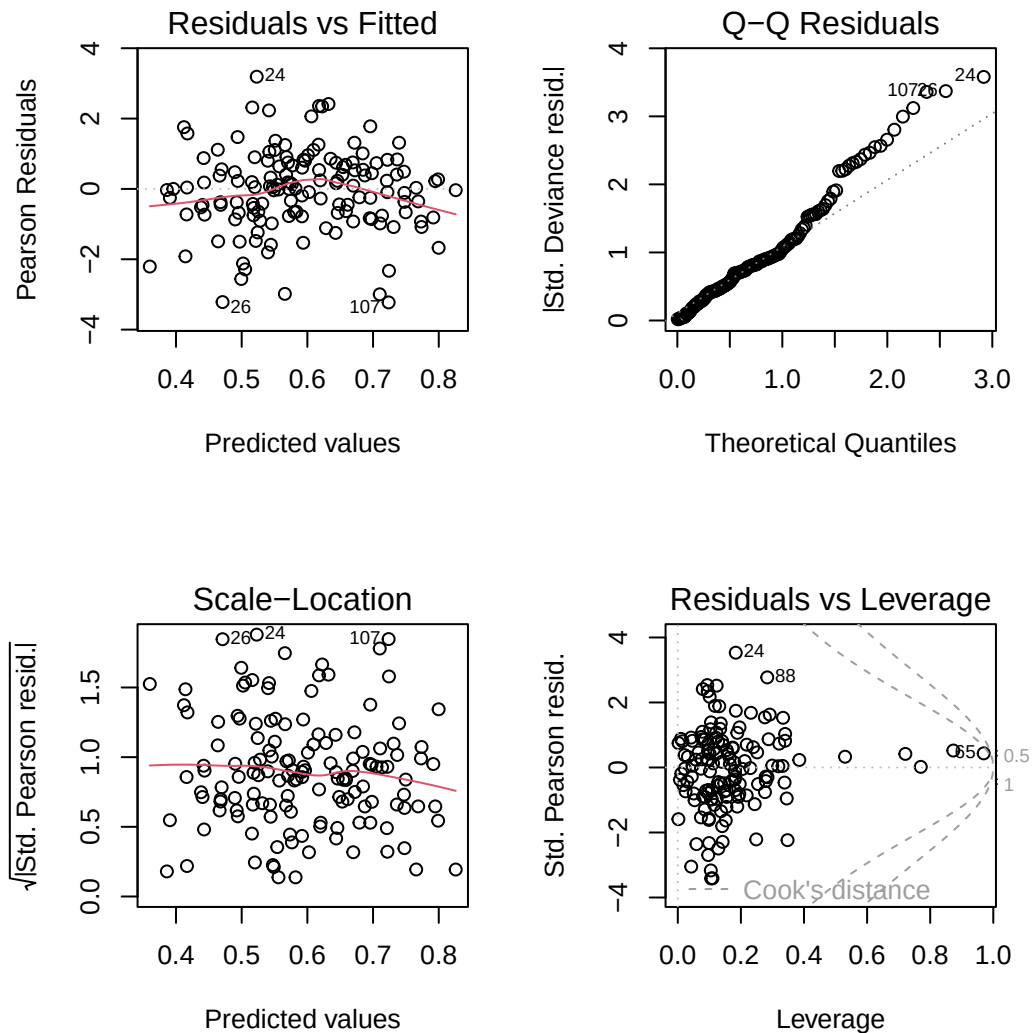
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1



(Dispersion parameter for binomial family taken to be 1)

Null deviance: 340.66 on 143 degrees of freedom  
Residual deviance: 190.82 on 118 degrees of freedom  
AIC: 1127.3

Number of Fisher Scoring iterations: 3



*What conclusions can you draw about the relationship between*

*years in the NFL and number of completed passes?* What considerations did we *not* consider here that is relevant for the interpretation of this coefficient?

## 5.3 Homework Assignment

[Dominick's Data](#) is a popular data set collected and shared by the University of Chicago Booth School of Business focused on understanding coupon and marketing trends. Dominick's Finer Foods is a former grocery store and for five years collected store-level data regarding product sales. We will use a *subset* of the customer count data set, which you can download directly [here](#) or load into R using:

```
load(url('https://byu.box.com/shared/static/q8joa24lz6y2vzigxze13g5ov8jo5i
```

The response variable, 'custcoun', is the number of customers purchasing a particular category of product during a store-week. Because stores differ in their total sales, consider using  $\log(\text{tot})$  as an offset, where 'tot' is the number of sales at the store during the week.

1. Exploratory analysis and model formulation.
  - i. Before fitting a model, examine the distribution of 'custcoun' and its relationship with 'tot'. Briefly explain why an offset might be appropriate for these data.
  - ii. Select at least 5 potential covariates from the more than 40 variables in the data set. Your selection should be motivated by a substantive question about customer purchasing behavior rather than simply choosing variables because they are statistically significant. (Read part (b) before selecting your variables.)
  - iii. Fit an initial Poisson regression model with 'custcoun' as the response and  $\log(\text{tot})$  as an offset.
2. Model building and comparison.
  - i. Fit several scientifically reasonable candidate models. At a minimum, consider: (a) a model with the offset and your selected main effects; (b)

- a model that includes at least one interaction or nonlinear effect that you have substantive reason to consider; (c) a reduced model obtained through some defensible variable-selection strategy.
  - ii. For each model, report the model specification and explain why you included or removed terms. Do not rely on  $p$ -values alone for variable selection.
  - iii. Compare the candidate models using appropriate measure including at least tests of deviance, AIC, and cross-validation.
3. Interpretation.
- i. Interpret the coefficients for at least three variables, including at least one variable whose interpretation is affected by the offset.
  - ii. At least one interpretation should be expressed in terms meaningful to a business manager. For variables ending in ‘coupon’, remember that negative values indicate great coupon use/heavy discounting, while positive values indicate fewer coupons and higher-priced items dominating sales.
4. Business inference. Choose a meaningful store-week scenario and use your preferred model to estimate the expected customer count (or rate) under that scenario. Report an appropriate interval for your estimate and explain what the interval means in business terms.
5. C-suite presentation. Create a slide deck (e.g., PowerPoint or Google Slides) presenting your findings to a C-suite audience that includes a CEO, CFO, and Chief Data Officer. When presenting to your company leaders, all information should be parsed down to only the essentials. Consider the following items:
- i. Which coupon variables appear most strongly related to customer counts?
  - ii. Is the relationship positive or negative, and how large is it?
  - iii. How much uncertainty is there in these conclusions?
  - iv. Does the Poisson model adequately describe the data?
  - v. Write the final model in a way that is easy for the Chief Data Officer to understand and evaluate.
  - vi. What is *one* substantive business conclusion that a manager could reasonably take from your analysis?
  - vii. What is *one* important limitation or caution the company should keep in mind?

- viii. Do not present software output. Translate model results into interpretable quantities, graphics, and business conclusions.

**Your final assignment should be submitted as a commented (not polished) R/qmd file and the presentation slides.**

Note: The original data is a designed experiment and thus analyses could be used to determine causality. However, this subset did not take into account the design and thus, our analyses can only examine relationships.

# 6 Overdispersion

This chapter will be very valuable as you model count data with GLM's. Very often we have data where the variance is larger than what the likelihood allows for; thus, our uncertainty estimates and inferences will be wrong. In this chapter, we first talk about overdispersion and what it is. In Section 6.2 we learn how to correct for the wrong uncertainty/inferences using the same models we've been fitting. In Section 6.3, we learn about actually changing the model so we can *model* the larger variance (for example: the Negative Binomial model!).

## 6.1 Overview of Overdispersion

### What is Overdispersion?

#### Note

**Definition:** Overdispersion occurs in GLM's when the variance of the data (after accounting for covariates) is larger than allowed by the model.

Examples:

- For  $Y_i \sim \text{Bin}(m_i, \pi_i)$  (with  $m_i > 1$ ), we have overdispersion when

$$\begin{aligned} E(Y_i) &= m_i \pi_i = \mu_i \\ \text{Var}(Y_i) &> m_i \pi_i (1 - \pi_i) = \frac{\mu_i(m_i - \mu_i)}{m_i} \end{aligned}$$

- For  $Y_i \sim \text{Poisson}(\mu_i)$ , we have overdispersion when

$$\begin{aligned} E(Y_i) &= \mu_i \\ \text{Var}(Y_i) &> \mu_i \end{aligned}$$

## Why is overdispersion bad?

In a GLM with overdispersion,

1. standard inferences do not apply, since
2. we underestimate standard errors and therefore
3. the tests are anti-conservative (i.e., we reject  $H_0$  too often).

## Simulation Study

Assume the following model:

$$Y_i \stackrel{iid}{\sim} \text{Poisson}(\mu_i), \text{ for } i = 1, \dots, n,$$
$$\log(\mu_i) = \beta_0 + \beta_1 x_i.$$

Simulate data such that:

- *Case 1:*  $\text{Var}(Y_i) = \mu_i$
- *Case 2:*  $\text{Var}(Y_i) > \mu_i$

For which case is our model inappropriate?

**Simulation setup:**

$$\beta_0 = 1$$

$$\beta_1 = 3$$

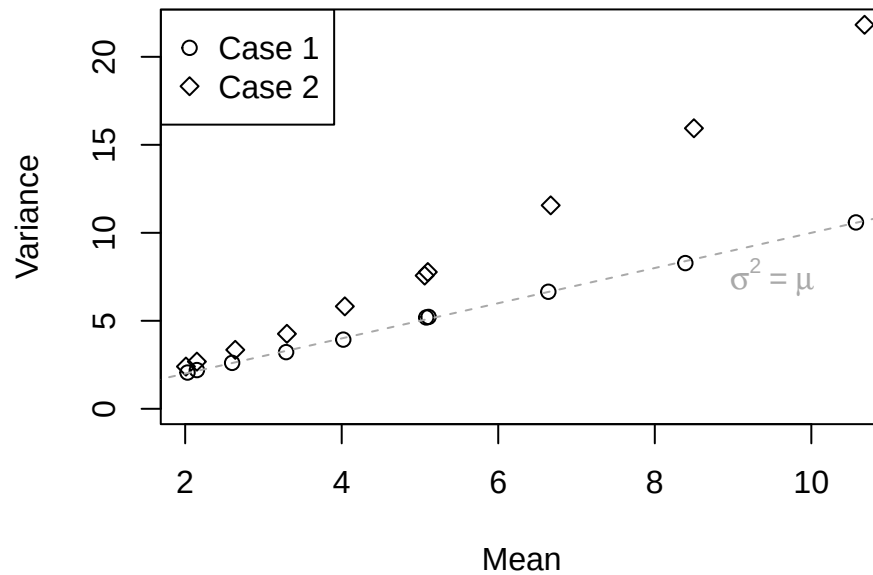
$$n = 10$$

$$x_i \sim \text{Unif}(-.5, .5), \text{ fixed (i.e., same for all repetitions and simulations)}$$

Generate  $Y_i$  ( $i = 1, \dots, n$ ) 5000 times and fit a poisson glm.

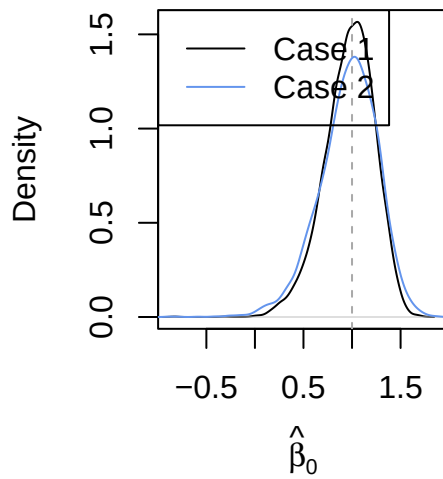
*How do the means and variances compare for each  $i$  for the different cases?*

## Variance vs. Mean

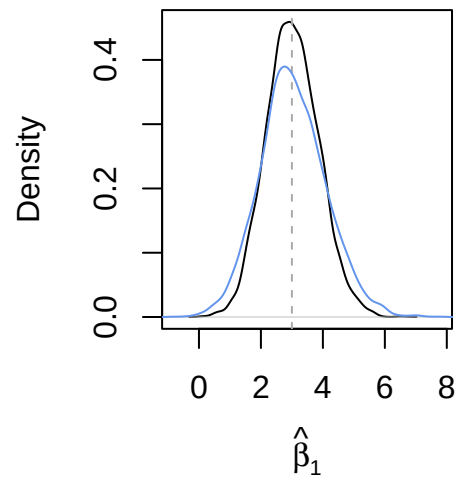


*How do the sampling distributions compare for the estimated coefficients?*

### Sampling Distribution

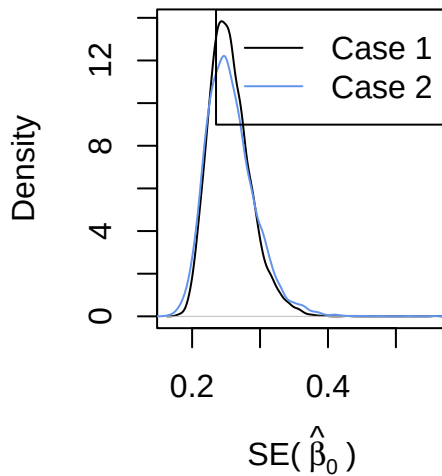


### Sampling Distribution

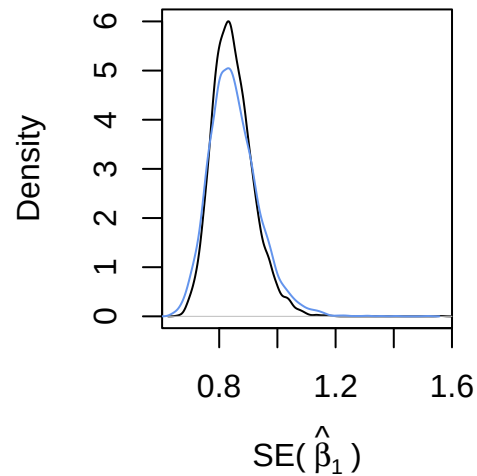


*How do the distributions of the standard errors compare for the estimated coefficients?*

**Sampling Distribution**

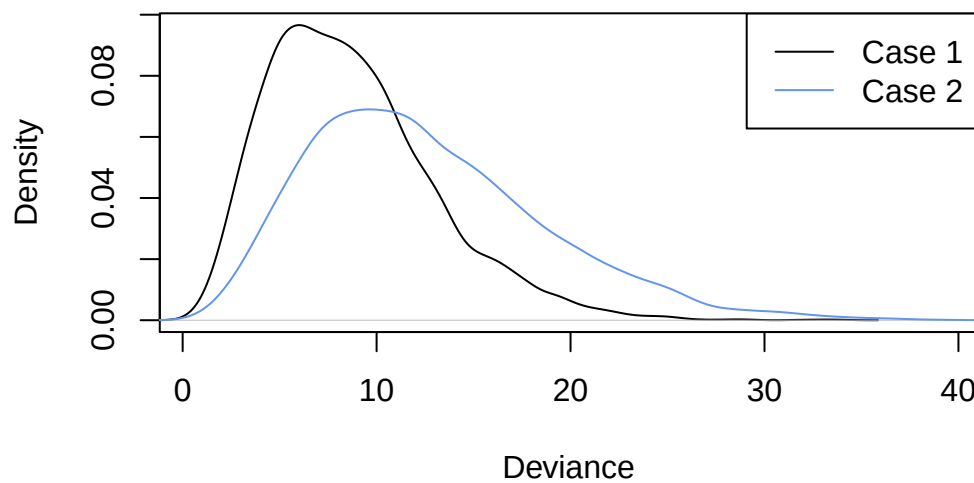


**Sampling Distribution**



*How do the distributions of the deviance compare?*

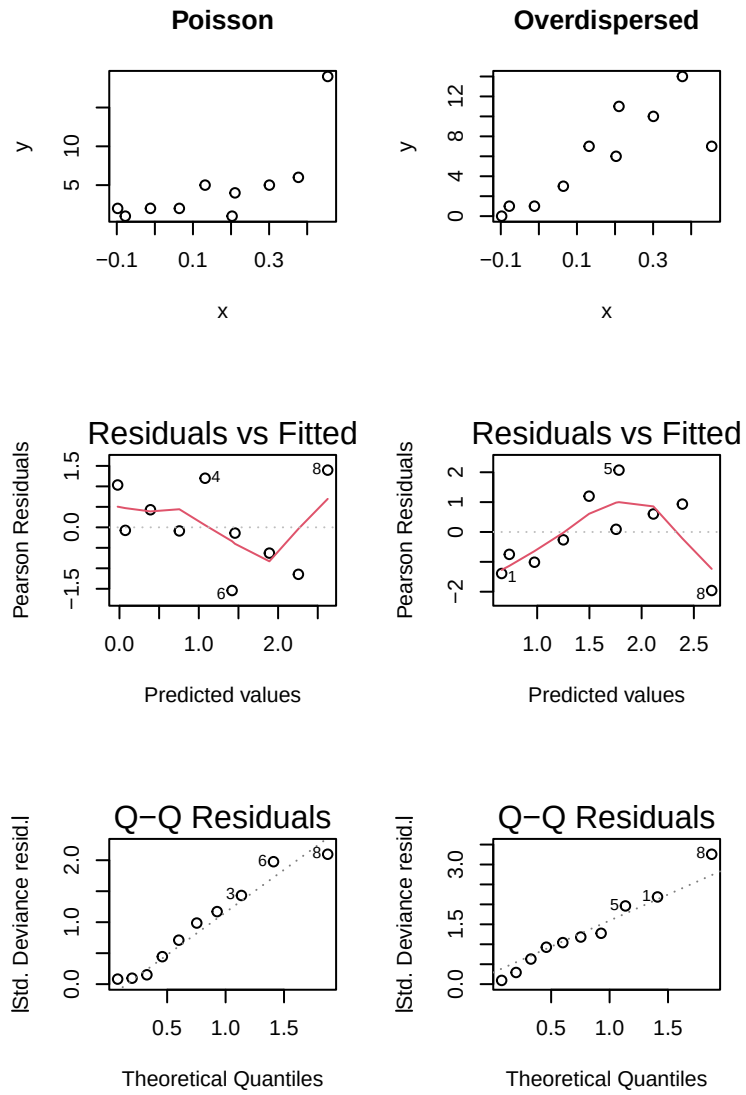
**Sampling Distribution**



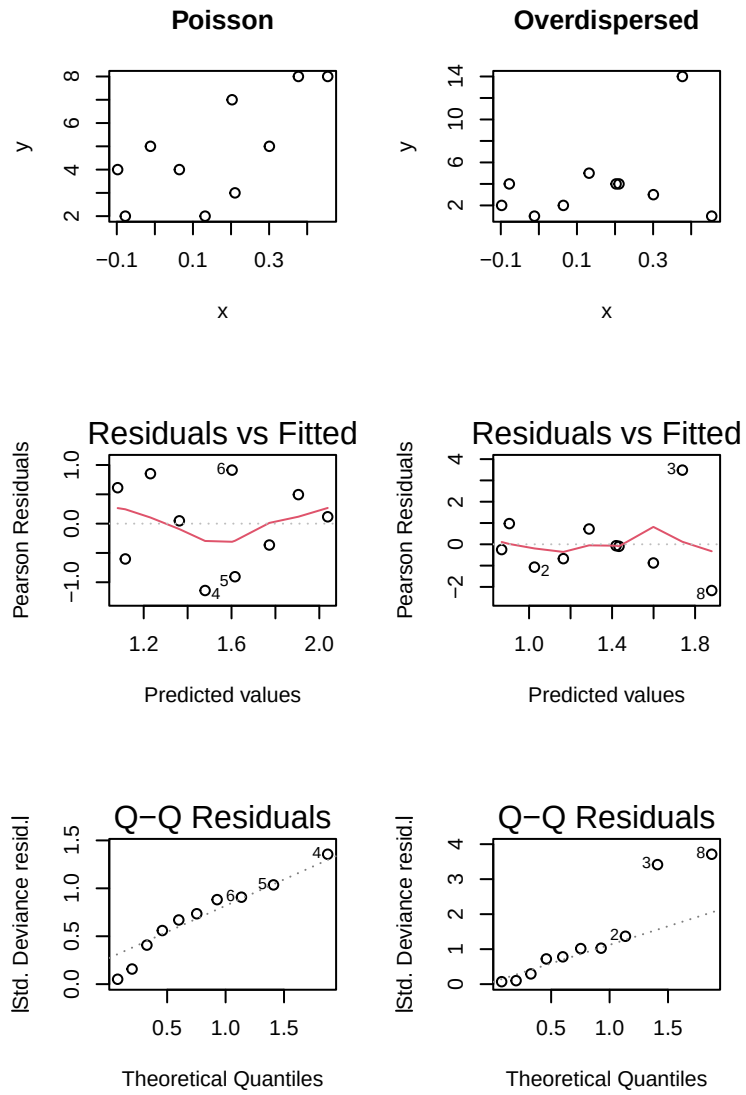
## Identifying Overdispersion

In practice, we often won't have repetitions to be able to compute sample means and variances for each  $i$ . Let's look at some of the simulated data and fitted models to see if we can identify some "rules of thumb" for identifying overdispersion in a single data set.

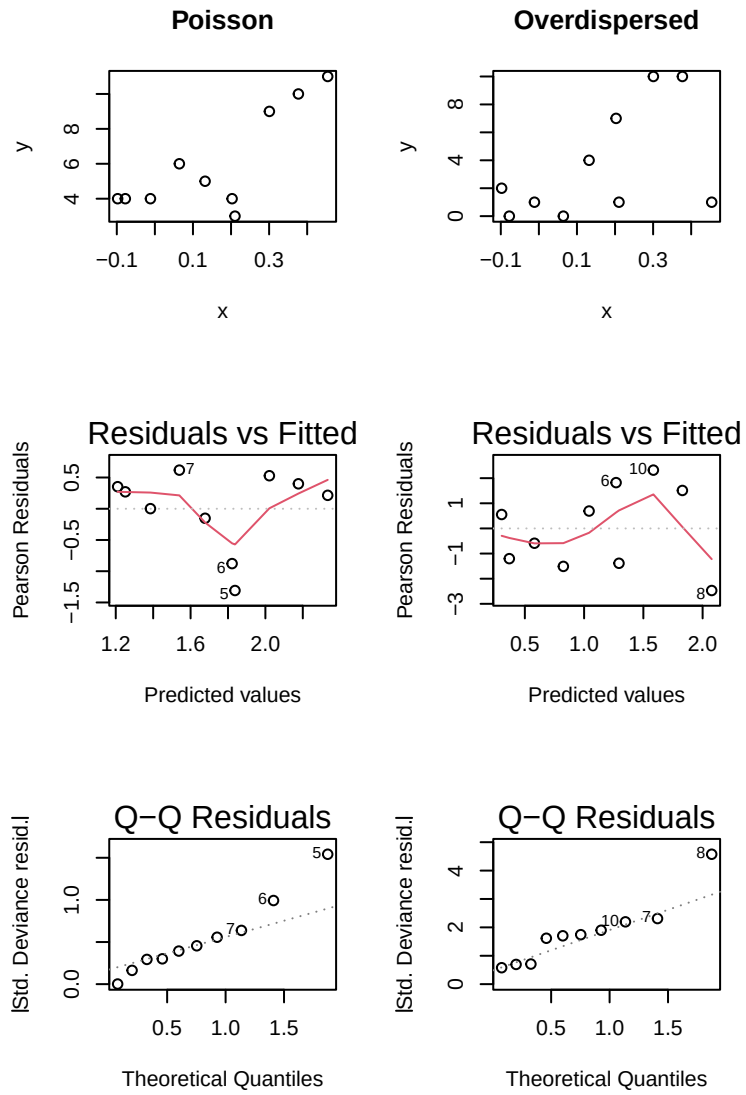




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## 6.2 Accounting for Overdispersion in our Model

Introduce a dispersion parameter  $\phi > 0$  and let

$$\text{Var}(Y_i) = \phi E(Y_i) \text{ (for Poisson GLM)}$$

$$\text{Var}(Y_i) = \phi m_i \pi_i (1 - \pi_i) \text{ (for Binomial GLM)}$$

...We then call the distribution of  $Y_i$  a “**quasi-likelihood**”.

In other words,  $Y_i$  would no longer have a Poisson likelihood, but would still have some Poisson characteristics, so we call it a “quasi-Poisson”.

### 6.2.1 Estimating $\phi$

McCullagh and Nelder (1989) suggest the estimator

$$\hat{\phi} = \frac{X^2}{n - k}$$

where

$$X^2 = \sum_{i=1}^n (r_P)_i^2,$$

and  $(r_P)_i$  is the  $i^{th}$  Pearson residual,

$$(r_P)_i = \frac{Y_i - \hat{\mu}_i}{\sqrt{\text{Var}(\hat{\mu}_i)}}.$$

If the Pearson residuals are approximately normal, then

$$\hat{\phi} \sim \phi \frac{\chi_{n-k}^2}{n - k}$$

This will be true when we are fitting the correct model, i.e.,  $\hat{\mu}_i \approx \mu_i$ .

## 6.2.2 Testing for Significance of Overdispersion

Using the distribution of  $\hat{\phi}$ , we can conduct the following hypothesis test:

$$H_0 : \phi = 1$$

$$H_1 : \phi > 1$$

Under  $H_0$ ,

$$\hat{\phi} \sim 1 \frac{\chi_{n-k}^2}{n-k}$$

Consider the third *Poisson-distributed* data set we looked at from the simulation study.

```
i <- 4999
#compute phi-hat using definition
glm1 <- glm(y1mat[,i]~x, family=poisson)
sum(resid(glm1, type="pearson")^2)/(n-2)
```

```
[1] 0.4465564
```

```
#compare to estimated dispersion from "quasipoisson"
glmq1 <- glm(y1mat[,i]~x, family="quasipoisson")
summary(glmq1)$dispersion
```

```
[1] 0.4465564
```

```
#Compare to chi-sq(df=8)/8
qchisq(.95, 8)/8
```

```
[1] 1.938414
```

Since  $\hat{\phi} < \chi_8^2(0.95)/8$ , we fail to reject  $H_0$  at the 0.05 level and conclude there is not enough evidence to say the data are overdispersed.

Now consider the third *overdispersed* data set from the simulation study.

```
i <- 4999
#compute phi-hat using definition
glm2 <- glm(y2mat[,i]~x, family=poisson)
sum(resid(glm2, type="pearson")^2)/(n-2)
```

```
[1] 2.984016
```

```
#compare to estimated dispersion from "quasi-poisson"
glmq2 <- glm(y2mat[,i]~x, family="quasipoisson")
summary(glmq2)$dispersion
```

```
[1] 2.984025
```

Since  $\hat{\phi} > \chi_8^2(0.95)/8$ , we reject  $H_0$  and conclude there is evidence of overdispersion.

### 6.2.3 Adjusting for Overdispersion in Inferences

How do the “summary” tables compare when we fit the Poisson vs. the quasi-Poisson models to the same data set?

```
summary(glm2) #Poisson model
```

```

Call:
glm(formula = y2mat[, i] ~ x, family = poisson)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.6184     0.3028   2.042  0.04112 *
x              3.2127     0.9915   3.240  0.00119 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 40.982  on 9  degrees of freedom
Residual deviance: 29.696  on 8  degrees of freedom
AIC: 57.697

Number of Fisher Scoring iterations: 5

```

```
summary(glmq2) #Quasi-Poisson fit to same data
```

```

Call:
glm(formula = y2mat[, i] ~ x, family = "quasipoisson")

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.6184     0.5231   1.182  0.2710
x              3.2127     1.7127   1.876  0.0975 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 2.984025)

    Null deviance: 40.982  on 9  degrees of freedom
Residual deviance: 29.696  on 8  degrees of freedom

```

AIC: NA

Number of Fisher Scoring iterations: 5

1. To adjust for overdispersion in the coefficients,

$$\hat{\beta} \sim \mathcal{N}(\beta, \phi(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}).$$

$\Rightarrow$  Our standard errors for  $\beta$  will be too small if we don't account for overdispersion!

2. In the presence of overdispersion, testing  $H_0 : \beta_j = 0$  vs.  $H_1 : \beta_j \neq 0$ , the test statistic should be,

$$\frac{\hat{\beta}_j}{\sqrt{\hat{\phi} SE(\hat{\beta}_j)}}$$

where  $SE(\hat{\beta}_j)$  is the standard error as estimated from the Poisson (or binomial) model. (When fitting the “quasi-Poisson” or “quasi-binomial” models, R automatically accounts for the dispersion in the summary table.)

3. Now consider how the overdispersion impacts **parameter interpretation**:
  - The standard error increases  $\Rightarrow$  the absolute value of the test statistic decreases  $\Rightarrow$  the p-values increase.
  - Therefore, the larger the dispersion parameter, the harder it is to reject  $H_0$ .
  - $\Rightarrow$  *The observed variation in the response variable is less likely due to the covariate and more likely due to the dispersion of the data.*

## 6.2.4 Analysis of Deviance with Overdispersion

Fit a model with  $q$  parameters...

- Estimate  $\mu$  with  $\hat{\mu}^{(1)}$ .
- Call this the “reduced” model.

$\Rightarrow$  Deviance is:

$$D(\mathbf{y}, \hat{\mu}^{(1)}) = 2\phi[\ell(\mathbf{y}) - \ell(\hat{\mu}^{(1)})]$$

Now fit a model with the same  $q$  parameters and added  $k - q$  parameters...



- Estimate  $\mu$  with  $\hat{\mu}^{(2)}$ .
- Call this the “full” model (but isn’t necessarily the saturated/full model).

$\Rightarrow$  Deviance is

$$D(\mathbf{y}, \hat{\mu}^{(2)}) = 2\phi[\ell(\mathbf{y}) - \ell(\hat{\mu}^{(2)})]$$

$\Rightarrow$  Difference in deviances is

$$D(\mathbf{y}, \hat{\mu}^{(1)}) - D(\mathbf{y}, \hat{\mu}^{(2)}) = -2\phi[\ell(\hat{\mu}^{(1)}) - \ell(\hat{\mu}^{(2)})] \sim \phi\chi^2(k - q)$$

Important Note: we must **assume the value of  $\phi$  is the same for both models**.

- Estimate  $\phi$ :  $\hat{\phi} = X^2/(n - k)$  from the *full model* (model with most variance accounted for with covariates).

Now, test

$$H_0 : \beta_{q+1} = \dots = \beta_k = 0$$

$$H_1 : \text{at least one } \beta_j \neq 0, j \in \{q + 1, \dots, k\}$$

that is, we are testing whether the parameters in the full model that are not in the reduced model are 0 or at least one is not.

An F-test on the deviance is,

$$F = \frac{\underbrace{D(y, \hat{\mu}^{(1)}) - D(y, \hat{\mu}^{(2)})/(k - q)}_{\sim \phi\chi^2(k-q)/(k-q)}}{\underbrace{\hat{\phi}}_{\sim \chi^2(n-k)/(n-k)}},$$

since,

- the numerator and denominator are independent, and
- $F \sim F(k - q, n - k)$  under  $H_0$ .

```
#using "anova" function on the "quasipoisson" model
anova(glmq2, test='F')
```

Analysis of Deviance Table

Model: quasipoisson, link: log

Response: y2mat[, i]

Terms added sequentially (first to last)

|      | Df | Deviance | Resid. Df | Resid. Dev | F      | Pr(>F)   |
|------|----|----------|-----------|------------|--------|----------|
| NULL |    |          | 9         | 40.982     |        |          |
| x    | 1  | 11.286   | 8         | 29.696     | 3.7821 | 0.0877 . |

---  
 Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
#computing by hand
```

```
Fstat <- ((glmq2$null.deviance - glmq2$deviance)/(1))/(summary(glmq2)$dispersi
print(Fstat)
```

```
[1] 3.782055
```

```
1-pf(Fstat, 1, 8)
```

```
[1] 0.08770049
```

Recall our beeltes and carbon disulfide data where we decided the complementary log-log link fit the data “best.” What if the “better” fit is just due

to chance and the variation is simply due to overdispersion? Consider the logistic binomial and quasi-binomial models.

Call:

```
glm(formula = pi.beetles ~ dose, family = binomial, weights = nbeetles)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -60.717  | 5.181      | -11.72  | <2e-16 *** |
| dose        | 34.270   | 2.912      | 11.77   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 284.202 on 7 degrees of freedom  
Residual deviance: 11.232 on 6 degrees of freedom  
AIC: 41.43

Number of Fisher Scoring iterations: 4

Call:

```
glm(formula = pi.beetles ~ dose, family = quasibinomial, weights = nbeetles)
```

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -60.717  | 6.697      | -9.066  | 0.000101 *** |
| dose        | 34.270   | 3.765      | 9.103   | 9.87e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 1.671141)

Null deviance: 284.202 on 7 degrees of freedom  
Residual deviance: 11.232 on 6 degrees of freedom

AIC: NA

Number of Fisher Scoring iterations: 4

*Is the overdispersion significant?*

Compare  $\hat{\phi} = 1.6711415$  to  $\chi^2(6)/6$ .

```
1-pchisq(summary(glm.qb)$dispersion*(8-2), 8-2)
```

```
[1] 0.1235259
```

*According to a test of deviance, should dose be included in the model?*

```
anova(glm.qb, test='F')
```

Analysis of Deviance Table

Model: quasibinomial, link: logit

Response: pi.beetles

Terms added sequentially (first to last)

|      | Df | Deviance | Resid. | Df | Resid. Dev | F      | Pr(>F)        |
|------|----|----------|--------|----|------------|--------|---------------|
| NULL |    |          |        | 7  | 284.202    |        |               |
| dose | 1  | 272.97   |        | 6  | 11.232     | 163.34 | 1.409e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## 6.3 Modeling Overdispersion

Mechanisms for overdispersion:

- The most common mechanism for overdispersion is due to cluster effects.
- Example of clusters in a population: families, households, litters, colonies, neighborhoods.
- We expect different clusters to behave differently in the population.
- However, what we *observe* is often a quantity which is aggregated over these clusters.
- What are possible clusters in the beetles data set?

### 6.3.1 Modeling Overdispersion in “Binomial” Data

- Suppose we have  $c$  individuals each sampled from  $\ell$  clusters in the population. We observe  $m = c\ell$  individuals in total.
- **The cluster model:** Let  $Z_j$  denote the number of successes in cluster  $j$ . Assume that  $\{Z_j : j = 1, \dots, \ell\}$  are a set of independent random variables with  $Z_j \sim \text{Bin}(c, \pi_j)$  for each  $j$ .
- **The cluster effects:**  $\{\pi_j : j = 1, \dots, \ell\}$  are iid RVs with  $E(\pi_j) = \pi$  and  $\text{Var}(\pi_j) = \tau^2 \pi(1 - \pi)$ , for some constants  $0 < \pi < 1$  and  $\tau^2 > 0$ .
- **What we observe:** Let  $Y = \sum_{j=1}^{\ell} Z_j$ .

A toy example of this latent data:  $\ell = 3$  and  $c = 2$  ( $m = 6$ ).

| Beetles | Coleopteran | Beatles |
|---------|-------------|---------|
| Chirp   | Camilla*    | John    |
| Squeek* | Gavin       | Paul    |
| Hiss    | Connor      | George  |
| Cluck*  | Evan        | Ringo*  |
|         | Grant*      | Stuart  |
|         | Patric      | Pete*   |
|         | Daisy       |         |

The \* represents that the beetle was collected in our sample of 6. Thus,  $Y$  = number of beetles sampled that are “in a band”.

What are the moments of  $Y$ ?

*The expectation of  $Y$*

$$\begin{aligned} E(Y) &= E\left(\sum_{j=1}^{\ell} Z_j\right) = E(E(\sum Z_j | p\pi_j)) \\ &= \sum E(c\pi_j) \\ &= \ell c\pi \\ &= m\pi \end{aligned}$$

The variance of  $Y$

$$\begin{aligned}
\text{Var}(Y) &= \text{Var} \left( \sum_{j=1}^{\ell} Z_j \right) = \sum_{j=1}^{\ell} \text{Var}(Z_j) \quad (\text{since indep}) \\
&= \sum_{j=1}^{\ell} [E(\text{Var}(Z_j|\pi_j)) + \text{Var}(E(Z_j|\pi_j))] \\
&= \sum_{j=1}^{\ell} [E(c\pi_j(1 - \pi_j)) + \text{Var}(c\pi_j)] \\
&= \sum_{j=1}^{\ell} [c(E(\pi_j) - E(\pi_j^2)) + k^2\tau^2\pi(1 - \pi)] \\
&= \sum_{j=1}^{\ell} [c(\pi - \underbrace{(\tau^2\pi(1 - \pi))}_{\text{Var}(\pi_j)} + \underbrace{\pi^2}_{E(\pi_j)^2}) + c^2\tau^2\pi(1 - \pi)] \\
&= \sum_{j=1}^{\ell} [c\pi - c\tau^2\pi(1 - \pi) - c\pi^2 + c^2\tau^2\pi(1 - \pi)] \\
&= \ell c\pi - \ell c\tau^2\pi + \ell c\tau^2\pi^2 - \ell c\pi^2 + \ell c^2\tau^2\pi - \ell c^2\tau^2\pi^2 \\
&= m\pi - m\tau^2\pi + m\tau^2\pi^2 + mc\tau^2\pi - mc\tau^2\pi^2 \\
&= m\pi(1 - \tau^2 + \tau^2\pi - \pi + c\tau^2 - c\tau^2\pi) \\
&= \vdots \\
&= m\pi(1 - \pi)[1 + \tau^2(c - 1)]
\end{aligned}$$

Note: If  $c = 1 \Rightarrow 1$  cluster  $\Rightarrow$  all same and  $\text{Var}(Y) = m\pi(1 - \pi)$ . Otherwise,  $\tau^2$  inflates variance (even if  $\tau^2 < 1$ ).

### 6.3.2 A Parametric Model for Binomial Overdispersion

Consider the following hierarchical model for  $Y$ :

$$\begin{aligned}
Y|\pi &\sim \text{Bin}(m, \pi) \\
\pi &\sim \text{Beta}(\alpha, \beta)
\end{aligned}$$

It can be shown that  $Y$  has a Beta-Binomial distribution with parameters  $m$ ,  $\alpha$ , and  $\beta$ .

The moments of  $Y$  are

$$E(Y) = m\pi$$
$$\text{Var}(Y) = m\pi(1 - \pi)[1 + (m - 1)\tau^2]$$

where  $\pi = \alpha/(\alpha + \beta)$  and  $\tau^2 = 1/(\alpha + \beta + 1)$

## The Beta-Binomial Model

When accounting for overdispersion, you can

1. Ignore the true dispersion. Fit a “quasi-binomial” GLM with an overdispersion parameter,  $\phi$  ( $\approx [1 + (m - 1)\tau^2]$  from the beta-binomial model).
2. Model the beta-binomial variables directly. Transform  $\pi_i$  using an appropriate link, then model the explanatory variables on the link space. Estimate the parameters via ML. (In R: “aod” and the “betabin” function.)

*Survival rate of rat litters from four different groups*

- 32 litters (ranging from 5 - 13 rats in each litter)
- Number dead for each litter
- Group of each litter (covariate)

```
library(aod)
data(rats)
bb.mod <- betabin(cbind(y, n-y)~group, ~1, link="logit", data=rats)
summary(bb.mod)
```

Beta-binomial model

-----

```
betabin(formula = cbind(y, n - y) ~ group, random = ~1, data = rats,
        link = "logit")
```

Convergence was obtained after 64 iterations.

Fixed-effect coefficients:

|             | Estimate  | Std. Error | z value   | Pr(>  z ) |
|-------------|-----------|------------|-----------|-----------|
| (Intercept) | 1.863e+00 | 3.652e-01  | 5.100e+00 | 3.401e-07 |



```
groupTREAT  -6.644e-01  4.596e-01 -1.446e+00 1.483e-01
```

Overdispersion coefficients:

|                 | Estimate  | Std. Error | z value   | Pr(>  z ) |
|-----------------|-----------|------------|-----------|-----------|
| phi.(Intercept) | 1.927e-01 | 7.258e-02  | 2.655e+00 | 3.969e-03 |

Log-likelihood statistics

| Log-lik    | nbpar | df res. | Deviance  | AIC       | AICc      |
|------------|-------|---------|-----------|-----------|-----------|
| -5.561e+01 | 3     | 29      | 6.605e+01 | 1.172e+02 | 1.181e+02 |

Note:  $\phi$  in the “betabin” model here is the same as the  $\tau^2$  in our notation. It is *not* the same as the overdispersion parameter from the quasi-binomial model.

Compare to the standard logit and quasibinomial logit model:

Call:

```
glm(formula = cbind(y, n - y) ~ group, family = binomial, data = rats)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(>  z )   |
|-------------|----------|------------|---------|-------------|
| (Intercept) | 2.1832   | 0.2637     | 8.280   | < 2e-16 *** |
| groupTREAT  | -0.9612  | 0.3298     | -2.915  | 0.00356 **  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 95.203 on 31 degrees of freedom  
Residual deviance: 86.187 on 30 degrees of freedom  
AIC: 135.35

Number of Fisher Scoring iterations: 4

Call:

```
glm(formula = cbind(y, n - y) ~ group, family = quasibinomial,  
     data = rats)
```

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | 2.1832   | 0.4323     | 5.050   | 2.02e-05 | *** |
| groupTREAT  | -0.9612  | 0.5407     | -1.778  | 0.0856   | .   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 2.687947)

Null deviance: 95.203 on 31 degrees of freedom

Residual deviance: 86.187 on 30 degrees of freedom

AIC: NA

Number of Fisher Scoring iterations: 4

How can we compare the quasi-binomial overdispersion to the beta-binomial overdispersion? (Useful: the average rat litter size is  $\approx 10$ .)

### 6.3.3 Modeling Overdispersion in “Poisson” Data

One example of Poisson overdispersion is when we observe a random sum (in terms of how many terms we include) of iid RVs.

- **The cluster effects:** Let  $\{Z_j : j = 1, \dots\}$  be a set of iid RVs where each  $Z_j$  has mean  $E(Z)$  and variance  $\text{Var}(Z)$ .
- **The clustering mechanism:** Suppose that  $N$  has a  $\text{Pois}(\lambda)$  distribution which is independent of each  $Z_j$ ,  $j = 1, \dots$ .
- **What we observe:** Let  $Y = \sum_{j=1}^N Z_j$ .

What are the first two moments of  $Y$ ?

*The expectation of  $Y$*

$$\begin{aligned}
 E(Y) &= E\left(\sum_{j=1}^N Z_j\right) = E(E(\sum_{j=1}^N Z_j | N)) \\
 &= E\left(\sum_{j=1}^N E(Z_j | N)\right) \\
 &= E\left(\sum_{j=1}^N E(Z)\right) = E\left(N \underbrace{E(Z)}_{\text{constant}}\right) \\
 &= \lambda E(Z)
 \end{aligned}$$

*The variance of  $Y$*

$$\begin{aligned}
 \text{Var}(Y) &= \text{Var}\left(\sum_{j=1}^N Z_j\right) = E\left(\text{Var}\left(\sum_{j=1}^N Z_j | N\right)\right) + \text{Var}\left(\underbrace{E\left(\sum_{j=1}^N Z_j | N\right)}_{NE(Z)}\right) \\
 &= E\left(\sum_{j=1}^N \text{Var}(Z_j | N)\right) + E(Z)^2 \text{Var}(N) \\
 &= E(N \text{Var}(Z)) + \lambda E(Z)^2 \\
 &= \lambda (\text{Var}(Z) + E(Z)^2) \\
 &= \lambda E(Z^2)
 \end{aligned}$$

Note: If  $E(Z) = 1$  and  $E(Z^2) = \phi > 1$ , then  $E(Y) = \lambda$  and  $\text{Var}(Y) = \phi\lambda$ . (Just like the poisson overdispersion we fit last time.)

### 6.3.4 A Parametric Model for Poisson Overdispersion

Consider the following hierarchical model for  $Y$ :

$$\begin{aligned} Y|\lambda &\sim \text{Pois}(\lambda) \\ \lambda &\sim \text{Gamma}(\alpha, \beta) \end{aligned}$$

What is the distribution of  $Y$ ? We have that

$$f(y|\lambda) = \frac{e^{-\lambda}\lambda^y}{y!} \text{ and } f(\lambda) = \frac{\beta^\alpha \lambda^{\alpha-1} e^{-\beta\lambda}}{\Gamma(\alpha)}.$$

So the joint distribution of  $Y$  and  $\lambda$  is

$$\begin{aligned} f(y, \lambda) &= f(y|\lambda)f(\lambda) \\ &= \frac{e^{-\lambda}\lambda^y}{y!} \frac{\beta^\alpha \lambda^{\alpha-1} e^{-\beta\lambda}}{\Gamma(\alpha)} \\ &= \frac{\beta^\alpha \lambda^{y+\alpha-1} e^{-(\beta+1)\lambda}}{\Gamma(\alpha)y!} \end{aligned}$$

Thus, the distribution of  $Y$  is given by

$$\begin{aligned} f(y) &= \int_0^\infty f(y, \lambda) d\lambda \\ &= \int_0^\infty \frac{\beta^\alpha \lambda^{y+\alpha-1} e^{-(\beta+1)\lambda}}{\Gamma(\alpha)y!} d\lambda \\ &= \int_0^\infty \frac{(\beta+1)^{y+\alpha} \lambda^{y+\alpha-1} e^{-(\beta+1)\lambda}}{\Gamma(y+\alpha)} d\lambda \\ &\quad \times \frac{\beta^\alpha \Gamma(y+\alpha)}{\Gamma(\alpha)(\beta+1)^{y+\alpha} y!} \\ &= \frac{\Gamma(y+\alpha)}{\Gamma(\alpha)\Gamma(y+1)} \left( \frac{\beta}{\beta+1} \right)^\alpha \left( \frac{1}{\beta+1} \right)^y. \end{aligned}$$

### Recognize this distribution?

$\Rightarrow$  It is a negative binomial distribution with  $\pi = \beta/(\beta + 1)$  and  $r = \alpha$ .

### Is this distribution a member exponential family?

- Suppose we have a set of independent RVs  $\{Y_i : i = 1, \dots, n\}$  that each have the following distribution: the pmf for each random variable is

$$f(y_i|p_i) = \binom{y_i + r - 1}{r - 1} \pi_i^r (1 - \pi_i)^{y_i}, 0 < \pi_i < 1.$$
$$= \exp\left\{ \underbrace{\underbrace{y_i}_{t(y)} \log(1 - \pi_i)}_{\theta} + \underbrace{r \log \pi_i}_{c(\theta)} + \underbrace{\log \binom{y_i + \theta - 1}{r - 1}}_{d(y)} \right\}$$

when  $r$  is fixed, then it *is* a member of the one-parameter exponential family.

**Important Note:** In R, the negative binomial model calls  $r$  “theta”. For this reason, we will use  $\theta$  instead of  $r$  for the remainder of this section.

### A Negative Binomial Model

The mean is

$$\mu_i = E(Y_i) = \frac{\theta(1 - \pi_i)}{\pi_i}.$$

The variance is

$$\text{Var}(Y_i) = \frac{\theta(1 - \pi_i)}{\pi_i^2}.$$

We can write the variance as

$$\text{Var}(Y_i) = \left[ \frac{\theta(1 - \pi_i)}{\pi_i} \right] \frac{1}{\pi_i} = \frac{\mu_i}{\pi_i}.$$

Rearranging the equation for  $\mu_i$  above, we see that

$$\frac{1}{\pi_i} = \frac{\mu_i + \theta}{\theta},$$

and so

$$\text{Var}(\mu_i) = \frac{\mu_i(\mu_i + \theta)}{\theta} = \mu_i + \frac{\mu_i^2}{\theta}.$$

Note:  $\mu_i$  is the variance for a Poisson RV; as  $\theta \rightarrow \infty$ ,  $\mu_i^2/\theta \rightarrow 0$ .

Consider a data set titled “Soldering of Components on Printed-Circuit Boards,” taken from “Statistical Models in S” (1992), by John M. Chambers and Trevor J. Hastie eds, Wadsworth and Brooks/Cole, Pacific Grove, CA. (“solder” data set in R package “rpart”).

The data set has 720 rows and 6 columns, representing a balanced subset of a designed experiment varying 5 factors on the soldering components on printed-circuit boards:

1. *Opening*: factor with levels L, M, and S, indicating the amount of clearance around the mounting pad.
2. *Solder*: factor with levels Thick and Thin, giving the thickness of the solder used.
3. *Mask*: factor with levels A1.5, A3, B3, and B6 indicating the type and thickness of mask used.
4. *PadType*: factor with levels D4, D6, D7, L4, L6, L7, L8, L9, W4, W9, giving the size and geometry of the mounting pad.
5. *Panel*: the panel on the board being tested (values 1, 2, or 3).

Research question and model:

- We are interested in modeling the number of visible solder “skips” in terms of these 5 factors.
- Since we are modeling counts, a Poisson GLM seems reasonable.
- Exercise:
  - Produce summaries of this data.
  - Find a suitable model for the number of visible solder skips in terms of the five factors.
- We will consider a simple model inspired by what Chambers and Hastie did.

**Poisson Model:**

```
Warning in data(solder.balance): data set 'solder.balance' not found
```

Call:

```
glm(formula = skips ~ ., family = poisson, data = solder.balance)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | -1.25506 | 0.10069    | -12.465 | < 2e-16  | *** |
| OpeningM    | 0.25851  | 0.06656    | 3.884   | 0.000103 | *** |
| OpeningS    | 1.89349  | 0.05363    | 35.305  | < 2e-16  | *** |
| SolderThin  | 1.09973  | 0.03864    | 28.465  | < 2e-16  | *** |
| MaskA3      | 0.42819  | 0.07547    | 5.674   | 1.40e-08 | *** |
| MaskB3      | 1.20225  | 0.06697    | 17.953  | < 2e-16  | *** |
| MaskB6      | 1.86648  | 0.06310    | 29.580  | < 2e-16  | *** |
| PadTypeD6   | -0.36865 | 0.07138    | -5.164  | 2.41e-07 | *** |
| PadTypeD7   | -0.09844 | 0.06620    | -1.487  | 0.137001 |     |
| PadTypeL4   | 0.26236  | 0.06071    | 4.321   | 1.55e-05 | *** |
| PadTypeL6   | -0.66845 | 0.07841    | -8.525  | < 2e-16  | *** |
| PadTypeL7   | -0.49021 | 0.07406    | -6.619  | 3.61e-11 | *** |
| PadTypeL8   | -0.27115 | 0.06939    | -3.907  | 9.33e-05 | *** |
| PadTypeL9   | -0.63645 | 0.07759    | -8.203  | 2.35e-16 | *** |
| PadTypeW4   | -0.11000 | 0.06640    | -1.657  | 0.097590 | .   |
| PadTypeW9   | -1.43759 | 0.10419    | -13.798 | < 2e-16  | *** |
| Panel       | 0.11818  | 0.02056    | 5.749   | 8.97e-09 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 6855.7 on 719 degrees of freedom  
Residual deviance: 1165.4 on 703 degrees of freedom  
AIC: 2781.6

Number of Fisher Scoring iterations: 5

Analysis of Deviance Table

Model: poisson, link: log

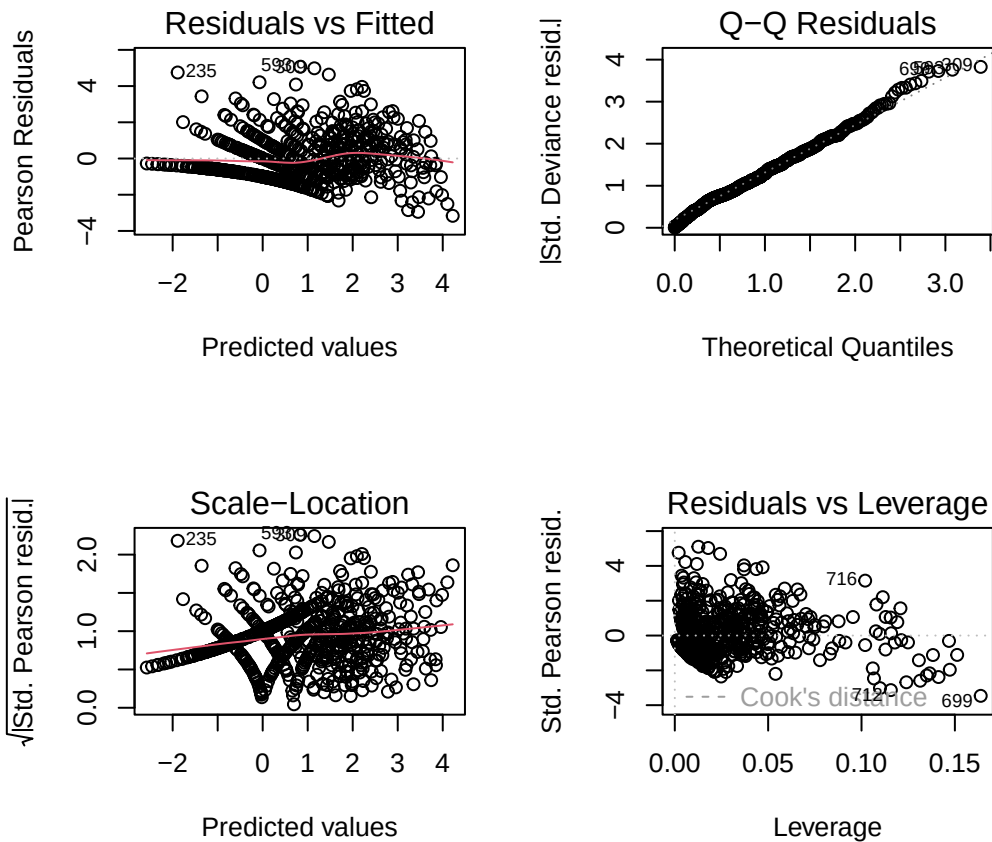
Response: skips

Terms added sequentially (first to last)

|         | Df | Deviance | Resid. Df | Resid. Dev | Pr(>Chi)      |
|---------|----|----------|-----------|------------|---------------|
| NULL    |    |          | 719       | 6855.7     |               |
| Opening | 2  | 2524.56  | 717       | 4331.1     | < 2.2e-16 *** |
| Solder  | 1  | 936.95   | 716       | 3394.2     | < 2.2e-16 *** |
| Mask    | 3  | 1653.09  | 713       | 1741.1     | < 2.2e-16 *** |
| PadType | 9  | 542.46   | 704       | 1198.6     | < 2.2e-16 *** |
| Panel   | 1  | 33.17    | 703       | 1165.4     | 8.449e-09 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1



**Quasi-Poisson Model** (test for overdispersion):

Call:

```
glm(formula = skips ~ ., family = quasipoisson, data = solder.balance)
```

Coefficients:



```

              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.25506    0.12483 -10.054 < 2e-16 ***
OpeningM      0.25851    0.08252   3.133 0.001805 **
OpeningS      1.89349    0.06649  28.477 < 2e-16 ***
SolderThin    1.09973    0.04790  22.959 < 2e-16 ***
MaskA3        0.42819    0.09357   4.576 5.59e-06 ***
MaskB3        1.20225    0.08303  14.480 < 2e-16 ***
MaskB6        1.86648    0.07823  23.859 < 2e-16 ***
PadTypeD6     -0.36865    0.08850  -4.166 3.49e-05 ***
PadTypeD7     -0.09844    0.08207  -1.199 0.230764
PadTypeL4      0.26236    0.07527   3.486 0.000521 ***
PadTypeL6     -0.66845    0.09721  -6.876 1.36e-11 ***
PadTypeL7     -0.49021    0.09182  -5.339 1.26e-07 ***
PadTypeL8     -0.27115    0.08604  -3.152 0.001692 **
PadTypeL9     -0.63645    0.09620  -6.616 7.31e-11 ***
PadTypeW4     -0.11000    0.08232  -1.336 0.181909
PadTypeW9     -1.43759    0.12917 -11.129 < 2e-16 ***
Panel         0.11818    0.02548   4.637 4.21e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 1.537103)

Null deviance: 6855.7  on 719  degrees of freedom
Residual deviance: 1165.4  on 703  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5

```

```
1-pchisq(summary(qmod)$dispersion*(703), 703)
```

```
[1] 0
```

Negative Binomial Model

- Fitting a negative binomial model to the data}
  - Now suppose  $\{Y_i : i = 1, \dots, n\}$  are a set of independent RVs with  $Y_i \sim \text{NegBin}(\theta, \pi_i)$  for each  $i$ .
  - Previously we saw that when  $\theta$  is fixed, this distribution is a member of the exponential family.
  - In R, load the “MASS” library, and use the function “glm”:  
`sold.nb1 <- glm(skips~Opening+Solder+Mask+PadType+factor(Panel), family=negative.binomial(theta=1), data=solder.balance)`
  - We set  $\theta = 1$ .
  - The default is a *log link*.

Call:

```
glm(formula = skips ~ Opening + Solder + Mask + PadType + factor(Panel),
     family = negative.binomial(theta = 1), data = solder.balance)
```

Coefficients:

|                | Estimate  | Std. Error | t value | Pr(> t ) |     |
|----------------|-----------|------------|---------|----------|-----|
| (Intercept)    | -1.838167 | 0.167163   | -10.996 | < 2e-16  | *** |
| OpeningM       | 0.429035  | 0.096955   | 4.425   | 1.12e-05 | *** |
| OpeningS       | 2.096484  | 0.089995   | 23.296  | < 2e-16  | *** |
| SolderThin     | 1.303673  | 0.073480   | 17.742  | < 2e-16  | *** |
| MaskA3         | 0.532464  | 0.114748   | 4.640   | 4.15e-06 | *** |
| MaskB3         | 1.313192  | 0.109465   | 11.997  | < 2e-16  | *** |
| MaskB6         | 2.120601  | 0.106597   | 19.894  | < 2e-16  | *** |
| PadTypeD6      | -0.449175 | 0.159909   | -2.809  | 0.005109 | **  |
| PadTypeD7      | 0.042471  | 0.154215   | 0.275   | 0.783087 |     |
| PadTypeL4      | 0.583751  | 0.149749   | 3.898   | 0.000106 | *** |
| PadTypeL6      | -0.399470 | 0.159249   | -2.508  | 0.012350 | *   |
| PadTypeL7      | -0.280122 | 0.157745   | -1.776  | 0.076201 | .   |
| PadTypeL8      | -0.012017 | 0.154760   | -0.078  | 0.938129 |     |
| PadTypeL9      | -0.480472 | 0.160336   | -2.997  | 0.002826 | **  |
| PadTypeW4      | 0.008515  | 0.154553   | 0.055   | 0.956078 |     |
| PadTypeW9      | -1.403285 | 0.177459   | -7.908  | 1.02e-14 | *** |
| factor(Panel)2 | 0.324748  | 0.088426   | 3.673   | 0.000258 | *** |
| factor(Panel)3 | 0.314383  | 0.088483   | 3.553   | 0.000406 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(1) family taken to be 0.5579472)

Null deviance: 1423.13 on 719 degrees of freedom

Residual deviance: 420.67 on 702 degrees of freedom

AIC: 2917.2

Number of Fisher Scoring iterations: 6

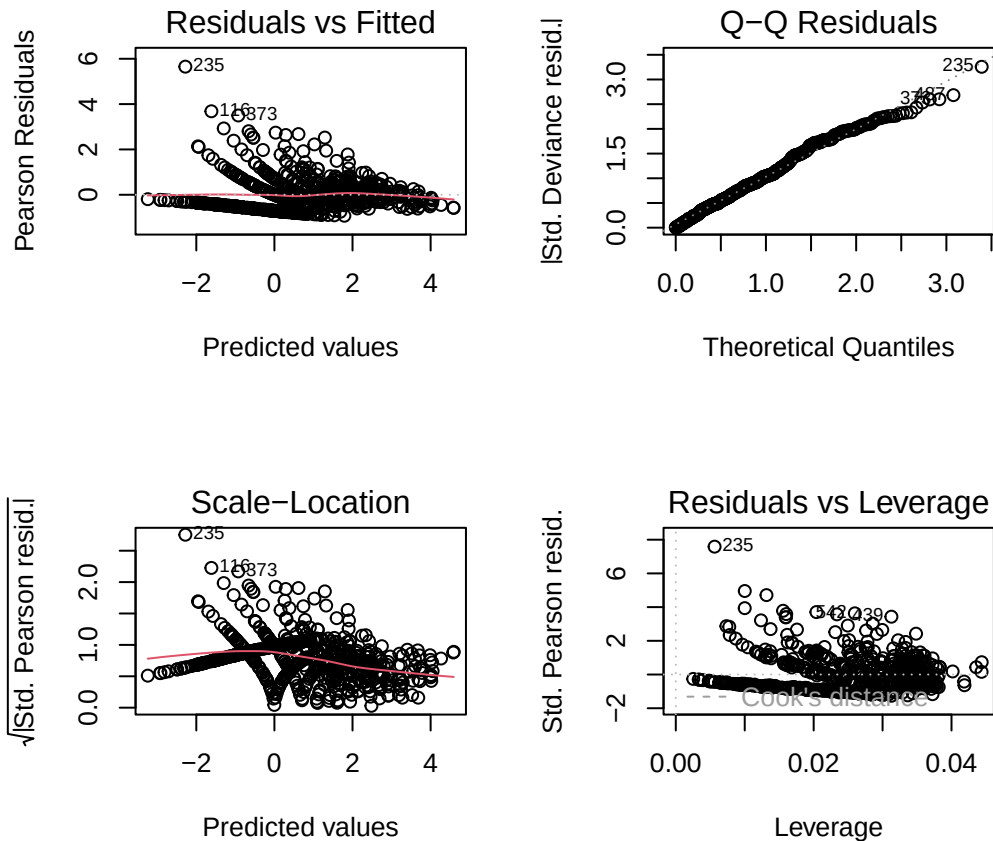
Analysis of Deviance Table

Model: Negative Binomial(1), link: log

Response: skips

Terms added sequentially (first to last)

|               | Df | Deviance | Resid. Df | Resid. Dev |
|---------------|----|----------|-----------|------------|
| NULL          |    |          | 719       | 1423.13    |
| Opening       | 2  | 435.34   | 717       | 987.79     |
| Solder        | 1  | 173.44   | 716       | 814.35     |
| Mask          | 3  | 292.97   | 713       | 521.38     |
| PadType       | 9  | 91.37    | 704       | 430.01     |
| factor(Panel) | 2  | 9.34     | 702       | 420.67     |



- We'd really like to be able to **estimate**  $\theta$ . The MASS package has a "glm.nb" function that optimizes over different values of  $\theta$ .

```
– sold.nb2 <- glm.nb(skips~Opening+Solder+Mask+PadType+factor(Panel), data=solder.balance)
```

Call:

```
glm.nb(formula = skips ~ Opening + Solder + Mask + PadType +
  factor(Panel), data = solder.balance, init.theta = 10.53445485,
  link = log)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -1.49846 | 0.12472    | -12.014 | < 2e-16 ***  |
| OpeningM    | 0.30623  | 0.07840    | 3.906   | 9.39e-05 *** |

|                |          |         |         |          |     |
|----------------|----------|---------|---------|----------|-----|
| OpeningS       | 2.00244  | 0.06686 | 29.950  | < 2e-16  | *** |
| SolderThin     | 1.21356  | 0.05236 | 23.179  | < 2e-16  | *** |
| MaskA3         | 0.45408  | 0.09112 | 4.983   | 6.25e-07 | *** |
| MaskB3         | 1.22575  | 0.08345 | 14.689  | < 2e-16  | *** |
| MaskB6         | 2.00400  | 0.07945 | 25.224  | < 2e-16  | *** |
| PadTypeD6      | -0.43954 | 0.10645 | -4.129  | 3.64e-05 | *** |
| PadTypeD7      | -0.05931 | 0.10023 | -0.592  | 0.5540   |     |
| PadTypeL4      | 0.39075  | 0.09470 | 4.126   | 3.69e-05 | *** |
| PadTypeL6      | -0.59935 | 0.10958 | -5.470  | 4.51e-08 | *** |
| PadTypeL7      | -0.41598 | 0.10602 | -3.924  | 8.72e-05 | *** |
| PadTypeL8      | -0.16791 | 0.10185 | -1.649  | 0.0992   | .   |
| PadTypeL9      | -0.58996 | 0.10938 | -5.394  | 6.91e-08 | *** |
| PadTypeW4      | -0.12083 | 0.10113 | -1.195  | 0.2322   |     |
| PadTypeW9      | -1.48691 | 0.13443 | -11.061 | < 2e-16  | *** |
| factor(Panel)2 | 0.34356  | 0.06057 | 5.672   | 1.41e-08 | *** |
| factor(Panel)3 | 0.30398  | 0.06087 | 4.994   | 5.92e-07 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(10.5345) family taken to be 1)

Null deviance: 4488.13 on 719 degrees of freedom  
 Residual deviance: 842.58 on 702 degrees of freedom  
 AIC: 2679.2

Number of Fisher Scoring iterations: 1

Theta: 10.53  
 Std. Err.: 2.02

2 x log-likelihood: -2641.24

Warning in anova.negbin(sold.nb2): tests made without re-  
 estimating 'theta'

Analysis of Deviance Table

Model: Negative Binomial(10.5345), link: log

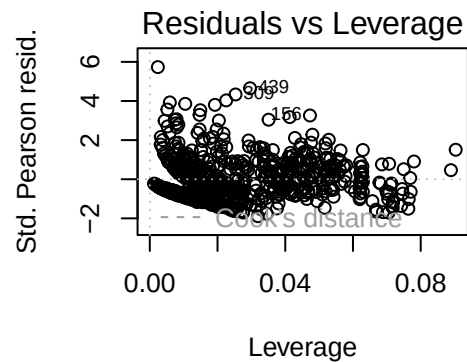
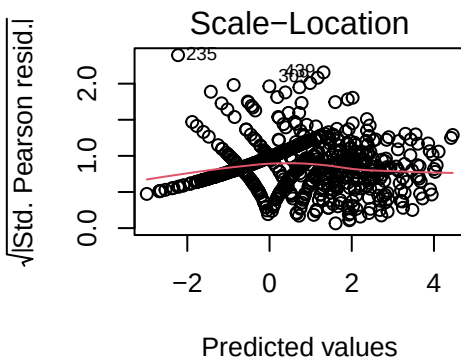
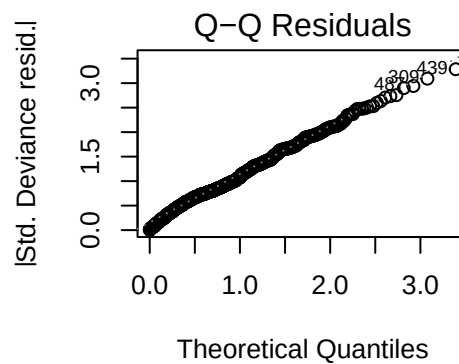
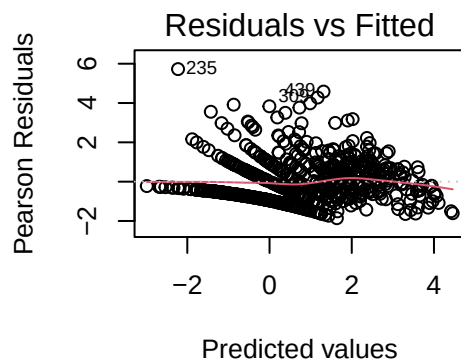
Response: skips

Terms added sequentially (first to last)

|               | Df | Deviance | Resid. Df | Resid. Dev | Pr(>Chi)      |
|---------------|----|----------|-----------|------------|---------------|
| NULL          |    |          | 719       | 4488.1     |               |
| Opening       | 2  | 1699.01  | 717       | 2789.1     | < 2.2e-16 *** |
| Solder        | 1  | 581.00   | 716       | 2208.1     | < 2.2e-16 *** |
| Mask          | 3  | 1014.20  | 713       | 1193.9     | < 2.2e-16 *** |
| PadType       | 9  | 313.65   | 704       | 880.3      | < 2.2e-16 *** |
| factor(Panel) | 2  | 37.70    | 702       | 842.6      | 6.526e-09 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1



Note 1:

$$\text{Var}(Y_i) = \underbrace{\mu_i}_{\text{Pois mean and var}} + \underbrace{\frac{\mu_i^2}{10.5344548}}_{\text{extra variability beyond Pois mean/var}}$$

Note 2: The estimate of  $\theta$  will change as we add more covariates.

| After adding covariate: | $\hat{\theta}$ value |
|-------------------------|----------------------|
| Opening                 | 0.6692532            |
| Solder                  | 0.9114626            |
| Mask                    | 2.7912396            |
| PadType                 | 8.0490995            |
| Panel                   | 10.5344548           |



Thus, the more covariates we add, the less overdispersion. In other words, the covariates are accounting for the extra variation in skips. If we think about our Stat 240 interpretation of the negative binomial distribution (“number of successes/failures/trials until the  $\theta$ th failure/success”), the covariates help us account for the variability and we have to “wait longer” (“ $\theta$  gets bigger”) to get the same number of failures/successes.

Note 3: The analysis of deviance table assumes the same value of  $\theta$  for all comparisons. We may want to allow for the changing estimates of  $\theta$  when we conduct a hypothesis test using the deviances, but we have to do this manually. Thus, in the analysis of deviance tables below, we focus on just the last variable added to the model.

```
anova(sold1)
```

```
Warning in anova.negbin(sold1): tests made without re-
estimating 'theta'
```

```
Analysis of Deviance Table
```

```
Model: Negative Binomial(0.6693), link: log
```

```
Response: skips
```

```
Terms added sequentially (first to last)
```

|                                                               | Df | Deviance | Resid. | Df  | Resid. | Dev     | Pr(>Chi)      |
|---------------------------------------------------------------|----|----------|--------|-----|--------|---------|---------------|
| NULL                                                          |    |          |        | 719 |        | 1085.26 |               |
| Opening                                                       | 2  | 310.83   |        | 717 |        | 774.44  | < 2.2e-16 *** |
| ---                                                           |    |          |        |     |        |         |               |
| Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1 |    |          |        |     |        |         |               |

```
anova(sold2)
```

```
Warning in anova.negbin(sold2): tests made without re-
estimating 'theta'
```

#### Analysis of Deviance Table

```
Model: Negative Binomial(0.9115), link: log
```

```
Response: skips
```

```
Terms added sequentially (first to last)
```

|                | Df | Deviance | Resid. | Df   | Resid. | Dev     | Pr(>Chi)           |
|----------------|----|----------|--------|------|--------|---------|--------------------|
| NULL           |    |          |        | 719  |        | 1338.78 |                    |
| Opening        | 2  | 403.54   |        | 717  |        | 935.24  | < 2.2e-16 ***      |
| Solder         | 1  | 162.43   |        | 716  |        | 772.81  | < 2.2e-16 ***      |
| ---            |    |          |        |      |        |         |                    |
| Signif. codes: | 0  | '***'    | 0.001  | '**' | 0.01   | '*'     | 0.05 '.' 0.1 ' ' 1 |

```
anova(sold3)
```

```
Warning in anova.negbin(sold3): tests made without re-
estimating 'theta'
```

#### Analysis of Deviance Table

```
Model: Negative Binomial(2.7912), link: log
```

```
Response: skips
```

```
Terms added sequentially (first to last)
```

|         | Df | Deviance | Resid. | Df  | Resid. | Dev     | Pr(>Chi)      |
|---------|----|----------|--------|-----|--------|---------|---------------|
| NULL    |    |          |        | 719 |        | 2606.49 |               |
| Opening | 2  | 913.63   |        | 717 |        | 1692.85 | < 2.2e-16 *** |
| Solder  | 1  | 326.86   |        | 716 |        | 1366.00 | < 2.2e-16 *** |

```
Mask      3    563.76      713    802.23 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(sold4)
```

```
Warning in anova.negbin(sold4): tests made without re-
estimating 'theta'
```

```
Analysis of Deviance Table
```

```
Model: Negative Binomial(8.0491), link: log
```

```
Response: skips
```

```
Terms added sequentially (first to last)
```

|         | Df | Deviance | Resid. | Df | Resid. | Dev       | Pr(>Chi) |
|---------|----|----------|--------|----|--------|-----------|----------|
| NULL    |    |          | 719    |    | 4115.8 |           |          |
| Opening | 2  | 1547.64  | 717    |    | 2568.2 | < 2.2e-16 | ***      |
| Solder  | 1  | 528.55   | 716    |    | 2039.6 | < 2.2e-16 | ***      |
| Mask    | 3  | 922.51   | 713    |    | 1117.1 | < 2.2e-16 | ***      |
| PadType | 9  | 284.09   | 704    |    | 833.0  | < 2.2e-16 | ***      |

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(sold5)
```

```
Warning in anova.negbin(sold5): tests made without re-
estimating 'theta'
```

```
Analysis of Deviance Table
```

Model: Negative Binomial(10.5345), link: log

Response: skips

Terms added sequentially (first to last)

|               | Df | Deviance | Resid. | Df | Resid. Dev | Pr(>Chi)      |
|---------------|----|----------|--------|----|------------|---------------|
| NULL          |    |          | 719    |    | 4488.1     |               |
| Opening       | 2  | 1699.01  | 717    |    | 2789.1     | < 2.2e-16 *** |
| Solder        | 1  | 581.00   | 716    |    | 2208.1     | < 2.2e-16 *** |
| Mask          | 3  | 1014.20  | 713    |    | 1193.9     | < 2.2e-16 *** |
| PadType       | 9  | 313.65   | 704    |    | 880.3      | < 2.2e-16 *** |
| factor(Panel) | 2  | 37.70    | 702    |    | 842.6      | 6.526e-09 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## 6.4 Varying Dispersion

Various models allow for dispersion to vary across observations, usually modeling such dispersion with an additional linear component. Here we introduce just one model that allows for varying dispersion. This model can account for both over- and under-dispersion.

### 6.4.1 Conway-Maxwell-Poisson Distribution

The COM-Poisson distribution is a flexible, two-parameter count distribution that allows for data dispersion relative to the Poisson model. This distribution was derived by Conway and Maxwell (1962). We follow Sellers (2023) in the distribution and GLM formation.

The pmf is of the form,

$$P(Y = y) = \frac{\lambda^y}{(y!)^\nu Z(\lambda, \nu)}, \text{ for } y = 0, 1, 2, \dots,$$

where  $Z(\lambda, \nu) = \sum_{j=0}^{\infty} (\lambda^j) / ((j!)^\nu)$  is the normalizing constant. The location parameter,  $\lambda = E(Y^\nu)$ , is a generalized form of the Poisson rate parameter. The dispersion parameter,  $\nu$ , must be greater than or equal to zero, and  $\nu = 1$  indicates equi-dispersion,  $\nu > 1$  indicates overdispersion (relative to the Poisson model), and  $\nu < 1$  indicates underdispersion (again, relative to the Poisson model). This parameterization is called the “CMP parameterization” of the generalized Poisson distribution.

This distribution does not have closed form expressions for the mean and variance. However, the asymptotic approximations are

$$E(Y) \approx \lambda^{1/\nu} + \frac{1}{2\nu} - \frac{1}{2}$$
$$\text{Var}(Y) \approx \frac{\lambda^{1/\nu}}{\nu}$$

where the approximations are accurate for  $\lambda^{1/\nu} > 10$ .

For fixed  $\nu$ , the CMP distribution is a member of the exponential family. Note the following relationships to other distributions for given values of  $\nu$ :

| Constraint               | Mean                        | Distribution                          |
|--------------------------|-----------------------------|---------------------------------------|
| $\nu = 1$                | $\lambda$                   | Poisson( $\lambda$ )                  |
| $\nu = 0, \lambda < 1$   | $\frac{\lambda}{1-\lambda}$ | Geometric( $\pi = 1 - \lambda$ )      |
| $\nu \rightarrow \infty$ | $\frac{\lambda}{1+\lambda}$ | Bern( $\pi = \lambda/(1 + \lambda)$ ) |

## 6.4.2 COM-Poisson Regression

Consider the count response variable,  $Y_i$ , where

$$Y_i \stackrel{ind}{\sim} CMP(\lambda_i, \nu_i), i = 1, \dots, n.$$

The CMP regression model includes linear components for both the location and dispersion parameters. Specifically,

$$\begin{aligned}\log(\lambda_i) &= \mathbf{x}_i' \beta, \\ \log(\nu_i) &= \mathbf{v}_i' \gamma,\end{aligned}$$

where  $\mathbf{x}_i$  and  $\mathbf{v}_i$  are vectors of covariates for the  $i$ th observation and  $\beta$  and  $\gamma$  are corresponding coefficients.

Sellers (2023) derives both maximum likelihood and moment-based estimators for both  $\beta$  and  $\gamma$ . (Note that other authors and packages derive regression model based on an approximate mean, which Sellers (2023) calls the “ACMP” parameterization, for interpretability).

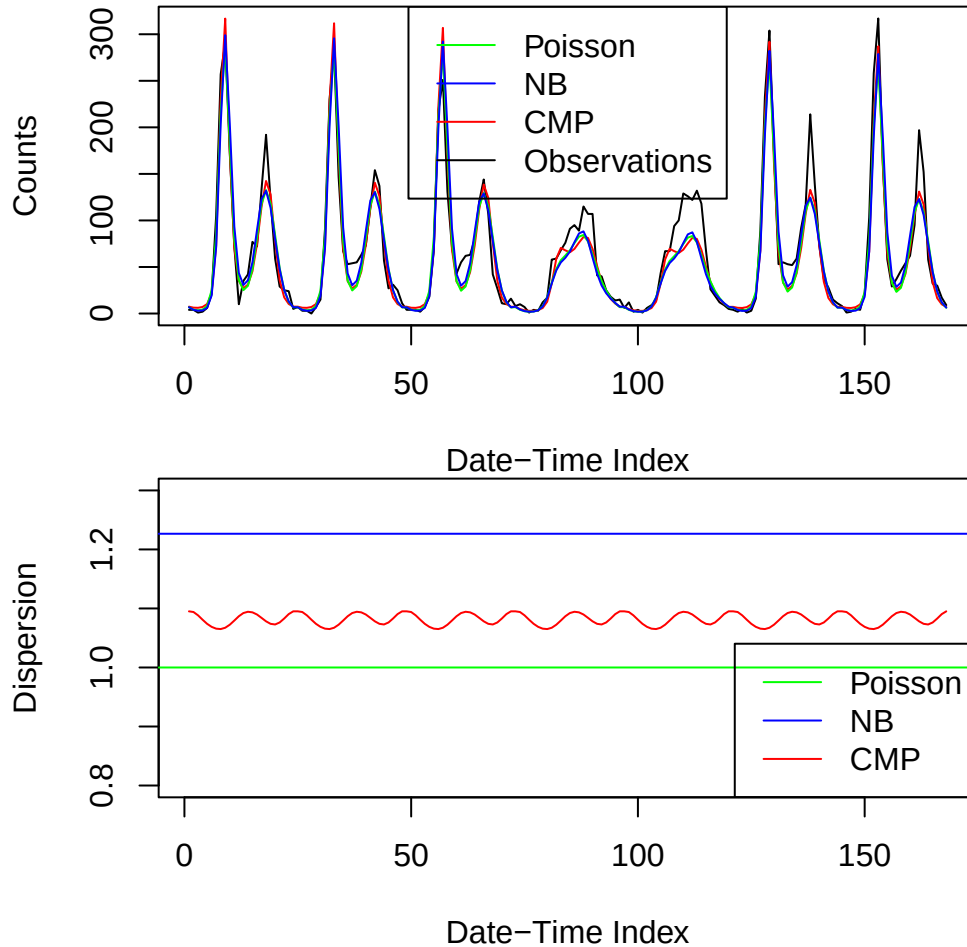
This [Kaggle data set](#) collects the number of bicyclists on the East sidewalk each hour from October 3, 2012 at midnight to October 31, 2019 at 11pm. We consider modeling the count of bicycles using the date and time of the count. Specifically, we use year (numeric), a weekend indicator variable, a circular spline for day of the year, and a circular spline for hour of the day as covariates in the model. We also consider a weekend-by-hour-of-the-day interaction.

We are interested in: 1. What can we learn about how the number of bicycles crossing the bridge behaved during this time period? 2. Is over- or under-dispersion present in this data? If so, how does inference change under the different model assumptions?

```
#library(devtools)
#install_github("thomas-fung/mpcmp")
#library(mpcmp) #mean-parameterized com-poisson regression
library(COMPoissonReg) #exact com-poisson regression
```

Warning: package 'COMPoissonReg' was built under R version 4.6.1

```
mod3 <- COMPoissonReg::glm.cmp(formula.lambda=Fremont.Bridge.East.Sidewalk~year
mod4 <- COMPoissonReg::glm.cmp(Fremont.Bridge.East.Sidewalk~year+csplinedoy+wk
```



```
summary(mod1)
```

Call:

```
glm(formula = Fremont.Bridge.East.Sidewalk ~ -  
1 + year + wknd +  
    csplinedoy + wknd * csplinehr, family = poisson, data = datsub)
```

Coefficients:

|                     | Estimate  | Std. Error | z value | Pr(> z ) |     |
|---------------------|-----------|------------|---------|----------|-----|
| year                | -0.023050 | 0.002072   | -11.122 | < 2e-16  | *** |
| wkndFALSE           | 5.437471  | 0.035260   | 154.210 | < 2e-16  | *** |
| wkndTRUE            | 5.029626  | 0.053557   | 93.912  | < 2e-16  | *** |
| csplinedoy1         | -0.959341 | 0.048210   | -19.899 | < 2e-16  | *** |
| csplinedoy2         | -0.909085 | 0.032038   | -28.375 | < 2e-16  | *** |
| csplinedoy3         | -0.884396 | 0.036821   | -24.019 | < 2e-16  | *** |
| csplinedoy4         | -0.304212 | 0.035780   | -8.502  | < 2e-16  | *** |
| csplinedoy5         | 0.077083  | 0.026291   | 2.932   | 0.003369 | **  |
| csplinedoy6         | -0.146902 | 0.039521   | -3.717  | 0.000202 | *** |
| csplinehr1          | -0.314332 | 0.052944   | -5.937  | 2.90e-09 | *** |
| csplinehr2          | -3.684217 | 0.045714   | -80.592 | < 2e-16  | *** |
| csplinehr3          | -4.601002 | 0.062093   | -74.098 | < 2e-16  | *** |
| csplinehr4          | 2.743155  | 0.026966   | 101.727 | < 2e-16  | *** |
| csplinehr5          | -3.468746 | 0.055742   | -62.229 | < 2e-16  | *** |
| wkndTRUE:csplinehr1 | -0.891556 | 0.119518   | -7.460  | 8.68e-14 | *** |
| wkndTRUE:csplinehr2 | 1.078515  | 0.101292   | 10.648  | < 2e-16  | *** |
| wkndTRUE:csplinehr3 | -0.602808 | 0.196675   | -3.065  | 0.002177 | **  |
| wkndTRUE:csplinehr4 | -3.385623 | 0.072636   | -46.611 | < 2e-16  | *** |
| wkndTRUE:csplinehr5 | 2.784155  | 0.110288   | 25.244  | < 2e-16  | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 413429.4 on 1000 degrees of freedom



Residual deviance: 9662.2 on 981 degrees of freedom  
AIC: 14541

Number of Fisher Scoring iterations: 5

`summary(mod2)`

Call:

```
glm.nb(formula = Fremont.Bridge.East.Sidewalk ~ -  
1 + year + wknd +  
      csplinedoy + wknd * csplinehr, data = datsub, init.theta = 4.413405556,  
      link = log)
```

Coefficients:

|                     | Estimate  | Std. Error | z value | Pr(> z ) |     |
|---------------------|-----------|------------|---------|----------|-----|
| year                | -0.023488 | 0.008437   | -2.784  | 0.00537  | **  |
| wkndFALSE           | 5.461760  | 0.147706   | 36.977  | < 2e-16  | *** |
| wkndTRUE            | 5.124571  | 0.198661   | 25.795  | < 2e-16  | *** |
| csplinedoy1         | -0.816539 | 0.180702   | -4.519  | 6.22e-06 | *** |
| csplinedoy2         | -0.948475 | 0.120196   | -7.891  | 3.00e-15 | *** |
| csplinedoy3         | -0.841985 | 0.145523   | -5.786  | 7.21e-09 | *** |
| csplinedoy4         | -0.329962 | 0.147747   | -2.233  | 0.02553  | *   |
| csplinedoy5         | 0.185074  | 0.114472   | 1.617   | 0.10593  |     |
| csplinedoy6         | 0.030866  | 0.174983   | 0.176   | 0.85998  |     |
| csplinehr1          | -0.537412 | 0.200305   | -2.683  | 0.00730  | **  |
| csplinehr2          | -3.592816 | 0.114494   | -31.380 | < 2e-16  | *** |
| csplinehr3          | -5.175910 | 0.194539   | -26.606 | < 2e-16  | *** |
| csplinehr4          | 2.721453  | 0.116160   | 23.429  | < 2e-16  | *** |
| csplinehr5          | -3.290032 | 0.205646   | -15.998 | < 2e-16  | *** |
| wkndTRUE:csplinehr1 | -1.144131 | 0.385296   | -2.969  | 0.00298  | **  |
| wkndTRUE:csplinehr2 | 1.036592  | 0.224434   | 4.619   | 3.86e-06 | *** |
| wkndTRUE:csplinehr3 | -0.287887 | 0.414088   | -0.695  | 0.48691  |     |
| wkndTRUE:csplinehr4 | -3.516504 | 0.230867   | -15.232 | < 2e-16  | *** |
| wkndTRUE:csplinehr5 | 2.354543  | 0.390412   | 6.031   | 1.63e-09 | *** |

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(4.4134) family taken to be 1)

    Null deviance: 165465.6  on 1000  degrees of freedom
Residual deviance:  1091.1  on   981  degrees of freedom
AIC: 7946.2

Number of Fisher Scoring iterations: 1

            Theta:  4.413
         Std. Err.:  0.250

2 x log-likelihood:  -7906.218

```

```
summary(mod3)
```

```

$DF

```

|                       | Estimate | SE     | z-value  | p-value    |
|-----------------------|----------|--------|----------|------------|
| X:(Intercept)         | 0.4324   | 0.0037 | 116.3125 | 0          |
| X:year                | -0.0020  | 0.0006 | -3.2524  | 0.001144   |
| X:wkndTRUE            | -0.0363  | 0.0152 | -2.3814  | 0.01725    |
| X:csplinedoy1         | -0.0856  | 0.0160 | -5.3631  | 8.182e-08  |
| X:csplinedoy2         | -0.0810  | 0.0104 | -7.8140  | 5.541e-15  |
| X:csplinedoy3         | -0.0782  | 0.0124 | -6.3227  | 2.57e-10   |
| X:csplinedoy4         | -0.0267  | 0.0114 | -2.3299  | 0.01981    |
| X:csplinedoy5         | 0.0087   | 0.0082 | 1.0654   | 0.2867     |
| X:csplinedoy6         | -0.0122  | 0.0135 | -0.9074  | 0.3642     |
| X:csplinehr1          | -0.0002  | 0.0204 | -0.0086  | 0.9932     |
| X:csplinehr2          | -0.4297  | 0.0188 | -22.9153 | 3.271e-116 |
| X:csplinehr3          | -0.4244  | 0.0225 | -18.8917 | 1.335e-79  |
| X:csplinehr4          | 0.2531   | 0.0100 | 25.3763  | 4.61e-142  |
| X:csplinehr5          | -0.3086  | 0.0172 | -17.9898 | 2.344e-72  |
| X:wkndTRUE:csplinehr1 | -0.0865  | 0.0269 | -3.2201  | 0.001281   |

|                       |         |        |           |           |
|-----------------------|---------|--------|-----------|-----------|
| X:wkndTRUE:csplinehr2 | 0.1361  | 0.0395 | 3.4440    | 0.0005731 |
| X:wkndTRUE:csplinehr3 | -0.2993 | 0.0749 | -3.9938   | 6.502e-05 |
| X:wkndTRUE:csplinehr4 | -0.2620 | 0.0242 | -10.8452  | 2.101e-27 |
| X:wkndTRUE:csplinehr5 | 0.2503  | 0.0335 | 7.4801    | 7.426e-14 |
| S:(Intercept)         | -2.5192 | 0.0110 | -229.7429 | 0         |

\$DF.lambda

NULL

\$DF.nu

|    | Estimate | SE    |
|----|----------|-------|
| nu | 0.0805   | 9e-04 |

\$n

[1] 1000

\$loglik

[1] -4010.408

\$aic

[1] 8060.816

\$bic

[1] 8158.972

\$optim.method

[1] "L-BFGS-B"

\$opt.message

[1] "NEW\_X"

\$opt.convergence

[1] 1

\$elapsed.sec

[1] 36.18511

```
summary(mod4)
```

```
Warning in sqrt(diag(vcov(object))): NaNs produced
```

```
Warning in sqrt(diag(vcov(object))): NaNs produced
```

```
$DF
```

|                       | Estimate | SE     | z-value   | p-value   |
|-----------------------|----------|--------|-----------|-----------|
| X:(Intercept)         | 0.4602   | NaN    | NaN       | NaN       |
| X:year                | -0.0020  | 0.0006 | -3.3814   | 0.0007211 |
| X:csplinedoy1         | -0.0855  | 0.0110 | -7.7558   | 8.78e-15  |
| X:csplinedoy2         | -0.0744  | 0.0094 | -7.9265   | 2.255e-15 |
| X:csplinedoy3         | -0.0756  | 0.0091 | -8.2790   | 1.243e-16 |
| X:csplinedoy4         | -0.0301  | 0.0095 | -3.1617   | 0.001568  |
| X:csplinedoy5         | 0.0058   | 0.0067 | 0.8661    | 0.3864    |
| X:csplinedoy6         | -0.0203  | 0.0097 | -2.0819   | 0.03735   |
| X:wkndTRUE            | -0.0210  | 0.0168 | -1.2504   | 0.2112    |
| X:csplinehr1          | -0.1220  | 0.0192 | -6.3375   | 2.335e-10 |
| X:csplinehr2          | -0.3795  | 0.0335 | -11.3197  | 1.049e-29 |
| X:csplinehr3          | -0.4517  | 0.0117 | -38.5396  | 0         |
| X:csplinehr4          | 0.0782   | 0.0054 | 14.6096   | 2.441e-48 |
| X:csplinehr5          | -0.2677  | 0.0334 | -8.0101   | 1.146e-15 |
| X:wkndTRUE:csplinehr1 | -0.0961  | 0.0368 | -2.6109   | 0.009029  |
| X:wkndTRUE:csplinehr2 | 0.1393   | 0.0387 | 3.5976    | 0.0003211 |
| X:wkndTRUE:csplinehr3 | -0.3505  | 0.0778 | -4.5036   | 6.681e-06 |
| X:wkndTRUE:csplinehr4 | -0.2163  | 0.0248 | -8.7199   | 2.786e-18 |
| X:wkndTRUE:csplinehr5 | 0.2138   | 0.0335 | 6.3832    | 1.734e-10 |
| S:(Intercept)         | -2.4153  | 0.0205 | -117.6555 | 0         |
| S:csplinehr1          | -0.4011  | 0.0396 | -10.1201  | 4.5e-24   |
| S:csplinehr2          | 0.1886   | 0.1577 | 1.1965    | 0.2315    |
| S:csplinehr3          | -0.2510  | 0.0603 | -4.1634   | 3.135e-05 |
| S:csplinehr4          | -0.4488  | NaN    | NaN       | NaN       |
| S:csplinehr5          | 0.0568   | 0.1095 | 0.5187    | 0.604     |

```
$DF.lambda
```

```
NULL
```

```
$DF.nu
```

```
NULL
```

```
$n
```

```
[1] 1000
```

```
$loglik
```

```
[1] -4001.897
```

```
$aic
```

```
[1] 8053.794
```

```
$bic
```

```
[1] 8176.488
```

```
$optim.method
```

```
[1] "L-BFGS-B"
```

```
$opt.message
```

```
[1] "NEW_X"
```

```
$opt.convergence
```

```
[1] 1
```

```
$elapsed.sec
```

```
[1] 48.7659
```

| Model   | In-Sample MSE | Out-of-Sample MSE |
|---------|---------------|-------------------|
| Poisson | 848.16        | 810.04            |
| NB      | 991.82        | 951.61            |
| CMP v1  | 851.9         | 826.93            |
| CMP v2  | 832.97        | 832.08            |

## 6.5 A Few Final Remarks

- Overdispersion introduces more modeling decisions.
- We still have to find a model for the mean of  $Y$ 
  - Choose a link function
  - Select the covariates to include
- In the presence of overdispersion either:
  1. Further parameterize our model with overdispersion parameter  $\phi$ . Very often we no longer have an exact distribution for  $Y$ . Sometimes we are only willing to specify a mean and variance for  $Y$  (in R: *family=quasi(link = “identity”, variance = “constant”)*).
  2. Change the distribution of  $Y$  by, e.g., proposing a mechanism for clustering that generates a different distribution. Depending on the choice of the distribution, MLE's can be difficult to calculate. Need to be more careful with interpretations.

## 6.6 Homework Assignment

The “aod” package in R contains a dataset, “dja”, used to analyze the mortality of Djallonke lambs in Senegal. Variables include: - group: a factor with 2 levels (CTRL and TREAT) indicating the treatment - village: a factor indicating the village of the herd - herd: a factor indicating the herd - n: the number of animals exposed to mortality - trisk: the exposition time to mortality (in years) - y: the number of deaths.

1. Fit a quasi-binomial model using group as a predictor and proportion of deaths as the response. (Note: Recall that for binomial regression, use the number of successes and failures as the response, i.e., `cbind(y, n-y)`.)
  - a. What is the estimate of the overdispersion parameter? Is it significant? How does it change the estimate of the variance of the  $Y_i$  from the binomial distribution?
  - b. Interpret the coefficient for group. Is it significant? Does the treatment affect the number of deaths in lambs as we would expect or hope?
2. Fit a beta-binomial model using group as a predictor and proportion of deaths as the response. (Again, be sure to use the number of successes and failures as the response.)
  - a. What is the estimate of the overdispersion parameter? Is it significant? How does it change the estimate of the variance of the  $Y_i$  from the binomial distribution?
  - b. Interpret the coefficient for group. Is it significant? Does the treatment affect the number of deaths in lambs as we would expect or hope and is this different from the quasi-binomial?
3. Fit a quasi-poisson model using group as a predictor and number of deaths as the response. Use exposition time to mortality as an offset (this is like controlling for the total number of lambs, but instead uses the total number of years of living available).
  - a. What is the estimate of the overdispersion parameter? Is it significant? How does it change the estimate of the variance of the  $Y_i$  from the poisson distribution?
  - b. Interpret the coefficient for group. Is it significant? Does the treatment affect the number of deaths in lambs as we would expect or hope?

4. Fit a negative-binomial model using group as a predictor and number of deaths as the response. Again, using exposition time to mortality as an offset.
  - a. What is the estimate of the overdispersion parameter? Is it significant? How does it change the estimate of the variance of the  $Y_i$  from the poisson distribution?
  - b. Interpret the coefficient for group. Is it significant? Does the treatment affect the number of deaths in lambs as we would expect or hope?
5. Which model to you prefer? Defend your choice by addressing, at a minimum, model diagnostics and similarities/differences in models (parameterizations, parameter estimates, parsimony).
6. Imagine you conducted this analysis for an organization deciding whether to continue using the treatment. Write a brief recommendation (100 words or less) for a non-statistician that addresses the items below. Do not use statistical jargon such as “quasi-likelihood”, “dispersion parameter”, or “negative binomial” without explaining what it means in plain language.
  - i. states whether the treatment appears effective,
  - ii. explains why accounting for overdispersion matters for this decision, and
  - iii. recommends which mdoel you would use for the final analysis.
7. (Bonus/not required) Try fitting a varying dispersion GLM.



# 7 GLM's for Nominal and Ordinal Data

Nominal response variables are categorical variables without an ordering. For example, laundry detergent brand a customer buys (Tide, Wisk, All), land covering type of a given location (wetland, forest, urban), mode of transportation to campus (bike, bus, walk), etc. Ordinal response variables are categorical variables whose order *does* matter. For example, a Likert survey (strongly agree to strongly disagree), year in school (freshman, sophomore, junior), etc.

Often, the categories (nominal or ordinal) are labeled using numbers, even if the order of the numbers doesn't matter. In the following sections, we take that approach because it is helpful for notation, but we'll replace the number labels when an alternative representation can help us more quickly interpret the quantities (such as a letter subscript).

## 7.1 Nominal Data

Instead of 2 possible outcomes as in standard logistic regression, suppose we now have 3 possible (unordered) outcomes for each  $Y_i$ ,  $i = 1, \dots, n$ .

Let,

$$P(Y_i = 1) = \pi_{i1},$$

$$P(Y_i = 2) = \pi_{i2},$$

$$P(Y_i = 3) = \pi_{i3},$$

such that

$$\pi_{i1} + \pi_{i2} + \pi_{i3} = 1.$$

Note that this allows for each observation/individual to have their own set of probabilities of each categorical outcome. For example, a customer choosing

laundry detergent will have their own set of probabilities for each detergent brand (and we will model these using a set of covariates); one customer may have the highest probability to buy Tide, while another customer tends to buy All.

When there were only two possible outcomes,

$$\pi_{i1} + \pi_{i2} = \pi_i + (1 - \pi_i) = 1,$$

and we modeled

$$g(\pi_i) = \mathbf{x}_i' \beta.$$

But what do we do with 3 (or  $J$ )  $\pi_{ij}$ 's?

Notice,

$$1 - \pi_{i1} = \pi_{i2} + \pi_{i3} \tag{7.1}$$

→ does it make sense to model  $g(\pi_{ij})$  without acknowledging Equation 7.1? (Of course not!)

How can we account for this linear constraint between the  $\pi_{ij}$ 's? (The answer is in the next section!)

### Data:

For a random sample of size  $n$ , let  $y_i$  be the observed outcome (category) for the  $i$ th observation.

Let  $\mathbf{x}_{ij}$  be the corresponding  $k \times 1$  vector of covariates for the  $i$ th observation and the  $j$ th category. Depending on the analysis,  $\mathbf{x}_{ij}$  may simplify to either  $\mathbf{x}_i$  or  $\mathbf{x}_j$ .

Consider the case where researchers are trying to examine the relationship between various covariates and a shopper's choice in laundry detergent.

- Observation-specific covariate ( $\mathbf{x}_{ij} = \mathbf{x}_i$ ): An example might be how much laundry the customer typically does each week or the type of laundry they are doing (thick/coarse or thin/smooth material). These values will be different for each customer, but will not change with the brand.

- Category-specific covariate ( $\mathbf{x}_{ij} = \mathbf{x}_j$ ): Examples might be brand smell or brightness of the branding. These are the same across all customers (the branding color can't change for each customer!).
- Observation-category-specific covariate ( $\mathbf{x}_{ij}$ ): Examples of this could be the amount of the coupon the customer has (e.g., they have a \$1.00 off coupon for Tide but a \$0.30 off coupon for All) or price (note that if the customers are all from the same store and the data was collected for a specific time period, the prices might be the same for each customer, but would obviously be different across brands; however, the more typical case is if the customers went to different stores and bought their detergents at different times so that the prices for the various detergents would be different across customers).

### Model:

First, we only consider pairs of response categories. Choosing a reference category – WLOG, choose category 1 – model the logit of category  $j$  with respect to the reference category:

$$\log \left( \frac{\pi_{ij}}{\pi_{i1}} \right) = \mathbf{x}'_{ij} \beta_j, \quad j = 2, \dots, J, \quad (7.2)$$

where  $\beta_j$  is a  $k \times 1$  category-specific coefficient. This is called **the baseline-category logit model** or **nominal logit model**. Because the coefficients are unique for each category, we have  $J - 1$  **logit equations**. We then estimate  $\beta_j, j = 2, \dots, J$  using all  $J - 1$  equations together.

To estimate our probabilities, notice that

$$\begin{aligned} \log \left( \frac{\hat{\pi}_{ij}}{\hat{\pi}_{i1}} \right) &= \mathbf{x}'_{ij} \hat{\beta}_j \\ \Rightarrow \hat{\pi}_{ij} &= \hat{\pi}_{i1} \exp\{\mathbf{x}'_{ij} \hat{\beta}_j\} \end{aligned} \quad (7.3)$$

Using the sum-to-one constraint,

$$\begin{aligned}
& \hat{\pi}_{i1} + \hat{\pi}_{i2} + \cdots + \hat{\pi}_{iJ} = 1 \\
& \Rightarrow \hat{\pi}_{i1} + \underbrace{\hat{\pi}_{i2} + \cdots + \hat{\pi}_{iJ}}_{=\sum_{j=2}^J \hat{\pi}_{i1} \exp\{\mathbf{x}'_{ij}\hat{\beta}_j\}} = 1 \\
& \Leftrightarrow \hat{\pi}_{i1}(1 + \sum \exp\{\mathbf{x}'_{ij}\hat{\beta}_j\}) = 1 \\
& \Leftrightarrow \hat{\pi}_{i1} = \frac{1}{1 + \sum \exp\{\mathbf{x}'_{ij}\hat{\beta}_j\}} \\
& \Rightarrow \hat{\pi}_{ij} = \frac{\exp\{\mathbf{x}'_{ij}\hat{\beta}_j\}}{1 + \sum_{r=2}^J \exp\{\mathbf{x}'_{ir}\hat{\beta}_r\}}.
\end{aligned}$$

### Interpretation:

To more fully explore interpretations, consider a binary explanatory variable  $x_i$  (a 2-level factor) representing some presence or absence of some exposure, e.g.,

$x_i = 1 \Rightarrow$  exposure present

$x_i = 0 \Rightarrow$  exposure absent

Suppose we are interested in the relationship between introversion/extroversion and music artist preference (among Taylor Swift, BTS, and Drake). We collect the following data from our class:

| $y_i$  | $x_i$           |
|--------|-----------------|
| TSwift | introverted (1) |
| Drake  | introverted (1) |
| ...    | ...             |

Take the survey [here!](#)

Let

$\pi_{j|1}$  = probability of outcome  $j$  given presence of exposure ( $x_i = 1$ )

$\pi_{j|0}$  = probability of outcome  $j$  given absence of exposure ( $x_i = 0$ )

Then,

- Odds of outcome  $j$  to response 1 when exposure is present is  $\pi_{j|1}/\pi_{1|1}$ .
- Odds of outcome  $j$  to response 1 when exposure is absent is  $\pi_{j|0}/\pi_{1|0}$ .

$$\log \left( \frac{\pi_{j|0}}{\pi_{1|0}} \right) = \beta_{0j} + 0 \beta_{1j} = \beta_{0j}$$

$$\log \left( \frac{\pi_{j|1}}{\pi_{1|1}} \right) = \beta_{0j} + 1 \beta_{1j} = \beta_{0j} + \beta_{1j}$$

$$\Rightarrow \log \left( \underbrace{\frac{\pi_{j|1}/\pi_{1|1}}{\pi_{j|0}/\pi_{1|0}}}_{\text{odds ratio for category } j \text{ to category } 1} \right) = \log \left( \frac{\pi_{j|1}}{\pi_{1|1}} \right) - \log \left( \frac{\pi_{j|0}}{\pi_{1|0}} \right)$$

$$= \beta_{1j}$$

$\Rightarrow$  Odds ratio for category/outcome  $j$  to category/outcome 1:

$$OR_j = e^{\beta_{1j}},$$

and we can compute confidence intervals on the odds ratios as before ( $e^{\hat{\beta}_{1j} \pm 1.96 SE(\hat{\beta}_{1j})}$ ).

## Law School Writing Speed Example

*1st year law students: Course and Writing Speed*

- Response variable: 1st Year Law School Course: Civil Procedure (1), Contracts (2), Criminal Law (3)
- Explanatory variable: Number of words written per minute on course final exam was larger than 30: True or False

**Table of counts:**

Table 7.1: Counts of individuals in each class and how fast they wrote on their final exam.

| WPM $\geq 25$ | Civil Proc. (P) | Contracts (T) | Crim. Law (L) | Total |
|---------------|-----------------|---------------|---------------|-------|
| True          | 18              | 43            | 191           | 252   |
| False         | 643             | 624           | 618           | 1885  |
| Total         |                 |               |               | 2137  |

Table of conditional proportions:



Table 7.2: Proportion of individuals in each class given the speed at which they wrote their final exam.

| WPM $\geq$ 25 | Civil Proc. (P) | Contracts (T) | Crim. Law (L) | Total |
|---------------|-----------------|---------------|---------------|-------|
| True          | 0.071           | 0.171         | 0.758         | 1     |
| False         | 0.341           | 0.331         | 0.328         | 1     |

### Model:

Notice that we can model these data at the individual level or the group level and would write the two different versions of the likelihood. The first way of writing the likelihood uses the individual responses, specifically,

$$Y_i \stackrel{ind}{\sim} \text{Mutlinom}(1, \pi_{P|x_i}, \pi_{T|x_i}, \pi_{L|x_i}),$$

where  $Y_i$  is the class for the  $i$ th respondent,  $x_i$  is an indicator for whether the  $i$ th respondent wrote at least 25 words per minute on their exam or not, and  $\pi_{P|x_i}$ ,  $\pi_{T|x_i}$ ,  $\pi_{L|x_i}$  represent the  $i$ th individual's probabilities for having taken the exam in the Civil **Procedures** class (P), **Contracts** class (T), or Criminal **Law** class (L), respectively.

Instead, because we have simple covariates, we can aggregate among the grouped covariates and write the likelihood for these groups,

$$Y_X \stackrel{ind}{\sim} \text{Mutlinom}(m_X, \pi_{P|X}, \pi_{T|X}, \pi_{L|X}),$$

where  $Y_X$  represents the  $3 \times 1$  vector of counts among each of the classes for all individuals' whose covariate is equal to  $X$  (one of  $T$  for “true” and  $F$  for “false”), and  $\pi_{P|X}$ ,  $\pi_{T|X}$ , and  $\pi_{L|X}$ , represent the probabilities that someone whose covariate is equal to  $X$  is in the Civil **Procedures** class (P), **Contracts** class (T), or Criminal **Law** class (L), respectively. We will use this latter approach to write our model in this example (but the code to fit the models uses the non-grouped data).

Thus, we have that  $\pi_{P|F}$  represents the probability that a student was in the Civil Procedures class given that they wrote fewer than 25 words per minute on their final exam. Likewise,  $\pi_{T|T}$  represents the probability that the student was in the Contracts class given that they *did* write at least 25 words per minute. Etc.

We have two equations for the **systematic component** of our model. Namely,

$$\begin{aligned} \log \left( \frac{\pi_{P|X}}{\pi_{L|X}} \right) &= \beta_{0P} + \beta_{1P}I(X) \\ \log \left( \frac{\pi_{T|X}}{\pi_{L|X}} \right) &= \beta_{0T} + \beta_{1T}I(X). \end{aligned}$$

Notice that for this simple case when  $X$  is a simple 2-level factor, the quantities that are estimated are:

$$\begin{aligned}\log\left(\frac{\pi_{P|F}}{\pi_{L|F}}\right) &= \beta_{0P} \\ \log\left(\frac{\pi_{P|T}}{\pi_{L|T}}\right) &= \beta_{0P} + \beta_{1P} \\ \log\left(\frac{\pi_{T|F}}{\pi_{L|F}}\right) &= \beta_{0T} \\ \log\left(\frac{\pi_{T|T}}{\pi_{L|T}}\right) &= \beta_{0T} + \beta_{1T}\end{aligned}$$

Finally, notice that just like our binary logistic regression models, the coefficients represent log odds ratios. Specifically, the odds that the student is in Civil Procedure relative to Criminal Law when the student wrote at least 25 words per minute on their exam are  $\exp\{\beta_{1P}\}$  times the same odds when the student wrote fewer than 25 words per minute. Likewise, the odds that the student is in Contracts relative to Criminal Law when the student wrote at least 25 words per minute on their exam are  $\exp\{\beta_{1T}\}$  times the same odds when the student wrote fewer than 25 words per minute.

**Fitted Model:**

```
library(VGAM) #for the vglm function
modv1 <- vglm(Course~WPM, family='multinomial', data=WordDat2)
summary(modv1)
```

Call:

```
vglm(formula = Course ~ WPM, family = "multinomial", data = WordDat2)
```

Coefficients:

|               | Estimate  | Std. Error | z value | Pr(> z )   |
|---------------|-----------|------------|---------|------------|
| (Intercept):1 | 0.039656  | 0.056332   | 0.704   | 0.481      |
| (Intercept):2 | 0.009662  | 0.056751   | 0.170   | 0.865      |
| WPMTRUE:1     | -2.401558 | 0.252858   | -9.498  | <2e-16 *** |
| WPMTRUE:2     | -1.500735 | 0.178067   | -8.428  | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,3]), log(mu[,2]/mu[,3])

Residual deviance: 4494.175 on 4270 degrees of freedom

Log-likelihood: -2247.088 on 4270 degrees of freedom

Number of Fisher scoring iterations: 4

Reference group is level 3 of the response

$$\log \left( \frac{\hat{\pi}_{P|F}}{\hat{\pi}_{L|F}} \right) = \hat{\beta}_{0P} = 0.0397$$

$$\log \left( \frac{\hat{\pi}_{P|T}}{\hat{\pi}_{L|T}} \right) = \hat{\beta}_{0P} + \hat{\beta}_{1P} = 0.0397 + -2.4016 = -2.3619$$

$(e^{0.0397}) = 1.0405$  and  $e^{-2.3619} = 0.0942$

$$\log \left( \frac{\hat{\pi}_{C|F}}{\hat{\pi}_{L|F}} \right) = \hat{\beta}_{0C} = 0.0097$$

$$\log \left( \frac{\hat{\pi}_{C|T}}{\hat{\pi}_{L|T}} \right) = \hat{\beta}_{0C} + \hat{\beta}_{1C} = 0.0097 + -1.5007 = -1.4910$$

$(e^{0.0097}) = 1.0097$  and  $e^{-1.491} = 0.2251$

**Model Significance:**

We'll test the significance of  $\text{WPM} \geq 25$  using the deviance statistic. To find the deviance for the null model, we fit the model with only an intercept.

```
mod0 <- vglm(Course~1, family="multinomial", data=WordDat2)
summary(mod0)
```

Call:

```
vglm(formula = Course ~ 1, family = "multinomial", data = WordDat2)
```

Coefficients:

|               | Estimate | Std. Error | z value | Pr(> z ) |     |
|---------------|----------|------------|---------|----------|-----|
| (Intercept):1 | -0.20205 | 0.05243    | -3.854  | 0.000116 | *** |
| (Intercept):2 | -0.19301 | 0.05230    | -3.690  | 0.000224 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,3]), log(mu[,2]/mu[,3])

Residual deviance: 4676.167 on 4272 degrees of freedom

Log-likelihood: -2338.084 on 4272 degrees of freedom

Number of Fisher scoring iterations: 3

Reference group is level 3 of the response

$$\begin{aligned}D_0 &= 4676.167 \\D_1 &= 4494.175 \\ \Rightarrow \Delta D &= D_0 - D_1 = 181.992\end{aligned}$$

```
pchisq(181.992, 2)
```

[1] 1

$\Rightarrow$  100%ile of  $\Delta D$  distribution, therefore WPM  $\geq 25$  is significant.

### Changing the base/reference category:

Instead of comparing ‘Civil Procedure’ and ‘Contracts’ to ‘Criminal Law’, we’d rather make our comparisons in reference to ‘Civil Procedure’. How can we do this without refitting the new model?

We want  $\log\left(\frac{\pi_{L|X}}{\pi_{P|X}}\right)$  and  $\log\left(\frac{\pi_{C|X}}{\pi_{P|X}}\right)$ .

$$\begin{aligned}\log\left(\frac{\pi_{C|X}}{\pi_{P|X}}\right) &= \log\left(\frac{\pi_{C|X}/\pi_{L|X}}{\pi_{P|X}/\pi_{L|X}}\right) = \log\left(\frac{\pi_{C|X}}{\pi_{P|L|X}}\right) - \log\left(\frac{\pi_{P|X}}{\pi_{L|X}}\right) \\ &= \underbrace{(\beta_{0C} - \beta_{0P})}_{\alpha_0} + \underbrace{(\beta_{1C} - \beta_{1P})}_{\alpha_1} I(x_i = X)\end{aligned}$$

and

$$\log\left(\frac{\pi_{L|X}}{\pi_{P|X}}\right) = \log\left(\frac{\pi_{L|X}/\pi_{L|X}}{\pi_{P|X}/\pi_{L|X}}\right) = \log(1) - \log\left(\frac{\pi_{P|X}}{\pi_{L|X}}\right) = -\log\left(\frac{\pi_{P|X}}{\pi_{L|X}}\right)$$

```
WordDat2$Course <- factor(WordDat2$Course, levels=sort(unique(factor(WordDat2$
modvs <- vglm(Course~WPM, family='multinomial', data=WordDat2)
summary(modvs)
```

Call:

```
vglm(formula = Course ~ WPM, family = "multinomial", data = WordDat2)
```

Coefficients:

|               | Estimate | Std. Error | z value | Pr(> z )    |
|---------------|----------|------------|---------|-------------|
| (Intercept):1 | -0.03966 | 0.05633    | -0.704  | 0.48145     |
| (Intercept):2 | -0.02999 | 0.05619    | -0.534  | 0.59351     |
| WPMTRUE:1     | 2.40156  | 0.25286    | 9.498   | < 2e-16 *** |
| WPMTRUE:2     | 0.90082  | 0.28625    | 3.147   | 0.00165 **  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,3]), log(mu[,2]/mu[,3])

Residual deviance: 4494.175 on 4270 degrees of freedom

Log-likelihood: -2247.088 on 4270 degrees of freedom

Number of Fisher scoring iterations: 4

Reference group is level 3 of the response

Note: Another popular package for fitting the multinomial model is the “nnet” package and “multinom” function. Be aware that it has different defaults (e.g., the base category is usually the first category instead of the last).

You try with the updated music artist and introversion data set [here](#). (This link will be updated so you can use this in class.)

Discussion questions:

1. How do we interpret each of the coefficients?
2. Is there a relationship?
3. How does this model compare to if you just fit two separate logistic models to the relevant subsets of data?
4. What about fitting other link functions?
5. Are you able to fit the model using a different base category? Do the inferences change?
6. Did you find any covariates that are related to music artist preference?



## 7.2 Ordinal Data

In the last section, we treated the response variable as nominal (or unordered categories). We could have rearranged the ordering of the response variables and not changed the essential meaning of the model.

In this section, we consider models which attempt to exploit the *order* of a categorical response variable. This is done by defining odds (or links) that capture the ordinal relationship of the categories.

Note that the likelihood is still the multinomial distribution:  $Y_i \overset{ind}{\sim} \text{Multinom}(1, \pi_{i1}, \pi_{i2}, \dots, \pi_{iJ})$ , where  $Y_i$  represents the category of the  $i$ th observation and  $\pi_{i1}, \dots, \pi_{iJ}$  the probabilities of categories  $1, \dots, J$  for the  $i$ th observation.

We consider three different types of odds for ordinal response variables which results in various logistic models:

### 1. Cumulative logit model:

*Cumulative odds for the  $j$ th category:*

$$\frac{P(Y_i \leq j)}{P(Y_i > j)} = \frac{\pi_{i1} + \dots + \pi_{ij}}{\pi_{ij+1} + \dots + \pi_{iJ}},$$

This represents the odds that the  $i$ th observation is in category  $j$  or less. In this section's example, our categorical response is severity of injury for a car crash. Thus, we would interpret these odds in context using, "The odds of the given severity level or a less severe injury for the  $i$ th observation." For a specific  $j$ , "The odds of being injured and taken to the hospital or a less severe injury." (And you can add, if it's helpful, "relative to a more severe injury...")

Notice how these odds *accumulate* the probabilities of the various categories as  $j$  increases:

$$\frac{\pi_{i1}}{\pi_{i2} + \dots + \pi_{iJ}}, \frac{\pi_{i1} + \pi_{i2}}{\pi_{i3} + \dots + \pi_{iJ}}, \frac{\pi_{i1} + \pi_{i2} + \pi_{i3}}{\pi_{i4} + \dots + \pi_{iJ}}, \dots, \frac{\pi_{i1} + \pi_{i(J-1)}}{\pi_{iJ}}$$

*Cumulative logit model:* Thus, the cumulative logit model uses the log of these odds for think link function:

$$\log \left( \frac{\pi_{i1} + \dots + \pi_{ij}}{\pi_{ij+1} + \dots + \pi_{iJ}} \right) = \mathbf{x}'_{ij} \beta_j = \beta_{0j} + \beta_{1j} x_{i1} + \dots + \beta_{kj} x_{ik}, \text{ for } j = 1, \dots, J-1.$$

(Just like the nominal model, there will be  $J - 1$  equations as part of the systematic component of the GLM.)

*Model fitting:*

- R: “vglm” function in the VGAM package (“family = cumulative”); see example below.
- SAS: “proc catmod; response clogits;”
- python: “statsmodels” library and “MNLogit” function

## 2. Adjacent categories logit model:

*Adjacent category odds:*

These odds use the ratios of single-category probabilities to its successive category probabilities:

$$\frac{\pi_{i1}}{\pi_{i2}}, \frac{\pi_{i2}}{\pi_{i3}}, \dots, \frac{\pi_{iJ-1}}{\pi_{iJ}}.$$

Thus, we interpret these odds as, “the odds of category  $j$  relative to category  $j + 1$ .” Again, let’s apply this to our severity of injury for a car crash example. Thus, we would say, “the odds of a car crash injury relative to the injury just one level more severe,” to keep it general, or “the odds of not being injured in a car crash relative to having a minor injury (but not taken to the hospital),” to describe  $\pi_{i1}/\pi_{i2}$ .

*Adjacent category logit model:*

$$\log \left( \frac{\pi_{ij}}{\pi_{i,j+1}} \right) = \mathbf{x}_{ij}' \beta_j = \beta_{0j} + \beta_{1j} x_{i1} + \dots + \beta_{kj} x_{ik}, \text{ for } j = 1, \dots, J - 1$$

*Connection to the baseline category logit model (multinomial logit model):*

Suppose we use the proportional odds assumption (discussed later in point number 4) so that our systematic component is simplified to,

$$\log \left( \frac{\pi_{ij}}{\pi_{i,j+1}} \right) = \beta_{0j} + \beta_1 x_{i1} + \beta_k x_{ik}, \text{ for } j = 1, \dots, J - 1.$$

In this case, the coefficients for all covariates are the same for all the odds considered. We’ll talk about interpretation later. However, mathematically, notice

then that if we want to consider the log odds from the nominal model using our adjacent categories model,

$$\begin{aligned}
\log\left(\frac{\pi_{ij}}{\pi_{iJ}}\right) &= \log\left(\frac{\pi_{ij}}{\pi_{ij+1}} \frac{\pi_{ij+1}}{\pi_{ij+2}} \dots \frac{\pi_{iJ-1}}{\pi_{iJ}}\right) \\
&= \log\left(\frac{\pi_{ij}}{\pi_{ij+1}}\right) + \log\left(\frac{\pi_{ij+1}}{\pi_{ij+2}}\right) + \dots + \log\left(\frac{\pi_{iJ-1}}{\pi_{iJ}}\right) \\
&= \sum_{\ell=j}^{J-1} (\beta_{0\ell} + \beta_1 x_{i1} + \dots + \beta_k x_{ik}) \\
&= \left(\sum \beta_{0\ell}\right) + \beta_1 (J-j)x_{i1} + \dots + \beta_k (J-j)x_{ik} \\
&= \beta_{0j}^* + \beta_{1j}^* x_{i1} + \dots + \beta_{kj}^* x_{ik},
\end{aligned}$$

where  $\beta_j^*$  are similar to the fitted coefficients from the nominal logistic regression model (baseline-category logit model). While there are additional estimation constraints in this adjacent categories proportional odds model, it does create a connection that can aid with interpretation or even model selection. In this version of the nominal model, the effect of  $x_1$  on category  $j$ ,  $\beta_{1j}^*$ , is  $(J-j)$  times the effect of  $x_1$  on category  $J$ ,  $(J-j)\beta_k$ .

*Model fitting:*

- R: “vglm” function using the “acats” option in VGAM package; (see example below)
- SAS: “proc catmod; response alogits;”
- python: “statsmodels” library and the “ACNLogit” function

### 3. Continuation ratio logit model:

*Continuation ratios:*

First, let

$$\omega_{ij} = P(Y_i = j | Y_i \geq j) = \frac{P(Y_i = j, Y_i \geq j)}{P(Y_i \geq j)} = \frac{\pi_{ij}}{\pi_{ij} + \pi_{ij+1} + \dots + \pi_{iJ}}.$$

Here,  $\omega_{ij}$  represents a conditional probability, the probability that the response of the  $i$ th observation is category  $j$  given that it is at least category  $j$ . A continuation ratio model will model the odds of these *conditional* probabilities.

*Continuation ratio logit model:*

We then model the logit of  $\omega_j$ , so that the systematic component is,

$$\begin{aligned}\text{logit}(\omega_{ij}) &= \log\left(\frac{\omega_{ij}}{1 - \omega_{ij}}\right) = \log\left(\frac{\frac{\pi_{ij}}{\pi_{ij} + \pi_{i,j+1} + \dots + \pi_{iJ}}}{\frac{\pi_{i,j+1} + \dots + \pi_{iJ}}{\pi_{ij} + \pi_{i,j+1} + \dots + \pi_{iJ}}}\right) \\ &= \log\left(\frac{\pi_{ij}}{\pi_{i,j+1} + \dots + \pi_{iJ}}\right) = \mathbf{x}'_{ij}\beta_j, \text{ for } j = 1, \dots, J - 1.\end{aligned}$$

Notice that we can interpret these odds two different ways:

- The odds of response  $j$  given the response is at least category  $j$ , or,
- The odds of response  $j$  relative to a response larger than  $j$ .

The latter interpretation corresponds to the “simplified ratios.”

*The simplified probability ratios are:*

$$\frac{\pi_{i1}}{\pi_{i2} + \dots + \pi_{iJ}}, \frac{\pi_{i2}}{\pi_{i3} + \dots + \pi_{iJ}}, \dots, \frac{\pi_{i,J-1}}{\pi_{iJ}}$$

*Other continuation ratios:*

Consider instead the conditional probability:  $\omega_{ij} = P(Y_i < j | Y_i \leq j)$ . In words, the probability of the  $i$ th response being less than category  $j$  given that it is at least category  $j$  or lower.

In this case, the simplified probability ratios are,

$$\frac{\pi_{i1}}{\pi_{i2}}, \frac{\pi_{i1} + \pi_{i2}}{\pi_{i3}}, \dots, \frac{\pi_{i1} + \pi_{i2} + \dots + \pi_{i,J-1}}{\pi_{iJ}}.$$

*Model fitting:*

- R: “vglm” function with “family=sratio(reverse=F)” (for first ratios) or “family=cratio(reverse=T)” (for the second ratios). Note that you could use “reverse=T” or “reverse=F” but just be careful which odds or ratios these are.
- SAS: “proc catmod;” and change probabilities in model statement
- python: “statsmodels” library and the “CMNLogit” function

#### 4. Proportional odds assumption:

The proportional odds assumption is to require equivalent covariate coefficients,  $\beta$ 's, for each  $j$ , but unique intercepts. In other words,

$$\mathbf{x}'_{ij}\beta_j = \beta_{0j} + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}, \text{ for } j = 1, \dots, J - 1,$$

→ different intercept for model  $j$ , but the same covariate relationships across categories for each covariate.

*Proportional odds model:*

The proportional odds assumption can be applied to any of the above logistic models; however, the “proportional odds model” usually refers to the proportional odds assumption being applied to the *cumulative logit model*. In this case, again, we assume equivalent covariate coefficients  $\beta$ 's for each  $j$ , but unique intercepts, e.g.,

$$\log \left( \frac{\pi_{i1} + \cdots + \pi_{ij}}{\pi_{i,j+1} + \cdots + \pi_{iJ}} \right) = \beta_{0j} + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}, \text{ for } j = 1, \dots, J - 1.$$

Let's look at a simple case (a single continuous covariate) to examine where the term “proportional odds” comes from. Consider  $J = 3$  and a systematic component of,

$$\begin{aligned} \log \left( \frac{\pi_{i1}}{\pi_{i2} + \pi_{i3}} \right) &= \beta_{01} + \beta_1 x_{i1}, \\ \log \left( \frac{\pi_{i1} + \pi_{i2}}{\pi_{i3}} \right) &= \beta_{02} + \beta_1 x_{i1}. \end{aligned}$$

The odds for a given value of  $x_{i1}$  are equal to,

$$\begin{aligned} \frac{\pi_{i1}}{\pi_{i2} + \pi_{i3}} &= e^{\beta_{01}} e^{\beta_1 x_{i1}}, \\ &\text{and} \\ \frac{\pi_{i1} + \pi_{i2}}{\pi_{i3}} &= e^{\beta_{02}} e^{\beta_1 x_{i1}}. \end{aligned}$$

Notice that the odds of category 1 relative to categories 2 and 3 are  $e^{\beta_{01}}/e^{\beta_{02}}$  times the odds for categories 1 or 2 relative to category 3, *for all* values of  $x_{i1}$ . Thus, the odds are proportional to each other and are the same proportion for every value of  $x_{i1}$ .

Let's see what this looks like in a plot of the log odds, odds, and probabilities.

```

beta1 <- c(-0.5, .5)
beta2 <- c(0, 0.5)
n <- 500
x <- seq(-1, 1, length=n)
X <- cbind(rep(1, n), x)
logodds1 <- X%*%beta1
logodds2 <- X%*%beta2
odds1 <- exp(logodds1)
odds2 <- exp(logodds2)
prob3 <- (1+odds1)/(odds1*odds2 + odds2 + odds1 + 1)
prob2 <- prob3*(odds2-odds1)/(1+odds1)
prob1 <- odds1*(prob2 + prob3)

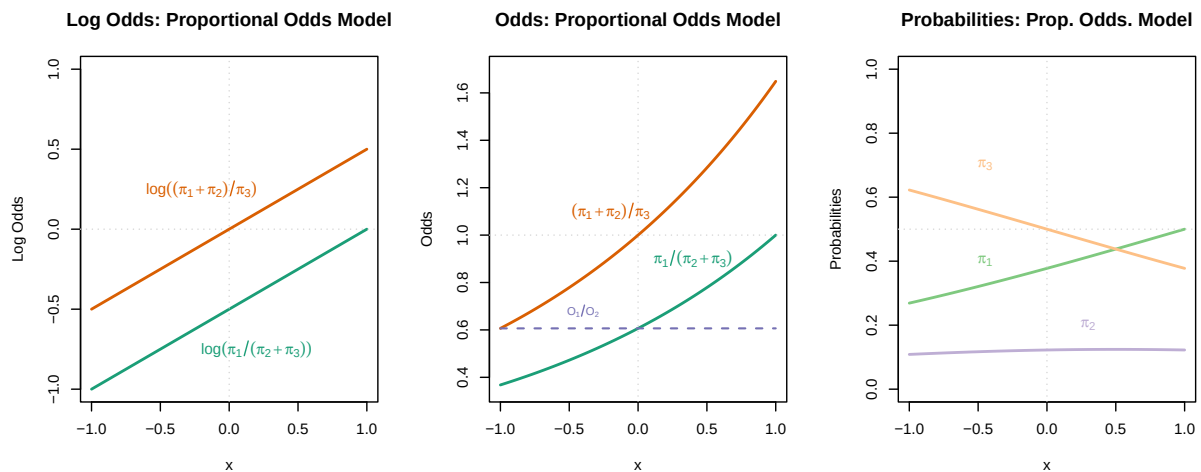
library(RColorBrewer)
mypalo <- brewer.pal(3, 'Dark2')
mypalp <- brewer.pal(3, "Accent")
par(mfrow=c(1,3))

#Log odds
plot(x, logodds1, type='l', col=mypalo[1], lwd=2, xlab="x", ylab="Log Odds", ylim=c(-1, 1))
abline(v=0, lty=3, col='lightgray')
abline(h=0, lty=3, col='lightgray')
lines(x, logodds2, col=mypalo[2], lwd=2)
text(0.2, -0.75, labels=expression(log(pi[1]/(pi[2]+pi[3]))), col=mypalo[1])
text(-0.2, 0.25, labels=expression(log((pi[1]+pi[2])/pi[3])), col=mypalo[2])

#Odds
plot(x, odds1, type='l', col=mypalo[1], lwd=2, xlab="x", ylab="Odds", ylim=c(0.1, 10))
abline(v=0, lty=3, col='lightgray')
abline(h=1, lty=3, col='lightgray')
lines(x, odds2, col=mypalo[2], lwd=2)
lines(x, odds1/odds2, col=mypalo[3], lwd=1.5, lty=2)
text(0.4, 0.9, labels=expression(pi[1]/(pi[2]+pi[3])), col=mypalo[1])
text(-0.2, 1.1, labels=expression((pi[1]+pi[2])/pi[3]), col=mypalo[2])
text(-0.4, 0.68, labels=expression(O[1]/O[2]), col=mypalo[3], cex=.7)

```

```
#Probabilities
plot(x, prob1, type='l', col=mypalp[1], lwd=2, xlab="x", ylab="Probabilities",
abline(v=0, lty=3, col='lightgray')
abline(h=0.5, lty=3, col='lightgray')
lines(x, prob2, col=mypalp[2], lwd=2)
lines(x, prob3, col=mypalp[3], lwd=2)
text(-0.45, 0.4, labels=expression(pi[1]), col=mypalp[1])
text(0.3, 0.2, labels=expression(pi[2]), col=mypalp[2])
text(-0.45, 0.7, labels=expression(pi[3]), col=mypalp[3])
```



*Impact of combining categories on parameter interpretation:*

- What happens to our estimates of  $\beta_1, \dots, \beta_k$  if we combine categories  $j$  and  $j+1$ ? Again, let's start with 3 categories and assume we combine categories 2 and 3 to be a new category 2\*. So the systematic component when we have the three categories is:

$$\log\left(\frac{\pi_{i1}}{\pi_{i2} + \pi_{i3}}\right) = \beta_{01} + \beta_1 x_{i1},$$

$$\log\left(\frac{\pi_{i1} + \pi_{i2}}{\pi_{i3}}\right) = \beta_{02} + \beta_1 x_{i1}.$$

When we combine categories 2 and 3 to make a new category  $2^*$ , where  $\pi_{i2^*} = \pi_{i2} + \pi_{i3}$ , then the systematic component becomes:

$$\log \left( \frac{\pi_{i1}}{\pi_{i2^*}} \right) = \beta_{01}^* + \beta_1 x_{i1}.$$

Since  $\beta_1$  was the same for all log odds, it remains the same for the combined log odds. (Obviously, the estimate for  $\beta_{0j}^*$ 's may change depending on the original number of categories and which categories were combined.) This property is called the **collapsibility property**. Note that in practice estimates can be slightly different (the difference in estimates often happens because of zeros for some of the covariate values).

### *Model Fitting:*

The cumulative logit proportional odds model is the “typical” model for ordinal logistic regression (specifically, in software).

- R: cumulative logit proportional odds model: “polr” function in “MASS” library, or “vglm” function with “family = cumulative(parallel=T)”
- R: proportional odds assumption for any of the logit models: “vglm” function with “parallel=T” command
- SAS: default of “proc logistic;” or “proc genmod; dist=mult and link=cum-logit;”
- python: default for the “MNLogit” function

## 7.2.1 Example Implementing and Comparing the Various Ordinal Logit Models

### Crash Injury Severity Example

Table 7.3: Injury level of 68,694 passengers in autos and light trucks involved accidents in the state of Maine in 1991. Agresti (2013)

| Gender | Location | Seat Belt | 1     | 2   | 3   | 4  | 5  |
|--------|----------|-----------|-------|-----|-----|----|----|
| Female | Urban    | No        | 7287  | 175 | 720 | 91 | 10 |
|        |          | Yes       | 11587 | 126 | 577 | 48 | 8  |



|      |       |     |       |     |     |     |    |
|------|-------|-----|-------|-----|-----|-----|----|
| Male | Rural | No  | 3246  | 73  | 710 | 159 | 31 |
|      |       | Yes | 6134  | 94  | 564 | 82  | 17 |
|      | Urban | No  | 10381 | 136 | 566 | 96  | 14 |
|      |       | Yes | 10969 | 83  | 259 | 37  | 1  |
|      | Rural | No  | 6123  | 141 | 710 | 188 | 45 |
|      |       | Yes | 6693  | 74  | 353 | 74  | 12 |

Table 7.3 summarizes observations of 68,694 passengers in autos and light trucks involved accidents in the state of Maine in 1991. The response categories are (1) not injured, (2) injured but not transported by emergency medical services, (3) injured and transported by emergency medical services but not hospitalized, (4) injured and hospitalized but did not die, and (5) injured and died.

```
response <- rep(1:5, each=8)
seatbelt <- factor(rep(c('no', 'yes'), 20), levels=c("yes", "no"))
location <- factor(rep(c('urban', 'urban', 'rural', 'rural'), 10), levels=c("u", "r"))
gender <- factor(rep(rep(c('female', 'male'), each=4), 5), levels=c("male", "female"))
counts <- c(7287, 11587, 3246, 6134, 10381, 10969, 6123, 6693, 175, 126, 73, 94, 136, 83, 141, 74, 720, 577, 710, 564, 566, 259, 710, 353, 91, 48, 159, 82, 96, 37, 188, 74, 10, 8, 31, 17, 14, 1, 45, 12)
```

## 1. Cumulative Logit Model

```
cummod1 <- vglm(response~factor(seatbelt), cumulative, weights=counts)
summary(cummod1)
```

Call:

```
vglm(formula = response ~ factor(seatbelt), family = cumulative,
     weights = counts)
```

Coefficients:

|                      | Estimate | Std. Error | z value | Pr(> z )    |
|----------------------|----------|------------|---------|-------------|
| (Intercept):1        | 2.68702  | 0.02106    | 127.611 | < 2e-16 *** |
| (Intercept):2        | 2.86781  | 0.02281    | 125.751 | < 2e-16 *** |
| (Intercept):3        | 4.90123  | 0.06009    | 81.566  | < 2e-16 *** |
| (Intercept):4        | 6.90126  | 0.16229    | 42.524  | < 2e-16 *** |
| factor(seatbelt)no:1 | -0.74178 | 0.02719    | -27.285 | < 2e-16 *** |
| factor(seatbelt)no:2 | -0.75734 | 0.02925    | -25.889 | < 2e-16 *** |

```

factor(seatbelt)no:3 -1.03543      0.07226 -14.330 < 2e-16 ***
factor(seatbelt)no:4 -1.17110      0.19071  -6.141 8.22e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: logitlink(P[Y<=1]), logitlink(P[Y<=2]),
logitlink(P[Y<=3]), logitlink(P[Y<=4])

Residual deviance: 52027.37 on 152 degrees of freedom

Log-likelihood: -26013.69 on 152 degrees of freedom

Number of Fisher scoring iterations: 4

Exponentiated coefficients:
factor(seatbelt)no:1 factor(seatbelt)no:2 factor(seatbelt)no:3
              0.4762677              0.4689110              0.3550724
factor(seatbelt)no:4
              0.3100270

```

*Fitted Model:*

$$\begin{aligned}
\log \left( \frac{\hat{\pi}_{i1}}{\hat{\pi}_{i2} + \dots + \hat{\pi}_{i5}} \right) &= 2.69 - 0.74I(\text{seatbelt}_i = \text{no}) \\
\log \left( \frac{\hat{\pi}_{i1} + \hat{\pi}_{i2}}{\hat{\pi}_{i3} + \dots + \hat{\pi}_{i5}} \right) &= 2.87 - 0.76I(\text{seatbelt}_i = \text{no}) \\
&\vdots \\
\log \left( \frac{\hat{\pi}_{i1} + \dots + \hat{\pi}_{i4}}{\hat{\pi}_{i5}} \right) &= 6.90 - 1.17I(\text{seatbelt}_i = \text{no})
\end{aligned}$$

*Interpret  $\hat{\beta}_{11} = -0.742$ :* the log odds of the passenger experienced an injury of 1 (or less; i.e., not injured or less) decreases by 0.742 when the passenger is not wearing a seatbelt.

## 2. (Cumulative Logit) Proportional Odds Model

```
propmod1 <- vglm(response~factor(seatbelt), cumulative(parallel=T),
  weights=counts)
summary(propmod1)
```

Call:

```
vglm(formula = response ~ factor(seatbelt), family = cumulative(parallel = T),
  weights = counts)
```

Coefficients:

|                    | Estimate | Std. Error | z value | Pr(> z )   |
|--------------------|----------|------------|---------|------------|
| (Intercept):1      | 2.68890  | 0.02107    | 127.63  | <2e-16 *** |
| (Intercept):2      | 2.86018  | 0.02174    | 131.58  | <2e-16 *** |
| (Intercept):3      | 4.70948  | 0.03730    | 126.27  | <2e-16 *** |
| (Intercept):4      | 6.61240  | 0.08685    | 76.13   | <2e-16 *** |
| factor(seatbelt)no | -0.74612 | 0.02716    | -27.47  | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: logitlink(P[Y<=1]), logitlink(P[Y<=2]),  
logitlink(P[Y<=3]), logitlink(P[Y<=4])

Residual deviance: 52047.97 on 155 degrees of freedom

Log-likelihood: -26023.99 on 155 degrees of freedom

Number of Fisher scoring iterations: 4

Exponentiated coefficients:

```
factor(seatbelt)no
      0.4742033
```

*Fitted Model:*

$$\begin{aligned}\log\left(\frac{\hat{\pi}_{i1}}{\hat{\pi}_{i2} + \dots + \hat{\pi}_{i5}}\right) &= 2.69 - 0.75I(\text{seatbelt}_i = \text{no}) \\ \log\left(\frac{\hat{\pi}_{i1} + \hat{\pi}_{i2}}{\hat{\pi}_{i3} + \dots + \hat{\pi}_{i5}}\right) &= 2.86 - 0.75I(\text{seatbelt}_i = \text{no}) \\ &\vdots \\ \log\left(\frac{\hat{\pi}_{i1} + \dots + \hat{\pi}_{i4}}{\hat{\pi}_{i5}}\right) &= 6.61 - 0.75I(\text{seatbelt}_i = \text{no})\end{aligned}$$

Interpret  $\hat{\beta}_1 = -0.746$ : the log odds that a passenger experienced a given level of injury or less, for any given injury level, decreases by 0.746 when the passenger is not wearing a seatbelt.

*Test the proportional odds assumption:*

Since the proportional odds model and the cumulative logit models are nested, we can conduct a test of deviance. Note the specific hypothesis test for this is slightly different than our typical “full vs. reduced” model comparison:

$$\begin{aligned}H_0 : \beta_{B1} &= \beta_{B2} = \beta_{B3} = \beta_{B4} \\ H_1 : &\text{At least one } \beta_{Bj} \neq \beta_{Bk} \text{ for } j \neq k.\end{aligned}$$

Using our test of deviance, we compute our p-value to be:

```
deltaD = deviance(propmod1) - deviance(cummod1)
deltaDF = propmod1@df.residual - cummod1@df.residual
1-pchisq(deltaD, deltaDF)
```

```
[1] 0.0001274816
```

And using a likelihood ratio test (which also requires nested models):

```
lrtest(cummod1, propmod1)
```

Likelihood ratio test

```
Model 1: response ~ factor(seatbelt)
```

```
Model 2: response ~ factor(seatbelt)
```

```
  #Df LogLik Df Chisq Pr(>Chisq)
1 152 -26014
2 155 -26024   3   20.6   0.0001275 ***
```

---

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Thus, we reject  $H_0$  that each equation in the systematic component has equivalent coefficient values, and conclude that there is evidence to suggest the coefficients are different for at least one of the equations in the systematic component. In other words, based only on these likelihood/deviance tests, we would choose the Cumulative Logit Model.

*Residuals:*

Take a minute to think about residuals here. What do they represent?

$$\mathbf{Y}_i \stackrel{ind}{\sim} \text{Multinom}(m_i, \pi_{i1}, \dots, \pi_{i5}),$$

where under this specification of the model,  $\mathbf{Y}_i$  is a  $5 \times 1$  vector of counts for each response category corresponding to the  $i$ th conditions, and  $m_i$  corresponds to the number of people under conditions  $i$  (in this case where we only have a single 2-level factor variable, it is either the individuals that were wearing a seatbelt or that were not). Our predicted values are  $\pi_i = (\hat{\pi}_{i1}, \dots, \hat{\pi}_{i5})'$  and our predicted counts are  $\hat{\mathbf{Y}} = (m_i \hat{\pi}_{i1}, \dots, m_i \hat{\pi}_{i5})'$ . Our Pearson residuals are then either:

$$\frac{(\mathbf{Y}_i - \hat{\mathbf{Y}}_i)}{\sqrt{\hat{\mathbf{Y}}_i(1 - \hat{\pi}_i)}},$$

or, if we divide all quantities by  $m_i$ ,

$$\frac{(\mathbf{Y}_i/m_i) - \hat{\pi}_i}{\sqrt{\hat{\pi}_i(1 - \hat{\pi}_i)}}.$$

Because we have 5 categories, many of our estimated probabilities will tend to be quite small. For example, these two models estimate fitted values as

small as 0.001. Thus, we will often be dividing by *very* small numbers. These residuals will be even *more* sensitive to small counts and small probabilities than the binomial models were. “Surrogate” and other residuals have been proposed to accommodate these types of models. If you’re interested in more model diagnostics for multinomial regression models, [this is a nice and recent overview paper](#).

### 3. Adjacent Categories Model with Proportional Odds Assumption

```
acatmod2 <- vglm(response~factor(seatbelt), acat(parallel=T,
reverse=T), weights=counts)
summary(acatmod2)
```

Call:

```
vglm(formula = response ~ factor(seatbelt), family = acat(parallel = T,
reverse = T), weights = counts)
```

Coefficients:

|                    | Estimate | Std. Error | z value | Pr(> z )   |
|--------------------|----------|------------|---------|------------|
| (Intercept):1      | 4.40779  | 0.03419    | 128.91  | <2e-16 *** |
| (Intercept):2      | -1.39563 | 0.03759    | -37.13  | <2e-16 *** |
| (Intercept):3      | 1.98248  | 0.04044    | 49.02   | <2e-16 *** |
| (Intercept):4      | 1.98544  | 0.09325    | 21.29   | <2e-16 *** |
| factor(seatbelt)no | -0.35724 | 0.01308    | -27.31  | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: loglink(P[Y=1]/P[Y=2]), loglink(P[Y=2]/P[Y=3]), loglink(P[Y=3]/P[Y=4]), loglink(P[Y=4]/P[Y=5])

Residual deviance: 52041.52 on 155 degrees of freedom

Log-likelihood: -26020.76 on 155 degrees of freedom

Number of Fisher scoring iterations: 7

Exponentiated coefficients:

```
factor(seatbelt)no  
0.6996022
```

*Fitted Model:*

$$\begin{aligned}\log\left(\frac{\hat{\pi}_{i1}}{\hat{\pi}_{i2}}\right) &= 4.41 - 0.36I(\text{seatbelt}_i = \text{no}) \\ \log\left(\frac{\hat{\pi}_{i2}}{\hat{\pi}_{i3}}\right) &= -1.40 - 0.36I(\text{seatbelt}_i = \text{no}) \\ &\vdots \\ \log\left(\frac{\hat{\pi}_{i4}}{\hat{\pi}_{i5}}\right) &= 1.99 - 0.36I(\text{seatbelt}_i = \text{no})\end{aligned}$$

*Interpretation of  $\hat{\beta}_1 = -0.357$ :* Not wearing a seatbelt decreases the log odds of category  $j$  to category  $j + 1$  by 0.357. Since the proportional odds assumption holds, the log odds of not injured compared to death decreases by  $\approx 4 \times 0.357 = 1.428$ . (This latter interpretation came from connecting the adjacent categories model to the multinomial model that we discussed above.)

#### 4. Continuation Ratio Model

```
contmod1 <- vglm(response~factor(seatbelt), sratio, weights=counts)  
summary(contmod1)
```

Call:

```
vglm(formula = response ~ factor(seatbelt), family = sratio,  
      weights = counts)
```



Coefficients:

|                      | Estimate | Std. Error | z value | Pr(> z ) |     |
|----------------------|----------|------------|---------|----------|-----|
| (Intercept):1        | 2.68702  | 0.02106    | 127.611 | < 2e-16  | *** |
| (Intercept):2        | -1.68453 | 0.05608    | -30.040 | < 2e-16  | *** |
| (Intercept):3        | 1.83787  | 0.06446    | 28.513  | < 2e-16  | *** |
| (Intercept):4        | 1.84721  | 0.17446    | 10.588  | < 2e-16  | *** |
| factor(seatbelt)no:1 | -0.74178 | 0.02719    | -27.286 | < 2e-16  | *** |
| factor(seatbelt)no:2 | -0.16580 | 0.07314    | -2.267  | 0.0234   | *   |
| factor(seatbelt)no:3 | -0.38669 | 0.07811    | -4.951  | 7.39e-07 | *** |
| factor(seatbelt)no:4 | -0.17198 | 0.20569    | -0.836  | 0.4031   |     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: logitlink(P[Y=1|Y>=1]), logitlink(P[Y=2|Y>=2]), logitlink(P[Y=3|Y>=3]), logitlink(P[Y=4|Y>=4])

Residual deviance: 52027.37 on 152 degrees of freedom

Log-likelihood: -26013.69 on 152 degrees of freedom

Number of Fisher scoring iterations: 6

Exponentiated coefficients:

|                      |                      |                      |
|----------------------|----------------------|----------------------|
| factor(seatbelt)no:1 | factor(seatbelt)no:2 | factor(seatbelt)no:3 |
| 0.4762677            | 0.8472180            | 0.6792987            |
| factor(seatbelt)no:4 |                      |                      |
| 0.8419923            |                      |                      |

*Fitted Model:*

$$\begin{aligned}
\log \left( \frac{\hat{\pi}_{i1}}{\hat{\pi}_{i2} + \dots + \hat{\pi}_{i5}} \right) &= 2.69 - 0.74I(\text{seatbelt}_i = \text{no}) \\
\log \left( \frac{\hat{\pi}_{i2}}{\hat{\pi}_{i3} + \dots + \hat{\pi}_{i5}} \right) &= -1.68 - 0.17I(\text{seatbelt}_i = \text{no}) \\
&\vdots \\
\log \left( \frac{\hat{\pi}_{i4}}{\hat{\pi}_{i5}} \right) &= 1.85 - 0.17I(\text{seatbelt}_i = \text{no})
\end{aligned}$$

*Interpret  $\beta_{14} = -0.172$ :* The log odds of injured and hospitalized given that the passenger was injured and hospitalized and either died or didn't die decreases by 0.172 when the passenger was not wearing a seatbelt.

## 5. Compare log likelihood of various models

We provide the log-likelihood estimates for various links and covariate combinations in the following table. Only models with the same links can be tested (and only if their covariates are nested, and in this case they are).

| Model                              | Covariates | Log-Likelihood |
|------------------------------------|------------|----------------|
| Cumulative Logit                   | B,L,G      | -25371.83      |
| Cumulative Logit with Prop Odds    | B,L,G      | -25439.74      |
| Adjacent Categories                | B,L,G      | -25372.97      |
| Adjacent Categories with Prop Odds | B,L,G      | -25416.33      |
| Continuation Ratio                 | B,L,G      | -25374.04      |
| Cumulative Logit                   | B*L,G      | -25369.40      |
| Cumulative Logit with Prop Odds    | B*L,G      | -25437.16      |
| Adjacent Categories                | B*L,G      | -25371.16      |
| Adjacent Categories with Prop Odds | B*L,G      | -25415.27      |
| Continuation Ratio                 | B*L,G      | -25371.22      |

Since the cumulative logit model has the smallest log-likelihood, let's do a hypothesis test on whether the interaction between seatbelt and location should be included in the model. *Remember to be careful about the different number of parameters being fitted between the two models.*

$$H_0 : \beta_{BL1} = \beta_{BL2} = \beta_{BL3} = \beta_{BL4} = 0$$

$$H_1 : \text{At least one } \beta_{BLj} \neq 0$$

$-2(-25371.83 + 25369.40) = 4.86$  and we compare this value to a  $\chi^2(4)$ . Thus, the p-value for this test is 0.302, indicating that we fail to reject the null hypothesis and conclude we should not include an interaction between seatbelt and location in our model.

Note that other metrics we can consider for model selection and diagnostics include: cross-validation mse/mae, multi-class ROC curves, sensitivity and specificity (either grouped or for each individual category), plots of predicted values for interesting covariate combinations or ranges, comparison to naive predictions (e.g., each category has same probability), etc.

## 7.3 Some Articles Using Multinomial Models

- A former MS student emailed me super excited by how often he uses multinomial models in his work. He sent [a paper](#) with some comments for me to share with you! See Goligher et al. (2023).
  - He states, “one more thing about the outcome or response variable in the PO model...it is common in medicine to ‘combine’ mortality and another clinically meaningful response measure. In this case, we combine death and organ support free days (OSFD) 0-22. Combining responses with different interpretations obviously warrants a lot of additional scrutiny from the larger community and doesn’t always ‘work’, but may be an interesting point for students to know that medical applications commonly mix and meld related but not equivalent measures into a single ordinal scale.”
  - He adds, “Frank Harrell at Vanderbilt (sort of an Andrew Gelman-like figure for Bayesian biostatistics) has discussed connections between proportional odds and time-to-event (analyzed with a non-parametric Wilcoxon rank-sum test, not Cox PH). I can’t remember if your students will have been exposed to survival models from other classes or not, but this seems to be a really relevant, and interesting(!), topic for students interested in biostatistics to be aware of. Frank Harrell’s blog post can be found at [this link](#).”
- A paper that reports the different coefficients for a multinomial model is [this one](#) regarding children hospitalizations due to asthma and the relationship with various demographics. See Grineski et al. (2010).

## 7.4 Homework Assignment

1. In class we fit a multinomial model examining the relationship between words written per minute of an exam (if they wrote at least 25 words per minute or not) and the course the exam was in. Instead of treating the explanatory variable as a binary variable, we fit the model using continuous number of words per minute (range: 0.08 and 46.49 words per minute). The fitted model is as follows, where the categories for the response values (courses) are as defined in the notes.

```
modvc <- vglm(Course~exam.word.permin, family="multinomial", data=WordDat2)
summary(modvc)
```

Call:

```
vglm(formula = Course ~ exam.word.permin, family = "multinomial",
     data = WordDat2)
```

Coefficients:

|                    | Estimate  | Std. Error | z value | Pr(> z )   |
|--------------------|-----------|------------|---------|------------|
| (Intercept):1      | -2.940476 | 0.169339   | -17.36  | <2e-16 *** |
| (Intercept):2      | -1.589217 | 0.147286   | -10.79  | <2e-16 *** |
| exam.word.permin:1 | 0.201776  | 0.010538   | 19.15   | <2e-16 *** |
| exam.word.permin:2 | 0.115571  | 0.009902   | 11.67   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,3]), log(mu[,2]/mu[,3])

Residual deviance: 4156.51 on 4270 degrees of freedom

Log-likelihood: -2078.255 on 4270 degrees of freedom

Number of Fisher scoring iterations: 5

Reference group is level 3 of the response

- a. Provide interpretations for  $\beta_{11}$  and  $\beta_{12}$  the coefficient for exam words per minute for categories 1 and 2, respectively.
  - b. Find the fitted model for the equation  $\log(\hat{\pi}_{i1}/\hat{\pi}_{i2})$  for a given value of  $x_i$ , where  $x_i$  is exam words per minute.
  - c. Without calculating estimated probabilities, explain why the intercept estimates indicate that for someone who writes 0 words per minute on all exams (yes, this is extrapolation!),  $\hat{\pi}_1 < \hat{\pi}_2 < \hat{\pi}_3$ . Around what value of exam words per minute does that relationship change?
  - d. Find and interpret a 95% confidence interval for a conditional odds ratio that can be estimated from this model. (There are several, I just want one.)
2. For data from a General Social Survey, the cumulative logit proportional odds model with  $Y$  = political ideology (very liberal, slightly liberal, moderate, slightly conservative, very conservative) and  $x = 1$  for the 428 Democrats and  $x = 0$  for the 407 Republicans has  $\hat{\beta}_1 = 0.975$  ( $SE = 0.129$ ) and  $\hat{\beta}_{01} = -2.469$ . Interpret  $\hat{\beta}_1$ . Find the estimated probability of a very liberal response for each party.
  3. Refer to the previous problem. With adjacent-category logits,  $\hat{\beta}_1 = 0.435$ . Interpret using odds ratios for adjacent categories and for the (very liberal, very conservative) pair of categories.

Table 7.4: Income, gender, and job satisfaction counts (from Agresti (2013))

| Gender | Income (\$) | 1 | 2 | 3  | 4 |
|--------|-------------|---|---|----|---|
| Female | < 5000      | 1 | 3 | 11 | 2 |
|        | 5000-15000  | 2 | 3 | 17 | 3 |
|        | 15000-25000 | 0 | 1 | 8  | 5 |
|        | > 25000     | 0 | 2 | 4  | 2 |
| Male   | < 5000      | 1 | 1 | 2  | 1 |
|        | 5000-15000  | 0 | 3 | 5  | 1 |
|        | 15000-25000 | 0 | 0 | 7  | 3 |
|        | > 25000     | 0 | 1 | 9  | 6 |

4. Refer to the data in Table 7.4.

- a) Fit a cumulative logit model with the proportional odds assumption to the data in Table 7.4 using gender and income to predict job satisfaction (1 represents ‘Dissatisfied,’ 2 represents ‘A Little Satisfied,’ 3 represents ‘Moderately Satisfied,’ and 4 represents ‘Very Satisfied’).
- b) Compare the estimated income effect ( $\hat{\beta}_I$ ) to the estimate after collapsing the response to three categories by combining categories (i) very satisfied and moderately satisfied, and (ii) very dissatisfied and a little satisfied. What property of the model does this reflect? (Don’t be too rigid here; this won’t be a perfect illustration of this property – i.e., the numbers won’t be exactly the same – but it will be close.)
- c) Consider  $\hat{\beta}_I/SE$  for the collapsing in part (a(i)). Usually, a disadvantage of collapsing multinomial responses is that the significance of effects diminishes. Is this a problem in this case? (Explain.)
- d) Check whether an improved model results from permitting interaction between income and gender. Interpret the interaction coefficient.

5. When  $J = 3$ , suppose that

$$\pi_j(x) = \frac{\exp\{\beta_{0j} + \beta_{1j}x\}}{1 + \exp\{\beta_{01} + \beta_{11}x\} + \exp\{\beta_{02} + \beta_{12}x\}},$$

for  $j = 1, 2$ . Show that  $\pi_j(x)$  is

- a) decreasing in  $x$  if  $\beta_{11} > 0$  and  $\beta_{12} > 0$ ,
- b) increasing in  $x$  if  $\beta_{11} < 0$  and  $\beta_{12} < 0$ , and
- c) nonmonotone when  $\beta_{11}$  and  $\beta_{12}$  have different signs.

## 7.5 Mini Project 3

### *The Casella-Berger Oscars*

#### Goal:

Find the best predicting model for the Amazon ratings of the Casella and Berger book, *Statistical Inference*.

#### Details:

1. Use the Rdata object obtained through <https://byu.box.com/v/CasBergTrain> which contains a data frame called “train” with 105 observations of the following variables:
  - a) stars (rating: 1, 2, 3, 4, or 5),
  - b) reviewer (reviewer’s name),
  - c) date (date of the review),
  - d) review.text (text of the review),
  - e) paperback (logical indicating whether the format of the book they are reviewing is paperback)
  - f) hardcover (logical indicating whether the format of the book they are reviewing is hardcover)
  - g) verified (logical indicating whether the reviewer is an Amazon verified purchaser)
  - h) helpful (number of people that voted the review as helpful; NA when it was not voted helpful by anyone)
2. I kept 45 (randomly selected) observations out of the original data set. These observations will be the evaluative data set. Although you could go to Amazon and possibly identify these 45 observations to improve your predictive model, that would be considered cheating and we would all sing the “party pooper” song to you.
3. You must use an ordinal (or nominal) regression model.
4. You can use any function/transformation of the covariates as predictors – splines, review text polarity, number of words in the review, etc.



5. **Your code must be able to run and return the MSE on my computer *within* 60 seconds.**
6. On the due date, each person will present their model (no presentation, just write the model on the board) and I will give out the awards.

### Awards:

There will be three awards. Awards 2 & 3 will be given by class vote.

1. Smallest out-of-sample MSE.
2. Most useful inferential model (for example, Amazon may find it useful for marketing textbooks or Casella and Berger may find it useful for a revision of their book).
3. Most creative model.

### What to turn in:

1. An R source code file named 'FirstNameCasellaBerger.R' (inserting your own first name for 'FirstName') that contains a function called 'casellaberger' that takes the 'test' and 'train' data frames (in that order) and returns the MSE for the test data:  $MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$ . In other words, I will already have the test and training data loaded. Then I will run the following lines of code:

```
> source(FirstNameCasellaBerger.R)
```

```
> casellaberger(test, train)
```

and it will return just the MSE for the test data. If you use any R packages, please specify that in your email so I know which packages I need to install.

Note: Your predictions for each reviewer must be a value of 1, 2, 3, 4, or 5.

Send this to me by *7:30am on the due date, at the latest*.

2. A 1-2 page report containing your full model and model diagnostics from your training data (for example, training error rate or 5-fold cross validation error rates). If possible, provide some intuition behind the interpretation of your model. If no interpretation is possible, discuss why the complex model might improve prediction.

Turn this in on *Learning Suite on the due date*.

# 8 Special Cases of the GLM

## 8.1 GLM's for Dependent Data

### *Motivating Data*

- Respiratory Data:
  - 111 patients with respiratory illness (2 different clinics)
  - randomly assigned placebo or active treatment
  - response variable: respiratory status (1 = good, 0 = poor)
  - 5 observations:
    - \* pre-treatment (baseline)
    - \* 4 visits during treatment
  - explanatory variables: center (clinic), id, age, treatment (active/placebo), gender, baseline (use as an explanatory variable)

Why can't we treat the 4 visit responses as  $\text{Bin}(4, \pi_i)$ ?

### **Repeated measures data:**

- Longitudinal data: repeated measurements over time on the same subjects
- Clustered data: repeated measurements from the same sampling unit/cluster (but on different individuals within that cluster). E.g., blood alcohol levels in LDS and Catholic church members, etc. will be different from each other. Note: This clustering also motivated our overdispersion, but now we're observing the individual data (not aggregated data).

→ Due to correlation, we want to drop our assumption of independence among the  $Y_i$

→ Another approach is to use multilevel modeling or random effects models (talk

about later...)

## 8.1.1 Generalized Estimating Equations

### 8.1.1.1 Normally-Distributed Data

How to handle repeated measures in a linear model for normally distributed data?

Let

$\mathbf{y}_i = (y_{i1}, \dots, y_{in_i})' \Rightarrow n_i$  observations on the  $i$ th individual...usually  $n_i = n \forall i$

$\mathbf{y} = (\mathbf{y}'_1, \dots, \mathbf{y}'_N)' \Rightarrow N$  individuals

Total number of observations:  $\sum_{i=1}^N n_i = n$

Maximum Likelihood Estimates for  $\beta$ :

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\beta, \mathbf{V}) \Rightarrow \mathbf{V}^{-1/2}\mathbf{Y} \sim \mathcal{N}(\mathbf{V}^{-1/2}\mathbf{X}\beta, I)$$

$\vdots$

$$\hat{\beta} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{Y} = \left(\sum_{i=1}^N \mathbf{X}'_i\mathbf{V}_i^{-1}\mathbf{X}_i\right)^{-1}\left(\sum_{i=1}^N \mathbf{X}'_i\mathbf{V}_i^{-1}\mathbf{Y}_i\right)$$

$$\text{Var}(\hat{\beta}) = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1} = \left(\sum_{i=1}^N \mathbf{X}'_i\mathbf{V}_i^{-1}\mathbf{X}_i\right)^{-1}$$

Let  $\mathbf{V}_i = \sigma^2\mathbf{R}_i$ , where  $\mathbf{R}_i$  is a correlation matrix. Common forms of  $\mathbf{R}_i$ :

1. Equicorrelated (a.k.a. exchangeable)

$$\mathbf{R}_i = \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix},$$

where  $\rho$  is an intra-class correlation coefficient and is the same for all classes.

2. Autoregressive

$$\mathbf{R}_i = \begin{bmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{n-1} \\ \rho & 1 & \rho & \cdots & \rho^{n-2} \\ \vdots & & & & \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \cdots & 1 \end{bmatrix}$$

where  $\rho$  is the lag-1 correlation parameter.

### 3. Unstructured

$$\mathbf{R}_i = \begin{bmatrix} 1 & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{12} & 1 & \cdots & \rho_{2n} \\ \vdots & & & \\ \rho_{1n} & \rho_{2n} & \cdots & 1 \end{bmatrix}$$

→ we can fit this and look for patterns that suggest a simpler correlation structure.

#### 8.1.1.2 Non-Normally Distributed Data

For GLM's,

$$E(Y_i) = \mu_i \quad \text{and} \quad g(\mu_i) = \mathbf{X}_i' \beta = \eta_i$$

Recall the score equation is

$$U_j = \sum_{i=1}^N \frac{(y_i - \mu_i)}{\text{Var}(Y_i)} \underbrace{x_{ij} \left( \frac{\partial \mu_i}{\partial \eta_i} \right)}_{= \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j} = \frac{\partial \mu_i}{\partial \beta_j}}$$

$$\Rightarrow U_j = \sum_{i=1}^N \frac{(y_i - \mu_i)}{\text{Var}(Y_i)} \frac{\partial \mu_i}{\partial \beta_j} = 0, j = 1, \dots, k$$

Let  $\mathbf{D}_i$  be the matrix of derivatives  $\frac{\partial \mu_i}{\partial \beta_j}$ .

Then, the **generalized estimating equations** (GEE's) are

$$\mathbf{U} = \sum_{i=1}^N \mathbf{D}_i' \mathbf{V}_i^{-1} (\mathbf{Y}_i - \mu_i) = \mathbf{0}.$$

These are also known as the quasi-score equations.

Writing  $\mathbf{V}_i$  as a variance matrix times a correlation function (with an optional dispersion parameter),

$$\mathbf{V}_i = \phi \mathbf{A}_i^{1/2} \mathbf{R}_i \mathbf{A}_i^{1/2},$$

where  $\mathbf{A}_i$  is a diagonal matrix with elements  $\text{Var}(y_{ik})$  ( $y_{ik}$  is the  $k$ th observation on the  $i$ th individual),  $\mathbf{R}_i$  is a correlation matrix for  $\mathbf{Y}_i$ , and  $\phi$  is a constant to allow for over- or under-dispersion, we can then adjust the correlation matrix to accommodate dependence. (The standard GLM assumes  $\mathbf{R}_i \equiv \mathbf{I}$ .)

If the  $\mathbf{R}_i$  are correctly specified,  $\hat{\beta}$  is consistent and asymptotically normal (Liang and Zeger (1986)), and  $\hat{\beta}$  is fairly robust against misspecification of  $\mathbf{R}_i$ .

Then, inferences for our coefficients follow, so that

$$\text{Var}(\hat{\beta}) = \mathfrak{J}^{-1} \mathbf{C} \mathfrak{J}^{-1},$$

where

$$\begin{aligned} \mathfrak{J} &= \sum_{i=1}^N \mathbf{D}_i' \hat{\mathbf{V}}_i^{-1} \mathbf{D}_i, \\ \mathbf{C} &= \sum_{i=1}^N \mathbf{D}_i' \hat{\mathbf{V}}_i^{-1} (\mathbf{Y}_i - \hat{\mu}_i)(\mathbf{Y}_i - \hat{\mu}_i)' \hat{\mathbf{V}}_i \mathbf{D}_i, \end{aligned}$$

and

$$\hat{\beta} \sim \mathcal{N}(\beta, \text{Var}(\hat{\beta})),$$

with inferences as usual.

## GEE's: Respiratory Data Example

### Model

First consider a binomial GLM using only “treatment” as a covariate. Using GEE to estimate parameters, allow for independent, equi-correlated/exchangeable, AR(1), and unstructured correlation structures. Parameter estimates for each type of correlation structure are shown in Table 8.1.

```
library(geepack)
data(respiratory)
model.glm <- glm(outcome~treat, family=binomial, data=respiratory)
model.ind <- geeglm(outcome~treat, id=id*center, family=binomial, data=respiratory)
model.ar1 <- geeglm(outcome~treat, id=id*center, family=binomial, data=respiratory)
model.exch <- geeglm(outcome~treat, id=id*center, family=binomial, data=respiratory)
model.un <- geeglm(outcome~treat, id=id*center, family=binomial, data=respiratory)
```

Table 8.1: Parameter estimates for each fitted model when only treatment is a covariate.

|                      | $\hat{\beta}_0$ (s.e.) | $\hat{\beta}_1$ (s.e.) |
|----------------------|------------------------|------------------------|
| GLM, pooled          | 0.756 (0.1459296)      | -0.985 (0.1976625)     |
| GEE,<br>independent  | 0.756 (0.228203)       | -0.985 (0.3113723)     |
| GEE,<br>exchangeable | 0.756 (0.228203)       | -0.985 (0.3113723)     |
| GEE, AR(1)           | 0.686 (0.2284994)      | -0.872 (0.3152724)     |
| GEE,<br>unstructured | 0.773 (0.2289619)      | -0.99 (0.3113529)      |

## Determining the Correlation Structure

1. Consider the correlation among the binary outcomes: e.g., for the estimated correlation between visits 1 and 2, i.e.,

```
respiratory %>%  
  pivot_wider(names_from=visit, values_from=outcome, names_prefix="Visit") %>%  
  select(starts_with("Visit")) %>%  
  cor()
```

|        | Visit1    | Visit2    | Visit3    | Visit4    |
|--------|-----------|-----------|-----------|-----------|
| Visit1 | 1.0000000 | 0.5087944 | 0.4431438 | 0.5139016 |
| Visit2 | 0.5087944 | 1.0000000 | 0.5821877 | 0.5301611 |
| Visit3 | 0.4431438 | 0.5821877 | 1.0000000 | 0.5871276 |
| Visit4 | 0.5139016 | 0.5301611 | 0.5871276 | 1.0000000 |

2. Consider the correlation among the residuals of the “pooled” model: e.g., for the estimated correlation between residuals from visits 1 and 2, i.e.,

```
respiratory %>%  
  add_column(residind = residuals(model.glm)) %>%  
  select(-c(outcome)) %>%  
  pivot_wider(names_from=visit, values_from=residind, names_prefix="Visit") %>%  
  select(starts_with("Visit")) %>%  
  cor()
```

|        | Visit1    | Visit2    | Visit3    | Visit4    |
|--------|-----------|-----------|-----------|-----------|
| Visit1 | 1.0000000 | 0.4771137 | 0.4137943 | 0.4961493 |
| Visit2 | 0.4771137 | 1.0000000 | 0.5500994 | 0.5035193 |
| Visit3 | 0.4137943 | 0.5500994 | 1.0000000 | 0.5655315 |
| Visit4 | 0.4961493 | 0.5035193 | 0.5655315 | 1.0000000 |

## Compare to the Estimated Correlation Parameters



- Exchangeable:

```
summary(model.exch)$corr
```

|       | Estimate  | Std.err    |
|-------|-----------|------------|
| alpha | 0.4953744 | 0.06321745 |

- AR(1):

```
summary(model.ar1)$corr
```

|       | Estimate  | Std.err    |
|-------|-----------|------------|
| alpha | 0.6484404 | 0.05210873 |

- Unstructured:

```
summary(model.un)$corr
```

|           | Estimate  | Std.err    |
|-----------|-----------|------------|
| alpha.1:2 | 0.4625069 | 0.09264674 |
| alpha.1:3 | 0.4107358 | 0.09404980 |
| alpha.1:4 | 0.5041009 | 0.09205986 |
| alpha.2:3 | 0.5264568 | 0.09082951 |
| alpha.2:4 | 0.5107025 | 0.09149420 |
| alpha.3:4 | 0.5575959 | 0.09046015 |

Based on these estimates, which correlation structure do you think is most appropriate for this data? Why?

→ While everyone may have different preferences, the exchangeable correlation structure seems to be the most appropriate for this data, as the estimated

correlation parameters are similar across all pairs of visits. The AR(1) structure is not appropriate, as the estimated correlation parameters do not decrease with increasing lag, and the AR(1) correlation is higher than any of the empirical correlation values. The unstructured correlation structure could be appropriate, however, accounting for the standard errors, the estimated correlation parameters are generally not different from each other in this case, and it costs many degrees of freedom to estimate them all. Thus, for parsimony and practicality, I would choose the exchangeable model.

### **Now Consider Using More Covariates**

In addition to treatment, now include center, age, gender, and baseline score in the model. Table 8.2 shows parameter estimates and standard errors.

```
mod.glm <- glm(outcome~treat+center+age+sex+baseline, family=binomial, data=re  
mod.ind <- geeglm(outcome~treat+center+age+sex+baseline, id=id*center, family=  
mod.ar1 <- geeglm(outcome~treat+center+age+sex+baseline, id=id*center, family=  
mod.exch <- geeglm(outcome~treat+center+age+sex+baseline, id=id*center, family=  
mod.un <- geeglm(outcome~treat+center+age+sex+baseline, id=id*center, family=b
```

Table 8.2: Parameter estimates for each fitted model with all 5 covariates.

|                      | GLM,<br>pooled | GEE,<br>independent | GEE,<br>exchangeable | GEE,<br>unstructured |
|----------------------|----------------|---------------------|----------------------|----------------------|
| $\hat{\beta}_0$      | -0.1034607     | -0.1034607          | -0.1034607           | -0.1825607           |
| (s.e.)               | (0.5516393)    | (0.8821336)         | (0.8821336)          | (0.8685088)          |
| $\hat{\beta}_{trt}$  | -1.2653555     | -1.2653555          | -1.2653555           | -1.2454772           |
| (s.e.)               | (0.2350005)    | (0.3466787)         | (0.3466787)          | (0.3455226)          |
| $\hat{\beta}_{cent}$ | 0.6494905      | -0.1034607          | 0.6494905            | 0.6555142            |
| (s.e.)               | (0.2382579)    | (0.3532235)         | (0.3532235)          | (0.3512794)          |
| $\hat{\beta}_{age}$  | -0.0187564     | -0.0187564          | -0.0187564           | -0.0175904           |
| (s.e.)               | (0.008816)     | (0.0129647)         | (0.0129647)          | (0.0128631)          |
| $\hat{\beta}_{sex}$  | -0.1367804     | -0.1367804          | -0.1367804           | -0.1143515           |
| (s.e.)               | (0.2933299)    | (0.4402452)         | (0.4402452)          | (0.4407357)          |
| $\hat{\beta}_{base}$ | 1.8457197      | 1.8457197           | 1.8457197            | 1.8944448            |
| (s.e.)               | (0.2392671)    | (0.3459784)         | (0.3459784)          | (0.3441201)          |

## Residual Correlation

Look at the correlation of residuals from the “pooled” GLM:

```
respiratory %>%  
  add_column(residind = residuals(mod.glm)) %>%  
  select(-c(outcome)) %>%  
  pivot_wider(names_from=visit, values_from=residind, names_prefix="Visit") %>%  
  select(starts_with("Visit")) %>%  
  cor()
```

|        | Visit1    | Visit2    | Visit3    | Visit4    |
|--------|-----------|-----------|-----------|-----------|
| Visit1 | 1.0000000 | 0.3575961 | 0.2467394 | 0.3490897 |
| Visit2 | 0.3575961 | 1.0000000 | 0.4519792 | 0.3850441 |
| Visit3 | 0.2467394 | 0.4519792 | 1.0000000 | 0.4466788 |
| Visit4 | 0.3490897 | 0.3850441 | 0.4466788 | 1.0000000 |

## Compare to Estimated Correlation Parameters

- Exchangeable:

```
summary(mod.exch)$corr
```

|       | Estimate  | Std.err    |
|-------|-----------|------------|
| alpha | 0.3285278 | 0.08381149 |

- AR(1):

```
summary(mod.ar1)$corr
```

|       | Estimate  | Std.err    |
|-------|-----------|------------|
| alpha | 0.4572727 | 0.08959821 |

- Unstructured:

```
summary(mod.un)$corr
```

|           | Estimate  | Std.err   |
|-----------|-----------|-----------|
| alpha.1:2 | 0.3214897 | 0.1178953 |
| alpha.1:3 | 0.2056238 | 0.1005731 |
| alpha.1:4 | 0.2836225 | 0.1102872 |
| alpha.2:3 | 0.4244609 | 0.1409635 |
| alpha.2:4 | 0.3433203 | 0.1188059 |
| alpha.3:4 | 0.3802187 | 0.1328983 |

For similar reasons as above, the exchangeable correlation structure seems to be the most appropriate here.

Note that, after accounting for center, a patient's age, sex, and baseline status, and correlation among a patient's observations, the treatment still significantly reduces the odds of a patient having respiratory illness. Thus, we conclude that there is evidence to suggest that treatment improves respiratory health.

## 8.1.2 Generalized Linear Mixed Models

In the last section, we specified a model for inducing dependence *directly* on the mean of response variable  $\mathbf{Y}_i$ . With generalized linear mixed models (GLMM), we place dependence on the response variable *conditionally* by adding a random component to  $\eta_i$ . By doing this, we **add stochasticity to the linear component**. Here we will consider allowing for variation among groups.

Recall:

- *fixed effect*: a factor effect that applies to *all* categories of interest, such as gender, age groupings, or treatments.
- *random effect*: a factor effect that applies to a *sample*. e.g., a “group” effect.

### 8.1.2.1 Normally-Distributed Random Effects Model

Consider the following simple normally-distributed random effects model:

$$Y_{ij} = \mu + a_j + \epsilon_{ij},$$

where  $Y_{ij}$  is the  $i^{th}$  observation on the  $j^{th}$  group,  $\mu$  is a constant mean,  $a_j$  is a random group effect, and  $\epsilon_{ij}$  is a random noise term independent of  $a_j$ .

Let

$$\begin{aligned} a_j &\sim \mathcal{N}(0, \sigma_a^2), \\ \epsilon_{ij} &\sim \mathcal{N}(0, \sigma_e^2), \end{aligned}$$

where  $\sigma_a^2$  is the across-groups variation and  $\sigma_e^2$  is the remaining within-groups variation.

Notice from this set-up of the model:

- Conditional on the random effect,  $a_j$ , the  $Y_{ij}$  are independent with constant variance. That is

$$Y_{ij}|a_j \sim \mathcal{N}(\mu + a_j, \sigma_e^2),$$

where

- $E(Y_{ij}|a_j) = \mu + a_j$ , and
- $\text{Var}(Y_{ij}|a_j) = \sigma_e^2$ .

Because of the normal distributions of the random effect **and** the error term, we can find the marginal distribution of  $Y_{ij}$ :

$$Y_{ij} \sim \mathcal{N}(0, \sigma_a^2 + \sigma_e^2).$$

⋮{callout-note ap} In a non-normal GLMM setting, however, the marginal distribution of  $Y_{ij}$  is often not available in closed form.

### 8.1.2.2 Non-Normally-Distributed Mixed Models (GLMM)

For a GLMM, we specify the distribution of  $Y_{ij}$  conditional on random effects. We consider two levels of randomness:

*1st Level: Conditional Likelihood*

$$Y_{ij}|a_j \sim f(y_{ij}|a_j)$$

*2nd Level: Link Function*

$$g(E(\mathbf{Y}|\mathbf{a})) = \mathbf{X}\beta + \mathbf{Z}\mathbf{a}$$

(random in the  $\mathbf{a}$ 's via  $f(\mathbf{a})$ ). Usually, we still assume normality in the random effects, i.e.,  $\mathbf{a} \sim \mathcal{N}(\mathbf{0}, \mathbf{V})$ .

#### GLMM's Respiratory Data Example

Let  $Y_{ijk}$  represent the  $i^{th}$  observation for individual  $j$  at the  $k^{th}$  center. Let  $g(\cdot)$  represent the logit link function. In the following linear component statements, we use lowercase Greek letters to represent fixed effects, and lowercase Latin letters to represent random effects.

##### Fixed Effects Models

1. Treatment Only:

$$g(E(Y_{ijk})) = \alpha + \beta_1 \text{trt}_{ijk}$$

2. Subject-Specific Intercept:

$$g(E(Y_{ijk})) = \alpha_{jk} + \beta_1 \text{trt}_{jk}$$

3. Multiple Covariates:

$$g(E(Y_{ijk})) = \alpha + \beta_1 \text{trt}_{jk} + \beta_2 \text{ctr}_k + \beta_3 \text{age}_{jk} + \beta_4 \text{sex}_{jk} + \beta_5 \text{baseline}_{jk}$$

## Mixed Models

4. Random Subject Effect; Fixed Treatment Effect:

$$g(E(Y_{ijk})) = \alpha + a_{jk} + \beta_1 \text{trt}_{ijk}$$

Therefore, if  $\text{trt} = 0$ , we get  $\alpha + a_j$ ; if  $\text{trt} = 1$ , we get  $\alpha + a_j + \beta_1$ .

5. Random Subject Treatment Effect:

$$g(E(Y_{ijk})) = \alpha + a_{jk} + (\beta_1 + b_{jk}) \text{trt}_{ijk}$$

Therefore, if  $\text{trt} = 0$ , we get  $\alpha + a_{jk}$ ; if  $\text{trt} = 1$ , we get  $\alpha + a_{jk} + \beta_1 + b_{jk}$ .

6. Random Subject Effect; Fixed Covariates:

$$g(E(Y_{ijk})) = \alpha + a_{jk} + \beta_1 \text{trt}_{jk} + \beta_2 \text{ctr}_k + \beta_3 \text{age}_{jk} + \beta_4 \text{sex}_{jk} + \beta_5 \text{baseline}_{jk}$$

7. All Random Subject Effects:

$$g(E(Y_{ijk})) = \alpha + a_{jk} + (\beta_1 + b_{1jk}) \text{trt}_{jk} + \beta_2 \text{ctr}_k + (\beta_3 + b_{3jk}) \text{age}_{jk} \\ + (\beta_4 + b_{4jk}) \text{sex}_{jk} + \beta_5 \text{baseline}_{jk}$$

8. Random Center Effect; Fixed Treatment Effect:

$$g(E(Y_{ijk})) = \alpha + a_k + \beta_1 \text{trt}_{ijk}$$

9. Random Center and Subject Effects; Fixed Treatment Effect:

$$g(E(Y_{ijk})) = \alpha + a_k + a_{jk} + \beta_1 \text{trt}_{ijk}$$

Fitting GLMM's in R:

- Package: lme4
- Function: glmer



- In the formula:

$$\underbrace{\text{design}}_{\text{(automatically fits)}} \quad | \quad \underbrace{\text{group variable random effects}}_{\text{(can specify separately)}}$$

- Notes:
  - (1+covariate | group) is the same as (covariate | group)
  - $\approx (1 | \text{group}) + (0 + \text{covariate} | \text{group})$  (same if covariate is continuous; different if covariate is a factor...it fits both factors, i.e., an intercept)

### GLMM's Respiratory Data Example (Cont.)

#### Model Summaries

*Model 3:*

```
fixed.3 <- glm(outcome~treat+factor(center)+age+sex+factor(baseline), family=b
summary(fixed.3)
```

Call:

```
glm(formula = outcome ~ treat + factor(center) + age + sex +
     factor(baseline), family = binomial, data = respiratory)
```

Coefficients:

|                   | Estimate  | Std. Error | z value | Pr(> z )     |
|-------------------|-----------|------------|---------|--------------|
| (Intercept)       | 0.546030  | 0.464268   | 1.176   | 0.23955      |
| treatP            | -1.265356 | 0.235000   | -5.384  | 7.27e-08 *** |
| factor(center)2   | 0.649490  | 0.238258   | 2.726   | 0.00641 **   |
| age               | -0.018756 | 0.008816   | -2.128  | 0.03337 *    |
| sexM              | -0.136780 | 0.293330   | -0.466  | 0.64100      |
| factor(baseline)1 | 1.845720  | 0.239267   | 7.714   | 1.22e-14 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 609.41 on 443 degrees of freedom  
Residual deviance: 487.49 on 438 degrees of freedom  
AIC: 499.49

Number of Fisher Scoring iterations: 4

*Model 6:*

```
library(lme4)
respiratory$id2 <- respiratory$center*100 + respiratory$id
mixed.6 <- glmer(outcome~treat+factor(center)+age+sex+factor(baseline) + (1|id2), data=respiratory)
summary(mixed.6)
```

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [glmerMod]  
Family: binomial ( logit )  
Formula: outcome ~ treat + factor(center) + age + sex + factor(baseline) + (1 | id2)  
Data: respiratory

| AIC   | BIC   | logLik | -2*log(L) | df.resid |
|-------|-------|--------|-----------|----------|
| 449.1 | 477.7 | -217.5 | 435.1     | 437      |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.7490 | -0.3718 | 0.1563 | 0.3927 | 2.1787 |

Random effects:

| Groups | Name        | Variance | Std.Dev. |
|--------|-------------|----------|----------|
| id2    | (Intercept) | 3.639    | 1.908    |

Number of obs: 444, groups: id2, 111

Fixed effects:

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | 0.79332  | 1.04262    | 0.761   | 0.44672  |

```

treatP          -2.04737    0.53498  -3.827  0.00013 ***
factor(center)2    0.98057    0.53083   1.847  0.06471 .
age              -0.02705    0.01957  -1.382  0.16687
sexM             -0.24636    0.65385  -0.377  0.70633
factor(baseline)1  2.94710    0.57797   5.099 3.41e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
              (Intr) treatP fct()2 age      sexM
treatP        -0.350
fctr(cntr)2   -0.125 -0.046
age           -0.712  0.058 -0.227
sexM          -0.734  0.228  0.145  0.263
fctr(bsln)1  -0.057 -0.282 -0.167  0.006 -0.111

```

*Model 7:*

```

mixed.7 <- glmer(outcome~treat+factor(center)+age+sex+factor(baseline) + (1+tr
summary(mixed.7)

```

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: outcome ~ treat + factor(center) + age + sex + factor(baseline) +
(1 + treat + sex + age | id2)
Data: respiratory

```

| AIC   | BIC   | logLik | -2*log(L) | df.resid |
|-------|-------|--------|-----------|----------|
| 462.1 | 527.6 | -215.0 | 430.1     | 428      |

Scaled residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -3.1920 | -0.4239 | 0.1917 | 0.3707 | 2.0139 |

Random effects:

| Groups | Name        | Variance  | Std.Dev. | Corr            |
|--------|-------------|-----------|----------|-----------------|
| id2    | (Intercept) | 24.630284 | 4.96289  |                 |
|        | treatP      | 5.434576  | 2.33122  | -0.79           |
|        | sexM        | 9.726680  | 3.11876  | -0.90 0.52      |
|        | age         | 0.004904  | 0.07003  | -1.00 0.81 0.89 |

Number of obs: 444, groups: id2, 111

Fixed effects:

|                   | Estimate | Std. Error | z value | Pr(> z )     |
|-------------------|----------|------------|---------|--------------|
| (Intercept)       | 1.16029  | 1.40635    | 0.825   | 0.40935      |
| treatP            | -1.90567 | 0.64328    | -2.962  | 0.00305 **   |
| factor(center)2   | 0.87751  | 0.63915    | 1.373   | 0.16977      |
| age               | -0.03849 | 0.02382    | -1.616  | 0.10609      |
| sexM              | -0.45525 | 0.77247    | -0.589  | 0.55563      |
| factor(baseline)1 | 2.97968  | 0.55796    | 5.340   | 9.28e-08 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

|             | (Intr) | treatP | fctr()2 | age    | sexM   |
|-------------|--------|--------|---------|--------|--------|
| treatP      |        |        |         |        |        |
| fctr(cntr)2 | -0.571 |        |         |        |        |
| age         | -0.345 | 0.191  |         |        |        |
| sexM        | -0.797 | 0.272  | 0.076   |        |        |
| fctr(bsln)1 | -0.785 | 0.386  | 0.113   | 0.516  |        |
|             | 0.009  | -0.182 | -0.142  | -0.117 | -0.131 |

optimizer (Nelder\_Mead) convergence code: 4 (failure to converge in 10000 evaluations)

Model failed to converge with max|grad| = 0.122215 (tol = 0.002, component 1)

See ?lme4::convergence and ?lme4::troubleshooting.

Model is nearly unidentifiable: large eigenvalue ratio

- Rescale variables?

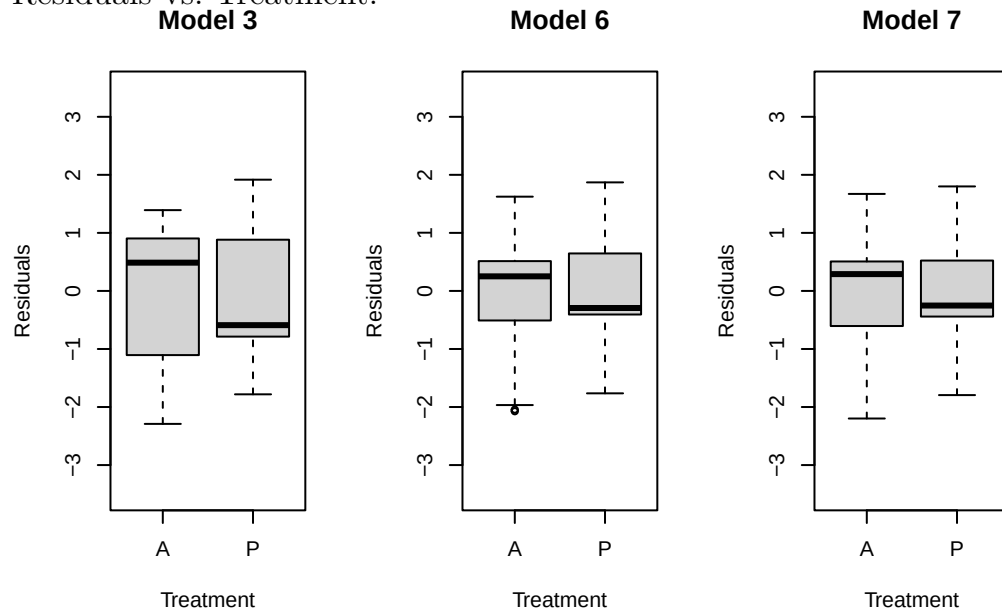
failure to converge in 10000 evaluations

## Residual Plots

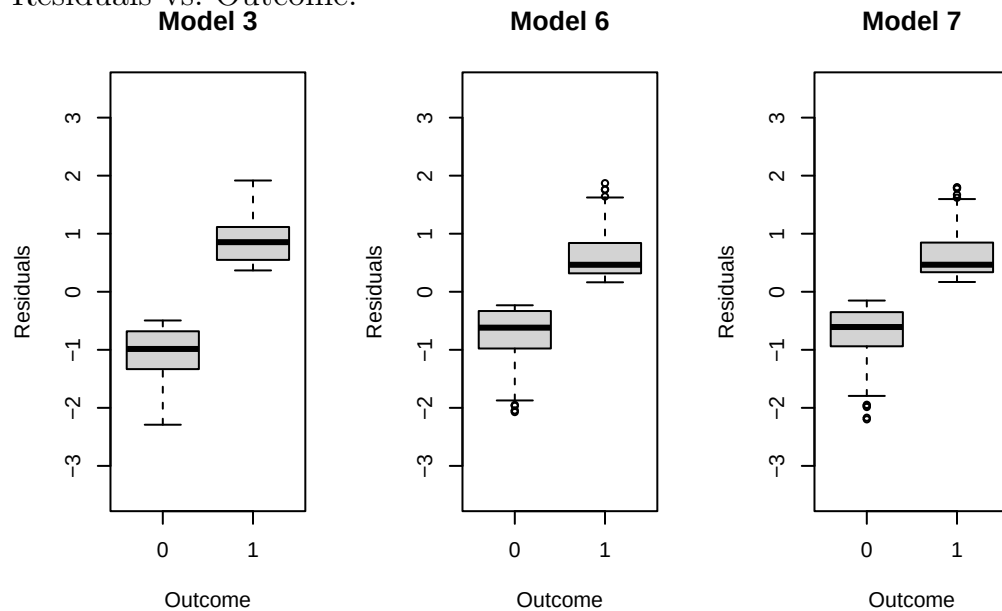
Because this is binary data, we use these plots mainly to look at the distribution of the size of the residuals. For example, residuals for Outcome 1 will always be greater than 0 and residuals for Outcome 0 will always be less than

0.

Residuals vs. Treatment:



Residuals vs. Outcome:



For the residuals, we see that adding the random effect(s) removes many of the larger (in absolute value) residuals observed in the fixed-effects-only model. This is what we hope to see.

### Mixed Effects Analysis of Deviance

```
anova(mixed.6, fixed.3)
```

Data: respiratory

Models:

fixed.3: outcome ~ treat + factor(center) + age + sex + factor(baseline)

mixed.6: outcome ~ treat + factor(center) + age + sex + factor(baseline) + (1

|  | npar | AIC | BIC | logLik | -2*log(L) | Chisq | Df | Pr(>Chisq) |
|--|------|-----|-----|--------|-----------|-------|----|------------|
|--|------|-----|-----|--------|-----------|-------|----|------------|

|         |   |        |        |         |        |  |  |  |
|---------|---|--------|--------|---------|--------|--|--|--|
| fixed.3 | 6 | 499.49 | 524.06 | -243.74 | 487.49 |  |  |  |
|---------|---|--------|--------|---------|--------|--|--|--|

|         |   |        |        |         |        |        |   |        |
|---------|---|--------|--------|---------|--------|--------|---|--------|
| mixed.6 | 7 | 449.06 | 477.73 | -217.53 | 435.06 | 52.425 | 1 | 4.47e- |
|---------|---|--------|--------|---------|--------|--------|---|--------|

13 \*\*\*

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
anova(mixed.6, mixed.7)
```

Data: respiratory

Models:

mixed.6: outcome ~ treat + factor(center) + age + sex + factor(baseline) + (1

mixed.7: outcome ~ treat + factor(center) + age + sex + factor(baseline) + (1

|  | npar | AIC | BIC | logLik | -2*log(L) | Chisq | Df | Pr(>Chisq) |
|--|------|-----|-----|--------|-----------|-------|----|------------|
|--|------|-----|-----|--------|-----------|-------|----|------------|

|         |   |        |        |         |        |  |  |  |
|---------|---|--------|--------|---------|--------|--|--|--|
| mixed.6 | 7 | 449.06 | 477.73 | -217.53 | 435.06 |  |  |  |
|---------|---|--------|--------|---------|--------|--|--|--|

|         |    |        |        |         |        |        |   |        |
|---------|----|--------|--------|---------|--------|--------|---|--------|
| mixed.7 | 16 | 462.05 | 527.58 | -215.03 | 430.05 | 5.0109 | 9 | 0.8334 |
|---------|----|--------|--------|---------|--------|--------|---|--------|

There is evidence that random effect in Model 6 improves the fit compared to the fixed-effects-only Model 3. However, there is not evidence that the additional random effects in Model 7 improve the fit compared to Model 6. Thus, of these three model, Model 6 is preferred.

## 8.2 Homework Assignment

Problem below partially taken from Dobson and Barnett (2018).

Table 8.3: Numbers of ears clear of acute otitis media at 14 days, tabulated by antibiotic treatment and age of the child.

|       | CEF |   |    |       | AMO |   |    |       |
|-------|-----|---|----|-------|-----|---|----|-------|
| Age   | 0   | 1 | 2  | Total | 0   | 1 | 2  | Total |
| < 2   | 8   | 2 | 8  | 18    | 11  | 2 | 2  | 15    |
| 2–5   | 6   | 6 | 10 | 22    | 3   | 1 | 5  | 9     |
| ≥ 6   | 0   | 1 | 3  | 4     | 1   | 0 | 6  | 7     |
| Total | 14  | 9 | 21 | 44    | 15  | 3 | 13 | 31    |

1. Data on the ears or eyes of subjects are a classical example of clustering—the ears or eyes of the same subject are unlikely to be independent. The data in Table 8.3 are the responses to two treatments – CEF and AMO – of children who had acute otitis media in both ears.

- a. Conduct an exploratory analysis to compare the effects of treatment and age of the child on the success of the treatments, ignoring the clustering within each child.
- b. Let  $Y_{ijkl}$  denote the response of the  $l$ th ear of the  $k$ th child in treatment group  $j$  and age group  $i$ . The  $Y_{ijkl}$ 's are binary variables with possible values of 1 denoting cured and 0 denoting not cured. A possible model is

$$\log \left( \frac{\pi_{ijkl}}{1 - \pi_{ijkl}} \right) = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{treatment}_j + b_k,$$

where  $b_k$  denotes the random effect for the  $k$ th child and  $\beta_0$ ,  $\beta_1$ , and  $\beta_2$  are fixed parameters. Fit this model to compare the two treatments. How well does the model fit? What do you conclude about the treatments? (Note that you will need to restructure the data to fit this model.)

- c. Decide on another random effects model to fit. How does it fit in comparison to the model fit in (b)? Which model do you prefer and why?
- d. An alternative approach, similar to the one proposed by the original researcher, is to use nominal logistic regression with response categories

0, 1, or 2 cured ears for each child. Fit a model of this type and compare the results with those obtained in (b). Which model do you prefer and why?

2. A colleague analyzes the ear-infection data by treating each ear as an independent observation in an ordinary logistic regression. They report the treatment coefficient, standard error, and p-value to their supervisor. In 3-5 sentences, explain the items below. Write your response as if your colleague understands basic regression but has never taken a course on correlated data.
  - a. what is potentially wrong with their analysis,
  - b. what consequences this could have for reported results, and
  - c. what type of model you would recommend instead.



## 8.3 Zero-Inflated Models

In some cases, we may observe more 0's than the likelihood allows. This is often motivated by a mixture of distributions – that some subjects have zero probability of a count greater than zero.

*Examples of excessive 0's in response variables:*

- number of times eat a certain food: e.g., allergies, food preferences
- number of unexcused absences in high school
- number of children born to 45-year-old adults
- your ideas?

### Note

We can model zero-inflated counts as a mixture of two sub-populations:

1. responders or “at risk” group
2. non-responders or “never at risk” group

Let  $C_i$  denote the class indicator for the  $i^{th}$  observation:

- $C_i = 0$  for **responders** (i.e., “at risk” of giving a value 0 or greater);
- $C_i = 1$  for **non-responders** (i.e., “never at risk” of giving a value larger than 0).

⇒ Mixture distribution:

$$P(Y_i = y) = P(Y_i = y|C_i = 0)(1 - \pi_i) + P(Y_i = y|C_i = 1)(\pi_i),$$

where  $\pi_i = P(C_i = 1)$ , or the probability of non-responders, are the mixing probabilities for each observation,  $i$ .

If we assume the counts within the responder sub-population follow a Poisson distribution and assume  $P(Y_i = 0|C_i = 1) = 1$ ,

$$\begin{aligned} P(Y_i = 0) &= P(Y_i = 0|C_i = 0)(1 - \pi_i) + \pi_i, \\ &= (1 - \pi_i)e^{-\mu_i} + \pi_i, \end{aligned}$$

and for  $y_i > 0$ ,

$$\begin{aligned} P(Y_i = y) &= P(Y_i = y | C_i = 0)(1 - \pi_i) + 0, \\ &= (1 - \pi_i)e^{-\mu_i} \frac{\mu_i^y}{y!}, \end{aligned}$$

where  $\mu_i$  is the mean of the Poisson distribution for observation  $i$ .

In this mixture case, what are the **mean** and **variance** of  $Y_i$ ?

$$\begin{aligned} E(Y_i) &= \sum_{y=0}^{\infty} yP(Y_i = y) = 0(\pi_i) + \underbrace{\sum_{y=1}^{\infty} yP(Y_i = y | C_i = 0)(1 - \pi_i)}_{=\sum_{y=0}^{\infty} yP(Y_i = y | C_i = 0)(1 - \pi_i)} \\ &= (1 - \pi_i)\mu_i \end{aligned}$$

$$\begin{aligned} E(Y_i^2) &= 0^2(\pi_i) + \sum_{y=0}^{\infty} y^2 P(Y_i = y | C_i = 0)(1 - \pi_i) \\ &= (1 - \pi_i)E(U^2), \text{ where } U \sim \text{Poisson}(\mu_i), \\ &= (1 - \pi_i)\mu_i(\mu_i + 1) \end{aligned}$$

$$\begin{aligned} \text{Var}(Y_i) &= E(Y_i^2) - E(Y_i)^2 = (1 - \pi_i)\mu_i(\mu_i + 1) - (1 - \pi_i)^2\mu_i^2 \\ &= (1 - \pi_i)\mu_i(\mu_i + 1 - (1 - \pi_i)\mu_i) \\ &= (1 - \pi_i)\mu_i(1 + \pi_i\mu_i) \end{aligned}$$

### Modeling $\pi_i$

- Along with using a linear component to model  $\mu_i$ ,

$$\log(\mu_i) = \mathbf{x}_i' \beta,$$

we can also use a linear component to model the  $\pi_i$ 's,

$$\text{logit}(\pi_i) = \mathbf{v}_i' \alpha,$$

where  $\mathbf{x}_i$  and  $\mathbf{v}_i$  may be different or the same set of covariates.

- The simplest mixture model assumes only an intercept in the binary model,

$$\text{logit}(\pi_i) = \alpha_0,$$

and as  $\alpha_0$  decreases,  $\pi_i \rightarrow 0 \Rightarrow$  model without zero inflation.

## Hospital Maternal Mortality Example

The US Centers for Medicare & Medicaid Services collects data regarding hospitals across the nation. They collect three variables to monitor “maternal mortality”; we will look at each hospital’s rate of “mothers whose deliveries were scheduled 1-2 weeks early when an early delivery wasn’t medically necessary.” These data are collected from April 1, 2022 through March 31, 2023.

*Variables:*

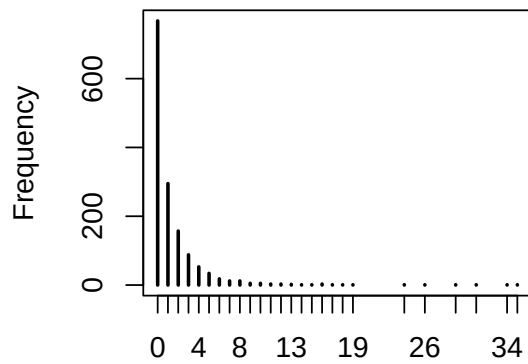
- count: the response variable (translated from the percentage and sample size)
- hospital overall rating: 1, 2, 3, 4, or 5 stars (treated as continuous)
- count of *mortality* measures that are better than the national value: integer from 0 to 6
- count of *safety* measures that are better than the national value: integer from 0 to 6
- count of *readmission* measures that are better than the national value: integer from 0 to 5
- emergency services: yes/no

Note: The codebook is available [here](#). The hospital data set is available [here](#). The maternal health data set is available [here](#).

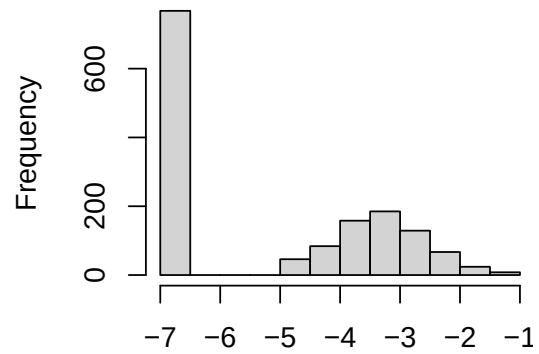
```
datm <- read.csv('../Lectures/Maternal_Health-Hospital.csv')
datm <- datm[which(datm$Measure.ID=='PC_01'),] #"Elective Delivery"
dath <- read.csv('../Lectures/Hospital_General_Information.csv')
dat <- merge(dath, datm)
dat$sample <- as.numeric(dat$Sample)
dat$score <- as.numeric(dat$Score)
dat$count <- round(dat$score*dat$sample/100)
```

## EDA

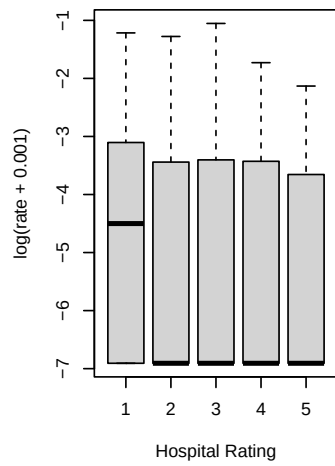
### Bar Plot of Counts



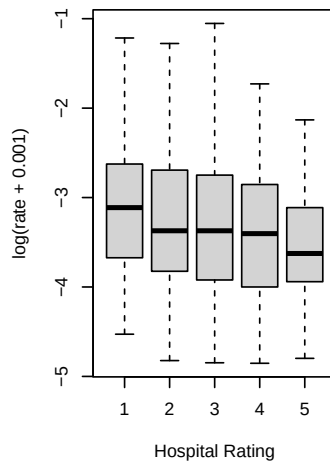
### Histogram of Log Rate



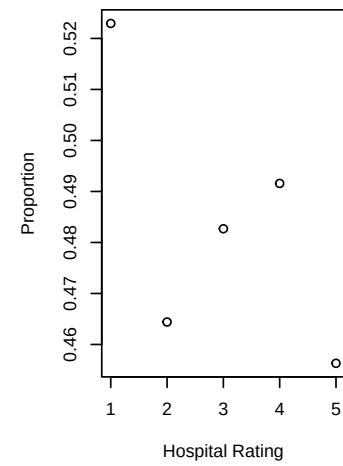
### Log Rate vs. Hospital Rating

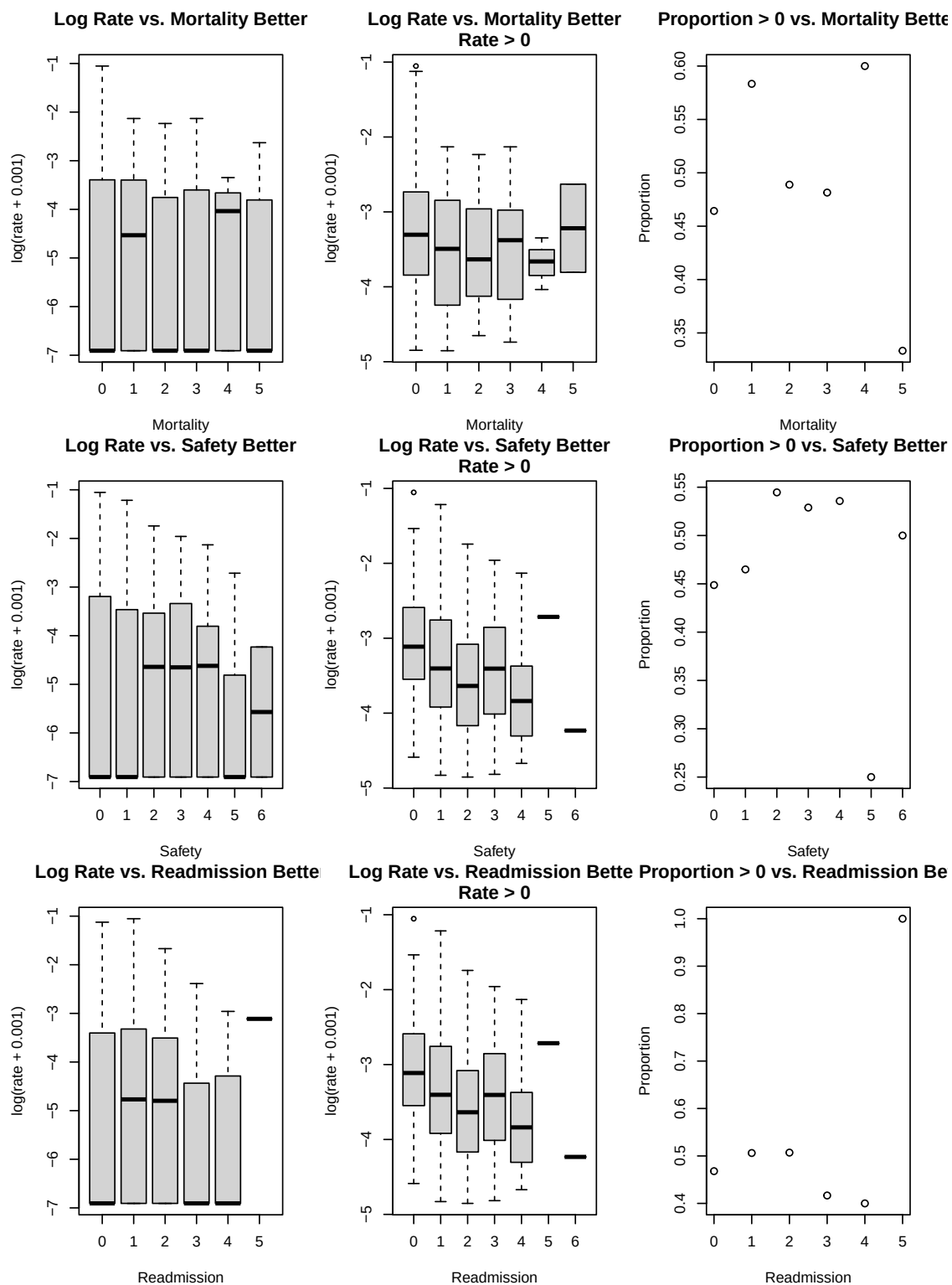


### Log Rate vs. Hospital Rating Rate > 0



### log(rate + 0.001) Proportion > 0 vs. Hospital Rating





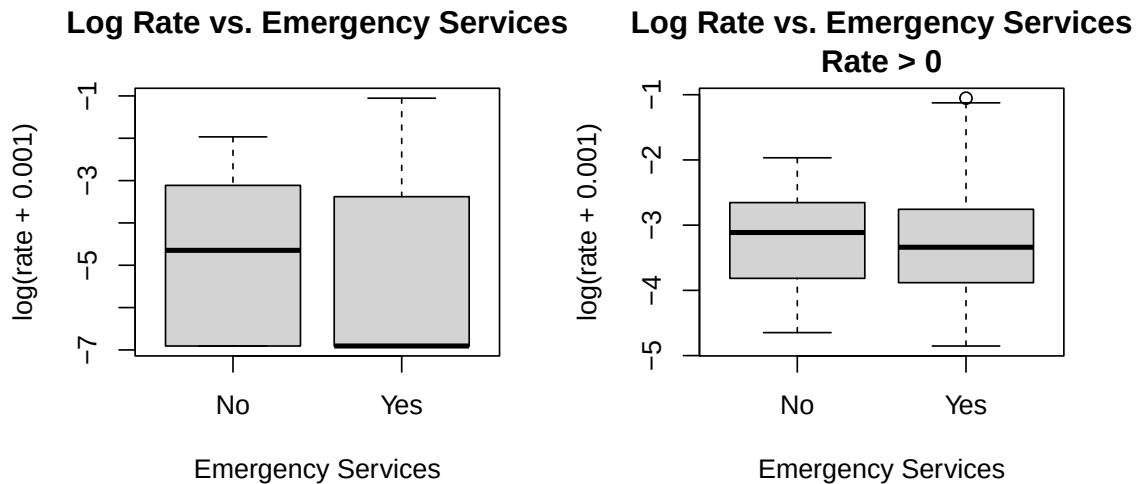


Table 8.4: Variables used in modeling of the US maternal health data set. Summary statistics are the ranges and median for continuous variables, and the proportions of the data for categorical variables.

| Variable                        | Description                                       | Summary           |
|---------------------------------|---------------------------------------------------|-------------------|
| Count                           | Number of <i>early</i> elective delivery births   | 0–35; median 0    |
| Hospital.overall.rating         | Rating                                            | 1–5; median 3     |
| Emergency.Services              | Yes/No                                            | (0.93/0.07)       |
| Count.of.Safety.Measures.Better | Safety measures better than national average      | 0–6; median 1     |
| safe2                           | Count.of.Safety.Measures.Better                   | 0–36; median 1    |
| Count.of.MORT.Measures.Better   | Mortality measures better than national average   | 0–5; median 0     |
| mort2                           | Count.of.MORT.Measures.Better                     | 0–25; median 0    |
| Count.of.READM.Measures.Better  | Readmission measures better than national average | 0–5; median 0     |
| read2                           | Count.of.READM.Measures.Better                    | 0–25; median 0    |
| sample                          | Hospital’s number of elective deliveries          | 11–867; median 44 |

## Models

### *Poisson Model*

Let  $Y_i$  represent the number of early elective deliveries at hospital  $i$ . We model,

$$Y_i \overset{ind}{\sim} \text{Pois}(\mu_i) \\ \log(\mu_i) = \mathbf{x}_i' \beta + \log(n_i),$$

where  $\mathbf{x}_i$  represents a set of eight covariates (found in Table 8.4),  $\beta$  the corresponding coefficients, and  $n_i$  represents the sample size at each hospital, whose log value is used as an offset.

```
mod.p <- glm(count~Hospital.overall.rating+Emergency.Services+Count.of.Safety.
summary(mod.p)
```

Call:

```
glm(formula = count ~ Hospital.overall.rating + Emergency.Services +
    Count.of.Safety.Measures.Better + safe2 + Count.of.MORT.Measures.Better +
    mort2 + Count.of.READM.Measures.Better + read2, family = poisson,
    data = dat, offset = log(sample))
```

Coefficients:

|                                 | Estimate     | Std. Error | z value | Pr(> z ) |
|---------------------------------|--------------|------------|---------|----------|
| (Intercept)                     | -3.286559    | 0.146203   | -       |          |
| 22.479                          | < 2e-16 ***  |            |         |          |
| Hospital.overall.rating         | -0.156367    | 0.024044   | -       |          |
| 6.503                           | 7.86e-11 *** |            |         |          |
| Emergency.ServicesYes           | -0.032874    | 0.124932   | -       |          |
| 0.263                           | 0.79245      |            |         |          |
| Count.of.Safety.Measures.Better | -0.136315    | 0.057693   | -       |          |
| 2.363                           | 0.01814 *    |            |         |          |
| safe2                           | -0.003233    | 0.014746   | -       |          |
| 0.219                           | 0.82646      |            |         |          |
| Count.of.MORT.Measures.Better   | 0.046876     | 0.069155   | 0.678   | 0.49788  |



```

mort2                0.019166    0.017614    1.088    0.27653
Count.of.READM.Measures.Better  0.239288    0.079923    2.994    0.00275 **
read2                -0.073690    0.035622    -
2.069    0.03858 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 3289.3  on 1307  degrees of freedom
Residual deviance: 3191.5  on 1299  degrees of freedom
(1748 observations deleted due to missingness)
AIC: 4880.1

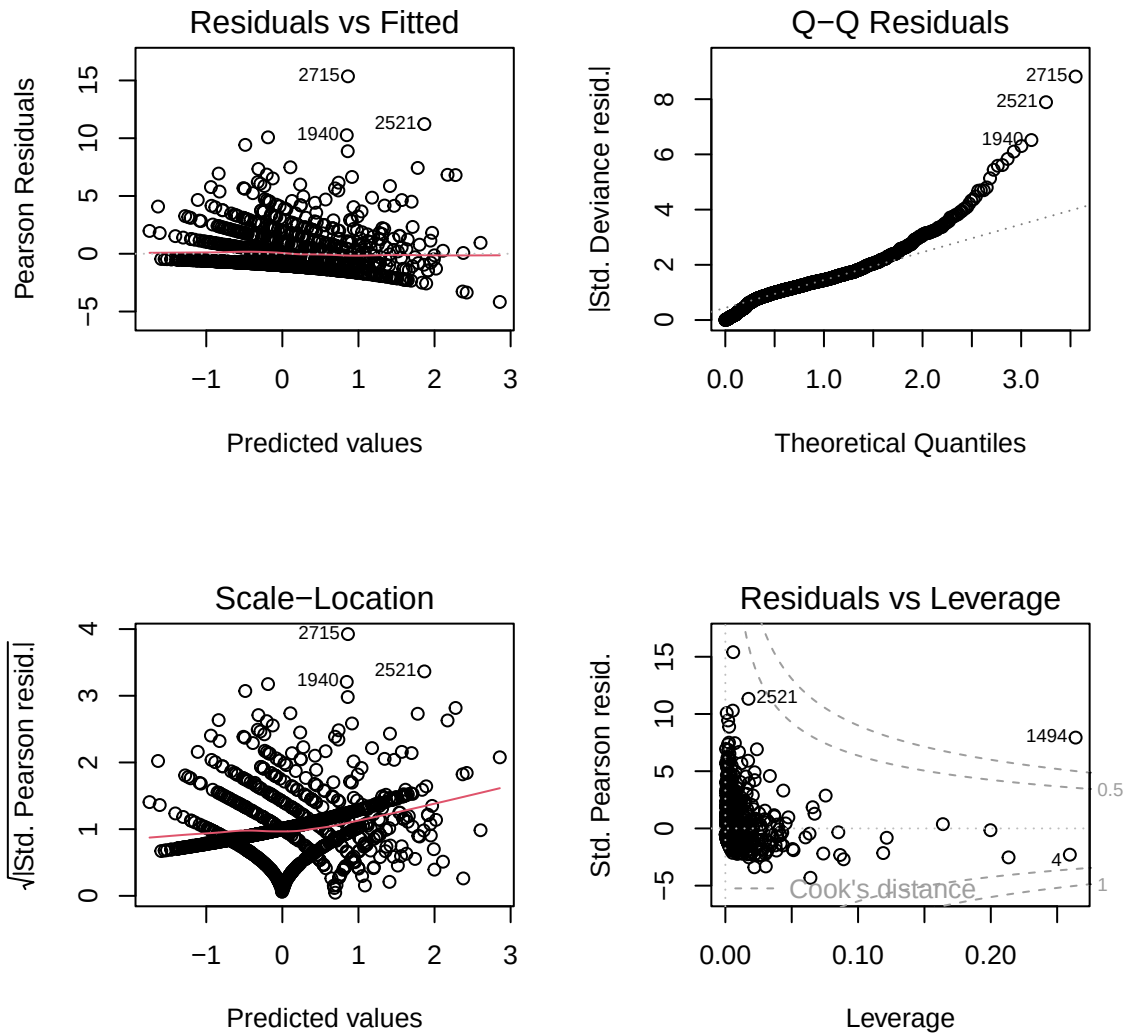
Number of Fisher Scoring iterations: 6

```

```

par(mfrow=c(2,2))
plot(mod.p)

```



Predict mean for the first observation:

$$\begin{aligned}\hat{\mu}_1 &= \exp(\mathbf{x}'_1 \hat{\beta} + \log(n_1)) \\ &= 1.4761058\end{aligned}$$

(Note the observed value is  $y_1 = 0$ .)

### *Negative Binomial Model*

Using the same response variable, covariate, and offset, fit a negative binomial model to account for overdispersion, i.e.,

$$Y_i \overset{ind}{\sim} \text{NB}(\mu_i, \theta),$$

$$\log(\mu_i) = \mathbf{x}_i' \beta + \log(n_i).$$

```
library(MASS)
mod.nb <- glm.nb(count~Hospital.overall.rating+Emergency.Services+Count.of.Saf
summary(mod.nb)
```

Call:

```
glm.nb(formula = count ~ Hospital.overall.rating + Emergency.Services +
  Count.of.Safety.Measures.Better + safe2 + Count.of.MORT.Measures.Better +
  mort2 + Count.of.READM.Measures.Better + read2 + offset(log(sample)),
  data = dat, init.theta = 0.6989449971, link = log)
```

Coefficients:

|                                 | Estimate  | Std. Error | z value | Pr(> z ) |
|---------------------------------|-----------|------------|---------|----------|
| (Intercept)                     | -3.098755 | 0.264401   | -       |          |
| 11.720 < 2e-16 ***              |           |            |         |          |
| Hospital.overall.rating         | -0.157093 | 0.043615   | -       |          |
| 3.602 0.000316 ***              |           |            |         |          |
| Emergency.ServicesYes           | -0.160964 | 0.226458   | -       |          |
| 0.711 0.477215                  |           |            |         |          |
| Count.of.Safety.Measures.Better | -0.182006 | 0.102267   | -       |          |
| 1.780 0.075121 .                |           |            |         |          |
| safe2                           | 0.016514  | 0.026758   | 0.617   | 0.537120 |
| Count.of.MORT.Measures.Better   | 0.054731  | 0.140740   | 0.389   | 0.697365 |
| mort2                           | -0.005102 | 0.040575   | -       |          |
| 0.126 0.899945                  |           |            |         |          |
| Count.of.READM.Measures.Better  | 0.211635  | 0.137577   | 1.538   | 0.123976 |
| read2                           | -0.051657 | 0.056123   | -       |          |
| 0.920 0.357348                  |           |            |         |          |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(0.6989) family taken to be 1)

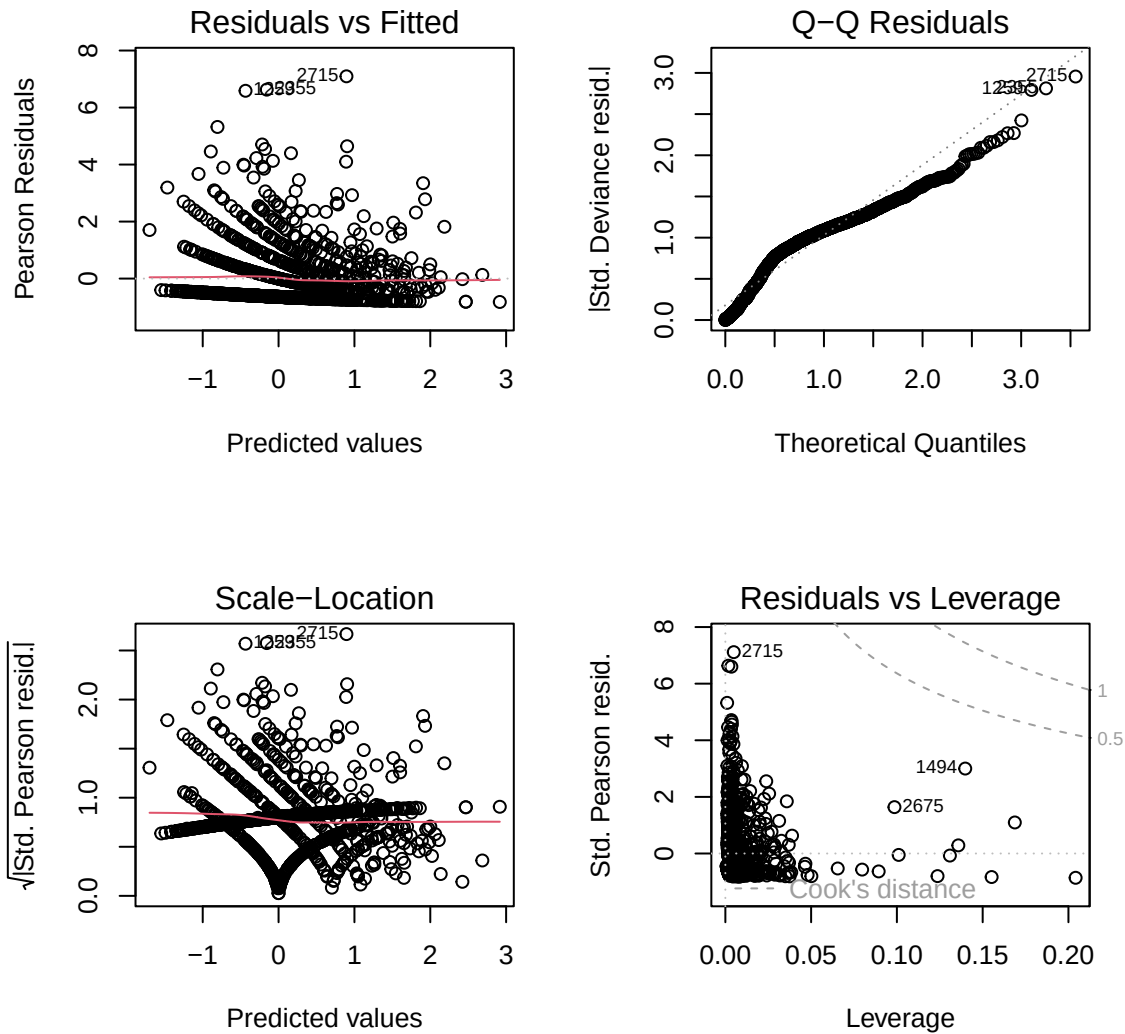
Null deviance: 1250.5 on 1307 degrees of freedom  
Residual deviance: 1226.1 on 1299 degrees of freedom  
(1748 observations deleted due to missingness)  
AIC: 3910.1

Number of Fisher Scoring iterations: 1

Theta: 0.6989  
Std. Err.: 0.0545

2 x log-likelihood: -3890.0820

```
par(mfrow=c(2,2))  
plot(mod.nb)
```



Predict mean for the first observation:

$$\begin{aligned}\hat{\mu}_1 &= \exp(\mathbf{x}'_1 \hat{\beta} + \log(n_1)) \\ &= 1.5234799\end{aligned}$$

### *Zero-Inflated Poisson Model*

Instead of accounting for the many zero's in our response variable with an overdispersion parameter, we'd like to directly model the probability that a hospital will “always” have 0 early elective births. We can model this with a zero-inflated Poisson model. First, we will model the zero-inflation

component with a constant probability (i.e., fitting just an intercept).

Use the “zeroinfl” function in the “pscl” package. (The “glmmTMB” package is another option.)

Model 1:

$$\begin{aligned} Y_i | C_i = 0 &\overset{ind}{\sim} \text{Pois}(\mu_i) \\ Y_i | C_i = 1 &= 0 \\ C_i &= \begin{cases} 1, w.p. \pi_i \\ 0, w.p. 1 - \pi_i \end{cases} \\ \log(\mu_i) &= \mathbf{x}'_i \beta + \log(n_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \alpha_0 \end{aligned}$$

```
library(pscl)
mod.zp0 <- zeroinfl(count~Hospital.overall.rating+Emergency.Services+Count.of.
summary(mod.zp0)
```

Call:

```
zeroinfl(formula = count ~ Hospital.overall.rating + Emergency.Services +
  Count.of.Safety.Measures.Better + safe2 + Count.of.MORT.Measures.Better +
  mort2 + Count.of.READM.Measures.Better + read2 | 1, data = dat, offset = 1,
  dist = "poisson")
```

Pearson residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.2626 | -0.7586 | -0.5391 | 0.4560 | 9.9957 |

Count model coefficients (poisson with log link):

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | -2.70930 | 0.15557    | -       | -        |
| 17.416      | < 2e-16  | ***        |         |          |

|                                 |          |         |       |         |    |
|---------------------------------|----------|---------|-------|---------|----|
| Hospital.overall.rating         | -0.16120 | 0.02615 | -     |         |    |
| 6.163                           | 7.12e-10 | ***     |       |         |    |
| Emergency.ServicesYes           | -0.04874 | 0.13207 | -     |         |    |
| 0.369                           | 0.71210  |         |       |         |    |
| Count.of.Safety.Measures.Better | -0.26449 | 0.06569 | -     |         |    |
| 4.027                           | 5.66e-05 | ***     |       |         |    |
| safe2                           | 0.02245  | 0.01674 | 1.341 | 0.17990 |    |
| Count.of.MORT.Measures.Better   | -0.04556 | 0.07288 | -     |         |    |
| 0.625                           | 0.53189  |         |       |         |    |
| mort2                           | 0.05024  | 0.01744 | 2.881 | 0.00396 | ** |
| Count.of.READM.Measures.Better  | 0.24099  | 0.08440 | 2.855 | 0.00430 | ** |
| read2                           | -0.07549 | 0.03710 | -     |         |    |
| 2.035                           | 0.04189  | *       |       |         |    |

Zero-inflation model coefficients (binomial with logit link):

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -0.55262 | 0.07888    | -7.006  | 2.45e-12 *** |

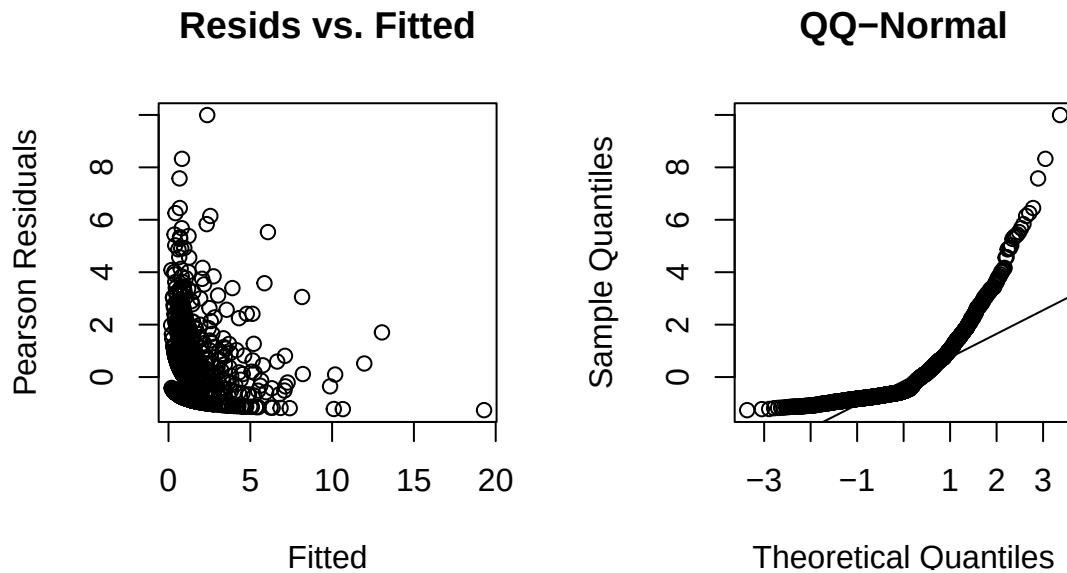
---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 17

Log-likelihood: -2103 on 10 Df

```
par(mfrow=c(1,2))
plot(fitted(mod.zp0), residuals(mod.zp0, type="pearson"), xlab="Fitted", ylab=
qqnorm(residuals(mod.zp0, type="pearson"), main="QQ-Normal")
qqline(residuals(mod.zp0, type='pearson'))
```



Note that this mixture likelihood is not a member of the exponential family. Thus, our asymptotic results will not hold and it is not surprising that our qq-plot of the residuals looks very bad.

Predict mean for the first observation:

$$\begin{aligned}
 E(Y_1) &= \mu_1(1 - \pi_i) \\
 \hat{\pi}_1 &= \frac{\exp(\alpha_0)}{1 + \exp(\alpha_0)} \\
 \hat{\mu}_1 &= \exp(\mathbf{x}'_1 \hat{\beta} + \log(n_1)) \\
 \widehat{E(Y_1)} &= 2.302079(1 - 0.3652556) \\
 &= 1.4612317
 \end{aligned}$$

Note: options for “predict” for a “zeroinfl” model object are “response” ( $\widehat{E(Y_i)}$ ; default), “count” ( $\hat{\mu}_i$ ), and “zero” ( $\hat{\pi}_i$ ).

Now let’s use our eight covariates to model the probability of a non-responder...

Model 2:



$$Y_i|C_i = 0 \overset{ind}{\sim} \text{Pois}(\mu_i)$$

$$Y_i|C_i = 1 = 0$$

$$C_i = \begin{cases} 1, w.p. \pi_i \\ 0, w.p. 1 - \pi_i \end{cases}$$

$$\log(\mu_i) = \mathbf{x}_i' \beta + \log(n_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \mathbf{v}_i' \alpha,$$

where  $\mathbf{x}_i = \mathbf{v}_i$ .

```
mod.zp1 <- zeroinfl(count~Hospital.overall.rating+Emergency.Services+Count.of.
summary(mod.zp1)
```

Call:

```
zeroinfl(formula = count ~ Hospital.overall.rating + Emergency.Services +
  Count.of.Safety.Measures.Better + safe2 + Count.of.MORT.Measures.Better +
  mort2 + Count.of.READM.Measures.Better + read2, data = dat, offset = log(s
  dist = "poisson")
```

Pearson residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.4030 | -0.7482 | -0.5393 | 0.4430 | 9.9384 |

Count model coefficients (poisson with log link):

|                         | Estimate     | Std. Error | z value | Pr(> z ) |
|-------------------------|--------------|------------|---------|----------|
| (Intercept)             | -2.68508     | 0.16165    | -       |          |
| 16.611                  | < 2e-16 ***  |            |         |          |
| Hospital.overall.rating | -0.16662     | 0.02737    | -       |          |
| 6.087                   | 1.15e-09 *** |            |         |          |
| Emergency.ServicesYes   | -0.02554     | 0.13763    | -       |          |
| 0.186                   | 0.85281      |            |         |          |

|                                 |          |         |       |            |
|---------------------------------|----------|---------|-------|------------|
| Count.of.Safety.Measures.Better | -0.29623 | 0.06824 | -     |            |
| 4.341                           | 1.42e-05 | ***     |       |            |
| safe2                           | 0.02725  | 0.01736 | 1.570 | 0.11644    |
| Count.of.MORT.Measures.Better   | -0.06000 | 0.07620 | -     |            |
| 0.787                           | 0.43106  |         |       |            |
| mort2                           | 0.05507  | 0.01797 | 3.065 | 0.00218 ** |
| Count.of.READM.Measures.Better  | 0.24554  | 0.08738 | 2.810 | 0.00495 ** |
| read2                           | -0.07976 | 0.03860 | -     |            |
| 2.066                           | 0.03883  | *       |       |            |

Zero-inflation model coefficients (binomial with logit link):

|                                 | Estimate   | Std. Error | z value | Pr(> z ) |
|---------------------------------|------------|------------|---------|----------|
| (Intercept)                     | -0.3913802 | 0.4983210  | -       |          |
| 0.785                           | 0.4322     |            |         |          |
| Hospital.overall.rating         | -0.0592807 | 0.0801891  | -       |          |
| 0.739                           | 0.4597     |            |         |          |
| Emergency.ServicesYes           | 0.2685723  | 0.4336218  | 0.619   | 0.5357   |
| Count.of.Safety.Measures.Better | -0.2324407 | 0.1856563  | -       |          |
| 1.252                           | 0.2106     |            |         |          |
| safe2                           | 0.0231938  | 0.0501913  | 0.462   | 0.6440   |
| Count.of.MORT.Measures.Better   | -0.3990890 | 0.2716244  | -       |          |
| 1.469                           | 0.1418     |            |         |          |
| mort2                           | 0.1513134  | 0.0739692  | 2.046   | 0.0408 * |
| Count.of.READM.Measures.Better  | -0.0008704 | 0.2428689  | -       |          |
| 0.004                           | 0.9971     |            |         |          |
| read2                           | -0.0217315 | 0.1011900  | -       |          |
| 0.215                           | 0.8300     |            |         |          |
| ---                             |            |            |         |          |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

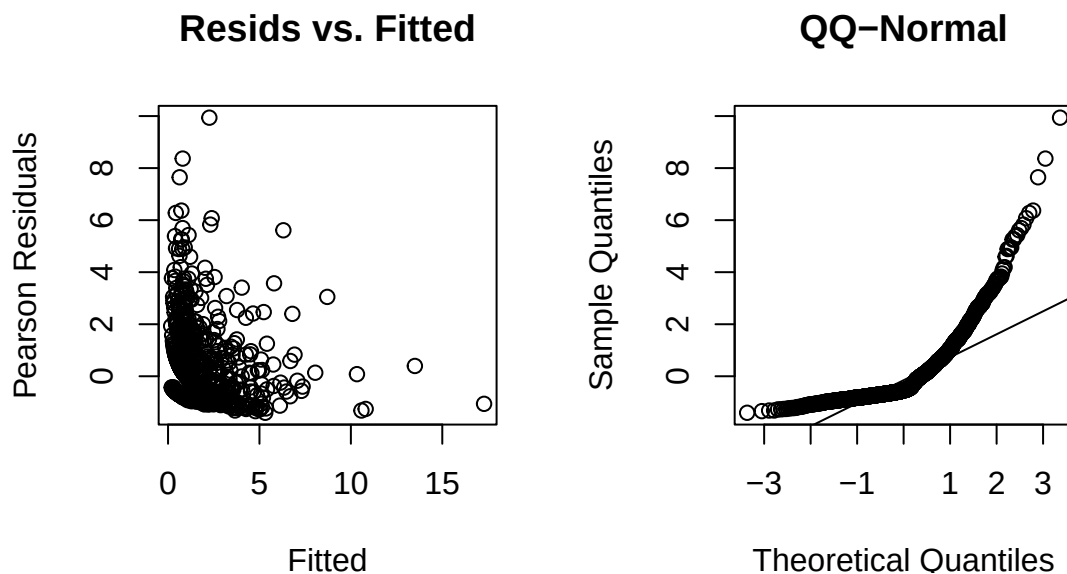
Number of iterations in BFGS optimization: 25

Log-likelihood: -2098 on 18 Df

```

par(mfrow=c(1,2))
plot(fitted(mod.zp1), residuals(mod.zp1, type="pearson"), xlab="Fitted", ylab="Pearson Residuals")
qqnorm(residuals(mod.zp1, type="pearson"), main="QQ-Normal")
qqline(residuals(mod.zp1, type='pearson'))

```



Predict mean for the first observation:

$$\begin{aligned}
 E(Y_1) &= \mu_1(1 - \pi_i) \\
 \hat{\pi}_1 &= \frac{\exp(\mathbf{v}_i \hat{\alpha})}{1 + \exp(\mathbf{v}_i \hat{\alpha})} \\
 \hat{\mu}_1 &= \exp(\mathbf{x}_1' \hat{\beta} + \log(n_1)) \\
 \widehat{E(Y_1)} &= 2.3118161(1 - 0.3752177) \\
 &= 1.4443817
 \end{aligned}$$

### *Zero-Inflated Negative Binomial Model*

Because we still saw large Pearson residuals in the zero-inflated Poisson model, there may still be overdispersion even after accounting for the excess zeros. Thus, we also consider two zero-inflated negative binomial models, one with a constant probability of non-responders and one with covariates to model

the probability of non-responders.

Model 1:

$$\begin{aligned} Y_i | C_i = 0 &\overset{ind}{\sim} \text{NB}(\mu_i, \theta) \\ Y_i | C_i = 1 &= 0 \\ C_i &= \begin{cases} 1, w.p. \pi_i \\ 0, w.p. 1 - \pi_i \end{cases} \\ \log(\mu_i) &= \mathbf{x}'_i \beta + \log(n_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \alpha_0 \end{aligned}$$

```
mod.znb0 <- zeroinfl(count~Hospital.overall.rating+Emergency.Services+Count.of
summary(mod.znb0)
```

Call:

```
zeroinfl(formula = count ~ Hospital.overall.rating + Emergency.Services +
  Count.of.Safety.Measures.Better + safe2 + Count.of.MORT.Measures.Better +
  mort2 + Count.of.READM.Measures.Better + read2 | 1, data = dat, offset = 1,
  dist = "negbin")
```

Pearson residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -0.8755 | -0.6494 | -0.4895 | 0.3236 | 7.3663 |

Count model coefficients (negbin with log link):

|                         | Estimate  | Std. Error | z value | Pr(> z ) |
|-------------------------|-----------|------------|---------|----------|
| (Intercept)             | -2.872019 | 0.263333   | -       |          |
| 10.906                  | < 2e-16   | ***        |         |          |
| Hospital.overall.rating | -0.165250 | 0.042695   | -       |          |
| 3.871                   | 0.000109  | ***        |         |          |

```

Emergency.ServicesYes      -0.160803    0.218847  -
0.735 0.462476
Count.of.Safety.Measures.Better -0.197537    0.102304  -
1.931 0.053497 .
safe2                      0.018961    0.027309    0.694 0.487481
Count.of.MORT.Measures.Better 0.003036    0.134085    0.023 0.981937
mort2                      0.016507    0.038972    0.424 0.671885
Count.of.READM.Measures.Better 0.216018    0.127972    1.688 0.091408 .
read2                     -0.053409    0.051453  -
1.038 0.299262
Log(theta)                 0.151819    0.181934    0.834 0.404015

Zero-inflation model coefficients (binomial with logit link):
      Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.5544      0.3222  -4.824 1.41e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

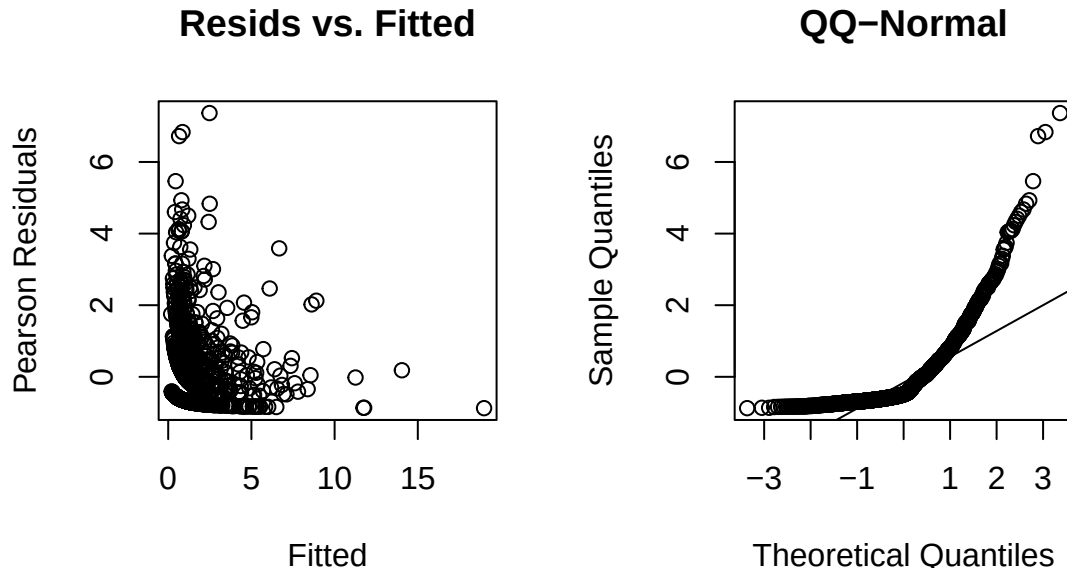
Theta = 1.1639
Number of iterations in BFGS optimization: 19
Log-likelihood: -1941 on 11 Df

```

```

par(mfrow=c(1,2))
plot(fitted(mod.znb0), residuals(mod.znb0, type="pearson"), xlab="Fitted", ylab="Residuals")
qqnorm(residuals(mod.znb0, type="pearson"), main="QQ-Normal")
qqline(residuals(mod.znb0, type='pearson'))

```



Predict mean for the first observation:

$$\begin{aligned}
 E(Y_1) &= \mu_1(1 - \pi_i) \\
 \hat{\pi}_1 &= \frac{\exp(\alpha_0)}{1 + \exp(\alpha_0)} \\
 \hat{\mu}_1 &= \exp(\mathbf{x}'_1 \hat{\beta} + \log(n_1)) \\
 \widehat{E(Y_1)} &= 1.8410551(1 - 0.1744525) \\
 &= 1.5198785
 \end{aligned}$$

Model 2:

$$\begin{aligned}
 Y_i | C_i = 0 &\overset{ind}{\sim} \text{NB}(\mu_i, \theta) \\
 Y_i | C_i = 1 &= 0 \\
 C_i &= \begin{cases} 1, w.p. \pi_i \\ 0, w.p. 1 - \pi_i \end{cases} \\
 \log(\mu_i) &= \mathbf{x}'_i \beta + \log(n_i) \\
 \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \mathbf{v}'_i \alpha,
 \end{aligned}$$

where  $\mathbf{x}_i = \mathbf{v}_i$ .

```
mod.znb1 <- zeroinfl(count~Hospital.overall.rating+Emergency.Services+Count.of
summary(mod.znb1)
```

Call:

```
zeroinfl(formula = count ~ Hospital.overall.rating + Emergency.Services +
  Count.of.Safety.Measures.Better + safe2 + Count.of.MORT.Measures.Better +
  mort2 + Count.of.READM.Measures.Better + read2, data = dat, offset = log(s
  dist = "negbin")
```

Pearson residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -0.8552 | -0.6280 | -0.4756 | 0.3107 | 6.7599 |

Count model coefficients (negbin with log link):

|                                 | Estimate | Std. Error | z value | Pr(> z ) |
|---------------------------------|----------|------------|---------|----------|
| (Intercept)                     | -2.90841 | 0.27089    | -       |          |
| 10.737 < 2e-16 ***              |          |            |         |          |
| Hospital.overall.rating         | -0.19993 | 0.04595    | -       |          |
| 4.351 1.36e-05 ***              |          |            |         |          |
| Emergency.ServicesYes           | -0.12377 | 0.22443    | -       |          |
| 0.551 0.58130                   |          |            |         |          |
| Count.of.Safety.Measures.Better | -0.29192 | 0.11423    | -       |          |
| 2.556 0.01060 *                 |          |            |         |          |
| safe2                           | 0.03858  | 0.02950    | 1.308   | 0.19101  |
| Count.of.MORT.Measures.Better   | 0.11816  | 0.15558    | 0.760   | 0.44754  |
| mort2                           | 0.01072  | 0.04785    | 0.224   | 0.82266  |
| Count.of.READM.Measures.Better  | 0.15870  | 0.13288    | 1.194   | 0.23236  |
| read2                           | -0.03905 | 0.05313    | -       |          |
| 0.735 0.46235                   |          |            |         |          |
| Log(theta)                      | -0.24858 | 0.09083    | -       |          |
| 2.737 0.00621 **                |          |            |         |          |

Zero-inflation model coefficients (binomial with logit link):

| Estimate | Std. Error | z value | Pr(> z ) |
|----------|------------|---------|----------|
|----------|------------|---------|----------|

|                                 |         |         |       |          |
|---------------------------------|---------|---------|-------|----------|
| (Intercept)                     | -3.2046 | 23.3859 | -     |          |
| 0.137 0.8910                    |         |         |       |          |
| Hospital.overall.rating         | -1.2227 | 0.5944  | -     |          |
| 2.057 0.0397 *                  |         |         |       |          |
| Emergency.ServicesYes           | 3.7083  | 23.2602 | 0.159 | 0.8733   |
| Count.of.Safety.Measures.Better | -4.0189 | 1.9082  | -     |          |
| 2.106 0.0352 *                  |         |         |       |          |
| safe2                           | 0.7141  | 0.3526  | 2.025 | 0.0429 * |
| Count.of.MORT.Measures.Better   | 2.9443  | 1.5704  | 1.875 | 0.0608 . |
| mort2                           | -0.1330 | 0.2693  | -     |          |
| 0.494 0.6215                    |         |         |       |          |
| Count.of.READM.Measures.Better  | -2.4046 | 1.9811  | -     |          |
| 1.214 0.2248                    |         |         |       |          |
| read2                           | 0.5943  | 0.4956  | 1.199 | 0.2305   |
| ---                             |         |         |       |          |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Theta = 0.7799

Number of iterations in BFGS optimization: 66

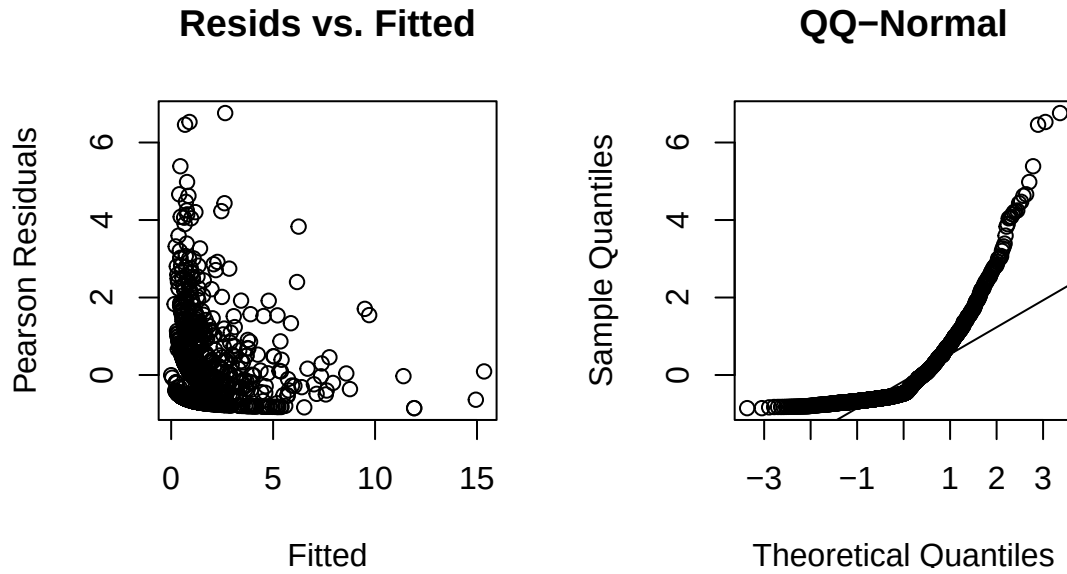
Log-likelihood: -1933 on 19 Df

```

par(mfrow=c(1,2))
plot(fitted(mod.znb1), residuals(mod.znb1, type="pearson"), xlab="Fitted", ylab="Residuals")
qqnorm(residuals(mod.znb1, type="pearson"), main="QQ-Normal")
qqline(residuals(mod.znb1, type='pearson'))

```





Predict mean for the first observation:

$$\begin{aligned}
 E(Y_1) &= \mu_1(1 - \pi_i) \\
 \hat{\pi}_1 &= \frac{\exp(\mathbf{v}'_1 \hat{\alpha})}{1 + \exp(\mathbf{v}'_1 \hat{\alpha})} \\
 \hat{\mu}_1 &= \exp(\mathbf{x}'_1 \hat{\beta} + \log(n_1)) \\
 \widehat{E(Y_1)} &= 1.5406269(1 - 0.0015483) \\
 &= 1.5382416
 \end{aligned}$$

### Vuong Non-Nested Models Hypothesis Test

Vuong (1989) proposed a likelihood ratio tests for non-nested hypotheses. This test statistic is provided in the “pscl” package.

Let  $p_i = f(\widehat{y_i} | \widehat{M_1})$ , the estimated probability of observing  $Y_i = y_i$  for the given model (Model 1) and  $f(\cdot)$ . Let  $q_i$  represent the same for the second model (Model 2). Let  $m_i = \log(p_i) - \log(q_i)$ . The Vuong test statistic is,

$$V = \frac{\sqrt{n} \bar{m}}{s_m},$$

where  $n$  is the sample size and  $\bar{m}$  and  $s_m$  are the mean and standard deviation of the  $m_i$ .

```
vuong(mod.nb, mod.p)
```

Vuong Non-Nested Hypothesis Test-Statistic:  
(test-statistic is asymptotically distributed  $N(0,1)$  under the  
null that the models are indistinguishable)

```
-----  
                Vuong z-statistic          H_A    p-value  
Raw                7.357995 model1 > model2 9.337e-14  
AIC-corrected      7.357995 model1 > model2 9.337e-14  
BIC-corrected      7.357995 model1 > model2 9.337e-14
```

```
vuong(mod.nb, mod.zp0)
```

Vuong Non-Nested Hypothesis Test-Statistic:  
(test-statistic is asymptotically distributed  $N(0,1)$  under the  
null that the models are indistinguishable)

```
-----  
                Vuong z-statistic          H_A    p-value  
Raw                4.289635 model1 > model2 8.9483e-06  
AIC-corrected      4.316749 model1 > model2 7.9172e-06  
BIC-corrected      4.386925 model1 > model2 5.7482e-06
```

```
vuong(mod.nb, mod.zp1)
```

Vuong Non-Nested Hypothesis Test-Statistic:  
(test-statistic is asymptotically distributed  $N(0,1)$  under the  
null that the models are indistinguishable)

```
-----  
                Vuong z-statistic          H_A    p-value  
Raw                4.136393 model1 > model2 1.7640e-05
```

|               |          |                 |            |
|---------------|----------|-----------------|------------|
| AIC-corrected | 4.380497 | model1 > model2 | 5.9204e-06 |
| BIC-corrected | 5.012270 | model1 > model2 | 2.6896e-07 |

```
vuong(mod.nb, mod.znb0)
```

Vuong Non-Nested Hypothesis Test-Statistic:  
(test-statistic is asymptotically distributed  $N(0,1)$  under the null that the models are indistinguishable)

|               | Vuong z-statistic | H_A             | p-value  |
|---------------|-------------------|-----------------|----------|
| Raw           | -1.466215         | model2 > model1 | 0.071295 |
| AIC-corrected | -1.091575         | model2 > model1 | 0.137510 |
| BIC-corrected | -0.121958         | model2 > model1 | 0.451466 |

```
vuong(mod.nb, mod.znb1)
```

Vuong Non-Nested Hypothesis Test-Statistic:  
(test-statistic is asymptotically distributed  $N(0,1)$  under the null that the models are indistinguishable)

|               | Vuong z-statistic | H_A             | p-value    |
|---------------|-------------------|-----------------|------------|
| Raw           | -2.656217         | model2 > model1 | 0.0039511  |
| AIC-corrected | -0.631208         | model2 > model1 | 0.2639523  |
| BIC-corrected | 4.609772          | model1 > model2 | 2.0155e-06 |

From these tests, either the negative binomial model or the zero-inflated negative binomial model with covariates in the zero-inflation component are preferred over the other models.

## 8.4 Gamma Regression

Why use the gamma distribution?

Examples of positive, skewed data with increasing variance:

- survival time (or other intervals of time)
- frequency severity model (of insurance claims)

### 8.4.1 The Gamma Distribution

Because we will take the link of the *mean*,  $\mu$ , we will use the following parametrization of the gamma distribution:

$$f(y) = \frac{1}{\Gamma(\nu)} \left( \frac{\nu y}{\mu} \right)^\nu \exp \left( -\frac{\nu y}{\mu} \right) \frac{1}{y} \quad (8.1)$$

Note that this corresponds to the typical  $\alpha$  (shape) and  $\beta$  (rate) parameterization, where  $\nu = \alpha$  and  $\mu = \alpha/\beta$ .

### 8.4.2 Coefficient of Variation

In linear regression, we assume constant variance. Instead, we might want to assume a constant *coefficient of variation*,

$$\begin{aligned} c_v &= \frac{SD(Y)}{E(Y)} \\ \Rightarrow \text{Var}(Y) &= c_v^2 \mu^2. \end{aligned}$$

Therefore, we will model the mean, but the variance will be a linear (quadratic) function of the mean, with a constant coefficient. The gamma distribution accommodates this assumption and from the notation in (Equation 8.1),

$$c_v = \frac{1}{\nu}.$$

Clearly  $c_v$  is just another dispersion parameter (and is treated as such in R).

### 8.4.3 Link Functions

The canonical link is,

$$g(\mu) = -\frac{1}{\mu}$$

(and results in  $t(y) = y$ ).

In practice, the canonical link is rarely used, but instead is replaced by the inverse link,

$$g(\mu) = \frac{1}{\mu},$$

which can be interpreted as the rate of a process. (The coefficients,  $\beta$ , of this model will be negative the coefficients in the canonical link model.) Note that this link function does not account for the fact that  $\mu$  must be  $> 0$ . Therefore, precautions must be taken when computing  $\beta$  to avoid negative values of  $\hat{\mu}$ .

Another link used is the identity link (although this has the same problem as the inverse link). The most popular link is the log link. This is actually very similar to log-normal distribution – particularly for small values of  $c_v$ . However, the gamma GLM tends to be more robust to misspecification than the log-normal regression model.

### 8.4.4 Gamma GLM Examples

(These examples are taken from Chapter 8 of McCullagh and Nelder (1989).)

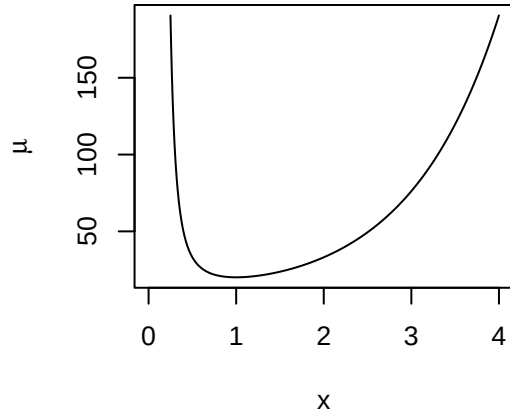
*Using weights (and non-linear parameters in the covariates)*

Powsner (1935) conducted an experiment to determine the effect of temperature on the duration of the developmental stages of the fruit fly *Drosophila melanogaster*. We will examine the data for the embryonic stage. In all cases the eggs were laid at approximately 25°C and remained at that temperature for 20-30 minutes. Subsequently, the eggs were brought to the experimental temperature (14.95-32°C) which was kept constant for the remainder of the experiment (all developmental stages). Powsner also provided batch size at

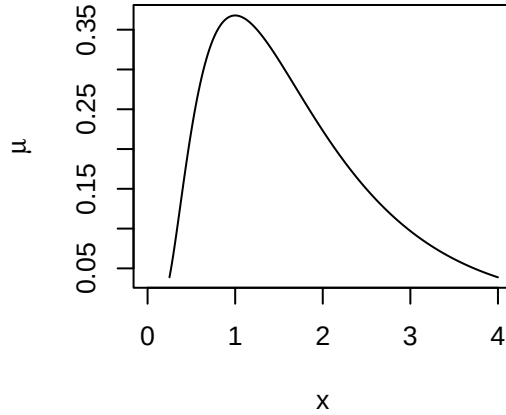
each temperature and the standard deviation of the duration of each stage.

It is known that no development takes place below a certain critical temperature. Therefore, there is probably a corresponding upper limit. In that case, we might want to consider functions of temperature that have asymptotes. The four figures below show different ways of creating asymptotes in the relationship between  $\log \mu$  and  $x$ .

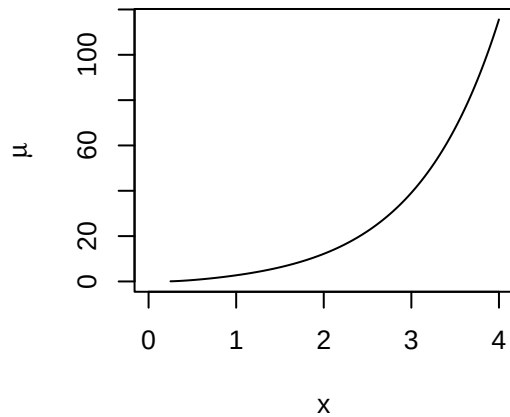
$$\log(\mu) = 1 + x + 1/x$$



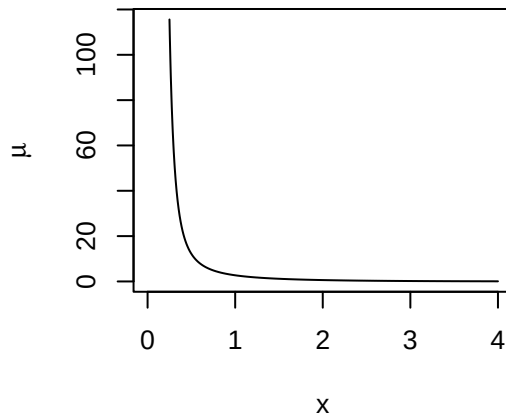
$$\log(\mu) = 1 - x - 1/x$$



$$\log(\mu) = 1 + x - 1/x$$



$$\log(\mu) = 1 - x + 1/x$$

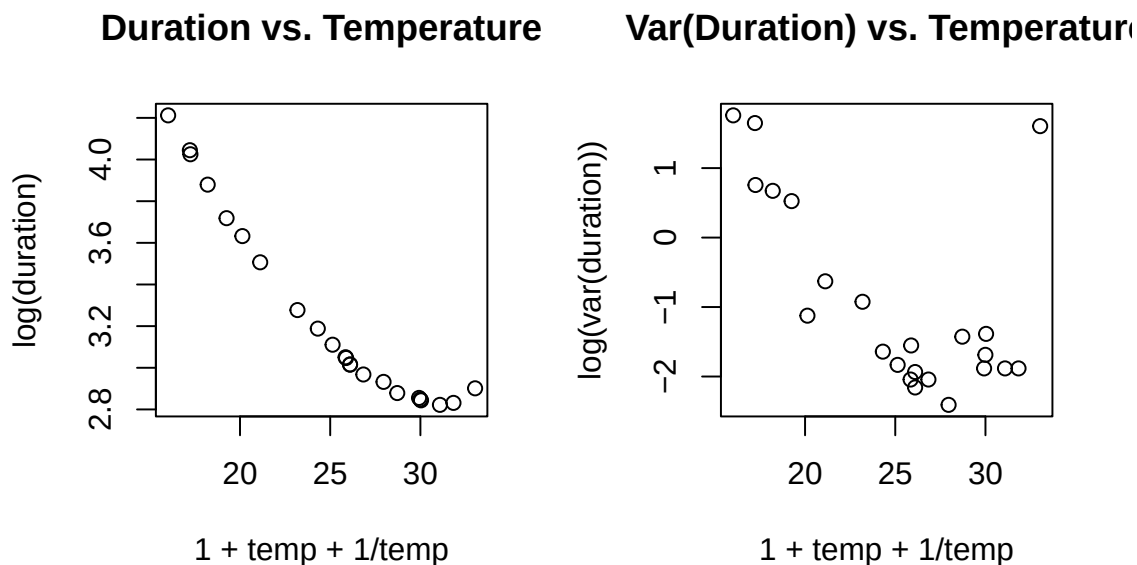


We consider the model

$$\log \mu_i = \beta_0 + \beta_1 T_i + \beta_2 \frac{1}{T_i - \delta},$$

where  $T_i$  represents temperature and  $\delta$  is a nonlinear parameter in the covariate.

```
temp <- c(14.95, 16.16, 16.19, 17.15, 18.20, 19.08, 20.07, 22.14, 23.27, 24.09)
duration <- c(67.5, 57.1, 56, 48.4, 41.2, 37.8, 33.33, 26.50, 24.24, 22.44, 21.44)
batch_size <- c(54, 182, 153, 129, 64, 94, 82, 57, 135, 188, 217, 141, 37, 84, 135)
dur_stdv <- c(2.41, 2.28, 1.46, 1.4, 1.3, .57, .73, .63, .44, .4, .36, .46, .36, .36, .36)
par(mfrow=c(1,2))
plot(1+temp+1/temp, log(duration), xlab="1 + temp + 1/temp", ylab="log(duration)")
plot(1+temp+1/temp, log(dur_stdv^2), xlab="1 + temp + 1/temp", ylab="log(var(duration))")
```



```
temp_inv <- 1/temp
gammod <- glm(duration~temp+ temp_inv, family=Gamma(link="log"), weights=batch_size)
summary(gammod)
```

Call:

```
glm(formula = duration ~ temp + temp_inv, family = Gamma(link = "log"),
```

```

weights = batch_size)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.332263   0.351616  -0.945   0.356
temp         0.037811   0.007807   4.843 9.87e-05 ***
temp_inv     60.826732   3.784146  16.074 6.68e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

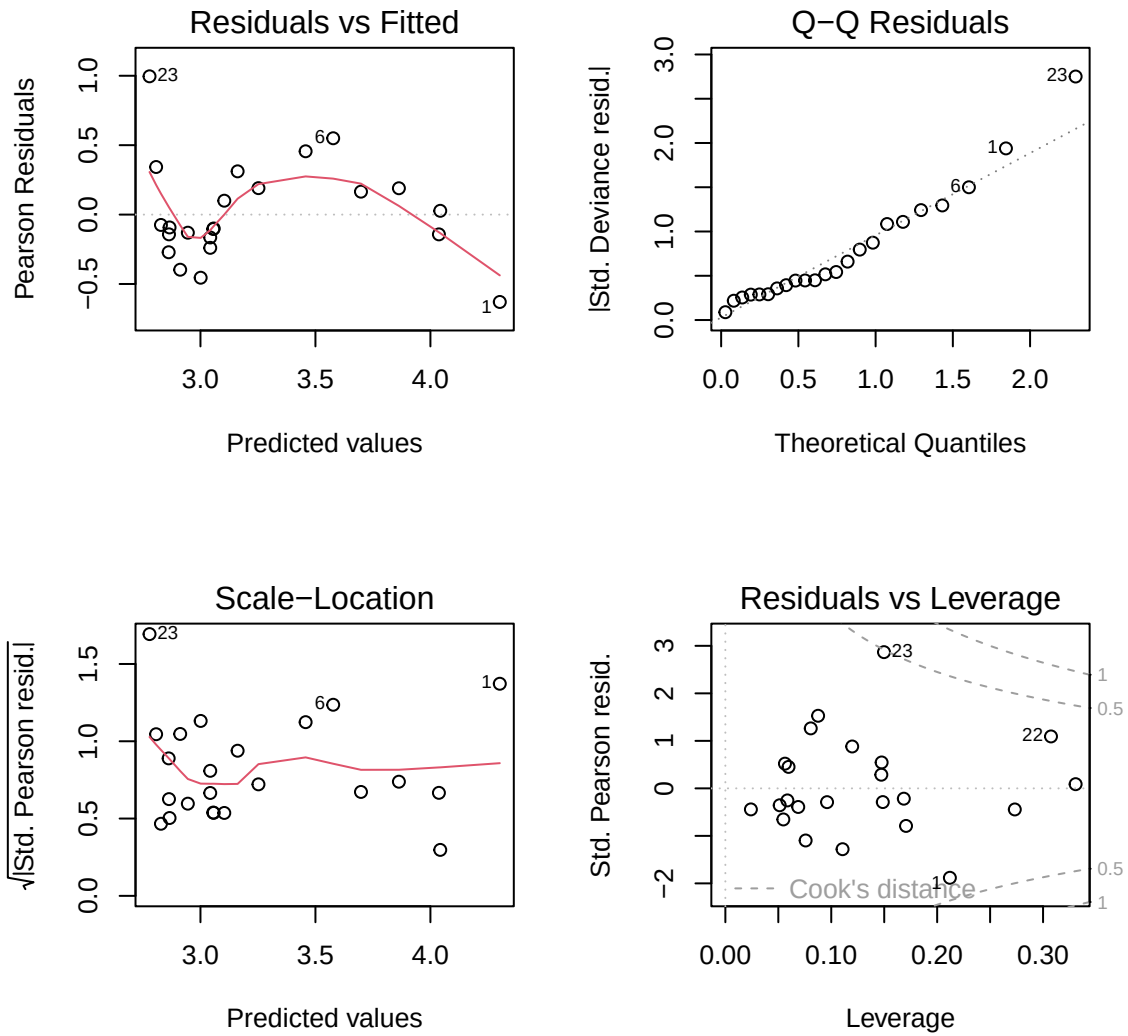
(Dispersion parameter for Gamma family taken to be 0.1419414)

Null deviance: 617.8643  on 22  degrees of freedom
Residual deviance:  2.7727  on 20  degrees of freedom
AIC: 6645.4

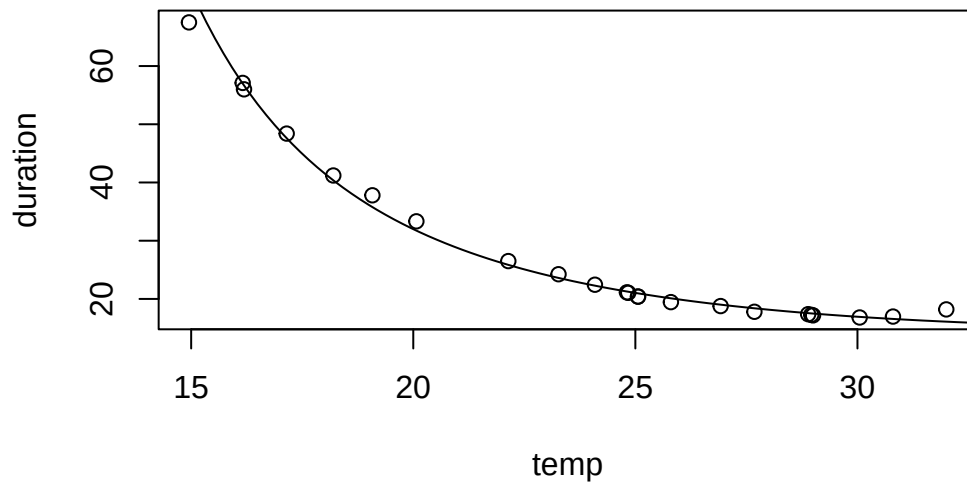
Number of Fisher Scoring iterations: 4

```



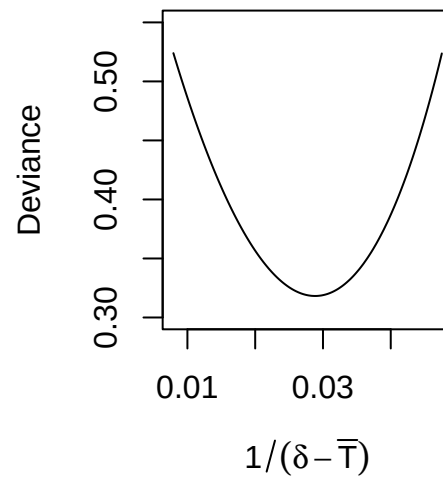
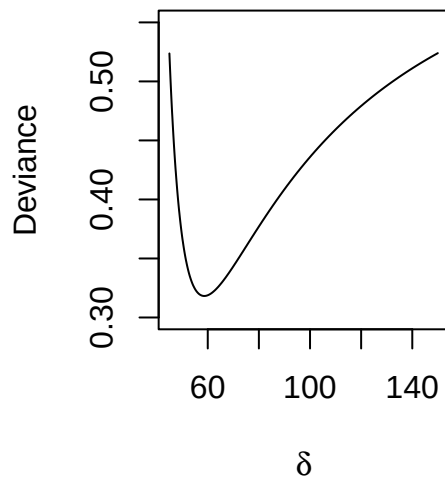


The solid line in the figure below shows the fitted values using the above model ( $\delta = 0$ ).



To find the optimal value of  $\delta$ , we will use deviance. The plots below show the deviance plotted against various values of  $\delta$ .

```
delta.seq <- seq(45, 150, length=1000)
dev.delta <- rep(0, 1000)
for(i in 1:1000){
  temp_inv <- 1/(temp - delta.seq[i])
  gammod <- glm(duration~temp+ temp_inv, family=Gamma(link="log"), weights=b
  dev.delta[i] <- gammod$deviance
}
par(mfrow=c(1,2))
plot(delta.seq, dev.delta, type='l', ylim=c(.3, .55), ylab="Deviance", xlab=ex
plot(1/(delta.seq-mean(temp)), dev.delta, type='l', ylim=c(.3, .55), ylab="Dev
```



The smallest deviance occurs when  $\delta \approx 58.66$ . The fitted line is seen in the dashed line in the plot of the fitted values and the model fit is provided below.

```
mydelta <- delta.seq[which.min(dev.delta)]
temp_inv <- 1/(mydelta - temp)
gammod <- glm(duration~temp+ temp_inv, family=Gamma(link="log"), weights=batch_size)
summary(gammod)
```

Call:

```
glm(formula = duration ~ temp + temp_inv, family = Gamma(link = "log"),
     weights = batch_size)
```

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )   |
|-------------|------------|------------|---------|------------|
| (Intercept) | 3.195901   | 0.044044   | 72.56   | <2e-16 *** |
| temp        | -0.264935  | 0.003662   | -72.34  | <2e-16 *** |
| temp_inv    | 217.460234 | 4.401258   | 49.41   | <2e-16 *** |

---

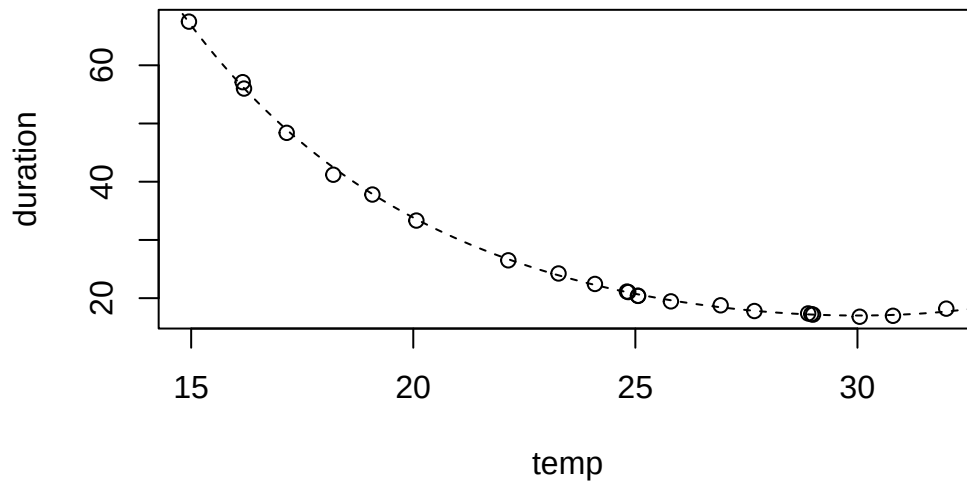
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

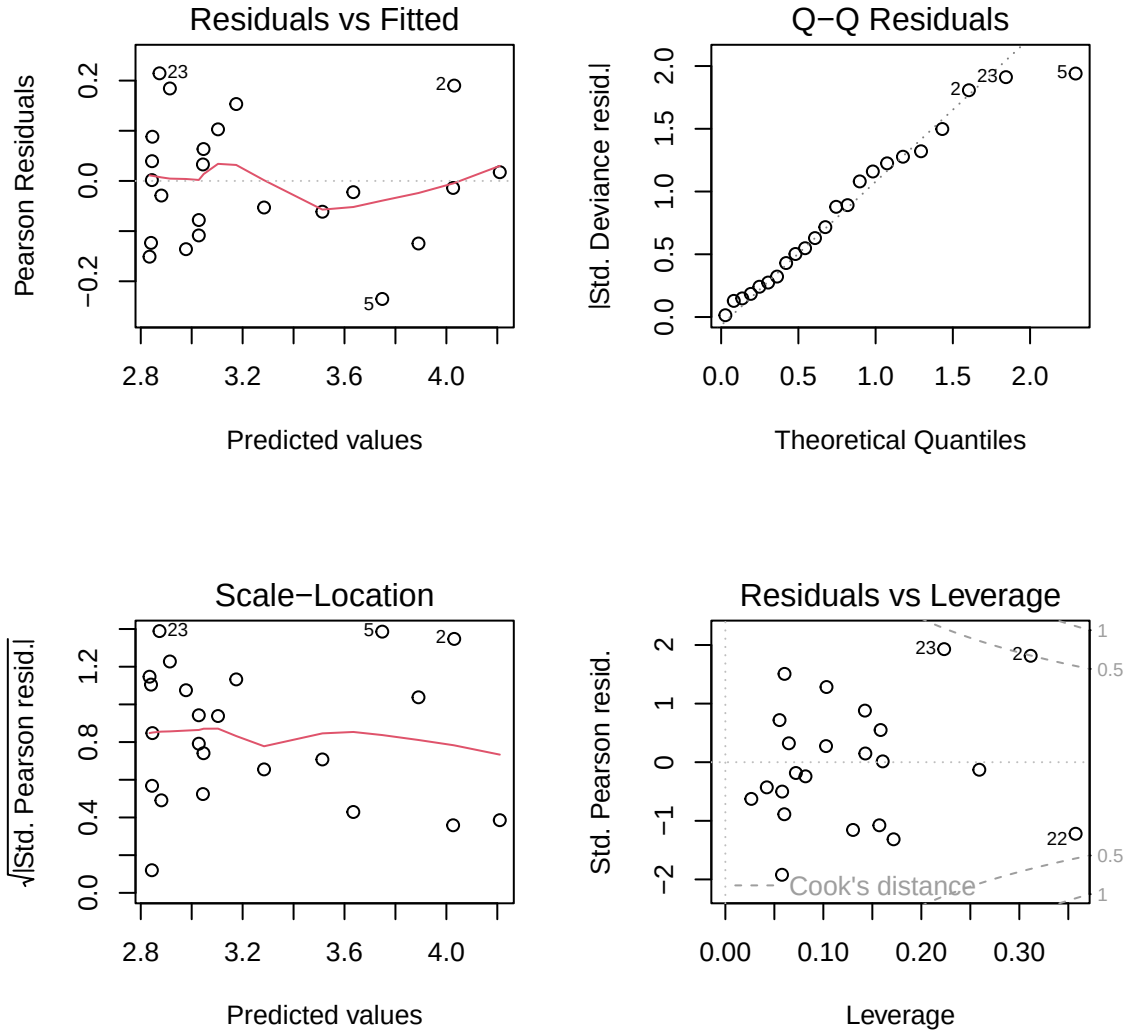
(Dispersion parameter for Gamma family taken to be 0.01591696)

Null deviance: 617.86435 on 22 degrees of freedom  
Residual deviance: 0.31823 on 20 degrees of freedom  
AIC: 418.98

Number of Fisher Scoring iterations: 3

```
par(mfrow=c(1,1))  
plot(temp, duration)  
x.seq <- seq(14, 33, length=100)  
lines(x.seq, predict(gammod, newdata=data.frame(temp=x.seq, temp_inv=(1/(mydel
```





Note that because we estimated  $\delta$ , the degrees of freedom should be 19 and not 20. The estimated dispersion parameter should therefore be  $0.31823/19 = 0.01674895$  and the estimated coefficient of variation is

$$c_v = \sqrt{0.01674895} = 0.1294177.$$

Because we used batch size as weights, this is the estimated coefficient of variation for the duration of the embryonic period of *individual eggs*. The estimated coefficient of variation of the batch means is then  $0.1294177/\sqrt{m_i}$ , where  $m_i$  is the batch size.

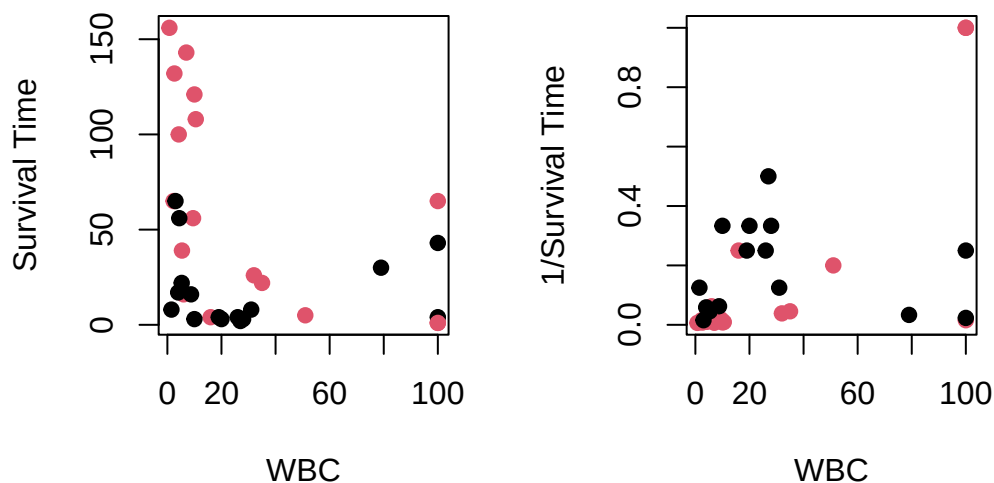
We can think of the above coefficient of variation as the between-batch variation. It is interesting to note that the within-batch variation (batch

standard deviation divided by the batch mean) is approximately 0.0278. Therefore, the between-batch coefficient of variation is much larger than the within-batch variation (4.65 times), implying that there were other differences than just temperature in the batches (perhaps humidity, lighting, time of day, etc.).

### *Inverse link function*

We examine data of survival times (in weeks) for leukemia patients. There are two covariates, white blood cell count in thousands (WBC) and the results of a test (AG positive or AG negative). The goal is to find a suitable model to predict survival time as a function of these covariates.

```
leuk <- read.csv('~Documents/Teaching/2024Stat637Winter/Notes/gamma/leukemia.
leuk$AG <- as.factor(leuk$AG)
par(mfrow=c(1,2))
plot(leuk$WBC, leuk$SurvTime, pch=19, col=leuk$AG, xlab="WBC", ylab="Survival
plot(leuk$WBC, 1/leuk$SurvTime, pch=19, col=leuk$AG, xlab="WBC", ylab="1/Survi
```



```
leukmod <- glm(SurvTime~WBC+AG, family=Gamma, data=leuk) #default link is inve
summary(leukmod)
```

```

Call:
glm(formula = SurvTime ~ WBC + AG, family = Gamma, data = leuk)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.0456971  0.0153962   2.968  0.00584 **
WBC          0.0004241  0.0002533   1.674  0.10445
AGPositive  -0.0359176  0.0153418  -2.341  0.02606 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 1.148902)

Null deviance: 57.852  on 32  degrees of freedom
Residual deviance: 40.776  on 30  degrees of freedom
AIC: 302.16

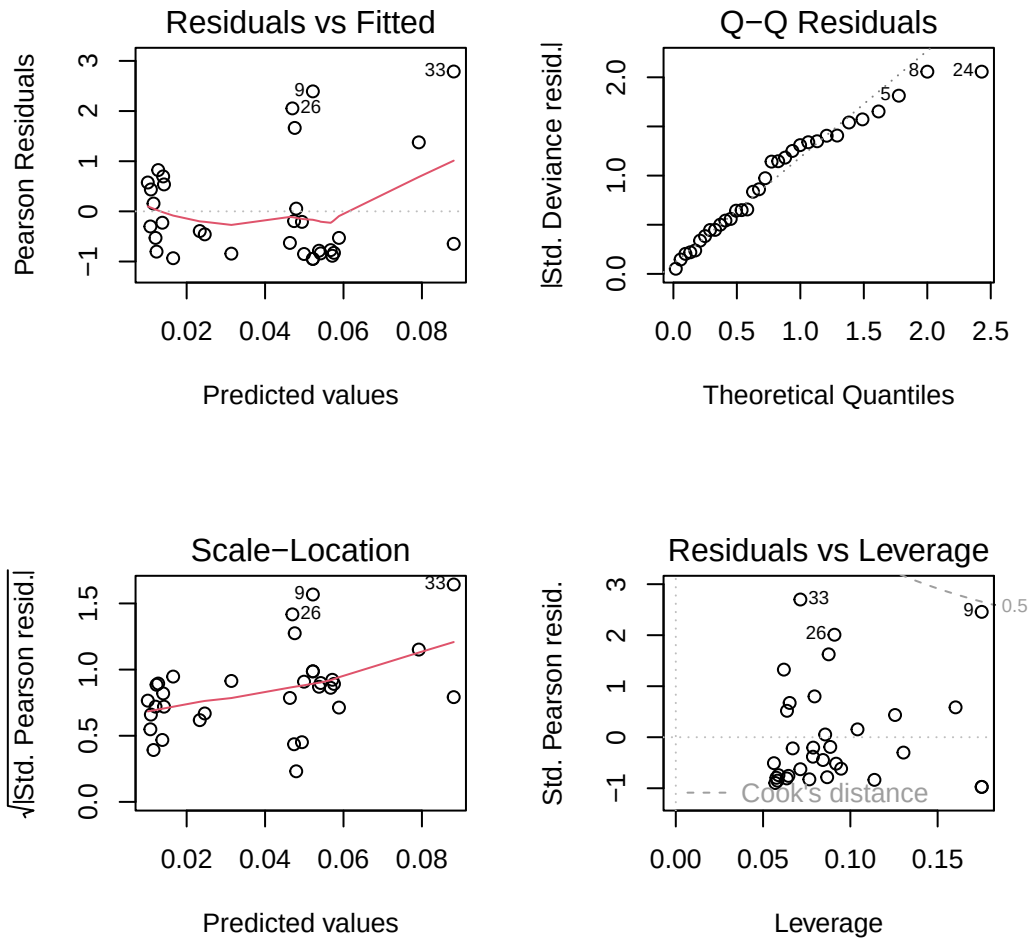
Number of Fisher Scoring iterations: 7

```

```

#interpretation of parameters?
par(mfrow=c(2,2))
plot(leukmod)

```



Compare to model with the log link:

```
leukmod1 <- glm(SurvTime~WBC+AG, family=Gamma(link="log"), data=leuk)
summary(leukmod1)
```

Call:

```
glm(formula = SurvTime ~ WBC + AG, family = Gamma(link = "log"),
    data = leuk)
```

Coefficients:

|  | Estimate | Std. Error | t value | Pr(> t ) |
|--|----------|------------|---------|----------|
|--|----------|------------|---------|----------|



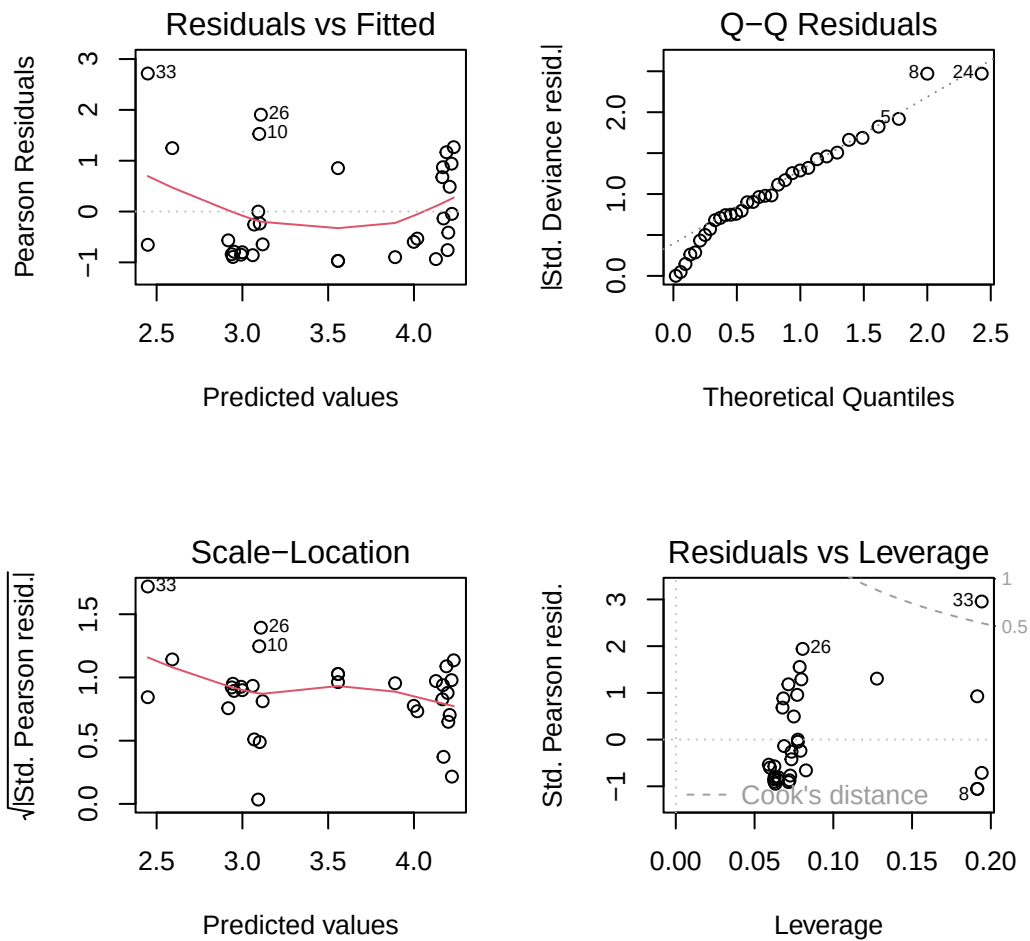
```
(Intercept)  3.128219    0.298200   10.490 1.48e-11 ***
WBC          -0.006797    0.005246   -1.296  0.20499
AGPositive   1.108718    0.356535    3.110  0.00408 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 1.047745)

Null deviance: 57.852  on 32  degrees of freedom
Residual deviance: 44.121  on 30  degrees of freedom
AIC: 305.26

Number of Fisher Scoring iterations: 12
```

```
par(mfrow=c(2,2))
plot(leukmod1)
```



### *Gamma vs. log-normal regression*

Consider the resistivity of a wafer using a full factorial design of four two-level factors.

```
library(faraway)
```

Attaching package: 'faraway'

The following object is masked from 'package:lme4':

```
toenail
```

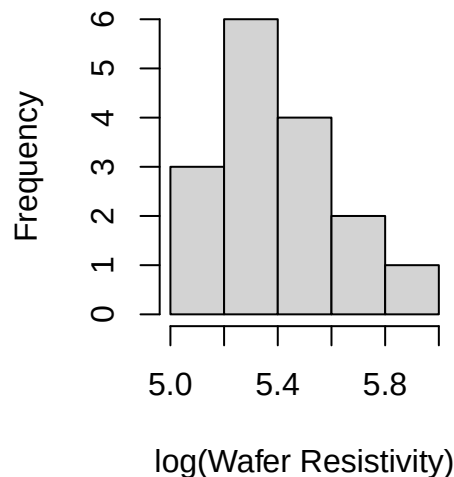
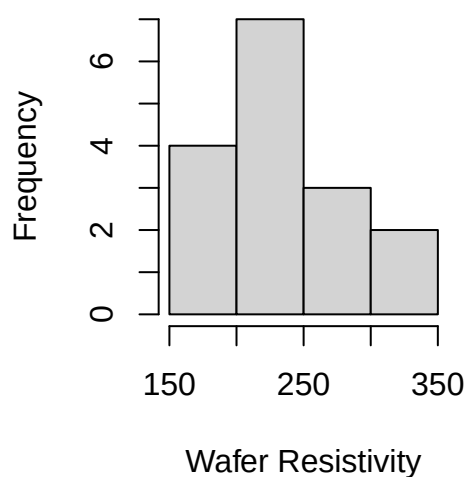
The following object is masked from 'package:geepack':

```
ohio
```

```
data(wafer)

attach(wafer)
par(mfrow=c(1,2))
hist(resist, xlab="Wafer Resistivity", main="Histogram of Wafer Resistivity")
hist(log(resist), xlab="log(Wafer Resistivity)", main="Histogram of log Wafer Resistivity")
```

### Histogram of Wafer Resistivity Histogram of log Wafer Resistivity



Gamma with log link model:

```
gmod <- glm(resist~x1+x2+x3+x4, family=Gamma(link='log'))
summary(gmod)
```

```

Call:
glm(formula = resist ~ x1 + x2 + x3 + x4, family = Gamma(link = "log"))

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.44552     0.05856  92.983  < 2e-16 ***
x1+          0.12115     0.05238   2.313  0.041090 *
x2+         -0.30049     0.05238  -5.736  0.000131 ***
x3+          0.17979     0.05238   3.432  0.005601 **
x4+         -0.05757     0.05238  -1.099  0.295248
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.01097542)

Null deviance: 0.69784  on 15  degrees of freedom
Residual deviance: 0.12418  on 11  degrees of freedom
AIC: 152.91

Number of Fisher Scoring iterations: 4

RMSE: 20.5260659

Log-normal model (identity link):

```

```

lnmod <- glm(log(resist)~x1+x2+x3+x4)
summary(lnmod)

```

```

Call:
glm(formula = log(resist) ~ x1 + x2 + x3 + x4)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.44048     0.05982  90.948  < 2e-16 ***

```

```

x1+          0.12277      0.05350    2.295 0.042432 *
x2+         -0.29986      0.05350   -5.604 0.000159 ***
x3+          0.17844      0.05350    3.335 0.006652 **
x4+         -0.05615      0.05350   -1.049 0.316515
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.01145078)

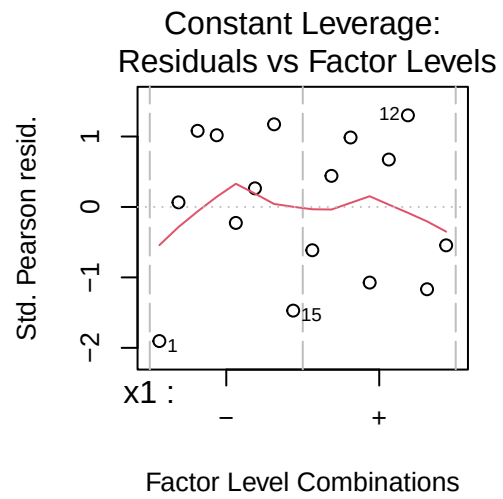
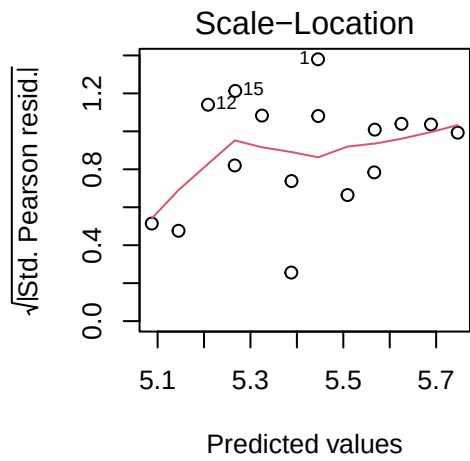
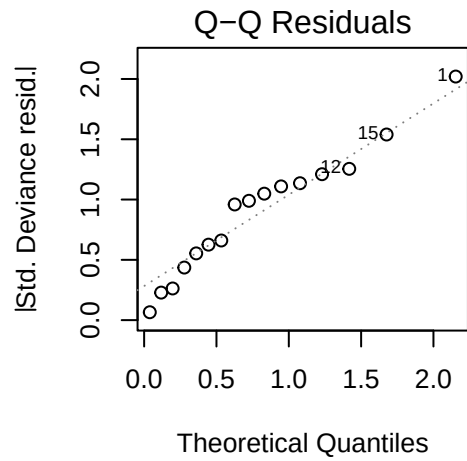
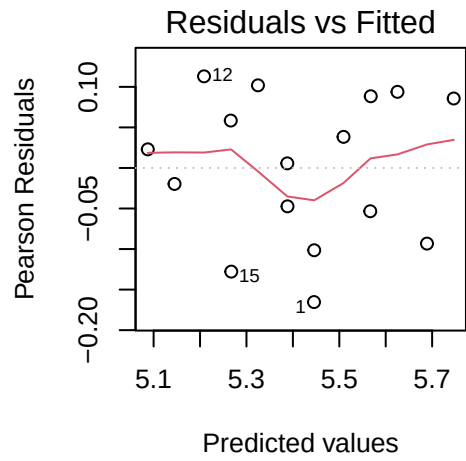
    Null deviance: 0.68587  on 15  degrees of freedom
Residual deviance: 0.12596  on 11  degrees of freedom
AIC: -20.104

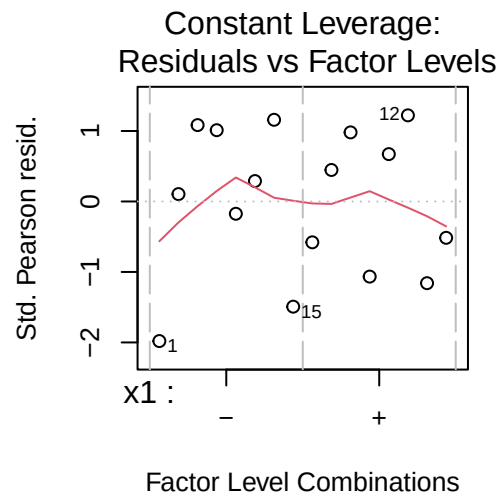
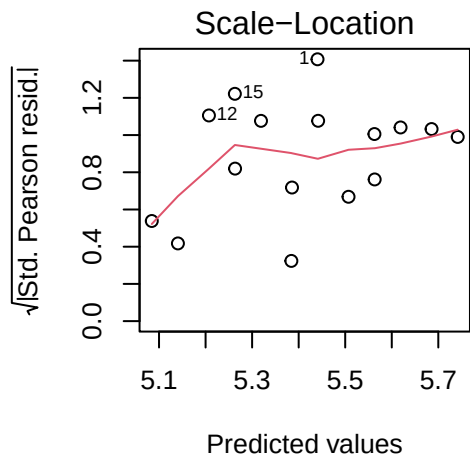
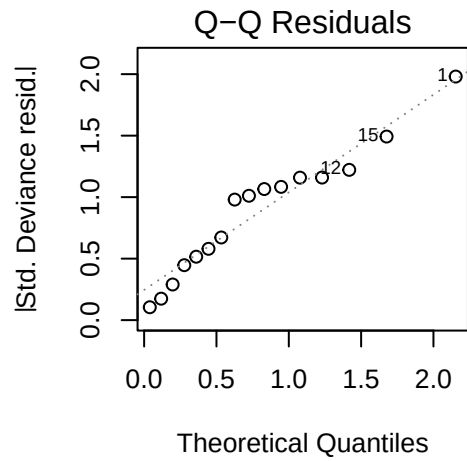
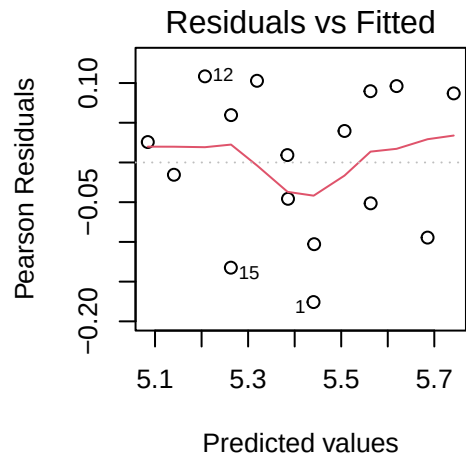
Number of Fisher Scoring iterations: 2

RMSE: 20.6196791

Comparison of residual plots:

```





## 8.5 Homework Assignment

1. Find a paper in a well-reputed journal that makes use of either a zero-inflated model or gamma regression model. Write an evaluation of the paper using the rubric below. Make sure you provide evidences for your evaluation in each area. This assignment should be around 3 pages.
  - a. Model Justification
    - Outstanding: The paper provides a clear, logical justification for using gamma regression or a zero-inflated model. Assumptions of the model are explicitly discussed and shown to align with the data characteristics (e.g., skewness, overdispersion, or excess zeros).
    - Proficient: The paper provides some justification for using the model, but details on why it was chosen over alternatives are incomplete or implied.
    - Developing: The justification is vague or unclear. Assumptions of the model are mentioned but not tied to the data.
    - Needs Improvement: The paper does not justify the choice of model or ignores its assumptions.
  - b. Data Description and Model Fit
    - Outstanding: The data are thoroughly described, including key characteristics (e.g., distributions, presence of zeros). Model fit is evaluated using appropriate metrics or diagnostics, and results are well-explained.
    - Proficient: The data are described, but the connection between data characteristics and model fit is less detailed. Model fit is evaluated, but some metrics or diagnostics are missing or unclear.
    - Developing: Data description is minimal or unclear, and model fit is only superficially evaluated.
    - Needs Improvement: Data characteristics are not discussed, and no model fit evaluation is presented.
  - c. Interpretation and Results
    - Outstanding: The results are clearly interpreted, with a focus on meaningful conclusions related to the research question. The implications of parameter estimates are discussed in context.
    - Proficient: Results are interpreted, but some conclusions are unclear or lack sufficient context.



- Developing: Results are minimally interpreted or lack clear connection to the research question.
- Needs Improvement: Results are presented without interpretation or discussion of their meaning.

d. Model Limitations and Alternative Approaches

- Outstanding: The paper discusses limitations of the chosen model and considers alternative approaches. Trade-offs between models are critically evaluated.
- Proficient: Limitations are acknowledged, but alternative models are only briefly mentioned or not well-discussed.
- Developing: Limitations are minimally discussed, and alternative approaches are not considered.
- Needs Improvement: Limitations and alternative models are not mentioned.

e. Statistical Rigor and Reproducibility

- Outstanding: The paper demonstrates strong statistical rigor, with clear reporting of methods, assumptions, and results. Sufficient detail is provided to ensure reproducibility.
- Proficient: Methods and results are clearly reported, but some details are missing for full reproducibility.
- Developing: Methods are vaguely described, and reproducibility would be difficult.
- Needs Improvement: The paper lacks statistical rigor, with poorly reported methods and insufficient detail.

f. Overall Quality of Writing and Presentation

- Outstanding: The paper is well-written, organized, and easy to follow, with clear tables and figures that enhance understanding.
- Proficient: The paper is generally clear and organized, but some sections or figures could be improved.
- Developing: The paper is difficult to follow in places, with unclear tables or figures.
- Needs Improvement: The paper is poorly organized and difficult to understand.

2. Reflect (1 paragraph): What did you learn about GLM's (and communication of GLM's) from this article and what will you change about your future analyses because of this paper?

## 8.6 Dynamic Generalized Linear Models

### 8.6.1 Dynamic Linear Models

DLM idea: coefficients/parameters change with time.

Simplest case: Let  $Y_t$  be an  $n \times 1$  vector of observations ( $n$  could be 1) for  $t = 1, \dots, T$ . We model

$$\begin{aligned} Y_t &= \mu_t + \epsilon_t \\ \mu_t &= \mu_{t-1} + \eta_t \\ \epsilon_t &\sim N(0, H_t) \\ \eta_t &\sim N(0, Q_t) \\ \mu_0 &\sim N(a_0, P_0) \end{aligned}$$

where

- $\mu_t$  is an unknown  $n \times 1$  mean vector,
- $\epsilon_t$  is Gaussian noise with unknown variance-covariance matrix  $H_t$  (if  $n = 1$ ,  $H_t$  is a scalar; the simplest model assumes  $H_t = h^2$ ),
- $\eta_t$  is Gaussian noise with unknown variance-covariance matrix  $Q_t$  (again, if  $n = 1$ ,  $Q_t$  is a scalar and the simplest model assumes  $Q_t = q^2$ ), and
- $a_0$  and  $P_0$  are the known (or assigned) mean and variance, respectively, of the initial mean state  $\mu_0$ .

$H_t$  and  $Q_t$  are the unknowns. They can be estimated using maximum likelihood or Bayesian methods. (Note:  $\mu_t$  is also unknown, but it doesn't have a likelihood so it can be found directly.)

*Simple IDE (integro-differential equation) case*

$$\begin{aligned} \underbrace{Y_t}_{n \times 1} &= \underbrace{\mu_t}_{n \times 1} + \underbrace{\epsilon_t}_{n \times 1} \\ \underbrace{\mu_t}_{n \times 1} &= \int \underbrace{k_t(\cdot, s)}_{1 \times n} \underbrace{\mu_{t-1}}_{n \times 1} ds + \underbrace{\eta_t}_{n \times 1} \\ &\approx \underbrace{K_t}_{n \times n} \mu_{t-1} + \eta_t \end{aligned}$$

where  $k_t(\cdot)$  is a smoothing kernel (usually in space).

$$\begin{aligned}\epsilon_t &\sim N(0, H_t) \\ \eta_t &\sim N(0, Q_t) \\ \mu_0 &\sim N(a_0, P_0)\end{aligned}$$

*Regression:*

$$\begin{aligned}\underbrace{Y_t}_{n \times 1} &= \underbrace{X_t}_{n \times p} \underbrace{\beta_t}_{p \times 1} + \epsilon_t \\ \beta_t &= \underbrace{T_t}_{p \times p} \beta_{t-1} + \underbrace{\eta_t}_{p \times 1}\end{aligned}$$

where  $T_t$  can be modeled but is often fixed. One example is

$$T_t = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

so that for the  $j$ th element of  $\beta_t$ ,

$$\beta_{tj} = \beta_{(t-1)j} + \eta_{tj}$$

Finally,

$$\begin{aligned}\epsilon_t &\sim N(0, H_t) \\ \eta_t &\sim N(0, Q_t) \\ \beta_0 &\sim N(a_0, P_0)\end{aligned}$$

*Seasonal Model:*

$$\begin{aligned}Y_t &= \alpha_t + \gamma_{tj}I(\text{month}(t) = j) + \epsilon_t \text{ (for identifiability, remove the intercept so that)} \\ &= \gamma_{tj}I(\text{month}(t) = j) + \epsilon_t\end{aligned}$$

Writing in matrix form, we have

$$Y_t = \underbrace{X}_{n \times 12} \underbrace{\gamma_t}_{12 \times 1} + \epsilon_t$$

$$\gamma_t = T\gamma_{t-1} + \eta_t$$

If  $T = I$ , then  $\gamma_{tj} = \gamma_{(t-1)j} + \eta_{tj}$ , but  $\gamma_{tj}$  probably also depends on  $\gamma_{(t-1),(j-1)}$ , so

$$T = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & 1 & 1 \end{bmatrix}$$

$$\Rightarrow \gamma_{tj} = \gamma_{(t-1),j} + \gamma_{(t-1),(j-1)} + \eta_{tj}$$

Finally,

$$\epsilon_t \sim N(0, H_t)$$

$$\eta_t \sim N(0, Q_t)$$

$$\gamma_0 \sim N(a_0, P_0)$$

## 8.6.2 Extending to the Exponential Family

| Normal Linear Regression                                                                                                         | GLM                                                             |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| $Y = x\beta + \epsilon$<br>$\epsilon \sim N(0, \sigma^2)$<br>$\Rightarrow Y \sim N(X\beta, \sigma^2)$<br>$g(\mu) = \mu = x\beta$ | $\rightarrow Y \sim f(y; \mu)$<br>$\rightarrow g(\mu) = x\beta$ |
| Normal Linear Mixed Model                                                                                                        | GLMM                                                            |
| $Y = x\beta + zu + \epsilon$<br>$\epsilon \sim N(0, \sigma^2)$<br>$\Rightarrow Y u \sim N(\mu, \sigma^2)$                        | $\rightarrow Y u \sim f(y; \mu)$                                |

| Normal Linear Mixed Model                                                                                                                                                                                                                                                                   | GLMM                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $g(\mu) = \mu = x\beta + zu$<br>$u \sim N(0, \sigma_u^2)$                                                                                                                                                                                                                                   | $\rightarrow g(\mu) = x\beta + zu$<br>$u \sim N(0, \sigma_u^2)$                                                                                                               |
| DLM                                                                                                                                                                                                                                                                                         | DGLM                                                                                                                                                                          |
| $Y_t = x_t\beta_t + \epsilon_t$<br>$\epsilon_t \sim N(0, H)$<br>$\beta_t = T\beta_{t-1} + \eta_t$<br>$\eta_t \sim N(0, Q)$<br>$\Rightarrow Y_t \sim N(\mu_t, H)$<br>$g(\mu_t) = \mu_t = x_t\beta_t$<br>$\beta_t = T\beta_{t-1} + \eta_t$<br>$\eta_t \sim N(0, Q), \beta_0 \sim N(a_0, P_0)$ | $\rightarrow Y_t \sim f(y; \mu_t)$<br>$\rightarrow g(\mu_t) = x_t\beta_t$<br>$\rightarrow \beta_t = T\beta_{t-1} + \eta_t$<br>$\eta_t \sim N(0, Q), \beta_0 \sim N(a_0, P_0)$ |

*Note on estimation:*

Singh and Roberts (1992) apply Kalman Filtering to get (posterior mode estimators) modified equation:

$$\begin{aligned}
\tilde{Y}_t &= F_t' \beta_t + \tilde{\epsilon}_t, \tilde{\epsilon}_t \sim N(0, \tilde{H}_t) \\
&= g(\mu_t) + (y_t - \mu_t)g'(\mu_t) \\
&= F_t' \beta_t \\
\text{and } \tilde{H}_t &= b''(\theta_t)[g'(\mu_t)]^2
\end{aligned}$$

Does this look familiar?!

*Tools/References:*

- R package: “KFAS” Helske (2017)
- West et al. (1985)
- Triantafyllopoulos (2009)
- Alves et al. (2022)

*Infant Bronchiolitis Counts: Poisson Likelihood Distribution*

- $Y_t$  represents the number of cases of infant bronchiolitis at time  $t$

aggregated over all the US

- $U_t$  represents the number of controls (“population size”) at time  $t$  (i.e., infants that could have contracted bronchiolitis)

**Model:**

$$Y_t \sim \text{Pois}(U_t e^{\theta_t})$$

$$\mu_t = U_t e^{\theta_t} \Rightarrow \log \mu_t = \theta_t + \underbrace{\log U_t}_{\text{offset}}$$

Notice:  $\log \frac{\mu_t}{U_t} = \theta_t$  so that  $\theta_t$  is a log rate.

Dynamic models we’ll consider:

1.  $\theta_t = \theta_{t-1} + \eta_t, \eta_t \sim N(0, Q)$
- 2.

$$\theta_t = X_t \underbrace{\gamma_t}_{\text{month effect}}$$

$$\gamma_t = \gamma_{(t-1)} + \eta_t,$$

where  $\eta_t \sim N(0, Q)$

3.  $\theta_t = X_t \beta_t$  and  $\beta_t = \beta_{t-1} + \eta_t$  where  $\eta_t \sim N(0, Q)$

*KFAS notation for the normal DLM:*

$$Y_t = Z_t \alpha_t + \epsilon_t$$

$$\alpha_{t+1} = T_t \alpha_t + R_t \eta_t$$

$$\epsilon_t \sim N(0, H_t) \text{ (not used for GLM)}$$

$$\eta_t \sim N(0, Q_t) \text{ (specify } Q \text{ to be NA to estimate it)}$$

$$\alpha_1 \sim N(a_1, P_1)$$

$$P_1 = \kappa P_\infty + P_{*,1}$$

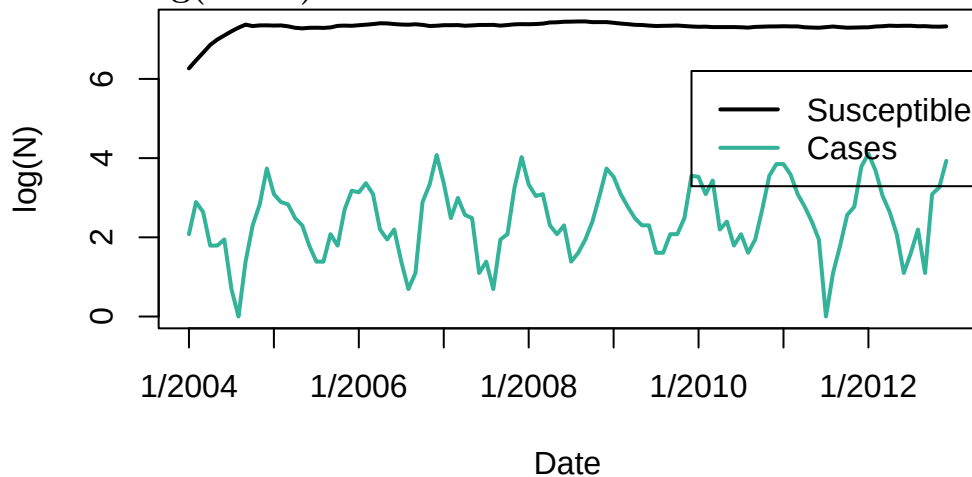
Note about the seasonality:  $\gamma_{t,11} = -\gamma_{(t-1),1} + \gamma_{(t-1),11}$

*Example: Bronchiolitis Incidence Proportion*

- This data is simulated (due to privacy of original data). Cases, popula-

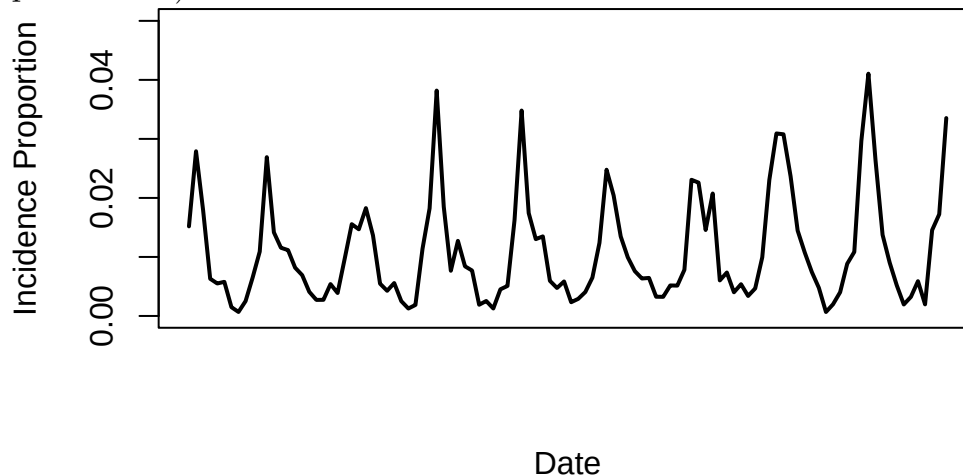
- tion, and rates are similar to observed data.
- US infant ( $< 2$  years old) bronchiolitis cases each month from January 2004 through December 2012 as reported by a nation-wide healthcare system
  - Number of susceptible infants (number of infants insured by the healthcare system) each month

**Plot of  $\log(\text{count})$**



**Plot of Bronchiolitis Incidence Proportions**

Incidence proportion is the population proportion of new cases of a disease for a given time. (Biostatistics/epidemiology terms to learn: incidence vs prevalence.)



**Model 0: Constant Mean**  
(GLM with intercept)



```
mod.glm <- glm(counts~1, family=poisson, offset=log(sus), data=dat)
summary(mod.glm)
```

```
Call:
glm(formula = counts ~ 1, family = poisson, data = dat, offset = log(sus))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -4.52796    0.02379  -190.3   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1088.7  on 107  degrees of freedom
Residual deviance: 1088.7  on 107  degrees of freedom
AIC: 1555.2

Number of Fisher Scoring iterations: 5
```

```
exp(coef(mod.glm))
```

```
(Intercept)
0.01080265
```

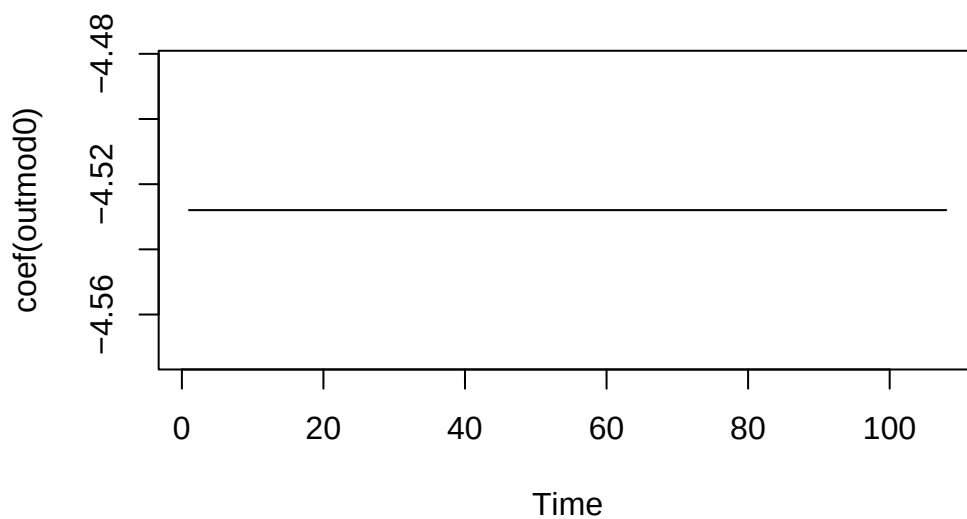
**Model 0 fit with KFAS package**

```
library(KFAS)
```

Please cite KFAS in publications by using:

Jouni Helske (2017). KFAS: Exponential Family State Space Models in R. *Journal of Statistical Software* 39. doi:10.18637/jss.v078.i10.

```
mod0 <- SSModel(counts~1, distribution="poisson", u=dat$sus, data=dat) #Fits a model
mod0fit <- fitSSM(mod0, inits=0, method="BFGS") #inits are the initial values
outmod0 <- KFS(mod0fit$model)
plot(coef(outmod0))
```



```
exp(coef(outmod0)) #compare to the model fit with glm()
```

Time Series:

Start = 1

End = 108

Frequency = 1

```
[1] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[7] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[13] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
```

```

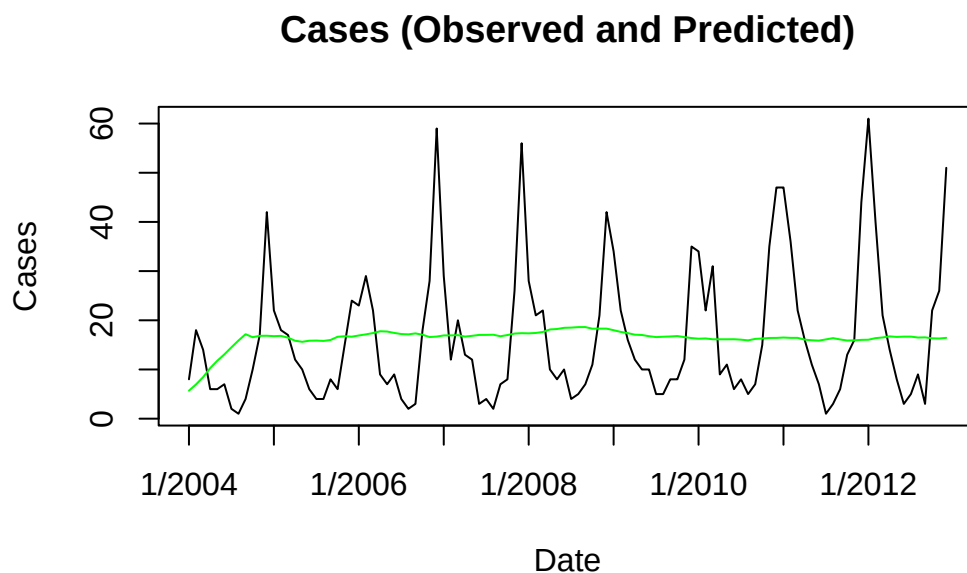
[19] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[25] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[31] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[37] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[43] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[49] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[55] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[61] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[67] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[73] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[79] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[85] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[91] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[97] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265
[103] 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265 0.01080265

```

```

plot(dat$counts, type='l', ylab="Cases", main="Cases (Observed and Predicted)"
axis(1, at=xx, labels=paste(dat$month[xx],"/", dat$year[xx] , sep="" ))
lines(predict(mod0fit$model), col='green')

```

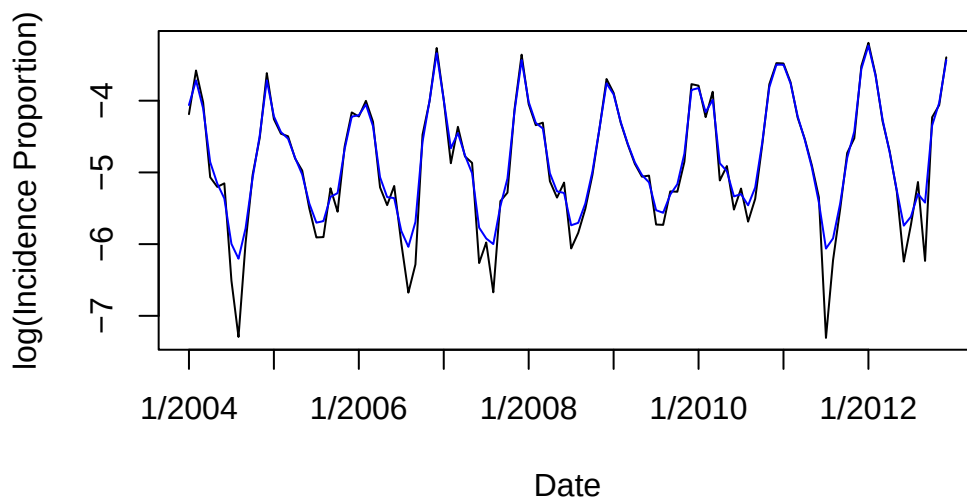


**Model 1: AR(1) on mean**

```

mod1 <- SSMModel(counts~-1+SSMcustom(Z=matrix(1), T=matrix(1), R=matrix(1), Q=m
mod1fit <- fitSSM(mod1, inits=1, method="BFGS")
outmod1.state <- KFS(mod1fit$model) #filters on the state
plot(log(dat$counts/dat$sus), type='l', ylab="log(Incidence Proportion)", xaxt
axis(1, at=xx, labels=paste(dat$month[xx],"/", dat$year[xx] , sep="" ))
lines(coef(outmod1.state), col='blue')

```

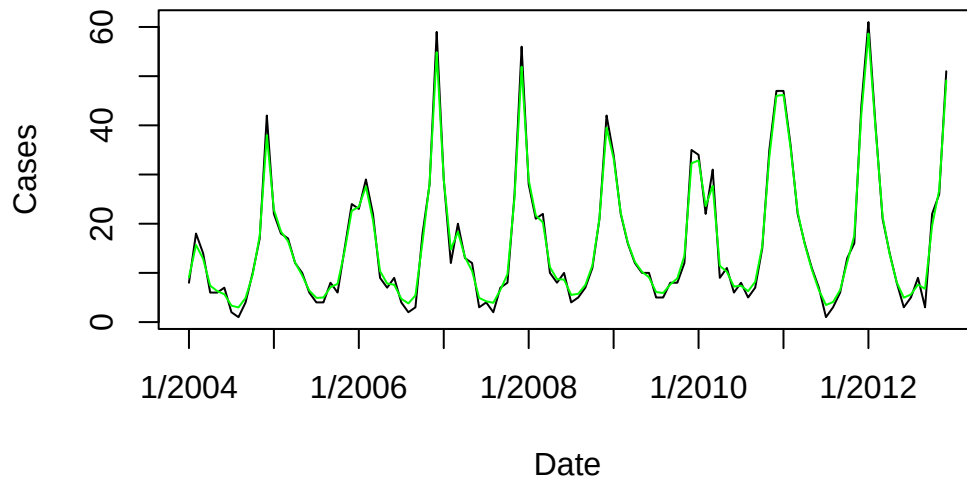


```

plot(dat$counts, type='l', ylab="Cases", main="Cases (Observed and Predicted)"
axis(1, at=xx, labels=paste(dat$month[xx],"/", dat$year[xx] , sep="" ))
lines(predict(mod1fit$model), col='green')

```

## Cases (Observed and Predicted)



Model 2: Seasonal component

```
mod2 <- SSMModel(counts~-1+SSMseasonal(period=12, Q=matrix(NA), sea.type="dummy",
mod2$T
```

```
, , 1
```

|             | sea_dummy1 | sea_dummy2 | sea_dummy3 | sea_dummy4 | sea_dummy5 | sea_dummy6 |
|-------------|------------|------------|------------|------------|------------|------------|
| sea_dummy1  | -1         | -1         | -1         | -          |            |            |
| 1           | -1         | -1         |            |            |            |            |
| sea_dummy2  | 1          | 0          | 0          | 0          | 0          | 0          |
| sea_dummy3  | 0          | 1          | 0          | 0          | 0          | 0          |
| sea_dummy4  | 0          | 0          | 1          | 0          | 0          | 0          |
| sea_dummy5  | 0          | 0          | 0          | 1          | 0          | 0          |
| sea_dummy6  | 0          | 0          | 0          | 0          | 1          | 0          |
| sea_dummy7  | 0          | 0          | 0          | 0          | 0          | 1          |
| sea_dummy8  | 0          | 0          | 0          | 0          | 0          | 0          |
| sea_dummy9  | 0          | 0          | 0          | 0          | 0          | 0          |
| sea_dummy10 | 0          | 0          | 0          | 0          | 0          | 0          |
| sea_dummy11 | 0          | 0          | 0          | 0          | 0          | 0          |

|             |    |    |    |   |   |
|-------------|----|----|----|---|---|
| sea_dummy1  | -1 | -1 | -1 | - |   |
| 1           | -1 |    |    |   |   |
| sea_dummy2  | 0  | 0  | 0  | 0 | 0 |
| sea_dummy3  | 0  | 0  | 0  | 0 | 0 |
| sea_dummy4  | 0  | 0  | 0  | 0 | 0 |
| sea_dummy5  | 0  | 0  | 0  | 0 | 0 |
| sea_dummy6  | 0  | 0  | 0  | 0 | 0 |
| sea_dummy7  | 0  | 0  | 0  | 0 | 0 |
| sea_dummy8  | 1  | 0  | 0  | 0 | 0 |
| sea_dummy9  | 0  | 1  | 0  | 0 | 0 |
| sea_dummy10 | 0  | 0  | 1  | 0 | 0 |
| sea_dummy11 | 0  | 0  | 0  | 1 | 0 |

```
mod2$Z
```

```
, , 1
      sea_dummy1 sea_dummy2 sea_dummy3 sea_dummy4 sea_dummy5 sea_dummy6
[1,]          1          0          0          0          0          0
      sea_dummy7 sea_dummy8 sea_dummy9 sea_dummy10 sea_dummy11
[1,]          0          0          0          0          0
```

```
mod2$Q
```

```
, , 1
      [,1]
[1,]    NA
```

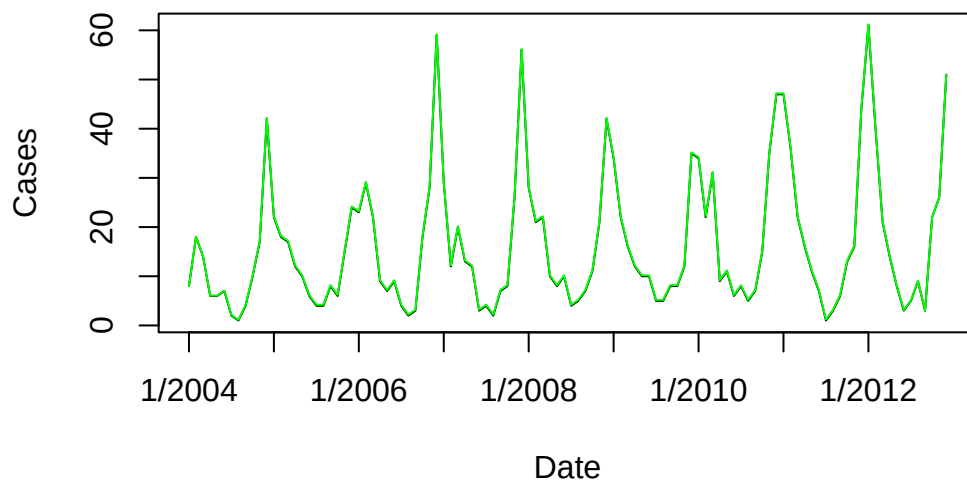
```
mod2$R
```

```
, , 1
```

```
      [,1]  
sea_dummy1      1  
sea_dummy2      0  
sea_dummy3      0  
sea_dummy4      0  
sea_dummy5      0  
sea_dummy6      0  
sea_dummy7      0  
sea_dummy8      0  
sea_dummy9      0  
sea_dummy10     0  
sea_dummy11     0
```

```
mod2fit <- fitSSM(mod2, inits=rep(0, 12), method="BFGS")  
outmod2.state <- KFS(mod2fit$model) #filters on the state  
preds2 <- predict(mod2fit$model)  
plot(dat$counts, type='l', ylab="Cases", main="Cases (Observed and Predicted)")  
axis(1, at=xx, labels=paste(dat$month[xx],"/", dat$year[xx] , sep="" ))  
lines(preds2, col='green')
```

### Cases (Observed and Predicted)



### Model 3: Seasonal component; different dependence

Not working...

```
# Not working
# mod3 <- SSMModel(counts~-1+SSMcustom(Z=array(rep(diag(1,12), length(2004:2012)
# mod3fit <- fitSSM(mod3, inits=rep(0,12), method="BFGS")
# outmod3.state <- KFS(mod3fit$model)
```

### Model 4: Regression

```
x <- 1:dim(dat)[1]
dat2 <- data.frame(counts=dat$counts, sus=dat$sus, x)
mod4 <- SSMModel(counts~-1+SSMregression(~x, Q=NA, remove.intercept=F), data=da
mod4$R
```

```
, , 1
```

```
      [,1]
(Intercept)  1
x             0
```

```
mod4$Q
```

```
, , 1
```

```
      [,1]
[1,]    NA
```

```
mod4$T
```



```
, , 1
```

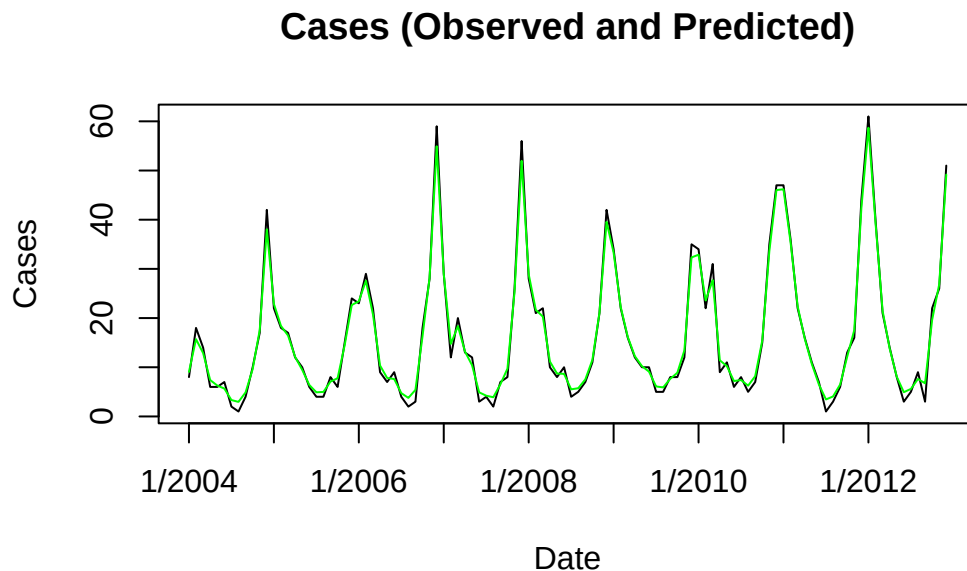
```
      (Intercept) x  
(Intercept)      1 0  
x                0 1
```

```
mod4$Z[, , 1]
```

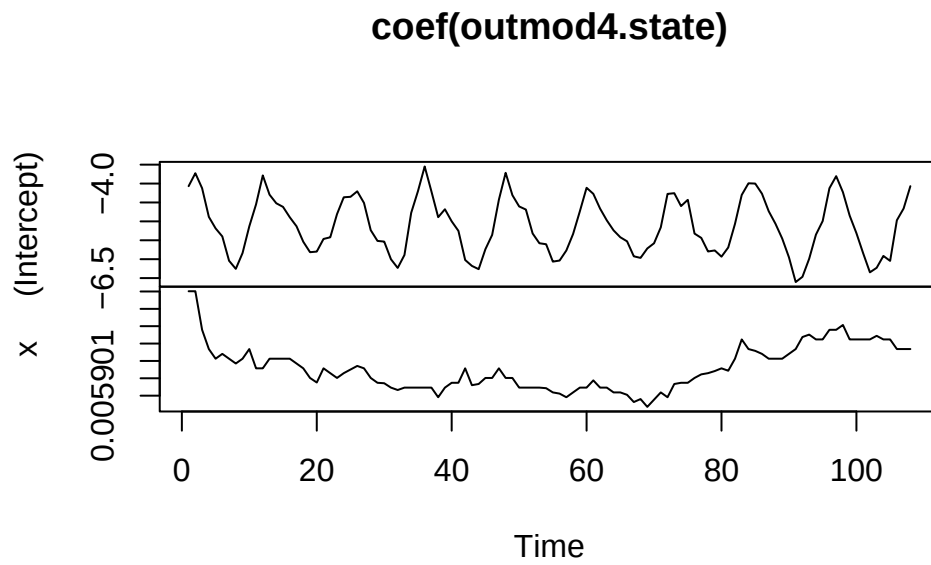
```
(Intercept)      x  
      1          1
```

```
mod4fit <- fitSSM(mod4, inits=1, method="BFGS")  
outmod4.state <- KFS(mod4fit$model)
```

```
preds4 <- predict(mod4fit$model)  
plot(dat$counts, type='l', ylab="Cases", main="Cases (Observed and Predicted)"  
axis(1, at=xx, labels=paste(dat$month[xx], "/", dat$year[xx] , sep="" ))  
lines(preds4, col='green')
```



```
plot(coef(outmod4.state))
```



**Forecasts** (still not working quite right)

```

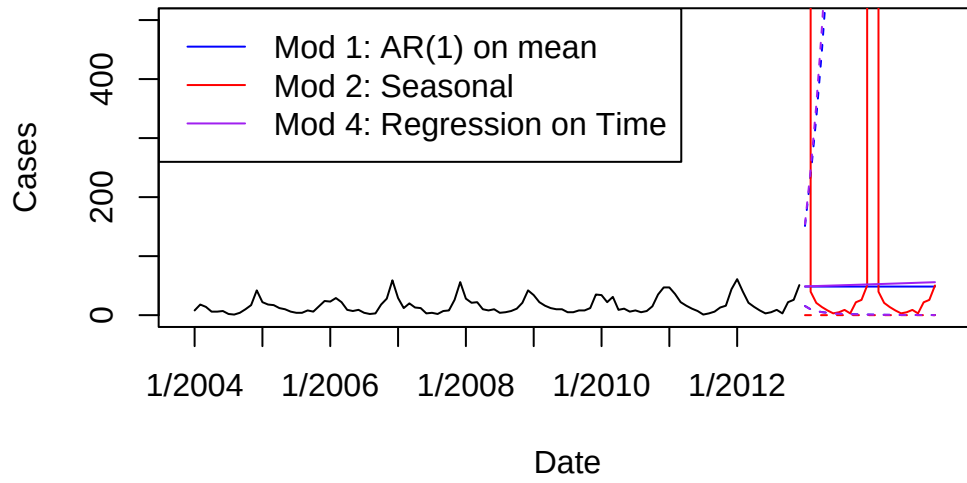
newcounts <- c(rep(NA, 24))
newsus <- c(rep(1500, 24))
newx <- 109:132
dat.pred <- data.frame(newcounts=rep(NA, 24), newsus = rep(1500, 24), newx=109:132)
mod1.f <- SSMModel(newcounts~-1+SSMcustom(Z=matrix(1), T=matrix(1), R=matrix(1)))
preds1 <- predict(mod1fit$model, newdata=mod1.f, interval="confidence")
mod2.f <- SSMModel(newcounts~-1+SSMseasonal(period=12, Q=matrix((mod2fit$model)$Q)))
preds2 <- predict(mod2fit$model, newdata=mod2.f, interval="confidence")

mod4.f <- SSMModel(newcounts~-1+SSMregression(~newx, Q=(mod4fit$model)$Q, remove=1))
preds4 <- predict(mod4fit$model, newdata=mod4.f, interval="confidence")

plot(dat$counts, type='l', ylab="Cases", main="Cases (Observed and Predicted)")
axis(1, at=xx, labels=paste(dat$month[xx],"/", dat$year[xx] , sep="" ))
lines(newx, preds1[,1], col="blue")
lines(newx,preds1[,2], col='blue', lty=2)
lines(newx, preds1[,3], col='blue', lty=2)
lines(newx, preds2[,1], col="red")
lines(newx, preds2[,2], col='red', lty=2)
lines(newx, preds2[,3], col='red', lty=2)
lines(newx, preds4[,1], col='purple')
lines(newx, preds4[,2], col='purple', lty=2)
lines(newx, preds4[,3], col="purple", lty=2)
legend("topleft", legend=c("Mod 1: AR(1) on mean", "Mod 2: Seasonal", "Mod 4:"))

```

### Cases (Observed and Predicted)



## 9 Considerations for Bayesian GLM's

### 9.1 Estimation, Inference, and Model Diagnostics for Bayesian GLM's

#### 9.1.1 Bayesian Modeling

- Under a *frequentist* paradigm, we estimate values for and make inferences about model parameters using the corresponding (approximate/asymptotic) *sampling distributions*.
- Under a *Bayesian* paradigm, inferences are based on *posterior probability distributions* of model parameters. Usually this is motivated by a belief that the parameter is a random variable, rather than a fixed value (as is done in the frequentist paradigm). Even if we believe the parameter has a fixed value, Bayesians view the uncertainty associated with that value as motivation for treating the parameter as a random variable.
- Posterior distributions are probability distributions on model parameters that quantify the *current state of knowledge* about the parameters, accounting for expert knowledge, past data, and current data.
- Expert knowledge and past data are added through a *prior distribution*, while the current data are added through the *likelihood function*.

Let  $L(\mathbf{y}|\theta)$  represent the likelihood function, a member of an exponential family. Let  $p(\theta)$  represent the prior distribution on  $\theta$ , the vector of parameters. Then, the posterior distribution is,

$$p(\theta|\mathbf{y}) = \frac{L(\mathbf{y}|\theta)p(\theta)}{\int L(\mathbf{y}|\theta)p(\theta)d\theta} = \frac{L(\mathbf{y}|\theta)p(\theta)}{m(\mathbf{y})},$$

where  $m(\mathbf{y})$  is the marginal likelihood.

For an observed set of data, the denominator is a constant, and therefore we often only use

$$p(\theta|\mathbf{y}) \propto L(\mathbf{y}|\theta)p(\theta).$$

### 9.1.2 Framework for a Bayesian GLM

- A first approach to developing a Bayesian GLM is to allow the random component to also include the prior distribution for the unknown model parameters:

- [random component]

$$\begin{aligned} Y_i &\sim f(y_i; \theta_i) \\ \beta &\sim p(\beta) \end{aligned}$$

- [link component]

$$\eta_i = g(\mu_i)$$

- [systematic component]

$$\eta_i = \mathbf{x}_i' \beta$$

- Bayesians will often refer to their model using the terms “Data,” “Process,” and “Prior,” meaning that within a Bayesian model, we have three model components: a model for the data (likelihood), a model for the process (link/systematic components), and a prior model for the parameters.
- Fit the model directly, or use data augmentation/latent variable:
  - a latent variable can allow for known full conditional distributions;
  - data augmentation can improve mixing and convergence (important for slow MCMC).

### 9.1.3 Bayesian Estimation

Estimation of the posterior distribution of model parameters depends on the model. There are two cases:

1. Posterior is known and available in closed form: (For GLM's, this is only the case for the normal linear regression model with conjugate priors.)
2. Posterior is unknown: approximate the posterior distribution
  - a. Asymptotic normal approximations
    - Suppose we have a  $\mathcal{N}(\beta_0, \Sigma_0)$  prior for  $\beta$ . The asymptotic normal approximation to the posterior is

$$\begin{aligned}
 p(\beta|\mathbf{y}, \mathbf{x}) &\propto \exp \left\{ -\frac{1}{2}(\beta - \beta_0)' \Sigma_0^{-1}(\beta - \beta_0) \right\} \\
 &\quad \times \exp \left\{ -\frac{1}{2}(\beta - \hat{\beta})' \mathcal{J}(\hat{\beta})(\beta - \hat{\beta}) \right\} \\
 &\propto \mathcal{N}(\tilde{\beta}, \tilde{\Sigma}_\beta)
 \end{aligned}$$

where  $\hat{\beta}$  is the mle of  $\beta$ , and the posterior mean and variance are

$$\begin{aligned}
 \tilde{\beta} &= \tilde{\Sigma}_\beta(\Sigma_0^{-1}\beta_0 + \mathfrak{J}(\hat{\beta})\hat{\beta}) \\
 \tilde{\Sigma}_\beta &= (\Sigma_0^{-1} + \mathfrak{J}(\hat{\beta}))^{-1}
 \end{aligned}$$

- Even for moderate sample sizes, asymptotic approximation may be inaccurate (just like frequentist asymptotics).
  - In, for example, logistic regression for rare outcomes or rare binary exposures, the actual posterior can be highly skewed.
  - In R, “bayesglm()” in the “arm” package uses this type of approach. (There are other packages that approximate posteriors for Bayesian GLM's such as “rstanarm”.)
  - Appealing to avoid any reliance on large sample assumptions and base inferences on *exact posteriors*.
- b. Approximate with samples from the (exact) posterior distribution: e.g., MCMC (Markov chain Monte Carlo)
    - If full conditional distributions of the parameters are known, use Gibbs sampling.
      - The full conditional distribution is the posterior distribution of a subset of parameters (often it's a single parameter), given the data and the current value(s) of the remaining parameter(s). For normal linear regression, {the unknown parameters are  $\beta$  and  $\sigma^2$ .

When the prior of  $\beta$  is a multivariate normal distribution and the prior of  $\sigma^2$  is an inverse gamma distribution,

- a. the full conditional distribution of  $\beta$  (often denoted by some version of “ $\beta|\mathbf{y}, \sigma^2$ ”) is a {multivariate normal distribution}, and\
- b. the full conditional of  $\sigma^2$  ( $\sigma^2|\mathbf{y}, \beta$ ) is an inverse gamma.
- Gibbs sampling then repeatedly samples from each of the full conditional distributions, each time conditioning on the current value of the other parameter(s).
- If full conditional distributions are not available, use Metropolis or Metropolis-Hastings (M-H) sampling:
  - A rejection sampling method that “accepts” or “rejects” proposed values of the parameter based on the model.

– Consider the simple case:

\* Let  $\beta$  be the parameter of interest with prior distribution  $p(\beta)$  defined for the likelihood  $f(\mathbf{y}|\beta)$ . Define a proposal distribution  $q(\beta|\zeta)$ , where  $q$  is a valid pdf (up to a constant) whose distribution depends on parameter(s)  $\zeta$ . Usually  $\zeta$  will be at least the current value of  $\beta$ , denoted by  $\beta^{[t]}$ , and a variance parameter,  $c^2$ .

\* The M-H sampling algorithm requires that at every iteration  $(t + 1)$ :

- i. Draw a proposal value of  $\beta$ , called  $\beta^*$ , from  $q(\beta|\beta^{[t]}, c^2)$ .
- ii. Compute the acceptance probability:

$$\alpha = \frac{f(\mathbf{y}|\beta^*)p(\beta^*)}{f(\mathbf{y}|\beta^{[t]})p(\beta^{[t]})} \frac{q(\beta^{[t]}|\beta^*, c^2)}{q(\beta^*|\beta^{[t]}, c^2)}$$

- iii. accept the value of  $\beta^*$  with probability  $\alpha$ . Usually this is done by drawing  $u \sim U(0, 1)$ . If  $u < \alpha$  (or equivalently, if  $\log u < \log \alpha$ ), accept set  $\beta^{[t+1]} = \beta^*$ , otherwise, set  $\beta^{[t+1]} = \beta^{[t]}$ .

– These methods are sensitive to starting values – requires *burn-in*. Depending on the complexity of the model, it can take many draws (and a lot of computation time) to burn-in to the joint posterior distribution.



- Must monitor convergence
  - \* Mixing:
  - \* Why do we care?

- \* Poor mixing:

- \* Good mixing:

- Convergence diagnostics:
  - \* Gelman-Rubin (multiple chains with dispersed starting values)
  - \* Geweke (test of stationarity to a single chain)
  - \* Heidelberger-Welch (stationarity – alternative to Geweke – and halfwidth – determines the number of samples needed)

- adequate for estimation of the posterior mean)
- \* Raftery-Lewis (determines the number of samples needed for adequate estimation of the percentiles)
- \* autocorrelation (high autocorrelation indicates slow mixing)
- \* effective sample size (low value relative to the actual number sampled indicates slow mixing)
- \* Monte Carlo standard errors (estimates the between batches standard deviation for a summary statistic of a parameter)
- \* Most of these can be implemented using the “coda” package in R. Plummer et al. (2006)

### 9.1.4 Bayesian Inference for Linear Regression Models

**Likelihood:**

$$y_i \sim \mathcal{N}(\mathbf{x}'_i \beta, \tau),$$

where  $\tau = \sigma^{-2}$ . The joint likelihood is therefore

$$\mathbf{y} \sim \mathcal{N}(\mathbf{x}\beta, \tau\mathbf{I})$$

**Priors:** Commonly used conjugate priors are

$$\begin{aligned}\beta &\sim \mathcal{N}(\beta_0, \Sigma_0) \\ \tau &\sim \text{Gamma}(a_\tau, b_\tau),\end{aligned}$$

where  $\beta_0$  is a  $k \times 1$  vector,  $\Sigma_0$  is a  $k \times k$  covariance matrix, and  $a_\tau$  (shape) and  $b_\tau$  (rate) are scalars such that  $E(\tau) = a_\tau/b_\tau$  and  $\text{Var}(\tau) = a_\tau/b_\tau^2$ .

- The *hyperparameters*,  $\beta_0$ ,  $\Sigma_0$ ,  $a_\tau$ , and  $b_\tau$ , quantify our state of knowledge about the regression parameters and precision prior to observing the data from the current study. For example,  $\beta_0$  is our best guess for  $\beta$  before looking at the current data and  $\Sigma_0$  expresses our uncertainty in this guess.
- The full conditional distribution for the regression coefficients can be derived

as follows:

$$\begin{aligned}
p(\beta|\mathbf{y}, \mathbf{x}, \tau) &\propto \exp \left\{ \frac{-1}{2} (\mathbf{y} - \mathbf{x}\beta)' (\tau \mathbf{I}) (\mathbf{y} - \mathbf{x}\beta) \right\} \exp \left\{ \frac{-1}{2} (\beta - \beta_0)' \Sigma_0^{-1} (\beta - \beta_0) \right\} \\
&\propto \exp \left\{ \frac{-1}{2} (\beta' \mathbf{x}' (\tau \mathbf{I}) \mathbf{x} \beta - 2\beta' \mathbf{x}' (\tau \mathbf{I}) \mathbf{y} + \beta' \Sigma_0^{-1} \beta - 2\beta' \Sigma_0^{-1} \beta_0) \right\} \\
&= \exp \left\{ \frac{-1}{2} (\beta' (\tau \mathbf{x}' \mathbf{x} + \Sigma_0^{-1}) \beta - 2\beta' (\tau \mathbf{x}' \mathbf{y} + \Sigma_0^{-1} \beta_0)) \right\}
\end{aligned}$$

from we recognize:

call  $(\tau \mathbf{x}' \mathbf{x} + \Sigma_0^{-1}) = \hat{\Sigma}_\beta^{-1}$  if this is the precision, it needs to be in the second part of the equation too

$$\begin{aligned}
&\propto \exp \left\{ \frac{-1}{2} (\beta' \hat{\Sigma}_\beta^{-1} \beta - 2\beta' \hat{\Sigma}_\beta^{-1} \underbrace{\hat{\Sigma}_\beta (\tau \mathbf{x}' \mathbf{y} + \Sigma_0^{-1} \beta_0)}_{\hat{\beta}}) \right\} \\
&\Rightarrow \beta|\mathbf{y}, \mathbf{x}, \tau \sim \mathcal{N}(\hat{\beta}, \hat{\Sigma}_\beta)
\end{aligned}$$

Therefore, the posterior mean is

$$\hat{\beta} = E(\beta|\tau, \mathbf{y}, \mathbf{x}) = \hat{\Sigma}_\beta (\tau \mathbf{x}' \mathbf{y} + \Sigma_0^{-1} \beta_0)$$

and posterior variance is

$$\hat{\Sigma}_\beta = \text{Var}(\beta|\tau, \mathbf{y}, \mathbf{x}) = (\tau \mathbf{x}' \mathbf{x} + \Sigma_0^{-1})^{-1}$$

Note that as the prior variance increases,  $\hat{\beta} \rightarrow (\mathbf{x}' \mathbf{x})^{-1} \mathbf{x}' \mathbf{y}$ , which is simply the least squares estimator or MLE. Hence, the posterior mean is *shrunk* back towards the prior mean to a degree dependent on the prior variance.

- The full conditional distribution of  $\tau$  is

$$\begin{aligned}
 p(\tau|\mathbf{y}, \mathbf{x}, \beta) &\propto \tau^{n/2} \exp \left\{ \frac{-1}{2} (\mathbf{y} - \mathbf{x}\beta)' (\tau \mathbf{I}) (\mathbf{y} - \mathbf{x}\beta) \right\} \tau^{a_\tau - 1} \exp \{-\tau b_\tau\} \\
 &= \tau^{n/2 + a_\tau - 1} \exp \left\{ -\tau \left( b_\tau + \frac{1}{2} (\mathbf{y} - \mathbf{x}\beta)' (\mathbf{y} - \mathbf{x}\beta) \right) \right\} \\
 &\Rightarrow \tau \sim \text{Gamma}(n/2 + a_\tau, b_\tau + 0.5(\mathbf{y} - \mathbf{x}\beta)' (\mathbf{y} - \mathbf{x}\beta))
 \end{aligned}$$

### 9.1.5 Prior Specification

Three common ways to select prior distributions:

1. Empirical
2. Objective
3. Subjective

We will mainly use subjective priors. Some considerations:

- Use a normal distribution (e.g.,  $\beta_j \sim \mathcal{N}(0, v_j)$  or  $\beta \sim \mathcal{N}_N(\mathbf{0}, \mathbf{V})$ )

We often think of normal distributions with “large” variances as being relatively non-informative. However, for GLM’s, this isn’t *necessarily* the case.

- May elicit prior probabilities associated with chosen predictor values.

### 9.1.6 Bayesian Model Inference

Cover in 651...

- DIC, WAIC, etc.
- Bayes Factors
- Bayesian Model Averaging
- CRPS
- Many more...

## 9.2 Bayesian GLM's for Binary Data

### 9.2.1 Bayesian Logistic Model

Model

$$\begin{aligned} Y_i &\sim \text{Bin}(m_i, \pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \mathbf{x}_i' \beta \\ \beta &\sim p(\beta) \end{aligned}$$

The posterior distribution for  $\beta$  is not available in closed form.

MCMC

*Beetles Data*

Recall:

- $Y_i$  = Number of beetles dead after 5 hours of exposure to gaseous carbon disulphide at various concentrations.
- $x_i$  = Concentration received, in  $\log_{10} CS_2$  mg/l.

Fit logistic regression model using a *Metropolis random walk*:

$$\begin{aligned} Y_i &\sim \text{Bin}(m_i, \pi_i), \text{ for } i = 1, \dots, n \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 x_i \\ \beta_j &\sim \mathcal{N}(0, 10), \text{ for } j = 0, 1 \end{aligned}$$

*Algorithm:*

Let  $\beta^{[t-1]} = (\beta_0^{[t-1]}, \beta_1^{[t-1]})'$  be the current value of  $\beta$ , or the value of  $\beta$  from the previous iteration. Draw proposal value of  $\beta_j^*$ :

$$\beta_j^* \sim N(\beta_j^{[t-1]}, c^2)$$

and accept proposal with probability

$$\frac{f(\mathbf{y}|\beta^*)p(\beta^*)}{f(\mathbf{y}|\beta^{[t-1]})p(\beta^{[t-1]})},$$

where  $f(\cdot)$  is the likelihood of  $\mathbf{y}$  evaluated at each value of  $\beta$ .  
 Use the log likelihood instead:

$$\log(p(\beta^*)) + \ell(\mathbf{y}|\beta^*) - \ell(\mathbf{y}|\beta^{[t-1]}) - \log(p(\beta^{[t-1]})).$$

To find  $\ell(\mathbf{y}|\beta^*)$ :

$$\begin{aligned} f(y_i|\pi_i) &\propto \pi_i^{y_i} (1 - \pi_i)^{m_i - y_i} \\ \Rightarrow \ell(y_i|\pi_i) &\propto y_i \log(\pi_i) + (m_i - y_i) \log(1 - \pi_i) \end{aligned}$$

And evaluate at  $\hat{\pi}_i$ :

$$\hat{\pi}_i = \frac{\exp\{\mathbf{x}_i\beta^*\}}{1 + \exp\{\mathbf{x}_i\beta^*\}}.$$

Example Code for Logistic Regression:

```

dose <- c(1.6907, 1.7242, 1.7552, 1.7842, 1.8113, 1.8369, 1.8610, 1.8839)
m.beetles <- c(59, 60, 62, 56, 63, 59, 62, 60)
y.beetles <- c(6, 13, 18, 28, 52, 53, 61, 60)

g.inv <- function(x, beta){
  exp(x%%beta)/(1+exp(x%%beta))
}
loglik <- function(p,n, y){
  sum(log(p)*y+log(1-p)*(n-y))
}
logprior <- function(beta){
  sum(dnorm(beta, 0, 10, log=T))
}
x <- cbind(rep(1, length(dose)), dose)
beta <- c(1,1) #starting value

full.loglik <- sum( (log(y.beetles/m.beetles)*y.beetles+
log(1-y.beetles/m.beetles)*(m.beetles-y.beetles))[1:7])
c.sq <- c(.6, 0.5) #proposal standard deviations

ITERS <- 10000
beta.save <- matrix(0, nrow=2, ncol=ITERS)
dev.save <- rep(0, ITERS)
accept <- c(0, 0)

beta.save[,1] <- beta
dev.save[1] <- 2*(full.loglik - loglik(g.inv(x, beta), m.beetles, y.beetles))

for(i in 2:ITERS){
  for(j in 1:2){
    beta.cur <- beta
    beta.samp <- rnorm(1, beta.cur[j], c.sq[j])
    beta.star <- beta.cur
    beta.star[j] <- beta.samp

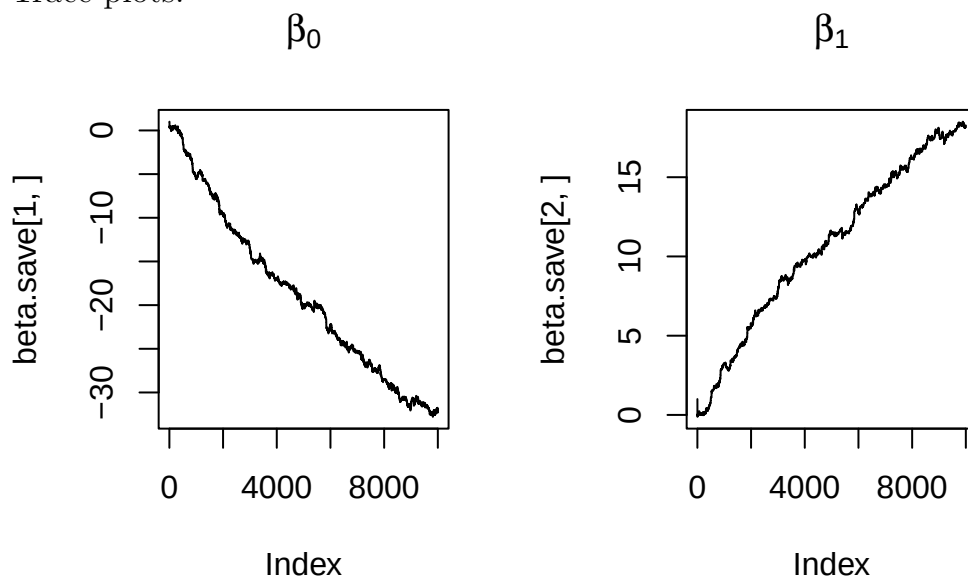
    logalpha <- logprior(beta.star) + loglik(g.inv(x, beta.star),
m.beetles, y.beetles) - (logprior(beta.cur) + loglik(g.inv(x, beta.cur),
m.beetles, y.beetles))

    logunif <- log(runif(1)) 371

    if(logunif < logalpha){
      beta <- beta.star
      accept[j] <- accept[j]+1
    } else{ beta <- beta.cur}
  }
}

```

Trace plots:



Problems?

Fit a logistic regression model using a Metropolis random walk, *centering the doses*:

$$Y_i \sim \text{Bin}(m_i, \pi_i), \text{ for } i = 1, \dots, n$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \tilde{\beta}_0 + \tilde{\beta}_1^*(x_i - \bar{x})$$

$$\beta_j \sim \mathcal{N}(0, 10), \text{ for } j = 0, 1$$



Notice:

$$\log \left( \frac{\pi_i}{1 - \pi_i} \right) = \tilde{\beta}_0 + \tilde{\beta}_1 (x_i - \bar{x})$$

$$\log \left( \frac{\pi_i}{1 - \pi_i} \right) = \tilde{\beta}_0 + \tilde{\beta}_1 x_i - \tilde{\beta}_1 \bar{x}$$

$$\log \left( \frac{\pi_i}{1 - \pi_i} \right) = (\tilde{\beta}_0 - \tilde{\beta}_1 \bar{x}) + \tilde{\beta}_1 x_i$$

$$\Rightarrow \tilde{\beta}_1 = \beta_1$$

$$\tilde{\beta}_0 = \beta_0 + \beta_1 \bar{x}$$

```

x <- cbind(rep(1, length(dose)), dose-mean(dose))
beta <- c(1,1) #starting value

full.loglik <- sum( (log(y.beetles/m.beetles)*y.beetles+
log(1-y.beetles/m.beetles)*(m.beetles-y.beetles))[1:7])
c.sq <- c(.6, 10) #proposal standard deviations

ITERS <- 10000
beta.save <- matrix(0, nrow=2, ncol=ITERS)
dev.save <- rep(0, ITERS)
accept <- c(0, 0)

beta.save[,1] <- beta
dev.save[1] <- 2*(full.loglik - loglik(g.inv(x, beta), m.beetles, y.beetles))

for(i in 2:ITERS){
  for(j in 1:2){
    beta.cur <- beta
    beta.samp <- rnorm(1, beta.cur[j], c.sq[j])
    beta.star <- beta.cur
    beta.star[j] <- beta.samp

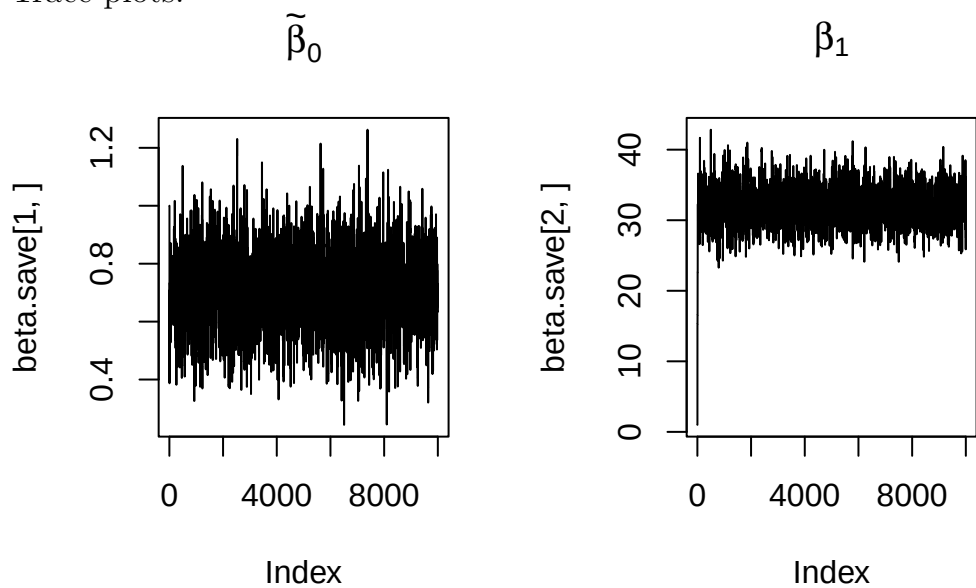
    logalpha <- logprior(beta.star) + loglik(g.inv(x, beta.star),
m.beetles, y.beetles) - (logprior(beta.cur) + loglik(g.inv(x, beta.cur),
m.beetles, y.beetles))

    logunif <- log(runif(1))

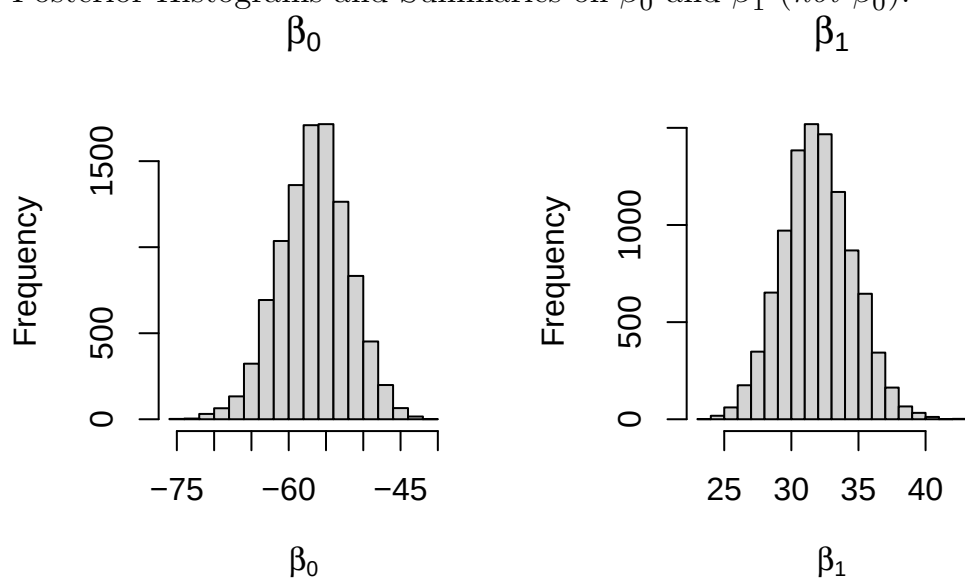
    if(logunif < logalpha){
      beta <- beta.star
      accept[j] <- accept[j]+1
    } else{ beta <- beta.cur}
  }
  beta.save[,i] <- beta
  dev.save[i] <- 2*(full.loglik - loglik(g.inv(x, beta), m.beetles, y.beetles))
}

```

Trace plots:



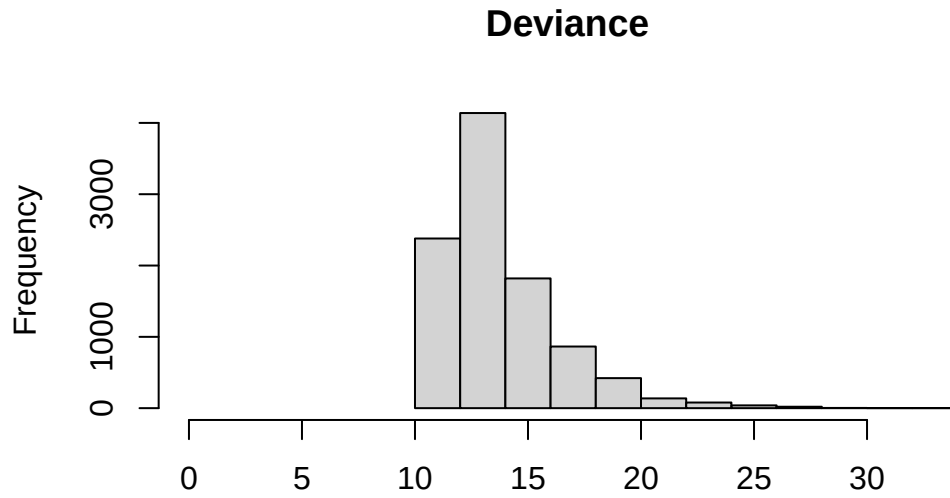
Posterior Histograms and Summaries on  $\beta_0$  and  $\beta_1$  (not  $\tilde{\beta}_0$ ):



| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| -75.92 | -59.67  | -56.43 | -56.61 | -53.46  | -40.91 |

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 23.28 | 30.20   | 31.86  | 31.96 | 33.67   | 42.83 |

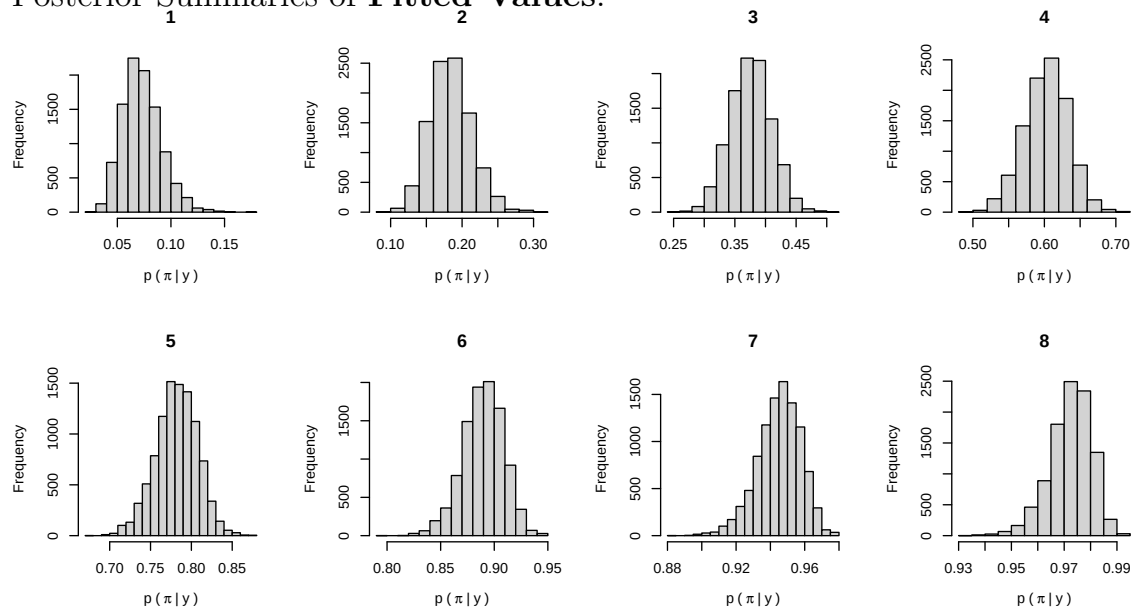
Posterior Deviance:

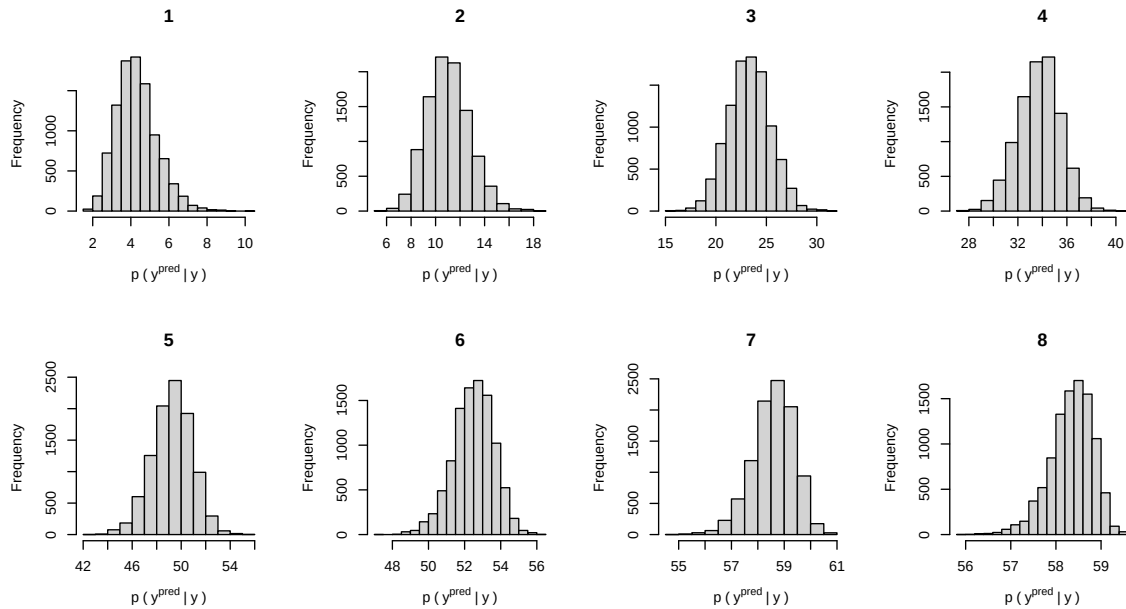


Deviance

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 11.23 | 12.03   | 13.04  | 13.82 | 14.86   | 33.51 |

Posterior Summaries of **Fitted** Values:





**Note:** What's the difference? How do we get the prediction intervals? Recall that we *can't* get prediction intervals with frequentist approach.

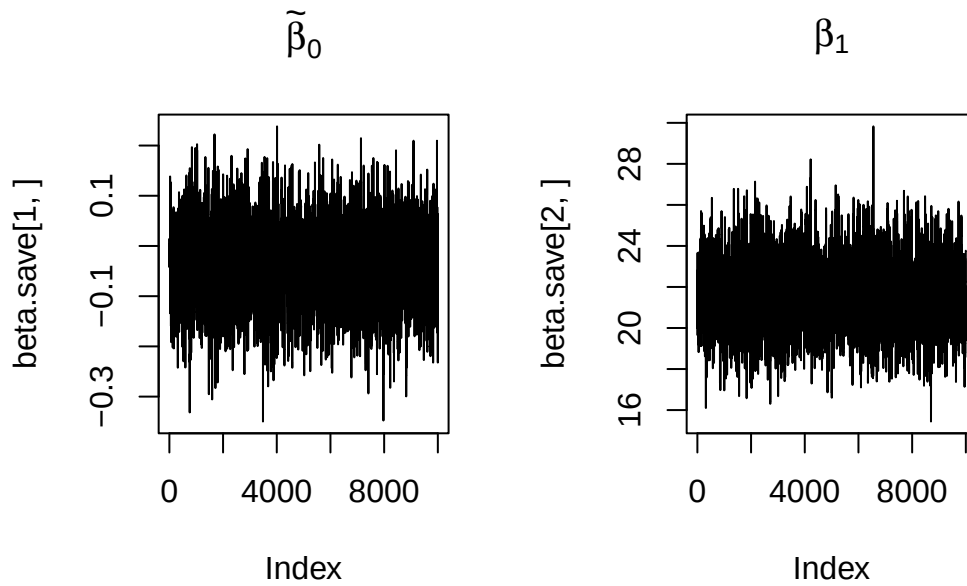
Fit model using the **c-log-log link** (with centered dose):

$$\begin{aligned}\log(-\log(1 - \pi_i)) &= \tilde{\beta}_0 + \tilde{\beta}_1(x_i - \bar{x}) \\ \Rightarrow \pi_i &= 1 - \exp\{-\exp\{\tilde{\beta}_0 + \tilde{\beta}_1(x_i - \bar{x})\}\}\end{aligned}$$

Only need to change “g.inv” function in our code:

```
g.inv <- function(x, beta){
  1 - exp( -exp(x%*%beta)) #cloglog link
}
```

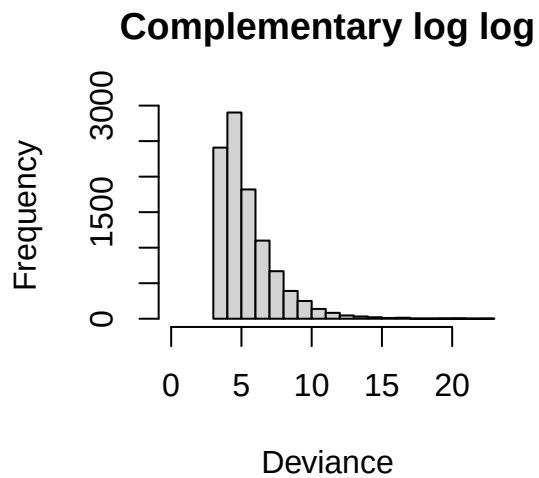
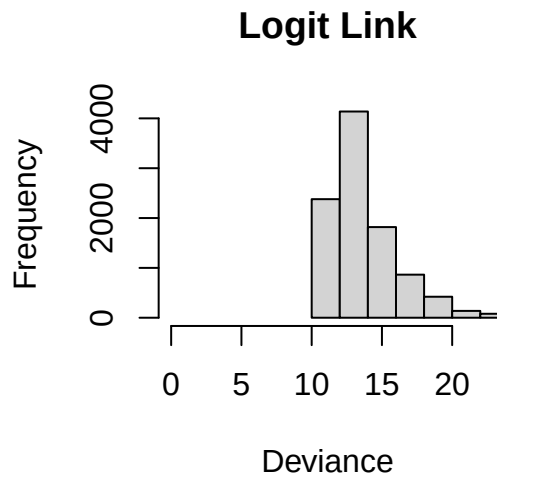
Trace Plots of  $\tilde{\beta}_0$  and  $\tilde{\beta}_1$ :



Posterior Histograms and Summaries of  $\beta_0$  and  $\beta_1$ :

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| -53.67 | -40.91  | -38.82 | -38.84 | -36.75  | -27.75 |
| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 15.44  | 20.47   | 21.63  | 21.64  | 22.79   | 29.83  |

*Posterior Deviance:* Since we have a posterior distribution on deviance we can “test” the deviance between link functions; in frequentist setting, can only compare the point estimates.



Which link do you prefer?

## 9.2.2 Approximating with JAGS

Fit the same models using Just Another Gibbs Sampler:

```
#####
# Download JAGS: http://mcmc-jags.sourceforge.net/
# R packages needed: R2jags, rjags (loads with R2jags)

library(R2jags)

dose <- c(1.6907, 1.7242, 1.7552, 1.7842, 1.8113, 1.8369, 1.8610, 1.8839)
n.beetles <- c(59, 60, 62, 56, 63, 59, 62, 60)
y.beetles <- c(6, 13, 18, 28, 52, 53, 61, 60)
data <- data.frame(cbind(dose, n.beetles, y.beetles))

##Logit Link
mdl.logit <- "
model {
  for(i in 1:8){
    y.beetles[i] ~ dbin(p[i], n.beetles[i]);
    logit(p[i]) <- beta[1]+dose[i]*beta[2];
  }
  for(j in 1:2){
    beta[j] ~ dnorm(0, .1);
  }
}"

writeLines(mdl.logit, 'logitbeetles.txt')
params <- c('beta', 'p')
inits <- list(list('beta'=rep(0,2)))
logit.sim <- jags(data=data, inits=inits, parameters.to.save=params,
model.file='logitbeetles.txt', n.iter=10000, n.chains=1, n.thin=1)

traceplot(logit.sim)
plot(logit.sim)

##Cloglog Link
mdl.cloglog <- "
model {
```



```

    for(i in 1:8){
      y.beetles[i] ~ dbin(p[i], n.beetles[i]);
      p[i] <- 1 - exp(-exp(beta[1]+dose[i]*beta[2]));
    }
    for(j in 1:2){
      beta[j] ~ dnorm(0, 0.1);
    }
  }"

writeLines mdl.cloglog, 'cloglogbeetles.txt')
params <- c('beta', 'p')
inits <- list(list('beta'=rep(0,2)))
cloglog.sim <- jags(data=data, inits=inits, parameters.to.save=params,
model.file='cloglogbeetles.txt', n.iter=200000, n.chains=1, n.thin=1)
traceplot(cloglog.sim)

```

### 9.2.3 Approximating with rstanarm

```
library(rstanarm)
```

Warning: package 'Rcpp' was built under R version 4.6.1

```

dat <- data.frame(dose, m.beetles, y.beetles)
mods <- stan_glm(cbind(y.beetles, m.beetles-y.beetles)~dose, data=dat, family=binomial)

```

SAMPLING FOR MODEL 'binomial' NOW (CHAIN 1).  
Chain 1:

```

Chain 1: Gradient evaluation took 0.00013 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 1.3
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:      1 / 2000 [  0%]   (Warmup)
Chain 1: Iteration:    200 / 2000 [ 10%]   (Warmup)
Chain 1: Iteration:    400 / 2000 [ 20%]   (Warmup)
Chain 1: Iteration:    600 / 2000 [ 30%]   (Warmup)
Chain 1: Iteration:    800 / 2000 [ 40%]   (Warmup)
Chain 1: Iteration:   1000 / 2000 [ 50%]   (Warmup)
Chain 1: Iteration:   1001 / 2000 [ 50%]   (Sampling)
Chain 1: Iteration:   1200 / 2000 [ 60%]   (Sampling)
Chain 1: Iteration:   1400 / 2000 [ 70%]   (Sampling)
Chain 1: Iteration:   1600 / 2000 [ 80%]   (Sampling)
Chain 1: Iteration:   1800 / 2000 [ 90%]   (Sampling)
Chain 1: Iteration:   2000 / 2000 [100%]   (Sampling)
Chain 1:
Chain 1:   Elapsed Time: 0.014 seconds (Warm-up)
Chain 1:                   0.015 seconds (Sampling)
Chain 1:                   0.029 seconds (Total)
Chain 1:

```

SAMPLING FOR MODEL 'binomial' NOW (CHAIN 2).

```

Chain 2:
Chain 2: Gradient evaluation took 3e-06 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.0
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:      1 / 2000 [  0%]   (Warmup)
Chain 2: Iteration:    200 / 2000 [ 10%]   (Warmup)
Chain 2: Iteration:    400 / 2000 [ 20%]   (Warmup)
Chain 2: Iteration:    600 / 2000 [ 30%]   (Warmup)
Chain 2: Iteration:    800 / 2000 [ 40%]   (Warmup)
Chain 2: Iteration:   1000 / 2000 [ 50%]   (Warmup)
Chain 2: Iteration:   1001 / 2000 [ 50%]   (Sampling)
Chain 2: Iteration:   1200 / 2000 [ 60%]   (Sampling)

```

```

Chain 2: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2:
Chain 2: Elapsed Time: 0.013 seconds (Warm-up)
Chain 2:           0.015 seconds (Sampling)
Chain 2:           0.028 seconds (Total)
Chain 2:

```

SAMPLING FOR MODEL 'binomial' NOW (CHAIN 3).

```

Chain 3:
Chain 3: Gradient evaluation took 3e-06 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.000003 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3: Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3: Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3: Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 3: Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 3: Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3: Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 0.014 seconds (Warm-up)
Chain 3:           0.014 seconds (Sampling)
Chain 3:           0.028 seconds (Total)
Chain 3:

```

SAMPLING FOR MODEL 'binomial' NOW (CHAIN 4).

```

Chain 4:
Chain 4: Gradient evaluation took 2e-06 seconds

```

```

Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.028 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4: Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4: Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4: Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4: Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4: Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4: Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 4: Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 4: Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 4: Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 4: Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 4: Iteration:  2000 / 2000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 0.013 seconds (Warm-up)
Chain 4:                   0.015 seconds (Sampling)
Chain 4:                   0.028 seconds (Total)
Chain 4:

```

```
summary(mods)
```

```

Model Info:
function:    stan_glm
family:      binomial [cloglog]
formula:     cbind(y.beetles, m.beetles - y.beetles) ~ dose
algorithm:   sampling
sample:      4000 (posterior sample size)
priors:      see help('prior_summary')
observations: 8
predictors:  2

```

Estimates:

|             | mean  | sd  | 10%   | 50%   | 90%   |
|-------------|-------|-----|-------|-------|-------|
| (Intercept) | -39.9 | 3.2 | -44.1 | -39.7 | -35.7 |
| dose        | 22.2  | 1.8 | 19.9  | 22.1  | 24.6  |

Fit Diagnostics:

|          | mean | sd  | 10%  | 50%  | 90%  |
|----------|------|-----|------|------|------|
| mean_PPD | 36.3 | 1.4 | 34.6 | 36.4 | 38.1 |

The mean\_ppd is the sample average posterior predictive distribution of the out

MCMC diagnostics

|               | mcse | Rhat | n_eff |
|---------------|------|------|-------|
| (Intercept)   | 0.1  | 1.0  | 2643  |
| dose          | 0.0  | 1.0  | 2648  |
| mean_PPD      | 0.0  | 1.0  | 3066  |
| log-posterior | 0.0  | 1.0  | 1844  |

For each parameter, mcse is Monte Carlo standard error, n\_eff is a crude measur

## 9.2.4 Approximating with arm

```
library(arm)
```

Warning: package 'lme4' was built under R version 4.6.1

```
modb <- bayesglm(cbind(y.beetles, m.beetles-y.beetles)~dose, data=dat, family=b  
summary(modb)
```

Call:

```
bayesglm(formula = cbind(y.beetles, m.beetles - y.beetles) ~
  dose, family = binomial(link = "cloglog"), data = dat, prior.scale = 10)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -39.530  | 3.235      | -12.22  | <2e-16 *** |
| dose        | 22.018   | 1.796      | 12.26   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 284.2024 on 7 degrees of freedom  
 Residual deviance: 3.4466 on 6 degrees of freedom  
 AIC: 111.52

Number of Fisher Scoring iterations: 5

## 9.2.5 Bayesian Probit Model

Albert and Chib (1993) first introduced a latent variable to facilitate a probit link function and known full conditional distributions. Knowing the full conditional distributions enables us to use a Gibbs sampling algorithm instead of a another approximation algorithm that can be more sensitive to starting values.

### Latent Variable/Data Augmentation

For the probit model,  $\pi_i = \Phi(\mathbf{x}_i'\beta)$ , or  $\Phi^{-1}(\pi_i) = \mathbf{x}_i'\beta$ . Suppose we set

$$Y_i = \begin{cases} 1, & Z_i \geq 0 \\ 0, & Z_i < 0 \end{cases}$$

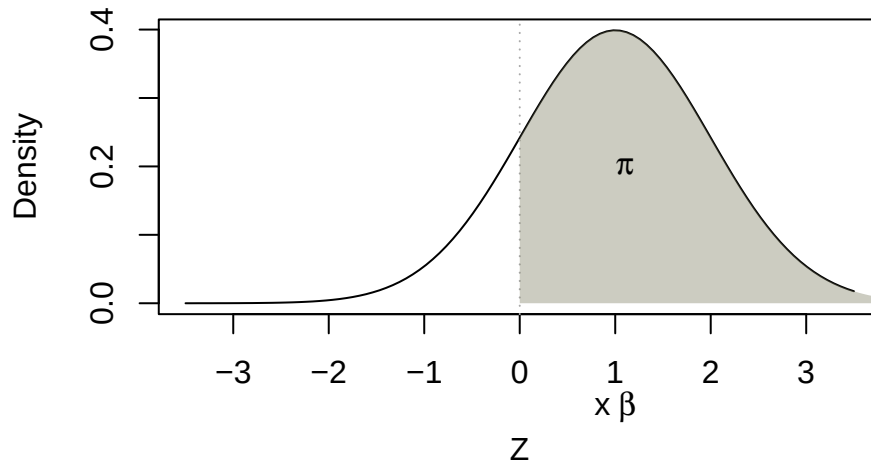
$$\text{and } Z_i = \mathbf{x}_i'\beta + \epsilon_i$$

$$\text{where } \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

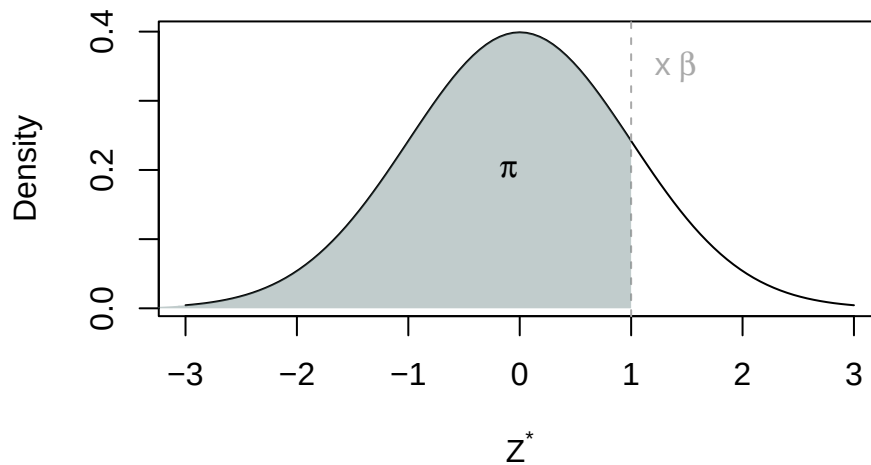
Then,

$$\begin{aligned} \pi_i &= P(Y_i = 1) = P(Z_i \geq 0) = P(\mathbf{x}_i'\beta + \epsilon_i \geq 0) \\ &= P(\epsilon_i \geq -\mathbf{x}_i'\beta) \\ &= P\left(\frac{\epsilon_i}{\sigma} \geq -\frac{\mathbf{x}_i'\beta}{\sigma}\right) \\ &= P\left(Z_i^* \geq -\frac{\mathbf{x}_i'\beta}{\sigma}\right) \\ &= 1 - P\left(Z_i^* \leq -\frac{\mathbf{x}_i'\beta}{\sigma}\right) \\ &= 1 - \Phi\left(-\frac{\mathbf{x}_i'\beta}{\sigma}\right) \\ &= \Phi\left(\frac{\mathbf{x}_i'\beta}{\sigma}\right) \end{aligned}$$

$$\pi = P(Y = 1) = P(Z > 0)$$



$$\pi = P(Y = 1) = P(Z^* < x \beta)$$



Note: the  $Z$  and  $\epsilon$  are only introduced to facilitate the probit link function and known full conditional distributions. Polson et al. (2013) introduced a similar type latent variable (a Polya-gamma instead of Gaussian latent variable) to facilitate the logit link.

$\beta$  and  $\sigma$  are not identifiable. We can consider two ways to accommodate:

1. Marginalize over  $\sigma$ . (This can improve convergence.)



2. Set  $\sigma = c$ . (WLOG, let  $c = 1$ , which corresponds to the standard normal distribution of the probit link.)

## Gibbs Sampling

*Likelihood:*

$$f(y_i|z_i) = I(z_i \geq 0) + I(z_i < 0)$$

$$\begin{aligned} f(z_i|\beta, \sigma = 1) &\propto \exp \left\{ -\frac{1}{2}(z_i - \mathbf{x}'_i\beta)^2 \right\} \\ \Rightarrow f(\mathbf{Z}|\beta) &\propto \exp \left\{ -\frac{1}{2}(\mathbf{Z} - \mathbf{X}'\beta)'(\mathbf{Z} - \mathbf{X}'\beta) \right\} \end{aligned}$$

*Prior:*

$$\pi(\beta) \propto \exp \left\{ -\frac{1}{2}\beta' \mathbf{V}^{-1} \beta \right\}$$

Then, the *full conditional distributions* for a Gibbs sampler are known:

1. Sample  $\beta|\mathbf{Z}, \mathbf{y}$

$$\begin{aligned} \pi(\beta|\mathbf{Z}, \mathbf{Y}) &\propto \exp \left\{ -\frac{1}{2}(\mathbf{Z} - \mathbf{X}'\beta)'(\mathbf{Z} - \mathbf{X}'\beta) - \frac{1}{2}\beta' \mathbf{V}^{-1} \beta \right\} \\ &\quad (\text{Don't need } Y \text{ since } f(Z, Y) = f(Z)) \\ &\propto \exp \left\{ -\frac{1}{2}(\beta - \hat{\beta})' \Sigma_{\beta} (\beta - \hat{\beta}) \right\}, \end{aligned}$$

where

$$\begin{aligned} \Sigma_{\beta} &= (\mathbf{X}'\mathbf{X} + \mathbf{V}^{-1})^{-1} \\ \hat{\beta} &= \Sigma_{\beta}(\mathbf{X}'\mathbf{Z}). \end{aligned}$$

$$\Rightarrow \beta \sim \mathcal{N}(\hat{\beta}, \Sigma_{\beta})$$

## 2. Sample $\mathbf{Z}|\beta, \mathbf{Y}$

$$\begin{aligned} Z_i|\beta, y_i &\propto \exp\left\{-\frac{1}{2}(z_i - \mathbf{x}'_i\beta)^2\right\} (I(z_i \geq 0) + I(z_i < 0)) \\ &\propto \begin{cases} \exp\left\{-\frac{1}{2}(z_i - \mathbf{x}'_i\beta)^2\right\} I(z_i \geq 0), & y_i = 1 \\ \exp\left\{-\frac{1}{2}(z_i - \mathbf{x}'_i\beta)^2\right\} I(z_i < 0), & y_i = 0 \end{cases} \\ \Rightarrow Z_i|\beta, y_i &\sim \begin{cases} \mathcal{N}(\mathbf{x}'_i\beta, 1)I(0, \infty), & y_i = 1 \\ \mathcal{N}(\mathbf{x}'_i\beta, 1)I(-\infty, 0), & y_i = 0 \end{cases} \end{aligned}$$

## 3. If we want to predict a new value:

- For every sampled value of  $\beta$ , draw  $Z^0 \sim \mathcal{N}(\mathbf{x}^{0'}\beta, 1)$ , and let  $Y^0$  be as defined above.
- Likewise, we could find  $\pi^0$  (and a credible interval) by finding  $\pi^0 = P(Z^0 > 0) = \Phi(\mathbf{x}^{0'}\beta)$ .

*Example: Banana Quality*

[Kaggle dataset](#)

We are interested in understanding the relationship between banana quality (“good” or “bad”) and the fruit’s:

- size (range: -8 to 8)
- weight (range: -8.3 to 5.7)
- sweetness (range: -6.5 to 7.6)
- softness (range: -7 to 8.3)
- harvest time (amount of time passed from harvesting the fruit; range: -7.6 to 6.3)
- ripeness (range: -7.5 to 7.3)
- acidity (range: -8.3 to 7.5)

```
dat <- read.csv(url("https://byu.box.com/shared/static/qrd3ahcjrsmwa6y05ae3i8
y <- dat$Quality=="Good"
X <- as.matrix(cbind(rep(1, length(y)), dat[,1:7]))
```

*Model:*

$$\begin{aligned}Y_i &\sim \text{Bin}(1, \pi_i), \text{ for } i = 1, \dots, 8000, \\ \Phi^{-1}(\pi_i) &= \mathbf{x}_i\beta, \\ \beta &\sim \mathcal{N}(\mathbf{0}, 10\mathbf{I}),\end{aligned}$$

where  $\mathbf{x}_i$  represents the  $8 \times 1$  vector of coefficients (including an intercept term),  $\beta$  is the corresponding vector of coefficients,  $\mathbf{0}$  is the vector of 0's, and  $\mathbf{I}$  is the identity matrix.

*Estimation:*

```

library(MASS) #to sample from mvnrm
library(truncnorm) #to sample from truncated normal dist.

#starting values
Z <- y - 0.5
beta <- rep(0, dim(X)[2])

#prior values
V.inv <- 0.1*diag(1, length(beta))

#MCMC values
iters <- 10000
beta.save <- matrix(0, nrow=dim(X)[2], ncol=iters)
z.save <- matrix(0, nrow=length(y), ncol=iters)
beta.var <- solve(t(X)%*%X + V.inv) #since X and V are constant, we don't have

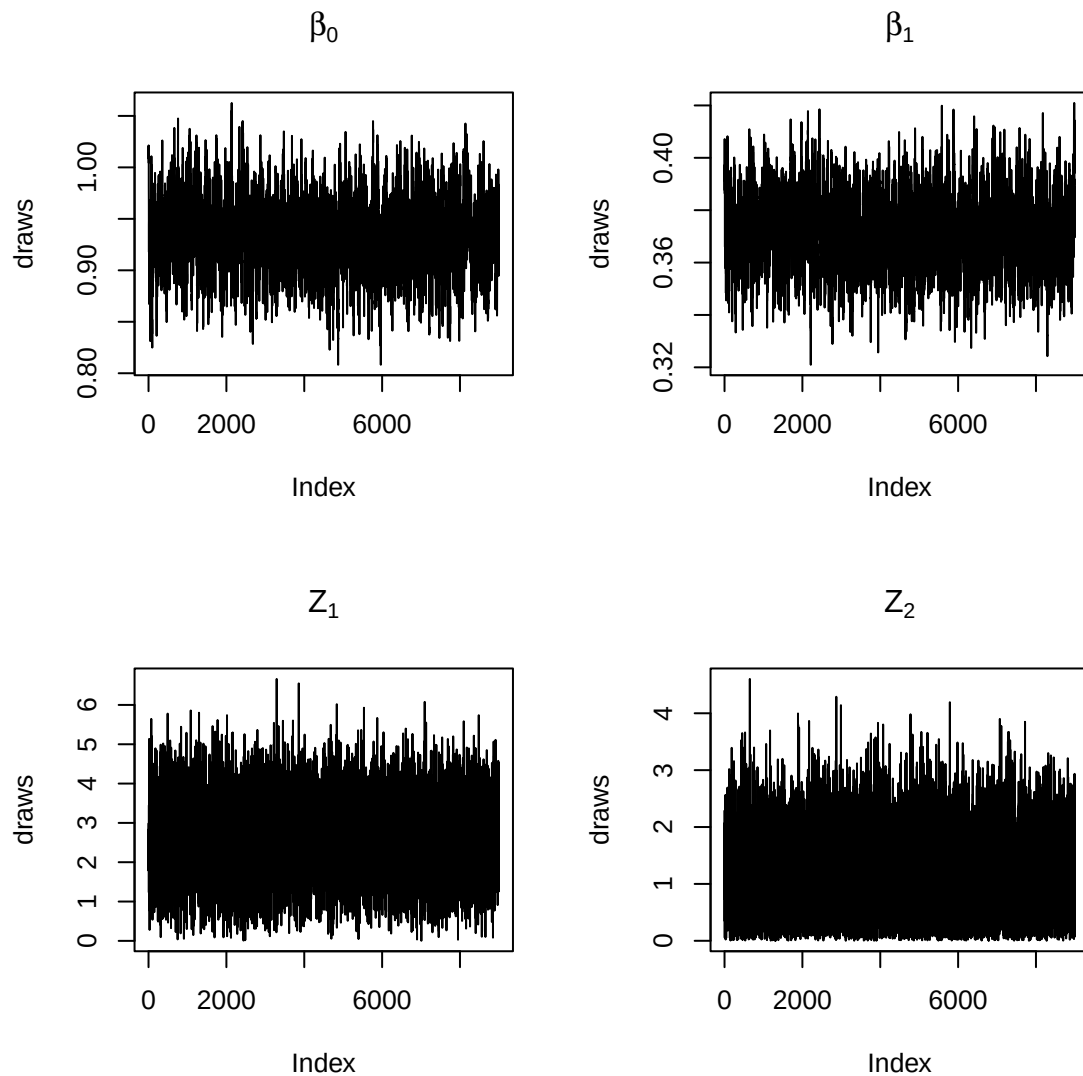
#MCMC
for(i in 1:iters){
  #sample beta
  beta.hat <- beta.var%*%(t(X)%*%Z)
  beta <- mvrnorm(1, beta.hat, beta.var)

  #sample z

  Z[which(y==1)] <- rtruncnorm(sum(y), a=0, b=Inf, X[which(y==1),]%*%beta, 1)
  Z[which(y==0)] <- rtruncnorm(sum(y==0), a=-Inf,b=0,X[which(y==0),]%*%beta, 1)

  #save the sampled values
  beta.save[,i] <- beta
  z.save[,i] <- Z
}

```



### *Coefficient Posterior Summaries*

|             | post.mean  | post.sd    | post.025    | post.975    |
|-------------|------------|------------|-------------|-------------|
| Intercept   | 0.9337604  | 0.03430543 | 0.86623149  | 1.00281792  |
| Size        | 0.3713981  | 0.01343921 | 0.34563372  | 0.39857474  |
| Weight      | 0.5321475  | 0.01596494 | 0.50004576  | 0.56292779  |
| Sweetness   | 0.4349181  | 0.01479273 | 0.40631843  | 0.46427966  |
| Softness    | 0.0451993  | 0.01081587 | 0.02364537  | 0.06632279  |
| HarvestTime | 0.3115037  | 0.01412450 | 0.28368313  | 0.33891500  |
| Ripeness    | 0.3279560  | 0.01338871 | 0.30268605  | 0.35474209  |
| Acidity     | -0.0572261 | 0.01101947 | -0.07862253 | -0.03595667 |

### *Compare to Frequentist Probit Model Summary*

Call:

```
glm(formula = Quality == "Good" ~ ., family = binomial(link = "probit"),  
    data = dat)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | 0.93232  | 0.03344    | 27.884  | < 2e-16  | *** |
| Size        | 0.37050  | 0.01345    | 27.541  | < 2e-16  | *** |
| Weight      | 0.53078  | 0.01594    | 33.305  | < 2e-16  | *** |
| Sweetness   | 0.43391  | 0.01463    | 29.666  | < 2e-16  | *** |
| Softness    | 0.04532  | 0.01080    | 4.196   | 2.72e-05 | *** |
| HarvestTime | 0.31109  | 0.01424    | 21.849  | < 2e-16  | *** |
| Ripeness    | 0.32742  | 0.01336    | 24.509  | < 2e-16  | *** |
| Acidity     | -0.05675 | 0.01089    | -5.211  | 1.88e-07 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 11090.3 on 7999 degrees of freedom  
Residual deviance: 4670.8 on 7992 degrees of freedom  
AIC: 4686.8

Number of Fisher Scoring iterations: 6

### *Predictions*

Consider two cases:

- A) All covariates are equal to 0.
- B) All covariates are equal to their median value.

```

xA <- c(1, rep(0, 7))
xB <- apply(X, 2, median)

#Approach 1 for predictions
zA <- rnorm(length(burn:iters), xA*beta.save[, burn:iters], 1)
zB <- rnorm(length(burn:iters), xB*beta.save[, burn:iters], 1)
yA <- zA >= 0
yB <- zB >= 0
mean(yA == 1) #pi-hat A

```

```
[1] 0.823353
```

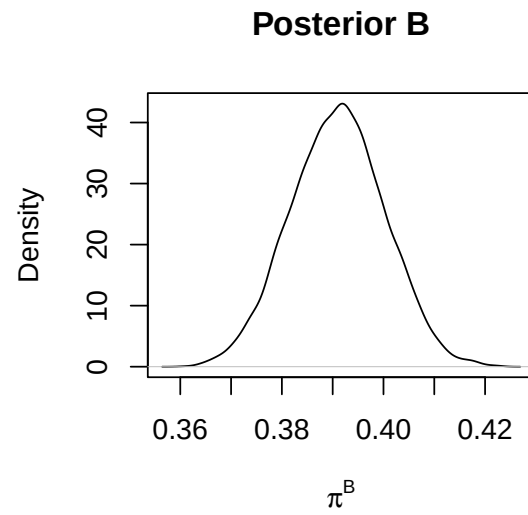
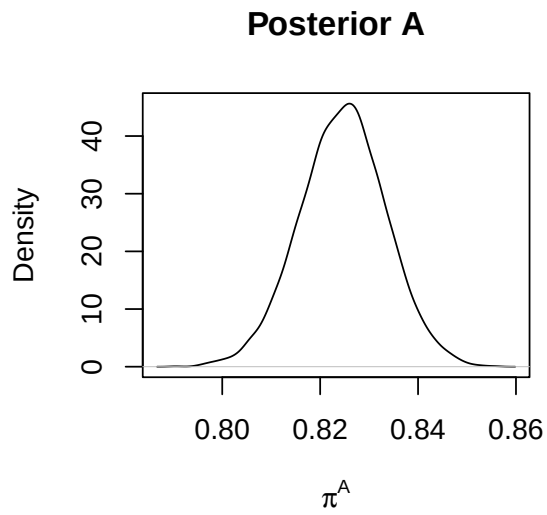
```
mean(yB == 1) #pi-hat B
```

```
[1] 0.3826242
```

```

#Approach 2 for predictions
piA <- pnorm(xA*beta.save[, burn:iters])
piB <- pnorm(xB*beta.save[, burn:iters])
par(mfrow=c(1,2))
plot(density(piA), xlab=expression(pi^A), main="Posterior A")
plot(density(piB), xlab=expression(pi^B), main="Posterior B")

```



```
mean(piA) #pi-hat A
```

```
[1] 0.8246446
```

```
mean(piB) #pi-hat B
```

```
[1] 0.3910085
```

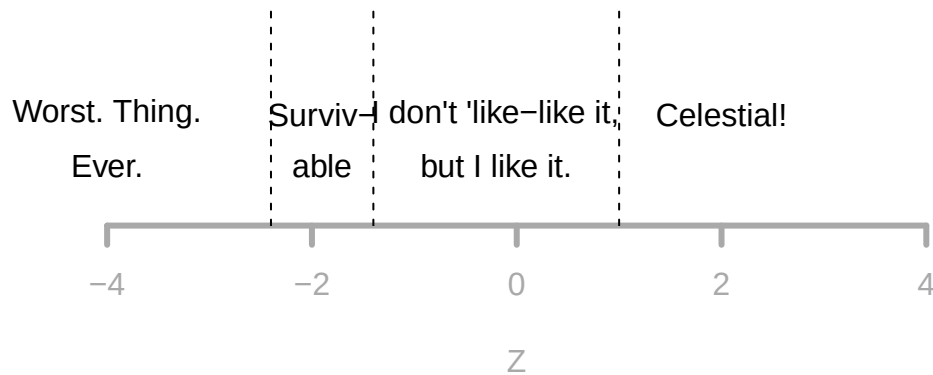


## 9.3 Bayesian GLM's for Nominal and Ordinal Data

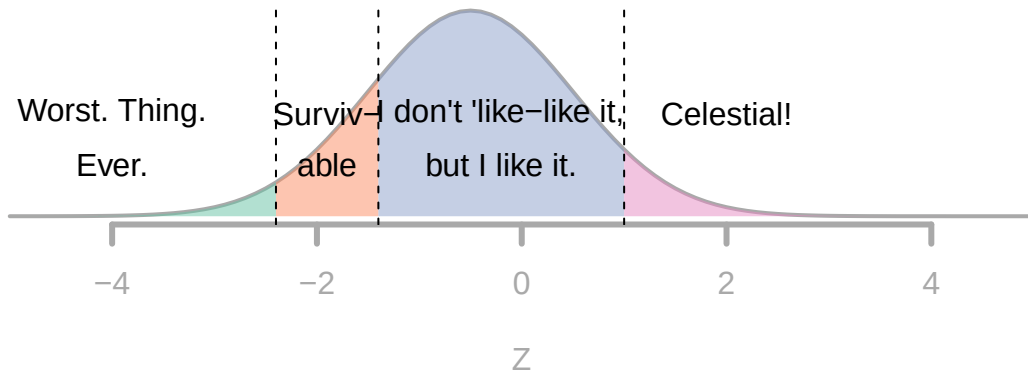
### 9.3.1 Bayesian Probit Ordinal Model

#### Motivation

- Often our discrete variables are crude estimates of a continuous variable. Consider a continuous response variable that may be difficult to measure, such as severity of disease or religious orthodoxy. While there is some underlying continuous scale, the values are actually reported discretely.
- Instead of measuring the continuous variable, we define cut points (or break points) along the domain of the variable to determine the ordered categories. (For example, when we define grades, we use a grading scale.)
- This continuous (unobserved) variable is called a *latent variable* (also called data augmentation), and we will denote it by  $\mathbf{Z}$ .
- If there are  $J$  ordered categories, it is necessary to define  $J - 1$  cut points,  $\gamma_j, j = 1, \dots, J - 1$ . Note that  $\gamma_1 < \dots < \gamma_{J-1}$ .



- The latent variable has a continuous probability distribution (can be any distribution, but a common assumption is to assume normality). Then, the probabilities associated within the cut points of the latent variable are the category-specific probabilities,  $\pi_j$ .

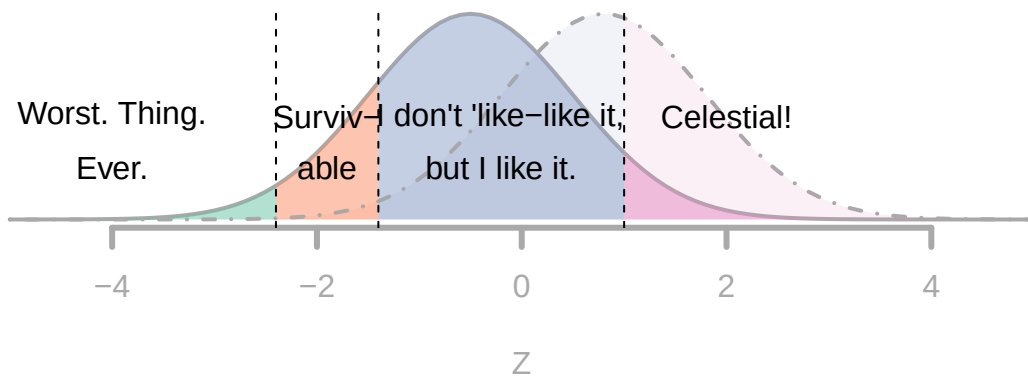


- The  $\pi_j$ 's can be found using the cdf of the normal distribution:

$$\pi_j = P(Y = j) = \Phi\left(\frac{\gamma_j - -0.5}{1}\right) - \Phi\left(\frac{\gamma_{j-1} - -0.5}{1}\right).$$

Assume  $\gamma_0 = -\infty$  and  $\gamma_J = \infty$ . (Note that the mean and standard deviation of the distribution above are -0.5 and 1, respectively.)

- Note: The cut point,  $\gamma_j$  can be anywhere between  $\gamma_{j-1}$  and  $\gamma_{j+1}$  and it is not necessary that the cut points are evenly spaced.
- However, we want a model that can account for different  $\pi_j$ 's for different observations ( $\pi_{ij}$ 's). Instead of changing the cut points for each observation, we'll consider a different mean for each observation.



- Unfortunately, we don't actually know the cut points. Therefore, in addition to modeling  $\mathbf{Z}$ , we must also model the  $\gamma_j$ 's.

## Latent Variable Bayesian Probit Model for Ordinal Data

Just like the latent variable model for binary response variables, we let

$$Z_i \sim \mathcal{N}(\mathbf{x}'_i \beta, 1).$$

Then,

$$Y_i = j, \text{ if } \gamma_{j-1} \leq Z_i \leq \gamma_j.$$

Notice:

$$\begin{aligned} P(Y_i \leq j) &= P(Z_i \leq \gamma_j) \\ &= P(\underbrace{Z_i - \mathbf{x}'_i \beta}_{\sim N(0,1)} \leq \gamma_j - \mathbf{x}'_i \beta) \\ &= \Phi(\gamma_j - \mathbf{x}'_i \beta) \\ \Rightarrow \Phi^{-1} \left( \underbrace{P(Y_i \leq j)}_{\text{cumulative probability}} \right) &= \gamma_j - \mathbf{x}'_i \beta \end{aligned}$$

## Model Fitting

For simplicity, we will use diffuse prior distributions:

$$\begin{aligned} \beta &\sim \mathcal{N}(\mathbf{0}, \infty \mathbf{I}) \\ \gamma_j &\sim U(\gamma_{j-1}, \gamma_{j+1}). \end{aligned}$$

The full conditional distributions are then:

$$\begin{aligned} \beta | \star &\sim \mathcal{N}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}, (\mathbf{X}'\mathbf{X})^{-1}) \\ Z_i | \star, Y_i = j &\sim \mathcal{TN}(\mathbf{x}'_i \beta, 1, \gamma_{j-1}, \gamma_j) \\ \gamma_j | \star &\sim U(\max\{\max\{Z_i : Y_i = j\}, \gamma_{j-1}\}, \min\{\min\{Z_i : Y_i = j+1\}, \gamma_{j+1}\}) \end{aligned}$$

Note: transformations on  $\gamma_j$ 's such as  $\alpha_j = \log(\gamma_j - \gamma_{j-1})$  with  $\gamma_1 = 0$  and prior distribution  $\alpha \sim N(\mathbf{a}, \mathbf{A})$  usually improves mixing.

### 9.3.2 Bayesian Probit Nominal Model

#### Motivation

The probit model for nominal data was developed by economists (see, Daganzo (1979)) to use in discrete choice theory. Discrete choice models are often used by economists and marketers to determine impact of changes to alternatives on consumer preferences.

For example, consider the case where there are two possible modes of transportation: **Red Bus** and Car. Assume that the probability of taking the red bus,  $\pi_{RB}$ , is 0.3, and the probability of taking a car,  $\pi_C$ , is 0.7. A logit model will model the odds:

$$\frac{\pi_{RB}}{\pi_C} = \frac{0.3}{0.7} = 0.4286.$$

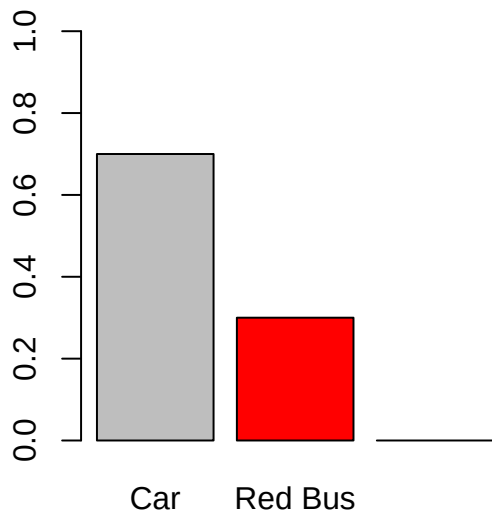
Suppose that the bus company paints one of its red buses blue, so that now there are three possible modes of transportation: **Blue Bus**, **Red Bus**, and Car. The multinomial logit model will model the odds of any one category with respect to the base category *without regard* to the other categories.

In other words, it fits these logits *independently* and treats the other categories as *irrelevant*. This is called the assumption of irrelevant alternatives (IIA).

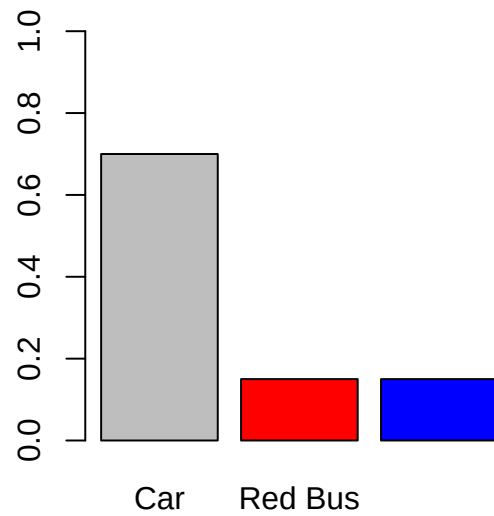
What this means in our example:

*What we expect to happen...*

**Probabilities before Blue Bus**



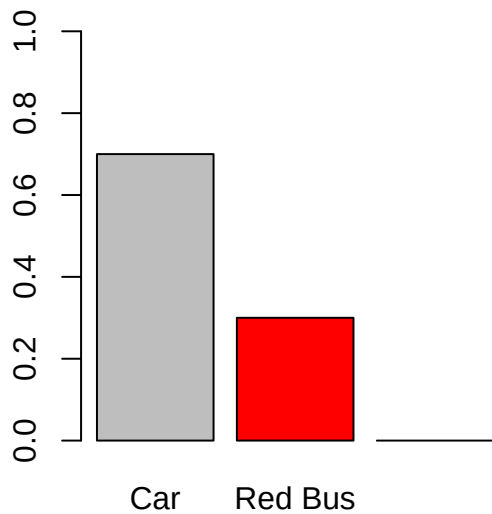
**Probabilities after Blue Bus**



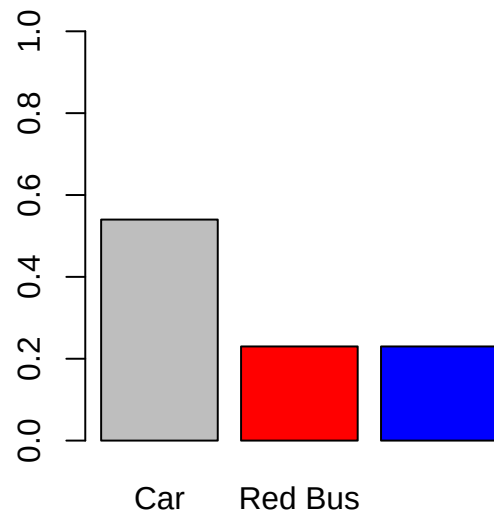
The red bus should split its riders with the blue bus (probably none from the car).

*What actually happens...*

**Probabilities before Blue Bus**



**Probabilities after Blue Bus**



The red bus and the car split (proportionally at  $\approx 77\%$ ) with the blue bus.

Why is this the case? Instead of matching the bus' probabilities, the independence

assumption will match their odds:

$$\frac{\pi_{RB}}{\pi_C} = 0.4286 = \frac{\pi_{BB}}{\pi_C}$$

$$\Rightarrow \begin{cases} \pi_{RB} = 0.4286 \pi_C \\ \pi_{BB} = 0.4286 \pi_C \end{cases}$$

and

$$\begin{aligned} & \pi_{RB} + \pi_{BB} + \pi_C = 1 \\ \Rightarrow & 0.4286\pi_C + 0.4286\pi_C + \pi_C = 1 \\ \Rightarrow & \pi_C = 0.5384 \end{aligned}$$

and therefore  $\pi_{RB} = \pi_{BB} = 0.2308$ .

In fact, this will be a problem whenever the  $\eta_j$ 's are assumed to be independent. Therefore, we need a model that can assume dependence between categories.

### Latent Variable Bayesian Probit Model for Nominal Data

For every observation,  $i$ , assume  $J - 1$  latent variables,  $\mathbf{Z}_i = (Z_{i1}, \dots, Z_{iJ-1})'$ . Let,

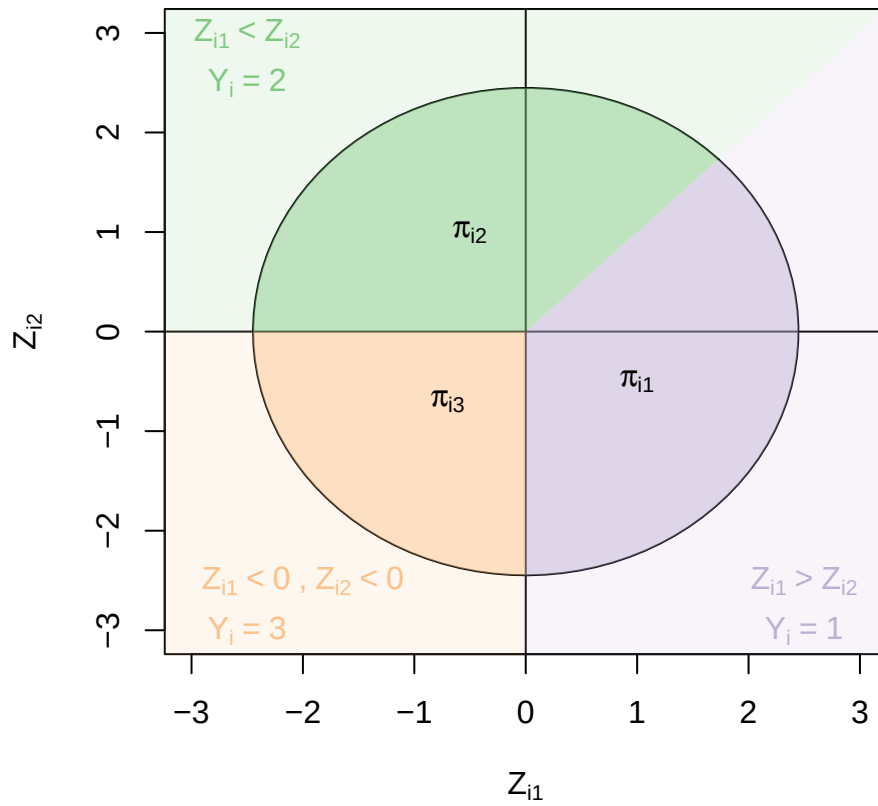
$$\mathbf{Z}_i \sim \mathcal{N}(\mathbf{X}_i\beta, \Omega),$$

where  $\mathbf{X}_i$  is the  $J - 1 \times k$  matrix of covariates and  $\Omega$  is the covariance matrix among the  $J - 1$  latent variables. Define

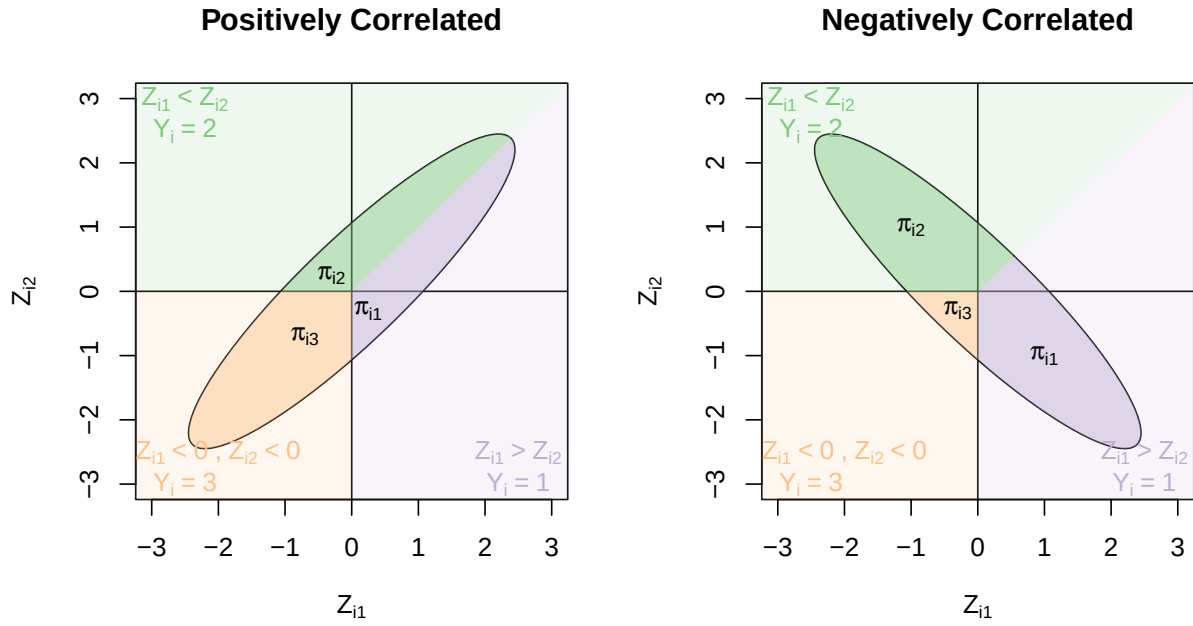
$$Y_i = \begin{cases} j, & \text{if } Z_{ij} > Z_{ik} \ \forall j \neq k \\ J, & \text{if } Z_{ij} < 0 \ \forall j \end{cases}$$

Let's take a second to examine the model. First we'll explore the covariance matrix,  $\Omega$ . Consider the case where  $J = 3$ . In this case, each  $Y_i$  will correspond to 2 latent variables,  $Z_{i1}$  and  $Z_{i2}$ . The plot below shows a circle representing the 95% probability ellipse of the bivariate normal distribution (centered at  $(0, 0)$ ) of  $Z_{i1}$  and  $Z_{i2}$ , when  $Z_{i1}$  and  $Z_{i2}$  are independent. The shaded areas represent the relative probabilities of each of the categories.

## Independent



In contrast, the plots below show 95% probability ellipses when the latent variables are positively and negatively correlated. Notice how the probabilities of each category changes with the correlation.

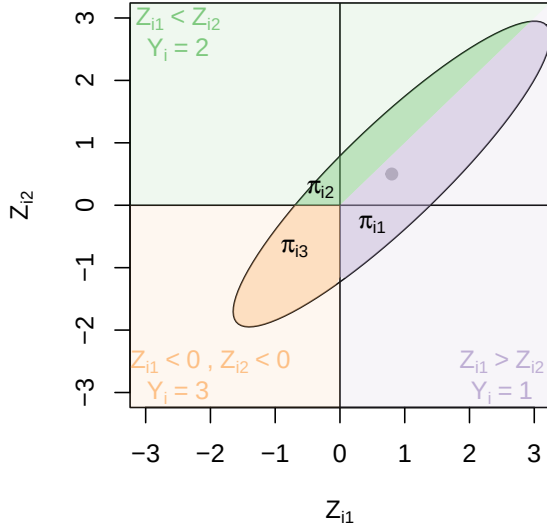


Positive correlation between  $Z_{i1}$  and  $Z_{i2}$  increases the probability of category 3, while negative correlation decreases the probability of category 3.

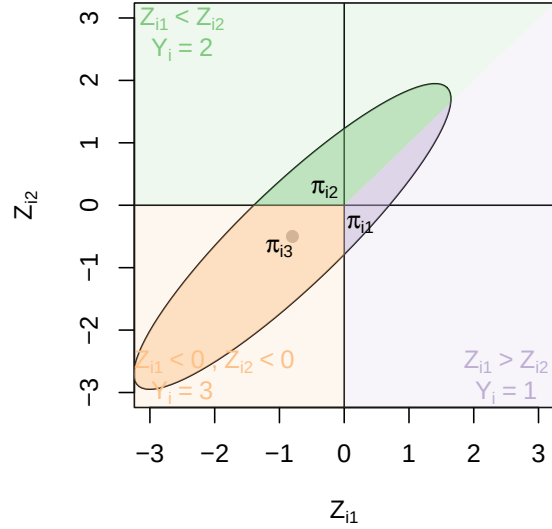
Consider now the mean,  $\mathbf{X}_i\beta$ . The model allows for each observation/individual to have its own mean. By changing the individual means, we can increase or decrease the individual probabilities for each category. The plots below show how the probabilities change for different means, but a constant covariance matrix.



Positively Correlated: shifted mean



Positively Correlated: shifted mean



Increasing the mean of  $Z_{i1}$  increases the probability of category 1, while increasing the mean of  $Z_{i2}$  increases the probability of category 2. Decreasing either variable's mean increases the probability of category 3.

### Model Fitting

To fit the model, we will write the latent variables of all observations as a single vector:

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Z}_1 \\ \mathbf{Z}_2 \\ \vdots \\ \mathbf{Z}_n \end{bmatrix}.$$

Then,  $\mathbf{Z} \sim \mathcal{N}(\mathbf{X}\beta, \Sigma)$ , where

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_n \end{bmatrix}$$

and

$$\Sigma = (I \otimes \Omega).$$

We assume a diffuse prior distribution on  $\beta$ , and  $\Omega \sim \text{InvWishart}(\nu, \mathbf{S})$ .

The full conditional distributions are then:

$$\begin{aligned}\beta|\star &\sim \mathcal{N}((\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{Z}, (\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}) \\ \mathbf{Z}_i|\star &\sim \mathcal{N}(\mathbf{X}_i\beta, \Omega) \text{ such that } \begin{cases} Z_{ij} > \max\{\max_{k \neq j}\{Z_{ik}\}, 0\} & \text{if } Y_i = j \\ Z_{ij} < \max\{\max_{k \neq j}\{Z_{ik}\}, 0\} & \text{if } Y_i \neq j \end{cases} \\ \Omega|\star &\sim \text{InvWishart}(n + \nu, \mathbf{S} + \sum_{i=1}^n (\mathbf{Z}_i - \mathbf{X}_i\beta)(\mathbf{Z}_i - \mathbf{X}_i\beta)') \text{ (with some identifiability constraints)}\end{aligned}$$

### *Detergent data*

1. Response: laundry detergent choice (All, Era, Solo, Surf, Tide, Wisk) by each household
2. 2,657 households
3. Covariates: for each household, the log price of each detergent

### *Table of Choices*

Warning: package 'MNP' was built under R version 4.6.1

MNP: Fitting the Multinomial Probit Model

Version: 3.1-6

Authors: Kosuke Imai [aut, cre],

David van Dyk [aut],

Hubert Jin [ctb]

| All | EraPlus | Solo | Surf | Tide | Wisk |
|-----|---------|------|------|------|------|
| 87  | 507     | 253  | 406  | 701  | 703  |

### *Fit Model Using 'MNP' Package*

```
library(MNP)
data(detergent)
mod2 <- mnp(choice~1, choiceX = list(Surf=SurfPrice, Tide=TidePrice,
    Wisk=WiskPrice, EraPlus=EraPlusPrice, Solo=SoloPrice,
    All=AllPrice), cXnames = "price", data = detergent, n.draws
    = 5000, burnin = 1000, thin = 3, verbose = TRUE)
```

The base category is `All'.

The total number of alternatives is 6.

The trace restriction is used instead of the diagonal restriction.

The choice-specific variables of the base category are subtracted from the corresponding base categories.

The dimension of beta is 6.

The number of observations is 2657.

Improper prior will be used for beta.

Starting Gibbs sampler...

```
10 percent done.
20 percent done.
30 percent done.
40 percent done.
50 percent done.
60 percent done.
70 percent done.
80 percent done.
90 percent done.
100 percent done.
```

```
summary(mod2)
```

Call:

```
mnp(formula = choice ~ 1, data = detergent, choiceX = list(Surf = SurfPrice,
  Tide = TidePrice, Wisk = WiskPrice, EraPlus = EraPlusPrice,
  Solo = SoloPrice, All = AllPrice), cXnames = "price", n.draws = 5000,
  burnin = 1000, thin = 3, verbose = TRUE)
```

Coefficients:

|                     | mean      | std.dev. | 2.5%      | 97.5%   |
|---------------------|-----------|----------|-----------|---------|
| (Intercept):EraPlus | 1.92113   | 0.08589  | 1.74937   | 2.087   |
| (Intercept):Solo    | 1.32717   | 0.12117  | 1.05552   | 1.540   |
| (Intercept):Surf    | 1.20287   | 0.07107  | 1.05389   | 1.328   |
| (Intercept):Tide    | 2.03329   | 0.08413  | 1.86568   | 2.184   |
| (Intercept):Wisk    | 1.24196   | 0.06110  | 1.11728   | 1.350   |
| price               | -59.47793 | 4.49076  | -69.11931 | -51.715 |

Covariances:

|                 | mean    | std.dev. | 2.5%     | 97.5% |
|-----------------|---------|----------|----------|-------|
| EraPlus:EraPlus | 0.61970 | 0.12009  | 0.41621  | 0.870 |
| EraPlus:Solo    | 0.49722 | 0.19945  | 0.13810  | 0.841 |
| EraPlus:Surf    | 0.15531 | 0.10117  | -0.06725 | 0.331 |
| EraPlus:Tide    | 0.19429 | 0.07324  | 0.06792  | 0.346 |
| EraPlus:Wisk    | 0.53219 | 0.09119  | 0.36377  | 0.709 |
| Solo:Solo       | 1.36411 | 0.32269  | 0.80817  | 1.917 |
| Solo:Surf       | 0.32947 | 0.26510  | -0.26533 | 0.700 |
| Solo:Tide       | 0.30015 | 0.17980  | -0.08975 | 0.587 |
| Solo:Wisk       | 0.80731 | 0.23661  | 0.37647  | 1.146 |
| Surf:Surf       | 0.93302 | 0.18244  | 0.62854  | 1.347 |
| Surf:Tide       | 0.45072 | 0.14377  | 0.22430  | 0.743 |
| Surf:Wisk       | 0.58329 | 0.12422  | 0.28524  | 0.802 |
| Tide:Tide       | 0.70415 | 0.16165  | 0.45024  | 1.027 |
| Tide:Wisk       | 0.57885 | 0.09468  | 0.39165  | 0.749 |

Wisk:Wisk            1.37901   0.13227   1.10697 1.647

Base category: All

Number of alternatives: 6

Number of observations: 2657

Number of estimated parameters: 20

Number of stored MCMC draws: 1000

When we interpret the coefficients, it is always with respect to the base category. So the ‘z-score’ for EraPlus is 1.9 higher than the ‘z-score’ for All  
⇒ On average, customers are more likely to buy EraPlus than All. When a detergent’s log price is higher than All’s log price by one, the z-score of any other detergent relative to All decreases by 59.

Interpreting the covariance values is even more difficult (impossible?). Can refer to our simple example for guidance, but may not help. Instead can just use it to account for the fact that alternatives/choices are relevant.

*Model with Different Coefficient of Detergent Prices for Each Category*

```
mod <- mnp(choice~TidePrice + WiskPrice + EraPlusPrice +  
  SurfPrice + SoloPrice + AllPrice, data = detergent, n.draws  
    = 5000, burnin = 1000, thin = 3, verbose = TRUE)
```

The base category is ‘All’.

The total number of alternatives is 6.

The trace restriction is used instead of the diagonal restriction.

The dimension of beta is 35.

The number of observations is 2657.

Improper prior will be used for beta.

Starting Gibbs sampler...

10 percent done.  
20 percent done.  
30 percent done.  
40 percent done.  
50 percent done.  
60 percent done.  
70 percent done.  
80 percent done.  
90 percent done.  
100 percent done.

```
summary(mod)
```

Call:

```
mnp(formula = choice ~ TidePrice + WiskPrice + EraPlusPrice +  
     SurfPrice + SoloPrice + AllPrice, data = detergent, n.draws = 5000,  
     burnin = 1000, thin = 3, verbose = TRUE)
```

Coefficients:

|                     | mean     | std.dev. | 2.5%     | 97.5%   |
|---------------------|----------|----------|----------|---------|
| (Intercept):EraPlus | -3.1051  | 0.5774   | -4.2703  | -2.051  |
| (Intercept):Solo    | -3.4589  | 1.0910   | -5.5920  | -1.289  |
| (Intercept):Surf    | -2.5731  | 0.7013   | -4.0997  | -1.422  |
| (Intercept):Tide    | 0.1752   | 0.6127   | -0.9792  | 1.476   |
| (Intercept):Wisk    | -3.2167  | 0.6657   | -4.8184  | -2.085  |
| TidePrice:EraPlus   | 15.3164  | 7.8436   | -1.2357  | 28.870  |
| TidePrice:Solo      | 32.6297  | 12.5424  | 7.1879   | 56.488  |
| TidePrice:Surf      | 14.0431  | 10.2580  | -5.6579  | 34.033  |
| TidePrice:Tide      | -46.4227 | 13.8217  | -75.0909 | -24.386 |
| TidePrice:Wisk      | 20.7959  | 5.9166   | 10.0905  | 33.775  |
| WiskPrice:EraPlus   | 27.3690  | 8.9142   | 10.5063  | 46.832  |
| WiskPrice:Solo      | 36.8980  | 10.2373  | 17.9742  | 56.775  |

|                      |           |         |           |         |
|----------------------|-----------|---------|-----------|---------|
| WiskPrice:Surf       | 41.8429   | 8.1517  | 25.8646   | 58.320  |
| WiskPrice:Tide       | 24.5624   | 7.9903  | 11.2881   | 40.874  |
| WiskPrice:Wisk       | -15.1791  | 4.9423  | -25.6273  | -6.787  |
| EraPlusPrice:EraPlus | -55.1446  | 10.1863 | -74.7523  | -36.779 |
| EraPlusPrice:Solo    | 0.2963    | 12.9484 | -24.1603  | 26.000  |
| EraPlusPrice:Surf    | 13.6555   | 8.1399  | -2.3248   | 29.184  |
| EraPlusPrice:Tide    | 8.3062    | 6.4849  | -4.4333   | 20.103  |
| EraPlusPrice:Wisk    | -2.9509   | 5.7000  | -14.1398  | 8.141   |
| SurfPrice:EraPlus    | -6.0639   | 5.0013  | -17.6338  | 2.247   |
| SurfPrice:Solo       | 5.8946    | 8.4255  | -10.3340  | 21.350  |
| SurfPrice:Surf       | -60.5954  | 8.7913  | -80.9337  | -47.187 |
| SurfPrice:Tide       | 3.9138    | 4.0113  | -3.5914   | 11.911  |
| SurfPrice:Wisk       | -3.3427   | 3.4168  | -9.7491   | 3.006   |
| SoloPrice:EraPlus    | 11.5550   | 5.2987  | 0.9219    | 20.464  |
| SoloPrice:Solo       | -105.1798 | 14.2092 | -134.4753 | -80.595 |
| SoloPrice:Surf       | 10.0994   | 5.7667  | -1.2870   | 22.167  |
| SoloPrice:Tide       | -9.0578   | 4.9369  | -19.0982  | -0.508  |
| SoloPrice:Wisk       | 10.8636   | 4.8378  | 2.2135    | 19.957  |
| AllPrice:EraPlus     | 102.7043  | 16.8457 | 69.6640   | 133.169 |
| AllPrice:Solo        | 101.1087  | 21.2047 | 58.4334   | 142.208 |
| AllPrice:Surf        | 29.8862   | 13.8442 | 3.7511    | 57.708  |
| AllPrice:Tide        | 38.5919   | 12.6500 | 15.6448   | 61.681  |
| AllPrice:Wisk        | 69.6712   | 14.1769 | 37.3256   | 94.546  |

Covariances:

|                 | mean      | std.dev. | 2.5%      | 97.5%  |
|-----------------|-----------|----------|-----------|--------|
| EraPlus:EraPlus | 0.449055  | 0.163089 | 0.167780  | 0.759  |
| EraPlus:Solo    | 0.213614  | 0.305303 | -0.350014 | 0.769  |
| EraPlus:Surf    | 0.116516  | 0.108400 | -0.075953 | 0.349  |
| EraPlus:Tide    | 0.047195  | 0.100251 | -0.108009 | 0.266  |
| EraPlus:Wisk    | 0.064863  | 0.065202 | -0.104559 | 0.181  |
| Solo:Solo       | 3.204199  | 0.328352 | 2.524358  | 3.769  |
| Solo:Surf       | -1.162513 | 0.347378 | -1.778884 | -0.305 |
| Solo:Tide       | -0.089550 | 0.182468 | -0.391260 | 0.341  |
| Solo:Wisk       | 0.021581  | 0.195514 | -0.327651 | 0.356  |
| Surf:Surf       | 0.818078  | 0.200073 | 0.523196  | 1.337  |
| Surf:Tide       | 0.090642  | 0.101804 | -0.146143 | 0.248  |

|           |           |          |           |       |
|-----------|-----------|----------|-----------|-------|
| Surf:Wisk | 0.006355  | 0.093711 | -0.176881 | 0.164 |
| Tide:Tide | 0.336978  | 0.156813 | 0.113418  | 0.707 |
| Tide:Wisk | -0.004902 | 0.043008 | -0.100572 | 0.075 |
| Wisk:Wisk | 0.191690  | 0.072845 | 0.067009  | 0.354 |

Base category: All

Number of alternatives: 6

Number of observations: 2657

Number of estimated parameters: 49

Number of stored MCMC draws: 1000



## 9.4 Homework Assignment

1. Fit a Bayesian latent-variable probit model to the data in Section 4.3 Problem 3.
  - a. Use a prior variance for  $\beta$  of 1. Now use a prior variance of 100,000. How do the posterior distributions for the coefficients change?
  - b. Interpret the coefficient for LI. Is there evidence that LI is related to the probability of remission?
  - c. Find a 95% credible interval for the probability of remission when LI is 22.
2. Fit a Bayesian latent-variable *multinomial* (not ordinal) probit model to the data in Section 7.4 Problem 4. You can use the ‘MNP’ package for this problem and use the proportional odds assumption.
  - a. Write out the full model.
  - b. Interpret the intercepts for this model.
  - c. How does the model estimate that income is related to job satisfaction?
  - d. What do the covariance estimates tell you about the relationships between the different response levels?

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